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Introduction

From sound law to derivation

Historical linguists ordinarily test an etymology by carrying a reconstructed form through the sound changes that separate it from its alleged reflex. Most such derivations remain implicit. A scholar knows that Proto-Germanic **p* yields Old English *f*, that West Germanic **z* became *r* under the appropriate conditions, and that one change must have preceded another because the reverse order produces the wrong form. For a single word this mental arithmetic presents little difficulty. Across hundreds of words and scores of interacting changes it becomes treacherous. Each step may look familiar while the derivation as a whole is false.

In computer-assisted phonological reconstruction (CAPR), I represent each sound law as a finite-state transducer and compose the transducers in the order argued for below. The resulting cascade applies every change without discretion: the same rule, with the same environment, reaches every eligible form. A proposed reconstruction either yields the Old English comparator or it does not. Moving a rule may repair one derivation and destroy another. Both results deserve attention, for a failure locates the point at which phonology ceases to suffice and philology must decide among analogy, borrowing, dialect mixture, uncertain attestation, or a faulty reconstruction.

No new principle is involved. The Neogrammarians already demanded exceptionless sound laws and an account of apparent exceptions (Osthoff and Brugmann 1878--1910). Their formula was the *Ausnahmslosigkeit der Lautgesetze*. Formalization merely enforces that demand mechanically: every eligible form bears the stated consequences. The computer cannot decide which reconstruction is historically defensible, but it prevents the investigator from forgetting what that reconstruction entails.

I apply this method to the development of Proto-Germanic and early West Germanic forms into Old English. Germanic makes a severe test. Its historical grammar rests on two centuries of philological labor, while Old English offers abundant but orthographically and dialectally varied testimony (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014; Fulk 2018). A computational account cannot plead scarcity of evidence. It must reproduce familiar developments, identify the evidence for their order, and explain why its input sometimes differs from the headword printed in an etymological dictionary.

The formal claim

A sound change defines a relation between strings. For each etymology I supply a reconstructed input to an ordered series of such relations and compare the result with an Old English form. Kaplan and Kay demonstrated that the familiar rewrite rules of phonology admit a finite-state interpretation (Kaplan and Kay 1994); Foma gives that interpretation executable form (Hulden 2009). A toy rule deleting final **z* may be written:

```
define ToyFinalZLoss [{*z} -> 0 || _ .#.];
```

The notation states a target, an output, and an environment. Actual rules also refer to segment classes, stress, syllable weight, word boundaries, and the output of earlier changes. A loose prose

formulation can conceal choices that code must resolve. Does the rule affect every vowel or only short vowels? Does a following consonant block it? Does it apply before or after the loss of a final syllable? A transducer forces each question into the open.

Order gives the cascade its historical content. An early change may create the environment for a later one and thus feed it; alternatively, it may remove that environment and bleed it. A word affected by both changes can therefore establish their relative chronology. Other changes commute across the available lexicon: reversing them alters no output in the active dataset. Such a result does not prove simultaneity or historical indifference. It shows only that the present evidence fails to order them. Throughout this book I distinguish chronology compelled by the tested forms from placement adopted on the authority of the historical grammars.

Backward reconstruction requires a further restriction. An unrestricted inverse transducer will propose formally possible strings that no Germanic language could have inherited. I therefore restrict backward reconstruction with an inventory and a statement of permissible ancestral forms. Reconstruction always combines correspondences with a theory of what could count as a word in the *Grundsprache*; this restriction states that theory rather than leaving it tacit.

Inputs, targets, and success

I compare a selected earlier Germanic form with a selected Old English target. Neither selection is innocent. Dictionaries cite lexemes, but sound change operates on word-forms. The ancestor of an Old English plural, preterite, or oblique case may differ from the reconstructed lemma in precisely the material on which a later sound law acts. Kroonen and Orel provide indispensable lexical reconstructions, while the grammars often supply the paradigm history needed to choose the actual input (Orel 2003; Kroonen 2013; Ringe and Taylor 2014).

I therefore distinguish the citation reconstruction from the transducer input. The former identifies the etymon; the latter represents the paradigm cell or remodeled stem whose history is at issue. Where the two differ, the lexical entry states the difference and argues for it. This prevents a convenient input from masquerading as a received reconstruction. An unmotivated alteration made solely to secure the desired output would empty the exercise of historical meaning.

The distinction is concrete in Old English *sċuldrum* ‘shoulders, dative plural’. Its inflectional history requires an input of the type **skúldramiz* ‘shoulders, reconstructed dative-plural input’; the singular headword represents a different paradigm cell.

The target also requires judgment. Old English spelling varies by date, dialect, manuscript, and editorial practice. The cascade may produce an internal phonological symbol that an orthographic transducer then maps to a normalized written form. A string match at this final stage cannot by itself establish an etymology; conversely, a superficial spelling mismatch need not disprove one. I treat phonological development and orthographic normalization as separate operations so that exact computation does not confuse notation with history.

Within these limits a successful derivation has three senses. It succeeds formally when the output string matches the target. It succeeds philologically when the chosen input and comparator are the proper forms to compare. It succeeds historically when the proposed path agrees with the wider Germanic evidence. The first kind of success is cheap. The argument of the book concerns the conjunction of all three.

The evidence of failure

Regular sound change makes irregularity legible. If an inherited form refuses to pass through an otherwise successful cascade, the mismatch demands a name. Analogy may have replaced the expected reflex with a form drawn from another paradigm cell. Borrowing may have introduced the word after the relevant changes. A dialectal form may lie outside the modeled West Saxon path. The target may be late, corrupt, or normalized beyond what the manuscript evidence warrants. Finally, either the reconstruction or the rule may be wrong.

These possibilities should not be suppressed by narrow, lexeme-specific "sound laws." A system that derives every target by multiplying exceptions has only encoded its answers. I instead distinguish several classes of non-regular result: attested variants, early and late analogy, reconstructed Old English comparators, known but unmodelled phonological and morphological developments, and unexplained exceptions. The categories are claims, not housekeeping labels. They identify where regular phonology ends and what additional history the evidence requires.

This treatment follows the original Burmish CAPR work, in which resistant forms often disclosed loans or mistaken cognate assignments. Old English shifts the balance toward morphology and analogy, but the methodological advantage remains the same. Failure concentrates inquiry. It tells us which assumption—input, target, environment, order, or lexical history—must bear the explanation.

Evidence for relative chronology

The chronology chapters combine three kinds of evidence. First come the statements of the standard historical grammars. These establish the received description and often the broad order of developments (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014; Fulk 2018). Second come individual witness words. A derivation that succeeds under one order and fails under the reverse order supplies direct lexical evidence for that relation. Third come exhaustive order tests across the active dataset. These reveal whether an apparently decisive relation is local, whether other words contradict it, and how far a rule can move without disturbing any output.

Suppose that a consonant change creates the environment for a later vowel change. Under the received order both apply and the Old English target emerges. Reverse them and the vowel change misses its environment. The word then supports the priority of the consonant change. The converse case is equally informative: if an early rule creates a segment that a later rule would wrongly alter, the creating rule must follow the other. When reversal changes no output, the lexical evidence leaves the order open even if philological considerations still favor one placement.

A note on notation: a superscript dagger placed immediately before an italicized form marks a concrete predicted output generated by a counterfactual manipulation of the rule order (for example, moving a change earlier or later than argued). This dagger indicates that the marked form is a model prediction, not an independently attested historical form; reconstructions remain starred in the usual way.

Sims-Williams argues for mechanizing precisely this kind of reasoning (Sims-Williams 2018). Computation does not replace the historical argument; it makes the extent of that argument measurable. A traditional chronology may prove correct but less tightly constrained than its customary presentation suggests. A relation described as local may in fact rest only on a broad terminus. Such negative results are salutary. They separate what the data demonstrate from what a convenient exposition merely presupposes.

Rules and words

The book accordingly moves twice through the same history. Part I begins with the ordered rules. For each development it states the historical problem, gives the rule in formal notation, and examines the evidence for placement. The code appears because it is part of the claim, but the code-name is never an explanation. `OEIUmlaut`, for example, is only a rule name; its linguistic content lies in the stated environment, the historical discussion, and the words whose derivations depend upon it.

Part II begins with the words. Each entry identifies the reconstruction, the selected input, the Old English comparator, and the derivational class. A trace shows which rules altered the form; the accompanying prose explains the philological decisions that the trace cannot make. Short entries record ordinary derivations. Longer ones treat doubtful reconstructions, variant attestations, paradigm-cell selection, or analogy. The asymmetry is intentional: equal database rows do not warrant equal historical discussion.

The reader can thus move in either direction. A chronology chapter names the lexical witnesses that constrain a rule; their entries display the complete derivations. A lexical entry invokes a change; the corresponding chapter explains its formulation and place in the cascade. The index verborum provides a third route through the material, gathering reconstructed and attested forms by language.

Reproducibility and disagreement

An executable derivation identifies the exact point of disagreement. One reader may accept a sound change but reject its environment; another may accept the rule and dispute its order; a third may object to the selected paradigm cell or to the normalization of the Old English target. Each objection addresses a recorded decision. Given the same inputs, rules, and order, the stated outputs follow. Reproducibility here concerns those consequences, not the surrender of philological judgment to an algorithm.

Prose can sound settled while concealing several incompatible derivations. Code is less accommodating. A rule that is too broad damages words it was never designed to explain; one that is too narrow leaves legitimate reflexes untouched. A reconstruction plausible in isolation may fail after three later changes. The resulting embarrassment is a virtue of the method. Applying every rule to the whole lexicon subjects local analyses to a global test.

The converse danger is false precision. A deterministic cascade may tempt the reader to mistake exact strings for exact history. Every result still depends on choices about segmentation, symbol inventory, reconstruction, morphology, dialect, chronology, and orthography. I therefore cite the philological sources, display the selected forms, preserve unresolved exceptions, and distinguish tested order from source-based order. Those choices remain accountable to philology.

The argument

A book must advance an argument. Mine is that traditional rule-based reconstruction becomes clearer when every proposed derivation can be executed, every rule must face the whole lexicon, and every failure is reported. The Germanic-to-Old-English case shows both the power and the boundary of that claim. Much of the history admits a coherent ordered account. The residue does not disappear: it resolves into morphology, analogy, variation, borrowing, imperfect attestation, and a small number of genuine problems.

The sound-change chapters ask how tightly the lexical evidence constrains the cascade. The lexical chapters ask whether individual etymologies survive that cascade without philological sleight of hand. Neither inquiry suffices alone. A formal system that reaches the targets by hiding doubtful inputs is bad historical linguistics; elegant prose that never reproduces its derivations is an incomplete formal account. Executable derivations make those two standards answer to one another.

Part I

Sound changes, formalization, and relative chronology

Sound-change overview

Scope and orientation

The sequence follows the executable reconstruction from its early Germanic developments through the Old English written surface.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

Numbering note

SC numbers remain stable legacy identifiers, not chronological positions. Each published entry has one authoritative source. The book automatically follows the individual stages of the canonical executable reconstruction; neither filenames nor chapter membership can override that order.

This is the order adopted by the executable reconstruction, not a claim that every pairwise historical chronology has been independently proved. Adjudication records distinguish demonstrated chronology, stage-entailed relations and computational serialization. Chapter boundaries and titles remain editorial; they divide the sequence without rearranging its entries.

Explicitly retained technical explanations are published at their executable positions without becoming adjudicated historical sound laws.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC113 annotates stress-foot context and is not a published law. SC077 is unused.

Chapter 1

From Proto-Germanic to Proto-Northwest Germanic

1.1 Historical interval

This chapter covers developments that took place within the Proto-Germanic period, from the inherited consonant system to the first changes that separate the Northwest Germanic line from the rest of the Germanic family.

The reconstruction labelled Proto-Germanic here is the common ancestor of Gothic, North Germanic, and West Germanic, reconstructed through the classical comparative method from attested descendant languages. The label Proto-Northwest Germanic designates the hypothetical node linking the ancestors of North Germanic (Old Norse and its relatives) and West Germanic (Old English, Old High German, Old Saxon, and Old Frisian among others), to the exclusion of Gothic and the other East Germanic varieties.

1.2 What this chapter contains

Chapter 1 contains one reader-facing sound-change section: the Proto-Germanic loss of a nasal before **x* (`SC103 PGmcNasalLossBeforeX`), with compensatory lengthening and nasalization of the preceding vowel. Its results are shared by every daughter language, Gothic included, so the change belongs to Proto-Germanic itself and precedes every Northwest Germanic and West Germanic development treated in the chapters that follow (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7).

Another historically Proto-Germanic rule, the positional allophony of Proto-Germanic **b* (`SC049 PGmcBAllophony`), executes late in the computational cascade because the alternation interacts with environments shaped by intermediate rule applications; its section therefore appears in Chapter 4, where the divergence between cascade placement and historical stage is noted explicitly.

One other historically Proto-Germanic change, Gm-simplification (`SC002 PGmcGmSimplification`), is documented in the book-entry plan and its literature dossier confirms the source base is narrow (two lexical families: **draugma-* ‘dream’ and **taugma-* ‘team’; (Kroonen 2013, 101, 511)). A reader-facing section for SC002 awaits a stronger explanatory source base and is not yet assembled in the reader-facing sequence.

1.3 Scope and genealogical context

Changes in this chapter are pan-Germanic in scope: they apply to ancestral forms that feed both the North Germanic and West Germanic descendants, or they represent internal Proto-Germanic

processes visible across the Germanic family.

The boundary between Proto-Germanic and Proto-Northwest Germanic is not sharp in the textbook literature. Ringe and Taylor treat many of the traditionally "Proto-Germanic" changes as part of a shared innovation package that is diagnostically older than North–West Germanic divergence but not necessarily earlier than the separation of the East Germanic line (Ringe and Taylor 2014, 1–30). For book purposes, the distinction matters primarily because it separates the features inherited uniformly from all Germanic from those shared selectively by North and West Germanic to the exclusion of Gothic.

1.4 A note on the rule names

The CAPR rules implemented in this chapter carry names beginning with `PGmc`. Those names are intended as stable internal identifiers, not as claims about the precise historical stage of every rule so labelled. A rule named `PGmcX` may in some cases be a later development that affects only the West Germanic or Northwest Germanic branch. Each entry's source-supported historical analysis takes priority over both its identifier and its chapter placement; the latter follows executable order, not an independent historical-stage map.

1.5 Proto-Germanic loss of a nasal before **x*

Historical discussion

The oldest change treated in this book is common to the whole family. In the group **-nx-* the nasal consonant was lost, the preceding vowel was lengthened in compensation, and that lengthened vowel was nasalized. Its results are shared by every daughter language: Gothic *þeihan* 'thrive', *brāhta* 'brought', *þūhta* 'seemed' stand beside Old High German *dīhan* 'thrive', *brāhta, fūht* 'damp' and Old English *þēon* 'thrive', *brōhte* 'brought', *þūhte* 'seemed', *fūht* 'damp' (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7). Because no daughter keeps the nasal, the change belongs to Proto-Germanic itself, and it precedes every Northwest Germanic, West Germanic, North Sea Germanic and Anglo-Frisian development described in the chapters that follow.

Only **a*, **i* and **u* occur in this position. Germanic had already raised **e* to **i* and **o* to **u* before a nasal followed by a consonant, so the mid vowels are absent from the input (Fulk 2018, 55, §4.1).

The vowel that the change creates is long and nasalized, and it is not yet rounded. Fulk emphasizes that the lengthened vowels remained nasalized for a considerable time, well past the close of the Northwest Germanic period, since the nasalized low vowel produced in this way went on to develop to *ō* in Anglo-Frisian and did not fall together with the Old English *ā* that came from **ai* (Fulk 2018, 55, §4.1). The comparative material makes the same point directly: Gothic, Old Norse, Old High German and Old Saxon all reflect the Proto-Germanic nasalized low vowel as unrounded *ā*, and only Old English and Old Frisian show *ō* (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). Ringe describes Proto-Germanic **hanhaną* as **[xą̄xaną]* and observes that its low vowel was rounded, along with the other nasalized low vowels, in the northernmost West Germanic dialects (Ringe 2017, 149–50, §3.2.7). That rounding is a separate and much later change, and it is treated in the chapter on the rounding of the long nasalized low vowel.

SC103. Proto-Germanic nasal loss before **x* (`PGmcNasalLossBeforeX`)

```
define PGmcNasalLossBeforeX Ctx([
    [
    {*a} -> {*ã} || _ EnglishStarNasal {*x},
    {*i} -> {*ī} || _ EnglishStarNasal {*x},
    {*u} -> {*ū} || _ EnglishStarNasal {*x},
    {*á} -> {*ã} || _ EnglishStarNasal {*x},
    {*í} -> {*ī} || _ EnglishStarNasal {*x},
```

```
    {*ú} -> {*ū} || _ EnglishStarNasal {*x}
] .o. [
    EnglishStarNasal -> 0 || _ {*x}
]
]);
```

The rule performs the three parts of the change together: it lengthens the vowel, it marks the lengthened low vowel as nasalized, and it then removes the conditioning nasal. The low vowel is written **ą̄* and the high vowels are written **ī* and **ū*, because the nasality of the high vowels has no further consequence: they go on to develop exactly as the inherited long **ī* and **ū* do (Campbell 1959, 47, §121). The nasality of the low vowel is carried forward because its later fate depends on it. In the executable definitions, ã is the single-symbol encoding of long nasalized **ą̄*, not a different or short vowel. The historical notation in the discussion retains its explicit length mark.

The corpus witnesses both branches of the rule. The high branch is **fúnxstiz* ‘fist’, which becomes **fúxsti* ‘fist’ and, after the loss of **x* before the cluster in SC028 PNWGmcPreconsonantalXLoss, gives Old English *fȳst* ‘fist’. The long vowel of Old High German *fūst* ‘fist’, Dutch *vuist* and German *Faust* shows that the word belongs here and to no later law (Kroonen 2013, 160). The rule also supplies the **xst* cluster on which the loss of **x* before a consonant operates, so the two stand in a feeding relation.

The low branch is **θánxtē* ‘thought’, the preterite of the verb ‘to think’, whose principal parts are reconstructed as **þankijaną* ‘to think’, **þanhtē* ‘thought’, **þanhtaz* ‘thought (past participle)’ with the nasal still standing before the fricative (Ringe 2017, 281; 2017, 136). Here the rule yields **θą̄xtē* ‘thought’, and the nasalized low vowel is later rounded by SC104 EAFNasalizedLowRounding to give Old English *þōhte* ‘thought’. This is the form the handbooks themselves cite for the Anglo-Frisian rounding of the vowel produced here (Fulk 2018, 55, §4.1). It does not enter the loss of **x* before a consonant, in which respect it differs from ‘fist’, so the fricative survives to the surface and the word displays the vowel history alone.

Chapter 2

From Proto-Northwest Germanic to Proto-West Germanic

2.1 Historical interval

This chapter covers the sound changes that took place in the proto-language shared by the West Germanic languages — Old English, Old Frisian, Old Saxon, Old High German, and Old Dutch — before the individual languages diverged. The starting reconstruction is Proto-Northwest Germanic (PNWGmc), the hypothetical common ancestor of North Germanic and West Germanic together; the ending reconstruction is Proto-West Germanic (PWGmc), the immediate common ancestor of the West Germanic languages specifically.

2.2 Scope and internal diversity

Changes in this chapter are not all equally pan-West-Germanic in scope. They may be grouped broadly as follows:

Northwest Germanic innovations (shared by both North and West Germanic): innovations in the unstressed vowel system, certain final-syllable vowel changes, and selected consonant cluster simplifications. Their historical scope must be established from comparative evidence, not from executable identifiers.

Proto-West-Germanic innovations (shared within West Germanic but not in North Germanic): the cluster of morphological and phonological changes that distinguish Old English, Old High German, Old Saxon, and Old Frisian from Old Norse. They include early apocope rules, certain consonant assimilations, and the West Germanic gemination of consonants before `*j`.

2.3 Major changes

The chapter opens with the root-noun nominative `*-z` loss (SC096), the generalization of endingless nominatives through the athematic consonant stems, complete before Proto-West Germanic: none of the West Germanic daughters shows any ending in this class (Ringe and Taylor 2014, 118). It is the earliest of the three historically distinct final-`*z` developments; the other two (SC020 and SC097) open Chapter 3.

The unstressed **ai* > **ē* development (SC014 PNWGmcUnstressedAiMonophthongization) is one of the widespread Northwest Germanic vowel shifts. Unstressed dative-singular and strong-adjective plural endings provide examples of monophthongization and merger with inherited unstressed **ē*. Ringe and Taylor distinguish this broad correspondence from the more questionable claim of a single completed Proto-Northwest Germanic event (Ringe and Taylor 2014, 25–26). The stressed monophthongization is treated separately under SC004 EAFAiMonophthongization.

The later SC015 PNWGmcILowering concerns stressed first-syllable **i*, not further unstressed-vowel reduction. Its following-vowel hypothesis has traditional authority, but its comparative regularity and precise consonantal conditioning remain disputed. The section compares vowel-conditioned and consonant-conditioned accounts and explains the provisional restrictions of the current implementation (Campbell 1959, 43, §§114–116; Fulk 2018, 56–57; Hill 2017, 149–58).

The West Germanic consonant changes of this chapter — j-gemination (SC010), early i-apocope (SC006), coronal-w assimilation (SC008), and related rules — represent the most productive phonological territory for the CAPR derivations. They feed a large proportion of the distinctive consonant clusters of Old English. Handbooks vary in exactly how they group and name these changes (Campbell 1959, §§ 404, 406; Hogg 1992a, 71, §4.11 (held 2011 reissue)).

Coronal assimilation also supplies the geminate in *fēower* 'four' and *ēow* 'you', before inherited short-vowel-plus-geminate-glide reanalysis. SC031 OEWWSimplification now represents that earlier development, not an unrestricted late deletion. The resulting diphthong-plus-glide sequence is inherited by later English realization; j-created and singleton paths remain distinct (Ringe and Taylor 2014, 41–42, 65–66; Campbell 1959, 45–47).

The quoted definitions use `Ctx(...)` to carry selected sentence context outside the segmental operation. It is computational transport, not a reconstructed segment or an additional sound law. Lexical accent, sentence stress and phonological-word finality are separate: the early high-vowel-loss account requires a heavy syllable in a sentence-unstressed, phonologically final word, not simply the absence of an acute. Ordinary evaluation selects strong-final citation context; the explicit weak-final selection for *ēow* is discussed with SC098 PWGmcUnstressedWordFinalIApocope (Ringe and Taylor 2014, 55, 57–58).

The nasal spirant corridor (SC026–SC027), treated in Chapter 3 at its cascade position, illustrates a type of change common in historical grammars of the "Ingvaeonic" or "North Sea Germanic" area: nasals disappear before voiceless fricatives, with compensatory vowel lengthening (Campbell 1959, §§ 462–463; Hogg 1992a, 56, §3.14 (held 2011 reissue)). The CAPR model splits this into two ordered steps to make the vowel effect computationally tractable; the book prose explains that split against the handbook tradition, which typically presents the change as a single process.

Chapters in this part of the book follow the executable reconstruction's adopted serialization, not an independently demonstrated total historical chronology. Several rules that carry `PWGmc` labels — final bare-`*a` loss (SC041), surviving bimoric `*ō` unrounding (SC042), and Sievers-law syncope (SC050) — execute later in the cascade and are therefore presented in Chapter 4, where their individual sections discuss their historical stage labels. Conversely, one rule with a West Saxon label, the palatal-glide rule (SC016), is an orthographic rule of the written surface: it executes after the Old English orthography stage and is presented in Chapter 5.

2.4 A note on source terminology and subgrouping

The literature uses several partly overlapping stage labels for this period:

- Northwest Germanic: the node uniting North and West Germanic.
- Proto-West Germanic: the node uniting only the West Germanic languages.
- North Sea Germanic or Ingvaeonic: a proposed subgroup within West Germanic covering Old English, Old Frisian, and Old Saxon (and sometimes Old Low Franconian), sharing certain innovations over a broader area.
- Anglo-Frisian: a narrower proposed subgroup linking only Old English and Old Frisian.

These labels are not always used consistently across sources. Ringe and Taylor are cautious about reconstructing a discrete Proto-West-Germanic node (Ringe and Taylor 2014, 50–55). Campbell notes that many of the "West Germanic" shared features could alternatively be treated as parallel developments rather than common inheritance (Campbell 1959, §§ 1–5).

CAPR uses `PWGmc` and `NWGmc` as organizing labels for this chapter without claiming to have settled all questions about West Germanic subgrouping. Changes that appear in the literature under "Ingvaeonic" labels but affect the Old English–to-Proto-Germanic derivation chain are treated

here as late expressions of the same West Germanic developmental period unless existing CAPR dossier research specifically argues for Anglo-Frisian or English-specific placement.

2.5 Rule names

Most CAPR rules in this chapter carry names beginning with `NWGmc` or `PWGmc`; the reformulated inherited-glide rule retains its older `OE` identifier. These names are stable internal identifiers. A name beginning with `NWGmc` does not guarantee that the change is exclusive to Northwest Germanic, and a name beginning with `PWGmc` does not guarantee that it is absent from North Germanic. The historical analysis in each sound-change section takes priority over the name prefix.

2.6 Early Northwest Germanic vowel developments

These two developments affect different parts of the vowel system. SC014 PNWGmcUnstressedAi-Monophthongization monophthongizes unstressed **ai*; SC015 PNWGmcILowering represents the disputed lowering of stressed short **i*. This thematic comparison does not make their separated rule sections adjacent or identify them as successive stages of unstressed-vowel reduction.

2.7 Root-noun nominative **-z* loss

Historical discussion

The athematic consonant stems — the "root nouns" of the handbooks — attached the nominative-singular marker directly to a consonant-final root, and the Proto-Germanic outcome of that collision is genuinely uncertain. Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states plainly that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306, §4.3.4). His own paradigm tables carry the uncertainty into print: the nominative of 'foot' appears as "fōts? (fōs?)", while 'mouse' is plain **mūs*, its expected extra sibilant already absorbed by degemination (Ringe 2017, 149, 313). Ringe and Taylor repeat the same three-way agnosticism — the root-noun nominative "either ended in **-s* or **-z*, or was endingless" (Ringe and Taylor 2014, 28, §2.3.1).

The dictionary traditions encode this situation in different notations, and the differences are conventions of citation rather than competing claims of fact. Orel prints morphologically explicit nominatives with final *-z* across the whole class: **bōkz* 'book', **gansz* 'goose', **lūsz* 'louse' (Orel 2003, 52, 126, 252). Kroonen cites the same words as bare stems or endingless forms, **bōk-* and **gans-* (Kroonen 2013, 71–72, 168–69), and Kluge/Seebold print a third variant with voiceless *-s*, as in **bōks* 'Buch' (Kluge 2011, 158). Bammesberger shows why the marker is nonetheless real: for voiced-final stems the overt nominative **-z* is positively reconstructible — **burgz* (Gothic *baúrgs*), **frijōndz* (Gothic *frijonds*) — even though **fōt-z* is "phonotaktisch kaum denkbar", and in West Germanic the ending simply "fiel es ab" (Bammesberger 1990, 190–92, §8.2.3.1). CAPR retains Orel's morphologically explicit forms as its inputs precisely because they record the inflectional marker whose fate this rule describes.

The three focal words are only superficially parallel. The root-final *s* of 'louse' is itself an extension of **luw-* on the model of 'mouse' (Bammesberger 1990, 195, §8.3). For 'goose', Szemerényi's lengthening would give a Proto-Indo-European nominative **ǵʰanss* > **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196, §8.3; Kroonen 2013, 168–69). 'Book' preserves a plain obstruent-final root. What unites them is not a single phonetic history but membership in a paradigm class that generalized the endingless nominative.

The branch evidence dates and localizes that generalization. Gothic keeps its sibilant throughout the class (*baúrgs*, *nahts*, *reiks*) (Fulk 2018, 165–66, §§7.26–7.27). Old Norse redistributes the ending morphologically: masculine root nouns keep *-r* (*fótr*), feminines are endingless (*nótt*, *geit*), while the vocalic-stem feminines *kýr*, *sýr*, *ær* retain *-r* from **-z* with R-umlaut (Fulk 2018, 167,

§7.28; Bammesberger 1990, 192–93). West Germanic alone is uniform: there was "no ending in PWGmc, as none of the daughters exhibits any" (Ringe and Taylor 2014, 118, §3.4). Fulk supplies the mechanism: Szemerényi's law removed the nominative sibilant after sonorant-final stems, and endinglessness then spread analogically through the class (Fulk 2018, 143, §7.2). The development is therefore best understood as a morphological generalization enacted differently in each branch — absolute in West Germanic, consonant- and gender-conditioned in North Germanic, absent in Gothic — rather than as one exceptionless sound law.

This change is distinct from the two later final-*z developments. It was complete before Proto-West Germanic, whereas the loss of *-z in unstressed syllables (SC020 EAFFinalZDeletion) is itself a Proto-West Germanic change: polysyllabic consonant-stem nominatives such as **fadurz* 'father' — and, in this corpus, **frijōndz* 'friend', **melukz* 'milk', and **mēnōþz* 'month' — kept their ending into Proto-West Germanic and lost it there in an unstressed syllable (Ringe and Taylor 2014, 44–45, §3.1.1). The still later northern loss of *-z in stressed monosyllables (SC097 MonosyllabicFinalZLoss) affects vowel-final monosyllables like **hwaz* and does not touch the consonant-final root nouns at all, whose ending was gone long before.

SC096. Root-noun nominative *-z loss (`RootNounNomZLoss`)

```
define RootNounNomZLoss Ctx([
    [{*z} -> 0 || .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
    ↪  EnglishStarVocalic+ [EnglishStarConsonant | EnglishPalatalConsonant]+ _
    ↪  .#.]
]);
```

The rule deletes word-final *z after a consonant in a monosyllable. Three claims of different kinds meet here and must be kept apart. The historical claim is morphological: the nominative-singular ending was lost in the athematic root-noun class, so that this one paradigm cell came to lack its marker — a development complete before Proto-West Germanic (Ringe and Taylor 2014, 118, §3.4). The lexical claim belongs to the dictionaries: Orel's citation forms, which supply the corpus inputs, print that marker explicitly as *-z* in **bōkz*, **flauxz*, **gansz*, and **lūsz* (Orel 2003, 52, 105, 126, 252). The executable statement is neither of these but a computational proxy for them: 'delete word-final *z after a consonant in a monosyllable'. It is not proposed as a Proto-Germanic sound law; it earns its place only because every form the corpus submits to the morphological development is a consonant-final monosyllable, so the narrow phonological statement covers the class exactly. Four corpus derivations witness the rule, each yielding its expected Old English outcome: PGmc **bōkz* 'book' yields OE *bōc* 'book', **gánsz* 'goose' yields *gōs* 'goose', **lūsz* 'louse' yields *lūs* 'louse', and **fláuxz* 'flea' yields *flēah* 'flea'. Should the corpus ever acquire a consonant-final monosyllable in *-z that is not a root-noun nominative, the proxy and the morphology would come apart, and the rule would need to be re-scoped; the project's regression tests pin the firing population to exactly these four words so that any fifth firing forces that adjudication rather than passing silently.

The rule applies at the head of the English line, before SC009 PWGmcIjContraction. That ordering is fixed by the identity of the process rather than by a wrong form: contraction turns the polysyllabic **fríjōndz* 'friend' into a monosyllable, and if the root-noun rule applied after contraction it would capture **friundz* — yet the ending of 'friend' survived into Proto-West Germanic and fell in an unstressed syllable, the change described under SC020 EAFFinalZDeletion (Ringe and Taylor 2014, 44–45, §3.1.1). Because both rules delete the same segment, moving this rule later changes no Old English output; the early placement keeps the derivation of 'friend' aligned with the historical account rather than with an accident of the cascade.

Negative controls behave as the morphology predicts. Stressed monosyllables whose *z follows a vowel — the domain of the later northern change — do not meet the post-consonantal environment. Medial *z is untouched and remains available for rhotacism (SC003 EAFRhotacism): PGmc **déuzą* 'deer' still yields OE *dēor* 'deer' with its rhotacized medial consonant.

2.8 Unstressed ai monophthongization

Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the widespread monophthongization of unstressed **ai* to a long mid vowel, largely merging with inherited unstressed **ē*. Their examples include final dative-singular and nonfinal inflectional syllables. They nevertheless question a single completed Proto-Northwest Germanic merger: Runic evidence suggests that some unstressed diphthongs survived later than others (Ringe and Taylor 2014, 25–26). The stressed development of **ái* to **ā* is treated separately as SC004 EAFAiMonophthongization.

SC014. Monophthongization of unstressed **ai* (PNWGmcUnstressedAiMonophthongization)

```
define PNWGmcUnstressedAiMonophthongization Ctx([
    [
    {*ai} -> {*ē}
]
]);
```

The dative-singular endings span **spánnai* ‘span’ and meed **mízdai* ‘meed’ carry this change; both give a final **ē*. If SC014 PNWGmcUnstressedAiMonophthongization is delayed until after SC072 OEUnstressedLongVowelShortening, the **ē* is no longer present for shortening, so PGmc **spánnai* ‘span’ yields †*spannē* rather than expected OE *spanne* ‘span’. This shows that SC014 PNWGmcUnstressedAiMonophthongization must come before SC072 OEUnstressedLongVowelShortening in the modeled sequence.

2.9 Unstressed **a*-raising before final **m*

Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

SC005. Unstressed **a*-raising before final **m* (PNWGmcAToUBeforeM)

```
define PNWGmcAToUBeforeM Ctx([
    [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
    ↪  ({*z})? .#.
]
]);
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 PNWGmcULowering, PGmc **skúldramiz* ‘shoulders’ yields †*sċoldrum* rather than expected OE *sċuldrum* ‘shoulders’; earlier placements converge on the expected output. The scope of the change is established by inflectional evidence across multiple paradigm types: a-stem dative plural ON *dǫgum* ‘days’, OE *dagum* ‘days’, OS *dagun* ‘days’, OHG *tagum* ‘days’, beside Gothic *dagam* ‘days’; strong-adjective dative singular ON *góðum* ‘good’, OE *gōdum* ‘good’, OS *gōdum* ‘good’, beside Gothic *godamma* ‘good’ (OS also shows variant forms gōdumu and -un); and first-plural present ON *berum* ‘we carry’, OHG *berumēs* ‘we carry’, beside Gothic *baíram* ‘we carry’. Across these sets, North/West Germanic shows unstressed *-um* where Gothic preserves

-am. The derivation of *sċuldrum* ‘shoulders’ supplies a CAPR ordering witness for the relative chronology, but the cognate set for ‘shoulder’ does not contribute comparative evidence for the rule’s historical scope.

2.10 Northwest Germanic lowering of long *ē*

Historical discussion

Proto-Germanic **ē₁* (the long mid vowel of PIE origin, as against the later **ē₂*) split East Germanic from the rest of the family. Gothic keeps a mid vowel, written ⟨e⟩, in *gadēþs* ‘deed’, *slēpan* ‘to sleep’, *mēna* ‘moon’, and *jēr* ‘year’, while Norse and all of West Germanic show a low vowel, as in Old Norse *ráða* ‘to consider, to advise’, *láta* ‘to let go, to allow’, *ár* ‘year’, *nál* ‘needle’, Old High German *tāt* ‘deed’, *slāfan* ‘to sleep’, *jār* ‘year’, *māno* ‘moon’, Old Saxon *dād* ‘deed’, *slāpan* ‘to sleep’, *jār*. Ringe and Taylor assemble the comparative set and reconstruct a Northwest Germanic sound change **ē₁ > *ā* (Ringe and Taylor 2014, 11–13). The change is directly dated by the earliest epigraphy. The Early Runic accusative *mākija* ‘sword’ (beside Gothic *mēkeis* ‘sword’) already shows ⟨a⟩ in the second half of the second century AD (Ringe and Taylor 2014, 12), and the a-rune likewise writes the stressed reflex in the Opedal stone’s *swestar* ‘sister’ (Stiles 2017, 4). Loanwords borrowed into Sami from early Proto-Scandinavian point the same way. Lid showed that the root vowel of Sami *mānno* ‘moon’, Proto-Germanic **ē* as in Gothic *mēna*, “had gone over to *ā* before the word was borrowed into Sami”, and adduced the parallel Sami *saððo* ‘bran’ beside Proto-Nordic **sāðō* from **sēðō-*, Old Norse *sáð* ‘bran’ (Lid 1952, 238). This is Sami borrowing from Proto-Scandinavian, so it dates and localizes the *ā* stage in the North; its application to West Germanic rests on the daughter correspondences above.

The change affected stressed syllables only. Ringe and Taylor restrict it explicitly: unstressed **ē* did not lower but was eventually shortened, as in **fadēr > *fader* ‘father’ (Ringe and Taylor 2014, 13, and p. 147); in the runic material the unstressed final vowel of *faþiR* ‘father’ has already merged with **-i* (Stiles 2017, 4). Within stressed syllables, however, the lowering was unconditioned: it applied before nasals exactly as elsewhere, and Ringe and Taylor accordingly print the intermediates **mānō* ‘moon’, **mānōþ-* ‘month’, and **spānuz* ‘spoon’ (Ringe and Taylor 2014, 11). The later, dialectally narrower fates of this **ā* — fronting in the North Sea area when oral, rounding when nasalized — are separate sound changes treated in their own chapters.

The reconstruction of the intermediate value is disputed. The objection is an old one: Bennett observed that the received account obliges Gothic to move *ē* to *ǣ* and back to *ē*, and the non-West-Saxon dialects and Old Frisian to run through *ē > ǣ > ā > ǣ > ē*, “with no apparent agreement among the languages and no discernible phonological trend”, and that the Proto-Germanic *ǣ* is on that account preserved directly nowhere (Bennett 1950, 232–33, 235). In its place he proposed that the backing to *ā* began in the north of the Germanic homeland and spread southward, never reaching Gothic in the east or Anglian and Frisian in the west, so that West Saxon *ǣ* is a retention of the intermediate stage rather than a re-fronting (Bennett 1950, 234–35). Fulk argues in the same direction on modern evidence: the Northwest Germanic vowel was a low front **ǣ*, retained unchanged in Anglo-Frisian, with the backing to *ā* of Norse and inner West Germanic a separate areal development spreading northward from Upper German territory; on that reading the runic ⟨a⟩ spellings write a front [æ:] for which the futhark had no better grapheme (Fulk 2018, 60–61, §4.6). Campbell is deliberately noncommittal, finding the **ā* stage “tempting to assume, though not definitely demonstrable” (Campbell 1959, 50–51, §§128–129). Ringe and Taylor answer with the lengthened place-adverbs *þǣr* ‘there’ and *hwǣr* ‘where’, whose front vowels are most naturally the output of fronting applied to a back **ā* (Ringe and Taylor 2014, 13–14). The present model adopts the two-step reconstruction while recording that the alternative remains live.

SC024. Lowering of stressed long *ē* (`PNWGmcLongELowering`)

```
define PNWGmcLongELowering Ctx([
    [
    {*é} -> {*ā}
]
]);
```

The rule reads the stressed tier **é* only, in keeping with the stress restriction; unstressed **ē*, as in **fadēr* 'father', is left for the unstressed-shortening rules of the Old English stage and never lowers. No segmental environment is imposed: nasal forms such as **mḗnōþz* 'month' and **spḗnuz* 'spoon' pass through **mānōþ-* and **spānuz* on their way to *mōnaþ* 'month' and *spōn* 'spoon', exactly as reconstructed by Ringe and Taylor (Ringe and Taylor 2014, 11).

The rule stands near the head of the cascade, before the genuinely West Germanic innovations such as early *i*-apocope, **ij*-contraction, and *j*-gemination — the placement follows from the dating: the second-century runic evidence puts the lowering among the earliest Northwest Germanic developments, well before the changes that separate West Germanic from Norse. Its output **ā* is consumed much later in the cascade by SC102 EAFHiatusWInsertion, SC025 EAFLongANasalRounding, and SC101 EAFLongAFronting: if the lowering is instead displaced after those rules, **mḗnōθz* 'month' yields †*mānaþ* rather than OE *mōnaþ* 'month', **skḗpą* 'sheep' yields †*sċāp* rather than *sċēap* 'sheep', and **sḗaną* 'to sow' never acquires the hiatus-filling *w* of *sāwan* 'to sow'. Within the cascade no earlier boundary has been demonstrated; the second-century runic attestation supplies the absolute dating.

2.11 Early i-apocope

Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geoguþ* 'youth' (Brunner 1965, §§ 145–146; Ringe and Taylor 2014, 141). Campbell's *dugup* 'troop' and *geogup* 'youth' examples belong to the same pattern (Campbell 1959, § 332).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

SC006. Early i-apocope (`PWGmcEarlyIApocope`)

```
define PWGmcEarlyIApocope Ctx([
    [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
]
]);
```

The absence of umlaut in *geoguþ* 'youth' provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLongDiphthong, PGmc **skáwōθi* 'shows' yields †*sċēaweþ* rather than expected OE *sċēawaþ* 'shows'.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

2.12 Final *ō-lowering before *r

Historical discussion

Ringe and Taylor separate two points here: a broader shortening of vowels before word-final **r* in unstressed syllables (for which kinship **r*-stems such as PGmc **fadér* > PWGmc **fader* are a key diagnostic), and the specific **ō*-before-**r* development needed here (Ringe and Taylor 2014, 58–59). The direct lexical witnesses for **ō* in that environment are two independent etyma: PGmc **fedwōr* 'four' with WGmc reflexes OE *fēower* 'four', OFris *fiuwer* 'four', OS *fiuwar* 'four'; and PGmc **watōr* 'water' with OE *wæter* 'water'.

The rule is historically secure but narrow: final or pre-final **ō* before word-final **r*. The clearest evidence remains concentrated in the `four` and `water` material. No broader environment for **ō* is attested.

SC007. Lowering of final bimoric **ō* before **r* (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering Ctx([
    [
    {*ō} -> {*a} || _ {*r} .#.
]
]);
```

OE *wæter* 'water' reveals why lowering must precede SC043 EAFBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* 'water' yields †*water* rather than expected OE *wæter* 'water': brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. Comparative support for the **ō*-before-**r* rule comes from the two lexical witnesses **fedwōr* 'four' and **watōr* 'water' and their WGmc reflexes; kinship **r*-stems belong to the broader pre-**r* shortening context, not to a direct **ō* > **a* control. Within CAPR, *wæter* 'water' is the form that establishes ordering before brightening. No broader lowering of **ō* is attested.

2.13 Coronal-w assimilation

Historical discussion

Ringe and Taylor treat the assimilation of **dw* and **zw* to **ww* as a shared Proto-West-Germanic innovation supported by one example of each input cluster (Ringe and Taylor 2014, 56–57; Stiles 1985, 89–94). The **dw* example is the numeral 'four': PGmc **feðwor* (Gothic *fidwor*) → WGmc **fewwar* → OE *fēower*, Old Frisian *fiuwer*, Old Saxon *fiuwar*. The **zw* example is the second-person plural pronoun, where two oblique case forms show the change: acc./dat. PGmc **izwiz* (Gothic *izwis*) → OE *eow*, Old Frisian *iu*, Old Saxon *iu*, OHG *iu*; and gen. Ringe and Taylor's PGmc **izweraz* (Gothic *izwara*) → OE *eower*, OHG *iuwer* (Ringe and Taylor 2014, 56). Stiles discusses the same pronominal material using his own reconstruction conventions and explicitly treats Gothic *izwara* among the relevant comparanda (Stiles 1985, 89–94). These two case forms belong to a single pronominal paradigm, not to two independent etyma.

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but the evidence for each cluster is confined: the numeral alone supplies the **dw* instance, and the oblique case forms of the second-person plural pronoun supply the **zw* instance. Both clusters are now witnessed in the corpus: 'four' for **dw*, and 'you' — selected in its dat.(-acc.) plural cell **izwiz* — for **zw*, deriving through **iwwi*, apocopated **iww*, to OE *ēow* 'you' (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, § 8.3, pp. 204–205).

SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation Ctx([
    [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
]
]);
```

OE *fēower* ‘four’ exposes a feeding relation: coronal assimilation creates the **ww* input of the early reanalysis in SC031 OEWWSimplification, yielding the diphthongal intermediate before later English realization. Ringe and Taylor explicitly give the coronal-assimilation and glide-reanalysis sequence (Ringe and Taylor 2014, 41–42). The earlier displacement result belonged to the superseded late **ww*-deletion proxy and does not by itself predict the result of displacing the reformulated early operation.

The numeral fixes that relative order. The pronoun now fixes a second one: assimilation must precede rhotacism (SC003 EAFRhotacism). The **z* of **izwiz* stands between vowel and **w*; had rhotacism applied first, it would have produced †*irwiz*, from which OE *ēow* ‘you’ can never be derived. The executable cascade composes the assimilation well before rhotacism, and the corpus derivation of *ēow* ‘you’ fails if the two are reversed. ‘Four’ remains the sole **dw* witness and the sole source of the coronal-assimilation → *ww*-simplification ordering constraint. The earlier boundary of the assimilation remains undetermined.

2.14 Inherited short-vowel-plus-geminate-glide reanalysis

Historical discussion

Inherited geminate glides after short vowels underwent reanalysis in Proto-West Germanic: the first glide became the offglide of a diphthong, while the second remained consonantal. This is not simply deletion of one *w*, and it is not the later English realization of the new diphthong. Ringe and Taylor distinguish those histories explicitly; Fulk specifies the short-vowel environment ((Ringe and Taylor 2014, 41–42, 65–66, 171–75; Fulk 2018, 117)).

Coronal assimilation feeds the change in *fēower* ‘four’ and *ēow* ‘you’: SC008 PWGmcCoronalWAssimilation supplies the geminate that becomes *eu*+*w* or *iu*+*w*. Inherited geminates also supply the histories of *ċēowan* ‘chew’, *dēaw* ‘dew’ and *hēawan* ‘hew’. The resulting diphthongs enter the separate English realization described under SC032 OEDiphthongLeveling ((Ringe and Taylor 2014, 41–42, 65–66, 172–75; Campbell 1959, 45–47)).

Homorganic *uww* requires a separate quantity decision. We adopt Ringe and Taylor’s preferred long-*u* account for *sċūwa* ‘shadow’, but retain their qualification of its evidence rather than treating the quantity as independently certain. Hypothetical inherited long-vowel-plus- *ww* sequences are not additional reconstructed witnesses: the held description specifies inherited geminates after short vowels. The parallel inherited *jj* development is independently governed and is not changed here ((Ringe and Taylor 2014, 65–66; Fulk 2018, 117)).

SC031. Inherited short-Vww reanalysis (`OEWWSimplification`)

```
define OEWWSimplification Ctx([
    [
    {*a} {*w} {*w} -> {*au} {*w},
    {*á} {*w} {*w} -> {*áu} {*w},
    {*e} {*w} {*w} -> {*eu} {*w},
    {*é} {*w} {*w} -> {*éu} {*w},
    {*i} {*w} {*w} -> {*iu} {*w},
    {*í} {*w} {*w} -> {*íu} {*w},
    {*u} {*w} {*w} -> {*ū} {*w},
```

```
    {*ú} {*w} {*w} -> {*ū} {*w}
]
]);
```

The transport wrapper carries an independently selected sentence context outside the segmental rewrite; it adds no reconstructed sound and does not condition the reanalysis. Acute and unmarked spellings are notation variants, not alternatives in sentence stress. The retained executable name likewise does not determine the historical stage.

The modeled position after coronal assimilation and before j-gemination is a representative serialization that keeps inherited/coronal-created geminates distinct from the subsequently created j-path. It is not a newly proved strict chronology between the whole events. *Hīeġ* 'hay' and reconstructed West Saxon strew retain the separately adjudicated SC029 OEAwwjResolution path. *Hīew* 'hue' has distinct j-created reanalysis at SC033 OEEwLongDiphthong and long-io realization at SC110 OEJGlideIO. The retained SC106 OEJWWSimplification compatibility operation is no longer fed by hue ((Campbell 1959, 45–47; Ringe and Taylor 2014, 173, 250)).

For you, early *iu*+*w* now reaches SC098 PWGmcUnstressedWordFinalIApocope. Its weak-final context, not the former presence of literal *ww*, licenses apocope before mutation. Earlier source-backed reanalysis thus coexists with genuine prosodic conditioning rather than an output-restoring segment proxy ((Ringe and Taylor 2014, 41–42, 55, 57–58)).

2.15 *ij*-contraction in *friend*

Historical discussion

Ringe and Taylor describe a change of **ijo* to **iu* in the ancestor of *friend*, with the pathway PGmc **frijōnd-* (Gothic *frijonds*) → PWGmc **friund* → OE *frēond*, Old Frisian *frīund*, Old Saxon *friund*, Old High German *friunt* (Ringe and Taylor 2014, 62). The same source immediately warns that the **ijo* sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62). Luick (printed p. 118) notes that *iu* generalised within several *j*-stem paradigms through a related but differently conditioned loss of *j*, but does not supply a second example of the exact stressed **ijo* sequence (Luick 1914--1940, 118).

The change concerns a rare sequence attested only in the **frijōnd-* etymon and cannot safely be generalized into a broadly productive rule.

SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

```
define PWGmcIjContraction Ctx([
    [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    # Stressed variant (stress preserved):
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
]
]);
```

Only the **frijōnd-* etymon tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* 'friend' yields †*friund* rather than expected OE *frēond* 'friend'; moving contraction earlier within the tested range changes no output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond the **frijōnd-* etymon, precisely the reservation made by Ringe and Taylor.

2.16 West Germanic j-gemination

Historical discussion

Fulk treats West Germanic consonant gemination before *j after a short vowel as a regular development and illustrates it with forms such as OE *settan* ‘set’ and *lecgan* ‘lay’ (Fulk 2018, 127, §6.15).

The change applies specifically after a short vowel before **j*, not to geminate consonants generally.

SC010. West Germanic j-gemination (`PWGmcJGemination`)

```
define PWGmcJGemination Ctx([
    [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j},
    # *w is an ordinary member of the same law. Campbell §407 p. 167:
    # "every consonant except r being affected after short syllables...
    # The forms in which w is doubled are dealt with in §120.2."
    # Ringe & Taylor p. 53: "it appears that that cluster too underwent
    # gemination in PWGmc." The geminate this creates is the input that
    # SC029 later resolves. Fulk §4.10 n. 1 dissents; see
    # audits/sc010-w-gemination-and-hay-depth-adjudication.md.
    {*w} -> {*w} {*w} || EnglishStarShortVowel _ {*j}
]
]);
```

OE *nett* ‘net’ fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátją* ‘net’ yields †*nete* rather than expected OE *nett* ‘net’. Earlier movement of gemination changes no output.

The chronology is phonologically transparent: the consonant must geminate before **j* ceases to be consonantal. The witness establishes no earlier boundary.

2.17 Syllabic j after final-vowel loss

Historical discussion

Ringe and Taylor state directly that after final unstressed *a and *ą were lost, postconsonantal *j became syllabic *i, with outcomes behind OE *here* ‘army’ and *rice* ‘kingdom’ (Ringe and Taylor 2014, 46).

The sources establish the development, although the lexical evidence supplies little independent support for its position. Its scope is postconsonantal j, not high-vowel vocalization generally.

SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

```
define PWGmcSyllabicJ Ctx([
    [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
]
]);
```

The same PGmc **nátją* ‘net’ witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGemination yields †*nete* rather than expected OE *nett* ‘net’; moving it later changes no output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* ‘army’ and *rice* ‘kingdom’ as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

2.18 Simplification of the cluster **xs* before a consonant

Historical discussion

The change treated here is narrow. When **x* stood before **s* and that **s* was itself followed by a further consonant, the **x* was lost and the cluster was reduced. Campbell states the rule in exactly these terms, that when a consonant follows, **xs* becomes *s*, and he illustrates it with *wæstm* ‘growth’ and *wæsma* ‘growth’ beside *weaxan* ‘to grow’, and with Northumbrian *sesta* ‘sixth’ beside West Saxon *siexta* (Campbell 1959, 170, §417). Brunner gives the same statement and adds that the following consonant may be *j* as well as a true obstruent, citing *nēosian* ‘to visit’, *þīsl* ‘pole’, *wæsma* and *wæstm* (Brunner 1965, 184, §221.2). Bülbring formulates it as a pre-English change and is careful to say that it held as a rule, with exceptions (Bülbring 1902, 215, §527).

The cluster that the change requires is a specific one, and two neighbouring clusters show that the restriction is real. Where **xs* stood with no consonant after it, the **x* was not lost at all. It survived long enough to cause breaking and then hardened to the sound written *x*, that is [ks], as in *fox* ‘fox’, *siex* ‘six’, *weaxan*, *oxa* ‘ox’ and *fleax* ‘flax’ (Campbell 1959, 170, §416). Where **x* stood before a single consonant it was likewise kept. Campbell observes that once **xs* had become [ks], the only group in which **x* still stood before a voiceless consonant in earliest Old English was **xt*, and that this group remained, as in *feohtan* ‘to fight’, *miht* ‘might’, *niht* ‘night’ and *sōhte* ‘sought’ (Campbell 1959, 186, §464). The contrast is attested outside English as well. Old High German and Old Saxon *lastar* ‘reproach’ comes from **laxstra-* with the **x* lost before **st*, while Old English *leahtor* comes from **laxtra-* with the **x* kept before **t* (Campbell 1959, 170, §417).

The change is not Proto-Germanic. Gothic keeps the **h* of this cluster in *bi-niuhsjan* ‘to spy out’ and *saihsta* ‘sixth’, so the loss must be later than the separation of Gothic (Ringe and Taylor 2014, 157–58). Campbell reports the loss from the whole West Germanic area and from North Germanic as well, citing Old Norse *ísl* ‘axle’ and *nýsa* ‘to investigate’, Old Saxon *weslon* ‘to exchange’, *wastum* ‘growth’ and *niustan*, and Old High German *niusen* ‘to try’ (Campbell 1959, 170, §417). Ringe and Taylor set out the same comparative material and reach a more guarded conclusion. They derive Proto-Germanic **niuhsijanǫ* through Proto-West Germanic **niusjan* to Old English *nēosan* ‘to seek out, to visit’, and **sehstō* to Northumbrian *sesta*, and they take the Old Saxon agreement in *wastum*, *thisla* and *niusian* to show a shared northern West Germanic change. Against that they weigh the competition between *þixl* ‘pole’ and *þīsl* ‘pole’ in early Mercian, the survival of **x* in *eaxl* ‘shoulder’ from **ahslu*, and the retention in Old High German *sehsto* and *dihsala*. Their conclusion is that the **h* was lost, possibly variably and possibly only in some dialects, when two or more consonants followed, and that the loss may have been in part a parallel development in the diverging Northwest Germanic dialects (Ringe and Taylor 2014, 157–58). The existence of the change is therefore secure while its exact date and extent are not, and the rating given here reflects that division.

Two chronological anchors are available. Ringe and Taylor place the loss after the Proto-West Germanic syncope of **-CijV-*, since it is that syncope which brings the **s* and the **j* of **niuhsjan* together (Ringe and Taylor 2014, 157). They place it before breaking, observing that the undiphthongized vowels of *wæstm* ‘growth’ and *þīsl* ‘pole’ can be accounted for only by supposing that these **h* were lost before breaking took place (Ringe and Taylor 2014, 158). The rule is stated between those two points.

One witness in this collection undergoes the change. Proto-Germanic **funxstiz* ‘fist’ reaches the rule as **fúxstiz*, the long vowel having been produced by the Proto-Germanic loss of a nasal before **x* described in SC103 PGmcNasalLossBeforeX. The cluster **xst* is then reduced to **st*, and the word continues to Old English *fȳst* ‘fist’. The comparative set for this particular word is West Germanic throughout, with Old Frisian *fēst* ‘fist’, Old Saxon and Old High German *fūst* ‘fist’, Dutch *vuist* and German *Faust* (Kroonen 2013, 160; Orel 2003, 157). Neither Gothic nor Old Norse preserves a reflex of it, so the word bears on the domain of the change rather than on its date.

The word *thought* is the control. Proto-Germanic **θánxtē* also passes through SC103 PGmcNasalLossBeforeX, which leaves **θāxtē*, but the **x* there stands before a single **t* and not before **s*, so the present rule does not touch it and Old English has *þōhte* ‘thought’ with its *h* intact. The pair *fist* and *thought* reproduces within this collection the same contrast that *lastar* and *leahtor* show across the West Germanic languages.

The relation between the two rules should not be overstated. The earlier rule does supply the cluster that this one simplifies, but the order of the two is established by their stages and not by any word in this collection. If the present rule were stated first, *fist* would still reach *fȳst* ‘fist’, because removing the **x* from **funxstiz* leaves a nasal before *s*, and that nasal is removed with compensatory lengthening by the North Sea Germanic nasal-spirant law. The outcome is overdetermined, and the chronology rests on Gothic *bi-niuhsjan* ‘to spy out’ and *saihsta* instead.

SC028. Simplification of **xs* before a consonant (PNWGmcPreconsonantalXLoss)

```
define PNWGmcPreconsonantalXLoss Ctx([
    [
    {*x} -> 0 || _ {*s} EnglishStarConsonant
]
]);
```

The **s* in the structural description carries the whole weight of the rule. Without it the rule would delete **x* before any two consonants, and it would then wrongly remove the first element of a geminate **xx* before **j*, where Old English in fact has *hliehhan* ‘to laugh’ with the geminate written *hh* (Campbell 1959, 186, §464). The rule fires on *fist* and on no other word in this collection, and it leaves *fox*, *six*, *wax*, *flax* and *ox* untouched, as it must, along with *thought*, *fight*, *night*, *light*, *might*, *knight*, *fright* and *wight*.

The implementation name retains an older description of the rule and will be brought into line in a separate pass.

2.19 *lþ*-voicing

Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan* ‘fold’, *beald* ‘bold’, *wuldor* ‘glory’, and *gylden* ‘golden’ (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold* ‘gold’, and *feld* ‘field’ (Campbell 1959, 169, §414).

The comparative evidence supports *lþ* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

SC012. Northern West Germanic *lþ*-voicing (`EAFLThVoicing`)

```
define EAFLThVoicing Ctx([
    [
    {*θ} -> {*d} || {*l} _
]
]);
```

The `field`, `fold`, `gold`, and `wold` families preserve **lþ* to **ld*, but none dates the change against a neighboring rule. Every output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lþ* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

2.20 Dental hardening

Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative `*ð` became stop `*d` in all positions (Ringe and Taylor 2014, 43).

The change is systemic across early West Germanic and extends beyond any one lexical family.

SC013. Dental hardening (`PWGmcDentalHardening`)

```
define PWGmcDentalHardening Ctx([
    [
    {*ð} -> {*d}
]
]);
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

2.21 Endingless *aw*: the regular straw branch

Historical discussion

The inherited nominative/accusative **stráwą* loses its final ending and vocalizes resulting final *w*, giving **strau*. Ordinary English diphthong development then produces reconstructed **strēa*. Attested *strēaw* ‘straw’ instead combines that long diphthong with *w* restored from inflected forms; the long nucleus also spreads into those forms. The regular free simplex and short obliques are explicitly unattested. The compound *strēaberige* ‘strawberry’ preserves the endingless branch, but is not substituted for an attested free noun (Campbell 1959, 46–47, 232–33; Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 172–73, 387).

The model keeps attested *strēaw* ‘straw’ as its comparison and produces regular reconstructed **strēa*: this is an explicit mismatch. The fragment below combines final-vowel loss and ensuing vocalization. Its narrow matcher does not make general final-vowel loss *w*-conditioned.

SC111. Endingless *aw* fragment (`PWGmcEndinglessAw`)

```
define PWGmcEndinglessAw [
    {*a} {*w} {*ą} -> {*au} || _ .#. ,
    {*á} {*w} {*ą} -> {*áu} || _ .#.
];
```

The ordinary *au* product feeds SC030 OEAuBrightening and SC032 OEDiphthongLeveling, not the retained show proxy SC034 OEAwLongDiphthong. The early holding site is compatible with the source's West Germanic history, not evidence for strict relative order against unrelated dental hardening (Ringe and Taylor 2014, 172–73).

2.22 Consonant-conditioned lowering of stressed short i

SC015. Consonant-conditioned lowering of stressed short **i* (PNWGmcILowering)

A disputed Germanic correspondence

Old English *nest* 'nest' and *wer* 'man' have long occupied a disproportionate place in the history of Germanic vowels. Their mid vowels appear to continue earlier short **i*, yet many words with apparently comparable endings retain a high vowel. The question is not whether lowering occurred somewhere, but which conditions explain its distribution, and whether the relevant developments belonged to a common ancestor or separate daughter histories. Campbell identifies only these nouns and isolated *spec* 'bacon', beside *spic* 'bacon', as clear English evidence for the traditional formulation (Campbell 1959, 43–44).

The account adopted here lowers stressed short **i* immediately before inherited **r*, tautosyllabic **z*, or **st*. It follows Hill's consonant-conditioned proposal, not a general assimilation to the following vowel. This is a working reconstruction with unequal evidential support for its components: the rhotic environment is strongest; the sibilant environments are more difficult. Its value lies in relating positive correspondences to independently important negative evidence, not in making every inherited formation unproblematic (Hill 2017, 153–58).

The vowel-conditioned tradition

Streitberg states a common-Germanic lowering before a following **a* or **ā*, blocked by intervening **j* or a nasal followed by another consonant. The latter qualification matters: a single nasal is not automatically a blocker. His presentation already acknowledges extensive analogical disturbance. Hirt likewise connects lowering with a following low or mid vowel, but qualifies its geographical unity and admits unknown conditions. Neither author offers an uncontested exceptionless distribution (Streitberg 1896, 56–57; Hirt 1931--1934, 45–46).

Luick is more guarded for English: lowering appears in only some formations and seems dependent on additional conditions not yet recovered. Campbell preserves the short statement while separating the much more regular lowering of **u* in SC017 PNWGmcULowering. The apparent parallel between the two vowels is historically important, but parallel notation does not establish identical conditions or chronology. Fulk retains a serious role for vowel-conditioned lowering and subsequent levelling, particularly the elimination of lowered vowels from paradigms (Luick 1914--1940, 108–9; Campbell 1959, 43–44; Fulk 2018, 56–59).

There was consequently no single traditional consensus subsequently overturned by one discovery. Lloyd's survey distinguishes authors who accepted a common change, those who restricted it to dialectal prehistories, and those who merely registered the forms or denied the law. That history cautions against treating every reconstructed **e* as an endorsement of the same vowel-harmony mechanism. The inference from a correspondence to a particular conditioner must be argued independently (Lloyd 1966, 738–39).

Why the following vowel is insufficient

Class I strong-verb participles supply a major difficulty. Short **i* survives before a non-high participial vowel where the parallel class II formation commonly shows lowered **u*. The contrast cannot be removed simply by noting that participles belong to paradigms: a proposed restoration must identify the source of the restored vowel and explain its distribution. Lloyd challenges precisely those morphological premises, rather than denying that analogy exists (Lloyd 1966, 739–40; Hill 2017, 150–52).

The uninflected adverbs are harder still. OHG *hina* ‘hence’, *hinana* ‘from here’, and *nidana* ‘from below’ retain high vowels before non-high continuations without an ordinary case paradigm from which **i* could be restored. Fish vocabulary supplies another independent class: PGmc **fiskaz* ‘fish’ and **wiskō* ‘wisp’ preserve **i* before **sk*. Changing their inflection after the event does not itself explain the inherited root vocalism (Lloyd 1966, 741–42; Hill 2017, 152, 154–55).

Cercignani’s defence of distinct **i* and **e* phonemes addresses a related but different proposition: phonemic opposition does not exclude a conditioned lowering. His criticism concerns the extension of one distribution and date across Germanic. In particular, the Norse restriction after preceding **k/g* cannot simply be transferred to West Germanic. Ringe and Taylor also separate the southern West Germanic lowering before labial and velar obstruents from the widely shared nest and man correspondences. An independent continental development therefore cannot be assumed to establish a general ancestral rule (Cercignani 1979, 75–76, 82; 1980, 129–31; Ringe and Taylor 2014, 33–36).

Hill’s consonantal environments

Hill redirects attention to the immediately following consonant. Before tautosyllabic **z*, the principal comparisons include PGmc **iz* ‘he’, **hiz* ‘this here’, and **mizdōn* ‘reward’. The last belongs with OE *meord* ‘reward’ and *mēd* ‘reward’, whose consonants and quantities subsequently diverge. Their mid vowels need not have been induced by a later low vowel: the monosyllabic pronouns are especially significant because they remove that presumed determinant (Hill 2017, 153).

Before **st*, **nistan* ‘nest’ supplies the central example. Its **e* is compatible with both accounts when a non-high ending is reconstructed, so it does not by itself choose the conditioner. The contrast with retained **sk* is more informative than nest alone. The comparative formation also supports an earlier high vowel: Kroonen relates the noun to the downward particle and the zero grade of ‘sit’. Nevertheless, one exceptionally clear noun is a thin foundation for the full productivity of an **st* law (Kroonen 2013, 391; Hill 2017, 153–55).

Before **r*, early **wiraz* ‘man’ is compared with OE *wer*, ON *verr* ‘man’, and Gothic *wair* ‘man’. Lloyd himself allows a special rhotic influence despite rejecting general a-umlaut. Howell’s account of Gothic breaking adds phonetic plausibility: rhotics, especially coda rhotics, commonly lower neighbouring vowels. But recurrence of a natural phonetic process is not proof of a single inherited event. Gothic could have lowered independently, and its breaking can obscure an older distinction between high and mid inputs (Lloyd 1966, 744, n. 9; Hill 2017, 156; Howell 2024, 81–82, 90–91).

Gothic consequently has a double evidential role. Its distribution can illuminate the phonetic environment of lowering, while its own breaking makes some comparisons historically ambiguous. A Gothic mid-vowel reflex before a rhotic is compatible with an earlier lowered vowel, but it is also compatible with an inherited high vowel affected by Gothic breaking. The correspondence cannot choose between those histories without additional chronological evidence. Hill’s proposal gains its force from combining the Gothic facts with West Germanic quantity and consonantal correspondences, not from treating Gothic spelling as a transparent record of one ancestral vowel. Conversely, the possibility of a Gothic innovation does not compel an independent innovation in every other daughter. A common development remains an explanatory hypothesis where its conditions and relative chronology can be defended. The inherited rhotic, the later rhotacized sibilant and the short versus long vowel must therefore be compared separately, rather than counted as interchangeable instances of the same environment (Hill 2017, 149, 156–58; Howell 2024, 81–82).

Hill further proposes that later i-mutation conceals some earlier lowering. His lowered Northwest Germanic **nẹstijana* 'to nest' and **lẹstiz* 'skill' can return to high vowels before a following high-front trigger. Here the later raising has a specific phonological cause; it is not an unspecified repair assigned separately to every exception. The same account must nevertheless establish that the trigger existed in the relevant historical formation. Identical final high vowels alone cannot demonstrate the two-stage history (Hill 2017, 153).

Lowering and the length of 'here'

The most consequential chronological proposal concerns OE *hēr* 'here'. Hill begins with early **hir*, parallel to the deictic adverbs 'where' and 'there', and argues for the sequence **hir* > **her* > **hēr*. His closer mid-vowel notation distinguishes the lowered result from an automatic identification with inherited **ē₁*. If monosyllabic lengthening had occurred first, an additional lowering of long **ī* would be needed, without the same independent support (Hill 2017, 148–49, 156–58).

On Hill's interpretation, lowering before **r* must therefore precede the lengthening. His broader argument treats West Germanic monosyllabic lengthening and aspects of Gothic breaking as potentially partly Proto-Germanic, making an early shared chronology a serious hypothesis. The conclusion is conditional, however: the source of the long vowel, the reconstruction of the short precursor and the identity of the lengthening must all be accepted. Ringe and Taylor discuss alternative ablaut and later-development interpretations. The modern correspondence does not uniquely force Hill's sequence (Hill 2017, 149, 156–58; Ringe and Taylor 2014, 36–37).

Nor does the argument date every consonantal component. Even if **hir* > *her* is established before lengthening, neither **z* nor **st* participates in that derivation. Their absolute dates require their own evidence. Likewise inherited **r* is distinct from **r* subsequently created by rhotacism: a later surface rhotic cannot simply be projected backwards as the conditioner of an earlier law (Hill 2017, 153–58; Crist 2002, 3–6).

Twine: a genuine difficulty for the sibilant condition

OE *twīn* 'twine' and High German *zwirn-* present the sharpest difficulty. Orel reconstructs masculine **twiznaz*; Kluge–Seebold gives **twizna-* and distinguishes the early verbal and nominal attestations. The modern long vowel is not evidence for inherited long **ī*. A short-vowel formation containing **zn* is supported independently of the lowering hypothesis (Orel 2003, 414; Kluge 2011, 1021).

Two familiar histories preserve its timbre: **twizna-* > *twirna-* by rhotacism, and **twizna-* > *twīna-* through loss and compensatory lengthening. De Vaan explicitly connects those treatments with assimilation as another outcome of **zn*; Campbell groups *twīn* 'twine' with lengthening after sibilant loss. If early lowering instead produces **twezna-*, neither rhotacism nor loss restores the high vowel: **twerna-* and **twēna-* remain different predictions. The conflict is not merely orthographical (Campbell 1959, 166, §404; Vaan 2016, unpaginated, paragraphs "Verwante vormen" and "Zowel").

Kilday offers a specific alternative: the English word may be borrowed, with a linen-related semantic history and possible verbal borrowing before nominal back-formation. That proposal is worth considering because it has an argument independent of the present rule. It is not thereby proved. Moreover, removing English from the inherited evidence would not explain away the High German high vowel. Crist's lowering-before-loss-before-rhotacism chronology makes the incompatibility particularly clear when applied to the same inherited **izn* sequence (Kilday 2024, 1, 4–5; Crist 2002, 3–6).

Twine must therefore remain counterevidence, not be assigned a convenient exceptional exclusion. A syllable-sensitive account might distinguish some sequences, but the proposed syllabification would need independent historical support. The fact that a consonant follows **z* is not itself proof that this sibilant had precisely the tautosyllabic status required by Hill's phonetic generalization (Hill 2017, 153; Orel 2003, 414).

Numeral adverbs: conditional rather than discharged

OHG *zwiro* ‘twice’, *zwiror* ‘twice’, and *driror* ‘thrice’ raise a related question. Their prehistory is less secure than that of twine. Kroonen gives **twis* and alternatives **twiswar twizwar*; Orel also preserves different sibilant formations. Fulk starts from **twis-wōz*. These cannot be reduced to one certain bare **twiz* simply because a later form contains a rhotic (Hirt 1931--1934, 116; Kroonen 2013, 530; Orel 2003, 414; Fulk 2018, 275).

The etymological discussion of *driror* explains final *-or* through comparative-adverbial remodelling and considers a **wa*-formation. That explains neither the root vowel nor automatically the first rhotic; its own **þrizwa-/twizwa-* retains a potentially relevant sibilant. A history with **sw*, one with intervocalic **z*, and one with coda **z* before **w* test different conditions. These adverbs are thus unresolved conditional controls, not three independent refutations and not three exceptions already disposed of by analogy (Lloyd et al. 1998, cols 784, s.v. driror; Noreen 1923, 309; Campbell 1959, 287).

The Norse participle and the disputed leather word

The participle of ON *bíða* ‘await’ is *beðinn* ‘awaited’, also cited as *beðenn* ‘awaited’, not an attested **biðinn* ‘awaited’. Noreen considers a lowered ending and influence from the participle of ‘bid’; Seebold allows regular lowering but prefers the latter connection. Fulk treats the lowering interpretation seriously and denies that the analogy has been demonstrated. This isolated low-vowel participle is therefore evidence the consonantal account must confront, but it does not establish a general inherited vowel-conditioned law for all class I participles (Noreen 1923, 326, 332; Seebold 1970, 94–95; Fulk 2018, 58–59, 286).

OHG *irh* ‘chamois leather’ potentially contradicts lowering before **r*. Hill favours an existing loanword interpretation; Kroonen reconstructs **irha-* ‘he-goat’ within a wider comparative family. The alternatives concern lexical ancestry, not merely the operation of a vowel law. If the relevant formation is inherited, its high vowel needs an explanation. Borrowing cannot be declared certain solely because it would remove a counterexample (Hill 2017, 156, n. 30; Kroonen 2013, 270).

Learn and the limits of paradigm reconstruction

OE *leornian* ‘learn’ alternates with *liornian* ‘learn’; OHG *lernēn* ‘learn’ likewise stands beside *lirnēn* ‘learn’. The comparative base is **lizn-*, but a single uncontroversial inherited paradigm has not been recovered. Jasanoff reconstructs a class III history and explicitly considers a lost third-singular **lizniþ* as a possible source of high-vowel forms. Hill instead emphasizes earlier lowering followed by mutation in appropriately extended stems. Both arguments depend on morphology not recoverable from the infinitive alone (Kroonen 2013, 340; Orel 2003, 247; Jasanoff 1973, 865, 868; Hill 2017, 153, n. 25).

The inherited high vowel and the later *io* spelling must be distinguished. Breaking can represent the continuation of a high vowel, but that high vowel could itself have resulted from mutation after an earlier lowering. Conversely, the existence of *leorn-* does not establish every reconstructed mutation trigger or levelling direction. Traditional dialect descriptions illuminate the distribution, not a simple opposition between conservative Northumbrian and innovative West Saxon: early West Saxon also has *liorn-*. Later weak-class remodelling further complicates the inherited cells (Campbell 1959, 48, 59, 125; Jasanoff 1973, 865, 868; Tarsi and Laker 2026, 42–43).

CAPR selects *leornian*, imperative *leorna* ‘learn!’, and third-singular *leornaþ* ‘learns’. These have independent textual support: Ælfric’s Treatise supplies an infinitive after *scealt*, Deuteronomy has the singular command, and Bede has *leornað* ‘learns’ with a singular ‘whoever’ subject. The dental spelling in the last is normalized without changing its vowel or cell. These are selections among attested forms, not three independent etymologies or proof that every *liorn-* history has been explained (Crawford 1922, 71, 351; Miller 1891, 380).

Adoption and its chronological limits

The evidential asymmetry is itself informative. A law whose only argument were the absence of exceptions in a small English sample would have little comparative force. Here the positive correspondences belong to several different formations: pronominal stems, a verbal family, a reward noun and the inherited nest word. Their consonantal overlap is more specific than the mere presence of a non-high vowel somewhere to the right. Conversely, the class I participles and the uninflected adverbs test the old following-vowel condition without automatically entering any of the proposed consonantal environments. The same examples thus have different diagnostic value under the two hypotheses (Lloyd 1966, 739–45; Hill 2017, 153–56).

Later raising must be argued with corresponding precision. A visible high-front trigger can make a low intermediate compatible with a later high reflex; it does not prove that lowering happened. Once the trigger has been lost or transferred across a paradigm, morphological evidence must establish its former presence. That is why a mutation-masked derivative can be a useful compatibility control while the proposed learn paradigm remains uncertain. Neither should be treated as an unlimited mechanism for restoring the high vowel of every counterexample. The distinction preserves testable content in the consonantal hypothesis rather than immunizing it against comparison (Hill 2017, 153; Jasanoff 1973, 865, 868).

The unified rule is adopted because it relates important positive and negative correspondences without requiring general vowel-conditioned lowering and unspecified restoration in each resisting class. Its strongest historical foundation remains the rhotic component. The reward and pronoun families support the sibilant component, but twine remains a real difficulty; the productivity of the *st* component is less securely demonstrated beyond nest. Explanatory economy is not permission to erase those differences (Hill 2017, 153–58; Lloyd 1966, 739–45).

```
define PNWGmcILowering Ctx([
    {*i} -> {*e} ||
        .#. EnglishStarConsonant* _
        [{*z} [EnglishStarConsonant | .#.]
         | {*s} {*t}
         | {*r}],
    {*í} -> {*é} ||
        .#. EnglishStarConsonant* _
        [{*z} [EnglishStarConsonant | .#.]
         | {*s} {*t}
         | {*r}]
]);
```

The definition affects the first short-vowel nucleus, with acute and unmarked forms treated in parallel; it does not lower long **ī*. Its following consonants, not the next vowel or the presence of an earlier velar, determine application. Preconsonantal or word-final **z* is an explicit approximation to tautosyllabicity. It must not be mistaken for independent evidence about every **zn* sequence. The Northwest Germanic placement is a qualified working reconstruction; the possible earlier date of the rhotic component remains separate.

The selected reward dative **mízdai* is not the nominative **mizdōn*; their shared root is the relevant lowering witness. In *nest*, the inherited formation and *st* are central. Selected thistle lowers before *st* and later returns to a high vowel through mutation, illustrating a concealed intermediate without proving all proposed concealed histories. Masculine PGmc **weþruz* ‘wether’ already has **e* and therefore supplies no *i*-lowering evidence (Hill 2017, 153; Kroonen 2013, 370, 391, 584; Orel 2003, 459–60).

Finally, *world* contains the inherited ‘man’ element: Ringe and Taylor’s **weraldiz* already has **e*. An earlier-vowel compound representation cannot independently prove lowering before SC036 OEInterStressRaising. The consonantal account has no following-vowel trigger for interstress raising to remove. The compound is compatible with the selected history, not a unique chronological discriminator. Neither it nor the conditional *hēr* ‘here’ argument establishes a total order for all three lowering environments (Ringe and Taylor 2014, 326; Orel 2003, 462; Hill 2017, 156–58).

2.23 Northwest Germanic u-lowering

Historical discussion

Northwest Germanic lowered **u* to **o* when the following syllable contained a non-high vowel. Campbell describes the change and lists *ġeoc* ‘yoke’ among its regular outcomes (Campbell 1959, 42–43, §115); Fulk gives the same word as a standard example — “OIcel. ok, OE geoc, OHG joh beside juh and OS juk” — and notes the paradigmatic alternation between lowered and unlowered stems that the conditioning produced (Fulk 2018, 56, §4.3). A word-initial **j* does not block the change: the blocking effect of *j* concerns only a consonantal *j* standing between the target vowel and the conditioning vowel, as in the class I weak verbs of the *cnyssan* ‘strike’ type (Fulk 2018, 56, §4.3). Ringe and Taylor accordingly reconstruct the Proto-West Germanic paradigm of ‘yoke’ with the lowering applied (Ringe and Taylor 2014, 129).

Regular corpus witnesses include *ġeoc* ‘yoke’, *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’. Where the following syllable kept a high vowel the lowering did not apply, as in *ġeoguþ* ‘youth’, whose root *u* survived (Brunner 1965, 64–65, §92.1).

Regular lowering does not imply that every attested inherited word preserves the lowered vowel. The masculine noun *rust* ‘rust’ is a lexical residue: selected PGmc **rústaz* ‘rust’ regularly predicts **rost* ‘rust’. Orel’s alternative **rustō* ‘rust’ gives the same lowered root; his masculine alternative is independently supported by Ringe (Orel 2003, 309; Ringe 2017, 247–48). Since **frústą* ‘frost’ regularly gives *forst* ‘frost’, a general *-st-* exemption is unwarranted. The retained *u* remains unexplained, alongside the other retained-*u* cases such as *bucc* ‘buck’, *fugol* ‘bird’, *wulf* ‘wolf’ and *wulle* ‘wool’; their consonantal environments do not establish a general replacement law (Campbell 1959, 43, §115; Ringe and Taylor 2014, 27–34). The lowering law is retained; its non-high conditioning class includes nasalized long **ǭ*, not just oral **ō*. The independently reconstructed accusative **wullǭ* therefore lowers just as the nominative **wullō* does under the preferred chronology (Ringe 2017, 242–43). This does not merge either ending with trimoric **ô*. The rust lexical discussion distinguishes the selected masculine citation, Orel’s alternative and diagnostic paradigm constructions.

SC017. Lowering of **u* before following non-high vowels (PNWGmcULowering)

```
define PNWGmcULowering Ctx([
    [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
    ]
]);
```

The implementation lowers the first nucleus through a consonantal interval without **j*; the immediately following consonant must not be nasal. This literal guard is not a claim that every nasal anywhere in the interval blocks the historical process. Fulk distinguishes tautosyllabic and heterosyllabic nasal environments (Fulk 2018, 56).

The lowering feeds the much later West Saxon palatal-glide spelling (SC016 OEWsPalatalGlide): the *o* that the scribes wrote in *ġeoc* ‘yoke’ is the output of this change, so PGmc **júką* ‘yoke’ passes through **joką* on its way to the attested spelling (Fulk 2018, 56, §4.3; Ringe and Taylor 2014, 129). If lowering is placed after SC019 PNWGmcFinalLongORaising, selected **núsō* ‘nose’ yields †*nusu* rather than expected *nosu*, PGmc **skúflō* ‘shovel’ yields †*sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* ‘sorrow’ yields †*surg* rather than expected *sorg*. These are conditional computa-

tional consequences, not independent proof that those paradigms underwent no levelling. CAPR provisionally prefers lowering before final raising because it regularly accounts for these low-root forms without additional, individually unproved analogical generalizations. Nose already involves earlier stem remodelling, and the retained high vowels of wool and love remain a genuine cost of that preference. The main comparison of Fulk, Ringe and Taylor, and Stiles, including the Runic evidence and the plausible but unproved nominative-generalization account of wool, is in SC019 PNWGmcFinalLongORaising.

2.24 Stressed monosyllable **ō*-raising

Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' (Campbell 1959, 47, §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

SC018. Raising of final stressed monosyllabic **ō* (PNWGmcStressedMonosyllableORaising)

```
define PNWGmcStressedMonosyllableORaising Ctx([
    [
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
]
]);
```

Campbell's *cū* 'cow', *hū* 'how', and *tū* 'two' establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 PNWGmcStressedMonosyllableORaising with neighboring changes leaves every output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

2.25 Raising of final unstressed long **ō*

Historical discussion

Final unstressed oral bimoric **ō* became short **u* in Northwest Germanic. The qualification matters: oral **ō*, nasalized **ǭ* and trimoric **ô* were not interchangeable endings. Ringe and Taylor distinguish the first two developments explicitly; Fulk also distinguishes the quantities when reconstructing the endings (Ringe and Taylor 2014, 15–16, 58–60; Fulk 2018, 82–86, 155–56).

The nominative singular **wullō* 'wool' and accusative singular **wullǭ* are independently reconstructed by Ringe. Their distinct endings must survive the comparison even when a later paradigm levels the root vowel (Ringe 2017, 242–43). Likewise, an oral vowel followed by **z* was not yet word-final. These distinctions explain both the scope of raising and its interaction with later loss.

SC019. Raising of final unstressed long **ō* (PNWGmcFinalLongORaising)

```
define PNWGmcFinalLongORaising Ctx([
    [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
    ]
]);
```

The preceding vowel and consonants select a polysyllabic input. The distinct stressed-monosyllable development is retained separately in SC018 PNWGmcStressedMonosyllableO-Raising; it is not absorbed into this unstressed ending law.

Lowering before raising: the preferred account

Fulk argues that *scolu* 'troop, shoal', from **skulō*, requires lowering of the root before final **ō* became **u*. If raising had already removed the non-high conditioner, an unchanged nominative would retain root **u* (Fulk 2018, 56). CAPR provisionally follows this order: SC017 PNWGmcU-Lowering, then final oral raising. It regularly gives the low roots of *nosu* 'nose', *sċofl* 'shovel' and *sorg* 'sorrow', rather than requiring a later root-vowel generalization for each selected form.

This is an argument from explanatory economy, not from the number of matching outputs. Fulk supplies the direct *scolu* argument; the comparison of its explanatory costs with the other selected paradigms is CAPR's assessment. The witnesses are not equally independent. *Nosu* already depends on an earlier remodeled zero-grade stem beside the comparative **nasō-*; it is not an untouched inherited nominative (Kroonen 2013, 383; Campbell 1959, 43, §115). For *sċofl* 'shovel', Luick also infers an earlier high-root doublet, **scufel*, beside *scofel* 'shovel'; that inference makes an alternating paradigm credible but does not establish its direction of levelling (Luick 1914--1940, 105–6, §78). The comparative formations of shovel and sorrow support the selected short-vowel stems, not a directly attested early English levelling operation (Orel 2003, 347; Kroonen 2013, 493). The preferred sequence avoids those additional operations, but it does not thereby explain every retained high vowel.

Raising before lowering: a coherent alternative

Stiles expressly places final **ō* raising before a-umlaut of **u* (Stiles 2012, 45). Ringe and Taylor explain how *wull* 'wool' and *lufu* 'love' could preserve a nominative root **u* by generalizing it through a paradigm whose other cells had lowered **o*; they also leave open genuine exceptions (Ringe and Taylor 2014, 32). With the nasal accusative correctly recognized, the alternative has the following regular consequences. The table gives reconstructed sound-law predictions, not unattested forms promoted to attestations.

Paradigm	Lowering before raising	Raising before lowering
Wool: nominative / accusative	**woll* / **wolle*	**wull* / **wolle*
Love: nominative / accusative	**lofu* / **lofe*	**lufu* / **lofe*
Shoal: nominative / accusative	**sċolu* / **sċole*	**sċulu* / **sċole*

The inherited ending opposition follows Ringe's reconstructed paradigm and the independently described ō-stem inflections; it is not a special ending invented to rescue wool (Ringe 2017, 242–43; Fulk 2018, 154–56). Under earlier raising, nominative-to-oblique generalization can account for attested *wull* and *lufu/lufe* 'love'. Conversely, generalization of the lowered oblique root can account for *scolu* without replacing its nominative ending. The same possibility exists for nose, shovel and sorrow. Neither particular direction is directly witnessed in the reconstructed early English paradigms.

There is independent evidence that levelling occurred: Ringe and Taylor discuss early *Holtijaz* and the daughter and ox paradigms, while Stiles distinguishes conditioned Runic forms from analogical generalizations (Ringe and Taylor 2014, 33–34; Stiles 2012, 47–48). That establishes the mechanism's credibility, not its occurrence in every inconvenient lexeme. Continental *skola* is not an untouched nominative control either: Fulk describes accusative-ending generalization into the nominative in Old Saxon and Old High German (Fulk 2018, 154–55). Thus his *scolu* argument is strong only conditional on the inherited nominative root not having been remodeled. It does not refute Ringe and Taylor's phonologically coherent wool account.

There is a further lexical caution: Fulk elsewhere calls *scolu* 'host' a loan from Latin *schola* 'school', whereas his chronology argument expressly starts from inherited **skulō* (Fulk 2018, 47, n. 1, 56). These analyses cannot silently supply independent confirmation of one another. The comparison here evaluates his inherited analysis.

The balance nevertheless favors retaining lowering first as a working account: several independently selected low-root forms follow without additional, individually unproved levelling. Wool and love remain its real explanatory cost. Ringe and Taylor's alternative is plausible but unproved, not a demonstrated error. Analogy belongs to the separately argued history of a paradigm, not to the exceptionless sound laws.

What the Runic evidence dates

Fulk cites *horna* 'horn' on the Gallehus horn, about 400, for lowering and *waritu* 'I write' on the Järsberg stone, about 450, for final oral raising (Fulk 2018, 56, 82). These are separate positive chronological bounds. An occurrence of raising around 450 does not establish that it began after 400. Nor does a written final *o* in another inflection establish retention of oral bimoric **ō*: nasalization, quantity and a following consonant must first be identified. Stiles's treatment of the Runic distribution also requires distinguishing conditioned realization, phonemicization and analogical extension (Stiles 2012, 47–48, 54–60). No new general revaluation law follows from those distinctions.

Fulk's note cites Ringe and Taylor on lowering and final **-az* loss, not as an explicit rebuttal of their wool explanation (Fulk 2018, 57, n. 1; Ringe and Taylor 2014, 27–34). Stiles predates their book. Luick discusses alternation and rejects a proposed labial protection law using regular low-root counterexamples; Campbell and Bülbring list retained-high-vowel cases (Luick 1914--1940, 104–6, §§77–79; Campbell 1959, 43, §115; Bülbring 1902, 45–46, §116). None of these checked passages supplies our exact comparative economy argument. No specific later criticism or defence was verified in the bounded literature search; this is not a claim of novelty or of its absence from all scholarship. The Campbell and Bülbring passages remain text-checked rather than newly certified against original images.

Final consonants and later ending reduction

In **rástōz* 'rest', **z* protects oral **ō* from this raising. After SC020 EAFFinalZDeletion, the newly final long vowel can undergo SC042 PWGmcSurvivingBimoricOUnrounding and later fronting and shortening, yielding *ræste* 'rest'. Moving raising after final consonant loss wrongly exposes that vowel and gives **rast* instead. This relation is retained separately; qualifying the lowering/raising comparison does not weaken it.

Nasal **ǭ* does not raise here. Its later reduction yields the accusative ending in **wolle*; the current serialization carries the nasal symbol to that reduction rather than claiming that nasality historically lasted until then. The genitive alternatives must also remain distinct: Ringe and Taylor reconstruct bimoric **-ōz*, where Fulk argues for trimoric **-ôz* (Ringe and Taylor 2014, 59, 299; Fulk 2018, 155). The trimoric genitive's complete pathway remains unmodeled; it is not silently replaced by a bimoric genitive or covered by the present nasal-vowel repair.

Chapter 3

From Proto-West Germanic to Anglo-Frisian

3.1 Historical interval

This chapter covers the sound changes that occurred after the Proto-West Germanic period and before, or during the emergence of, the specifically English line. The starting reconstruction is Proto-West Germanic; the end point is the Anglo-Frisian ancestor required by the model's historical tree. Its existence and the identification of its innovations are separate questions. Its inventory must be reconstructed from conditioned laws and daughter chronologies, not from the traditional subgroup label alone.

3.2 A necessary terminological caution

The title of this chapter uses "Anglo-Frisian" as an organizing historical concept. That choice requires an explicit qualification.

The scholarly literature uses several overlapping terms for this developmental period:

- North Sea Germanic and Ingvaeonic: labels used by some scholars for a proposed subgroup comprising Old English, Old Frisian, and Old Saxon (or more narrowly, Old English and Old Frisian only). The innovations associated with this label — especially the nasal spirant changes and certain vowel developments — are sometimes described as diffusion rather than shared inheritance (Bremmer 2009, 24–27, 126–28).
- Anglo-Frisian: a label used specifically for the Old English / Old Frisian branch, or for innovations shared between the two languages. Its use presupposes a tighter relationship between English and Frisian than between either and Old Saxon.
- Proto-Anglo-Frisian (PAF): the strongest interpretation, positing a discrete reconstructed common ancestor for Old English and Old Frisian specifically. This is the position of Ringe and Taylor, who reconstruct a PAF stage between Proto-West Germanic and Proto-Old-English (Ringe and Taylor 2014, 54–68).

CAPR requires a discrete Anglo-Frisian ancestral node. This does not guarantee that every traditionally associated innovation is inherited from it, or exclusive to English and Frisian. A broader innovation can already be present in its input; similar daughter outcomes can also result from independent events. Authors' diffusion interpretations remain accurately attributed, but they cannot resolve an incompatible chronology within CAPR's ordered tree (Versloot 2017, 302–19; Bremmer 2008, 284, 292–93).

3.3 A worked historical tree test

If a supposed shared innovation obligatorily follows a daughter-only change in Frisian which English never underwent, it cannot occupy the common stem under those premises. Either the ordering or scope premise must change, the innovation happened independently, or the apparently identical events are historically different. The required ancestor is not removed.

Campbell reconstructs Frisian au contraction before ordinary fronting; Goblirsch instead proposes a shared diphthong followed by Frisian stress shift and contraction (Campbell 1939, 91, 104; Goblirsch 1991, 17, 20–21). The daughter-only event is a predecessor in the first account and a follower in the second. The second is compatible with sharing, but does not attest the reconstructed Frisian intermediate. Surface agreement, compatibility with inheritance, compelled parallelism under premises, and unresolved history must therefore remain distinct judgments.

3.4 The working account and why it is used

The English reconstruction adopted as a working scaffold is the conventional account developed by Campbell and critically refined by Ringe and Taylor. It distinguishes inherited long vowels from the long vowels produced by diphthong contraction, and it treats fronting, breaking, restoration and mutation as conditioned phonological developments. That scaffold is useful because it specifies contrasting inputs and intermediate states, not because one author's authority decides the history (Campbell 1959, 52–53, 60–72; Ringe and Taylor 2014, 170–73, 215–37).

The comparative specialists test particular load-bearing premises. Campbell's argument about English and Frisian fronting, Versloot's runic interpretations and Goblirsch's proposed lost Frisian diphthong do not form one automatically compatible package (Campbell 1939, 90–91, 104; Versloot 2017, 295–97, 318; Goblirsch 1991, 17–21). Where an alternative changes the inherited inventory or the identity of an event, it must be evaluated as that alternative, not inserted into the conventional chain without its premises.

CAPR implements regular sound laws. Prosodic and phonological conditions may distinguish environments, but lexical diffusion, grammatical categories and individually selected exceptions cannot repair a contradiction. Paradigm remodeling and borrowing may affect which historical form a lexical entry represents; they are not thereby conditions on a sound law. This methodological choice is distinct from faithfully reporting an author's different theoretical account.

3.5 Long vowels, diphthongs and the ancestral inventory

The inherited-long contrast is central. On the conventional reconstruction, inherited oral long **ā* has already acquired a front quality when English stressed **ai* completes its development to a new back **ā*. Otherwise the new vowel should share the inherited vowel's fronting. Ringe and Taylor formulate this as a relation between temporal milestones: inherited-long fronting must be well under way before contraction completes. They allow overlap; a serial implementation is not proof that the changes occupied disjoint periods (Campbell 1939, 90; Ringe and Taylor 2014, 170).

The inherited vowel itself is disputed. Fulk's retained-front reconstruction does not require precisely the same earlier backing and re-fronting as Ringe and Taylor's account. That is a disagreement about the input inventory, not a small adjustment to the position of an otherwise identical rule (Fulk 2018, 60–61; Ringe and Taylor 2014, 10–13). Nielsen also proposes a different causal relation between contraction and long-vowel restructuring, allowing co-occurrence rather than imposing the conventional strict sequence (Nielsen 2001, 514–16, 521).

Luick's proposed fronted-ai path is a further alternative, not an observed intermediate series. His conjecture is **æe* > **æə* > **æa* > **ā*; the middle offglide is schwa. Campbell rejects that explanation, but the rejection is not an independent observation that such a phonetic path is impossible (Luick 1914--1940, 132–33; Campbell 1939, 90, n. 3). The inherited-long and secondary-long histories must consequently remain separate even where later mutation makes their final vowels

converge.

Nasal environments require another distinction. Nasalization, loss of a nasal with compensatory lengthening, rounding and eventual merger are not one event. The nasalized input can escape an oral-fronting law before its later rounded outcome is established. A shared final vowel therefore does not place every substep at the same ancestral node (Bremmer 2009, 24–27; Ringe and Taylor 2014, 142–46).

The following table specifies the relevant inventory questions at a conservative ancestral cut. It is a working reconstruction, not a newly adjudicated full phoneme inventory. Earlier West Germanic developments can be inherited at this cut without being exclusively Anglo-Frisian innovations.

Input class	Conservative working ancestral state	English daughter consequence	Remaining premise
Inherited oral long vowel	Front long vowel under the conventional account	Subsequent dialectal and palatal treatment	Earlier back versus retained-front input
Nasal long vowel	Distinguished from the oral long vowel	Rounding and merger must be dated separately	Which nasal substeps precede the cut
Stressed ai	Retained diphthong before completed English contraction	Secondary back long vowel, eligible for later mutation	Runic interpretation and any limited ancestral onset
Ordinary short non-nasal a	Unfronted at this conservative cut	Ordinary English fronting	Daughter contraction before fronting
Inherited au	Not identified with completed English or Frisian outcome	English front element and later offglide development	Identity with ordinary short fronting
Supported inherited Vww classes	Adopted West Germanic a/e/i-ww reanalysis; separate homorganic uww quantity	Later English diphthong realization, not inherited English surface vowels	Qualified long-u shadow account; j-created classes remain separate
Velars and palatal tendencies	No blanket completed assibilation is stipulated	Class-specific daughter histories	Articulation, cutoff and merger are distinct

The evidence underlying these qualifications is component-specific (Ringe and Taylor 2014, 41–42, 65–66, 170–73; Versloot 2017, 295–97, 318; Laker 2007, 167–84). The table does not assign inherited-short fronting to the stem merely because the node is called Anglo-Frisian.

The inherited-glide component is now implemented rather than left as a possible future decomposition. SC031 OEWWSimplification belongs to the earlier West Germanic account discussed in Chapter 2: short nonhomorganic **Vww* gives **Vuw*, with retained consonantal **w*; homorganic **uww* separately gives long **ūw*. The latter follows Ringe and Taylor's preferred but qualified shadow analysis. These earlier products can be inherited at the Anglo-Frisian node without placing the later English long diphthongs there. The selected sentence context for early apocope is independent of the segmental reconstruction and of lexical accent (Ringe and Taylor 2014, 41–42, 55, 57–58, 65–66, 171–75; Fulk 2018, 117).

3.6 Alternative cuts and their tree consequences

A more expansive common-stem reconstruction is possible only with different premises. In Luick's account, participation of the diphthong nucleus in fronting can coexist with a conjectured later Frisian development. In Goblirsch's account, a common fronted au diphthong precedes Frisian stress shift and contraction. Both require historical intermediates whose existence

must be argued, not obtained from the name of the subgroup (Luick 1914--1940, 130–33; Goblirsch 1991, 17, 20–21).

Proposed common-stem content	Required premise	Conditional consequence
Inherited-long restructuring, but retained ai and short a	Completed English contraction is later	Compatible conservative cut
Ordinary short fronting	Its secure predecessors can also occur before the split	Excluded if daughter-only English contraction obligatorily precedes it
Shared fronted-au development	Frisian stress shift/contraction follows that development	Compatible in Goblirsch's account, not proof of inheritance
Shared consonantal palatalization	Corresponding consonant classes and necessary predecessors are genuinely identical	Unresolved; apparent similarities do not settle all layers

Campbell's English chain makes the second row especially consequential: if completed contraction is English-only and ordinary fronting must follow it, the latter cannot be inherited from the common stem. The corresponding English and Frisian frontings are then parallel under those premises. Changing a disputed premise can change that conclusion, but cannot remove the required ancestor (Campbell 1939, 90–91; Versloot 2017, 295–97, 318).

3.7 Palatalization is not a single subgroup character

A palatal articulation can precede phonemic contrast, and the productive cutoff of a conditioning process can precede its eventual assibilated reflex. Fricative-g merger with j is different again. Geminate and postnasal stops, singleton fricatives, k, h and sk must not be treated as one dated change simply because later spellings look palatal (Hogg 1979, 100–111; Laker 2007, 167–68, 175–84).

The shared-palatalization hypothesis must therefore identify which layer is inherited. A daughter-specific predecessor can exclude a completed component from the stem without excluding every earlier articulatory tendency. Conversely, removing an argument against sharing establishes compatibility, not that one common event actually occurred (Laker 2007, 175–84). The vowel diphthongization following initial palatals is another event; Luick explicitly distinguishes the similar Frisian and English outcomes (Luick 1914--1940, 162–63).

The English implementation now distinguishes singleton palatal fricatives from gg/ng stops and adopts postmutation fricative merger as a working account (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32). Hogg's objection remains explicit (Hogg 1979, 102–11). This daughter account does not select a complete ancestral inventory, exclude every shared early articulatory tendency, or date stop affrication from an eventual written reflex.

3.8 What the runes date

An inscription dates a written object. A sound-change terminus additionally requires the reading, etymology, linguistic affiliation and relation of the grapheme to phonetic or phonemic structure. The interpretation of the Caistor inscription as retaining ai and the interpretation of Frisian retention are consequential but conditional (Versloot 2017, 295–97, 318). Waxenberger likewise distinguishes early allophonic mutation from later phonemicization and lacks immediate runic evidence for every short-vowel split (Waxenberger 2019, 63–74).

No one calendar date is therefore attached here to every change described as Anglo-Frisian. The English discussion that follows starts from the working inventory and makes its necessary daugh-

ter developments explicit. Canonical classification, preferred reconstruction and an unresolved component are kept distinct until individual adjudication.

3.9 Major changes and their historical basis

West Germanic rhotacism (SC003)

The medial change of `*z` to `*r` in environments such as **déuzaz* 'deer', **xúrdaz* 'hoard', and **líznōjaną* 'learn' is historically a post-Proto-West-Germanic development. Ringe and Taylor argue that rhotacism was not inherited from Proto-Northwest Germanic and was not uniform within West Germanic (Ringe and Taylor 2014, 52, 98, 102). Crist separates this change explicitly from the deletion of word-final `*z` and argues that rhotacism must follow the deletion rules (Crist 2001, 104–6; 2002, 1, 4). Bammesberger gives the standard Old English–facing summary: `*z` yielded `*r` in intervocalic position but was generally lost in final position (Bammesberger 1992, 39).

The executable identifier `EAFRhotacism` is retained for stability, not as independent evidence of historical stage or scope. The preceding source discussion motivates the post-Proto-West-Germanic account; the chapter's placement follows the adopted executable sequence.

Word-final *z deletion (SC020) and the three final-*z developments

The loss of word-final `*z` is not one process but three historically distinct developments, and this chapter contains two of them. The central one, SC020, is the Proto-West Germanic loss of `*z` in unstressed syllables, seen in forms such as **rástōz* 'rest (nom.sg.)' and stated for the whole branch by Ringe and Taylor (Ringe and Taylor 2014, 44–45); Crist's analysis distinguishes it both from the earlier NWGmc changes and from the later narrower Ingvaeonic deletion rules (Crist 2002, 1, 4). Earlier still, the consonant-stem root nouns had generalized endingless nominatives before Proto-West Germanic (SC096, Chapter 2). Later, and only in the north, `*z` was lost in stressed monosyllables with compensatory lengthening (SC097, this chapter); the southern dialects instead retained and rhotacized it. The standard handbooks confirm the West Germanic deletion in general terms: Bammesberger gives a clean statement that Germanic `*z` is generally lost in final position (Bammesberger 1992, 39); the three-way division refines those summaries rather than contradicting them.

The CAPR rule for the unstressed loss is still named `EAFFinalZDeletion`, an identifier that predates the restaging of the change to Proto-West Germanic; the name is retained as a stable identifier pending a global renaming pass, and the historical stage recorded in the staging metadata takes priority over the name prefix. SC020 and rhotacism jointly account for the fate of the remaining `*z`, but their sections are separated by intervening executable stages. Their thematic connection does not determine their publication positions.

Completed English ai contraction (SC004, treated in Chapter 4)

The English outcome of stressed/root **ai* is **ā*. Versloot distinguishes an early velar-conditioned Frisian contraction from later non-velar treatment, with mutation and delabialization intervening (Versloot 2017, 302–9, 316). His wave account is an author's interpretation, not CAPR's own genealogy. CAPR now places completed English contraction on the daughter branch, while preserving the distinct possibility of a narrower conditioned ancestral onset (Campbell 1939, 90–91; Versloot 2017, 295–97, 318). This local decision does not select every component of the ancestral inventory.

The current SC004 EAFAiMonophthongization models English contraction after SC101 EAFLongAFronting. The new ai-derived **ā* escaped the earlier fronting of inherited oral **ā*; Campbell states that inference and Ringe and Taylor retain it (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70). Soul provides another implementation boundary at interstress raising, not the sole historical argument or proof of inherited contraction.

The unstressed development `*ai > *ē` (in final and nonfinal syllables) is a separate and earlier Proto-Northwest Germanic change (SC014), discussed in Chapter 2; its corpus witnesses are the dative-singular endings of `span` (**spánnai* ‘span’ > *spanne* ‘span’) and `meed` (**mízdai* ‘meed’ > *meorde* ‘meed’).

Anglo-Frisian brightening (SC043, treated in Chapter 4)

The fronting of low **a* to **æ* outside nasal environments is traditionally called Anglo-Frisian brightening. The name does not establish one inherited event. It executes later in the cascade than the changes of this chapter, so its full section appears in Chapter 4; it is introduced here because it anchors the “Anglo-Frisian” label that names this period. Campbell gives the classical statement: “By a very early change Prim. Gmc. `a > æ` in OE and OFris. when not followed by a nasal consonant” (Campbell 1959, 52, §131). The traditional name also appears in modern comparative treatments; it does not independently establish a shared event (Fulk 2018, 72–73).

The change is notable for what follows it: OE Breaking presupposes the fronted input; OE a-Restoration partially undoes it in back-vowel environments. The three-change sequence is the conventional English working account, but its components and conditioning remain individually arguable. Versloot proposes restricted early fronting and late e-breaking; final forms can coincide under front-and-restore and never-front histories (Versloot 2025, 123, 125–28, 131–35).

Campbell notes that English and Frisian may not simply reflect one undifferentiated shared prehistoric event, and Ringe and Taylor leave open whether the wider spread of fronted outcomes happened mainly on the continent or in Britain (Ringe and Taylor 2014, 60–62). CAPR’s implementation treats the stressed English component separately from the retained unstressed and final-vowel representations.

The completed ordinary stressed component is now characterized on the English daughter under Campbell’s conventional working chain (Campbell 1939, 90–91; 1959, 52–53). Ringe and Taylor’s diphthong-nucleus objection remains explicit (Ringe and Taylor 2014, 170–75); earlier restricted ancestral tendencies are not excluded. English plain-a/au process identity is distinct from inherited stem identity, and their current serialization is retained. The unstressed and final-vowel contributions are not assigned the same date from this argument. This local adoption does not select the complete ancestral inventory.

3.10 Cascade vs. historical order in this chapter

The sections follow executable serialization so that derivations are inspectable. That order is not an independently demonstrated total historical chronology. The broader final-z loss, later northern loss and rhotacism are distinguished by their actual environments (Ringe and Taylor 2014, 44–45; Crist 2002, 1, 4).

Earlier-stage proxies remain in this editorial corridor. The cluster change represented by SC022 PNWGmcMnDissimilation and final-n loss represented by SC023 PNWGmcNStemNLoss are classified as Common Germanic/Proto-Germanic despite retained name prefixes (Fulk 2018, 121; Ringe 2017, 101–3). The retired unstressed-o raising is not an active member. Book order, computational position and historical stage must remain separate.

3.11 West Germanic final **z*-deletion

Historical discussion

Word-final **z* in unstressed syllables was lost in Proto-West Germanic. Ringe and Taylor state the change for the whole branch and illustrate it with the nominative plural **dagōz* > **dagō* and the consonant-stem nominative **fadurz* > **fadur*, noting that the ending is lost after consonants as well as after vowels (Ringe and Taylor 2014, 44–45, §3.1.1). Crist’s handout formulates the same development and its Ingvaeonic sequels (Crist 2002, 2, §§5–6). The change is pan-West-Germanic, not specifically Ingvaeonic: every West Germanic daughter shows the loss, and the Frienstedt

comb inscription *kaba* < **kambaz* ‘comb’ (c. 250–300 CE) supplies early epigraphic confirmation (Fulk 2018, 25, n. 1).

The conditioning segment is specifically the voiced sibilant **z*, never **s*: Ringe and Taylor’s near-minimal pair of nominative singular **dagaz* > **dag* beside genitive singular **dagas*, which keeps its sibilant into Old English *dæġes*, shows that the change reads the Verner voicing distinction (Ringe and Taylor 2014, 212, §6.1). Where the handbooks disagree about whether a given ending had **-s* or **-z* — as for the nominative plural **-ōz* — the disagreement matters directly to whether this rule applies (Ringe and Taylor 2014, 115–16, §4.2.1).

This is the middle of three historically distinct final-**z* developments, and Ringe and Taylor explicitly separate it from the later loss in stressed monosyllables, citing Crist’s demonstration that they are two changes (Ringe and Taylor 2014, 44–45, §3.1.1). Earlier, the consonant-stem (root-noun) nominatives of monosyllables had already generalized endinglessness before Proto-West Germanic (SC096 RootNounNomZLoss), so forms like **bōkz* ‘book’ never reach this rule with their marker intact. Later, and only in the north, **z* was lost in stressed monosyllables with compensatory lengthening (SC097 MonosyllabicFinalZLoss); the present rule leaves stressed monosyllables untouched. Older accounts that grouped all of these under one loss of final **z*, such as Campbell’s, are superseded by this three-way division (Campbell 1959, 166).

At the boundary with Chapter 2’s Northwest Germanic sequence, the derivation of *ræste* ‘rest’ shows that final **ō*-raising (SC019 PNWGmcFinalLongORaising) must precede this rule: raising applies to **-ō* but not to **-ōz*, whose final vowel is still sheltered by the sibilant when raising runs (Ringe and Taylor 2014, 15–16, 24). On the later side, Ringe and Taylor order the loss of **z* before the loss of word-final bare **-a*, since **dagaz* first becomes **daga* and only then **dag* (Ringe and Taylor 2014, 45–46, §3.1.2).

SC020. West Germanic final **z*-deletion (`EAFFinalZDeletion`)

```
define EAFFinalZDeletion Ctx([
    [{*z} -> 0 ||
    .#. ?* EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+
     ↪  EnglishStarVocalic ?* _ .#.,
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]* EnglishStarVocalic+
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.]
]);
```

The rule deletes word-final **z* in unstressed syllables, stated through two environments. The first clause covers polysyllables, where the final syllable of these corpus forms is unstressed: this is the ordinary case, with 110 corpus derivations, such as PGmc **bárdaz* ‘beard’ on its way to OE *beard* ‘beard’ and **rástōz* ‘rest’ on its way to *ræste* ‘rest’. The second clause covers post-consonantal **z* in monosyllables. By the time this rule runs, SC096 RootNounNomZLoss has already removed the genuine root-noun nominative endings, so the only form reaching the second clause is **frījōndz* ‘friend’, contracted to monosyllabic **frīundz* by SC009 PWGmcIjContraction; its ending, like that of **fadurz*, stood in an unstressed syllable when the Proto-West Germanic change applied and so belongs here rather than to the root-noun development (Ringe and Taylor 2014, 44–45, §3.1.1). Stressed monosyllables ending in vowel plus **z* meet neither clause and are left for SC097 MonosyllabicFinalZLoss.

The chronology of word-final **z*-loss is unusually well delimited: *ræste* ‘rest’ supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 PNWGmcFinalLongORaising, PGmc **rástōz* ‘rest’ yields †*rast* rather than expected OE *ræste* ‘rest’. After SC040 OEMedUnstressedULowering, PGmc **bébruz* ‘beaver’ yields †*befro* rather than expected *befer* ‘beaver’, PGmc **kwéðuz* ‘cud’ yields †*cwedo* rather than expected *cwedu* ‘cud’, and PGmc **félθuz* ‘field’ yields †*feldo* rather than expected *feld* ‘field’, alongside eight other newly failing rows. Final *z*-loss therefore follows SC019 PNWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The **rástōz* ‘rest’ derivation fixes the local relation to SC019 PNWGmcFinalLongORaising. The

distant boundary at SC040 OEMedUnstressedULowering shows only that word-final *z-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final *ō-raising.

3.12 Early apocope in unstressed words

Historical discussion

Alongside the regular loss of word-final short high vowels in third syllables (SC006 PWGmcEarlyIApocope), Ringe and Taylor identify a second, earlier apocope: “Short high vowels were also lost after heavy syllables in unstressed words” (Ringe and Taylor 2014, 57–58, §3.1.4). The two laws must not be conflated. Fully stressed disyllables kept their final *-*i* long enough to cause i-umlaut — OE *ġiest* ‘guest’ < **gastiz* and *fȳr* ‘fire’ < **fūri* require exactly that survival (Ringe and Taylor 2014, 55, §3.1.4) — whereas words that carried no sentence stress lost the vowel already in Proto-West Germanic.

The conditioning is prosodic. Like Verner’s law, the change is governed by accent: it applied in words unstressed in the sentence, and its apparent exceptions are systematic, not sporadic. Forms such as OE *ymbe* ‘around’ and OHG *umbi* kept their final vowel because, as Ringe and Taylor observe, proclitics “were not phonologically word-final” and so stood outside the environment altogether (Ringe and Taylor 2014, 57–58, §3.1.4). Sentence-level accent placement therefore decides which sandhi variant each daughter language continues, and doublets across the family reflect the stressed and unstressed sentence forms of the same word — regular sandhi, not lexical diffusion.

The diagnostic witness is the second-person plural pronoun. Ringe and Taylor print the Proto-West Germanic form as a doublet: PGmc **izwiz* (Gothic *izwis*) → **iwwi* (by SC008 PWGmcCoronalWAssimilation and the loss of final *z, SC020 EAFFinalZDeletion) → PWGmc **iuwi* ~ **iuw* (Ringe and Taylor 2014, 41–42, §3.1.1). Old English continues the apocopated, unstressed variant, and Ringe and Taylor’s proof is the vocalism itself: “OE iow ‘you (dat. pl.)’ definitely does [show early apocope] (since it does not exhibit i-umlaut)” (Ringe and Taylor 2014, 57–58, §3.1.4). Had the *-*i* survived, i-umlaut (SC055 OEIUmlaut) would have fronted the diphthong; West Saxon *ēow* ‘you’ beside early West Saxon and Northumbrian *īow* ‘you’ shows the normal unumlauted development (Campbell 1959, § 702, p. 283).

SC098. Early apocope in unstressed words (PWGmcUnstressedWordFinalIApocope)

```
define PWGmcUnstressedWordFinalIApocope [
    [ [{*i}|{*u}] -> 0 || .#. {*ᵘ} ?* [
        [EnglishStarLongVowel | EnglishStarLongDiphthong] EnglishStarConsonant*
        | EnglishStarShortVowel EnglishStarConsonant EnglishStarConsonant+
    ] _ .#. ]
    .o. [{*ᵘ} -> 0]
    .o. [
        [{*ᶜ}:0 [EnglishStarAlphabet - {*ᶜ}]* 0:{*ᶜ}]
        | [EnglishStarAlphabet - {*ᶜ}]*
    ]
];
```

Sentence stress and phonological finality are now explicit, independently selected context. They do not alter the PGmc reconstruction, the selected segmental input or its historical stage. The computational marks in the rule distinguish weak-final context from weak-nonfinal context; they are not reconstructed phonemes and are absent from displayed forms. Ordinary evaluation explicitly chooses strong-final citation context. You selects the independently source-discussed weak-final variant, not an unstressed classification inferred from a pronoun label or missing acute.

The worked history is **izwiz* → **iwwiz* by assimilation → **iuwiz* by SC031 OEWWSimplification → **iuwi* by final *z*-loss → weak-final **iuw* by this rule → OE *ēow*. The operation requires a heavy syllable and a phonologically final short high vowel: long vowel/diphthong weight or a short vowel followed by a closing consonant cluster, not literal *ww*. Its condition therefore survives glide reanalysis. Loss of final *z* still feeds it, and removal of *i* bleeds SC055 OEIUmlaut. Its PWGmc placement precedes the later northern loss of stressed-monosyllabic final *z* (SC097 MonosyllabicFinalZLoss) ((Ringe and Taylor 2014, 41–42, 55, 57–58)).

The staged strong contexts retain the high vowel in the histories of *ġiest* 'guest' and *fȳr* 'fire'. A weak-final and context yields *and* 'and', whereas the proposed proclitic context retains the vowel in *ymbe* through the later apocope corridor. The right-boundary annotation is removed only after SC063 OEHighVowelApocope has had the opportunity to apply. These controls implement the defended exceptionless account; they do not claim direct observation of prehistoric sentence stress or lexical optionality ((Ringe and Taylor 2014, 55, 57–58)).

The corresponding English realization of *iu+w* belongs to SC032 OEDiphthongLeveling, not a late vocalization of retained *ww*. The model's retained-i counterfactual produces an umlauted result after intervening w-loss; its exact terminal spelling is a computational prediction, not an attested strong OE form. The attested weak-final target and the source's mutation argument must not be replaced by that counterfactual ((Ringe and Taylor 2014, 41–42, 57–58, 173–75; Campbell 1959, 283)).

3.13 Northern monosyllabic final **z*-loss

Historical discussion

Long after the Proto-West Germanic loss of final **z* in unstressed syllables (SC020 EAFFinalZDeletion), the northern West Germanic dialects lost word-final **z* in stressed monosyllables as well, with compensatory lengthening of a short nucleus. Ringe and Taylor's witness set is OE *mā* 'more' < **maiz*, the pronouns *wē* 'we', *ġē* 'you', *mē* 'me', *þē* 'thee', *hē* 'he', *hwā* 'who' < **hwaz*, and — hedged in their own print with question marks — *cū* 'cow' (Ringe and Taylor 2014, 86, §3.3.1). The southern dialects retained the sibilant and rhotacized it: Old High German *mir*, *wir*, *mēr*, *er* answer the Old English endingless forms, which is why Fulk counts this loss among the diagnostic Ingvaeonic features (Fulk 2018, 18, n. 6). Ringe and Taylor explicitly treat this as a change separate from the Proto-West Germanic unstressed loss, citing Crist's demonstration that the two must be distinguished (Ringe and Taylor 2014, 44–45, §3.1.1).

The scholarship disagrees about the exact conditioning, and the disagreement is worth recording. An older account, represented by Campbell and going back to Luick, derived the endingless pronouns from unaccented sentence variants rather than from a regular sound change (Campbell 1959, 166; Luick 1914--1940, 819). Ringe and Taylor reject that analysis because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86, §3.3.1). Crist formulates an Ingvaeonic rule in which **z* is lost after front vowels, with compensatory lengthening, covering preconsonantal cases as well; his data contain no word-final back-vowel monosyllables, so forms like **hwaz* and the ancestor of *cū* fall outside what his statement can decide — a documented gap rather than a refutation (Crist 2002, 1, 4, §§1, 10). Kilday narrows the preconsonantal subcase to Old Saxon and Old Frisian while accepting the word-final monosyllabic loss for Old English, contrasting regular *meord* 'reward' with the loanword-influenced *mēd* 'reward' (Kilday 2024, 1–3). CAPR adopts Ringe and Taylor's quality-neutral formulation because it alone generates the back-vowel witnesses, while noting that the front-vowel forms are compatible with both analyses.

Apparent counterexamples are analogical, not phonological: OE *dēor* 'deer', *ār* 'oar', and *gār* 'spear' show final *-r* from levelling out of inflected forms where the sibilant was word-internal and regularly rhotacized, not from retention of word-final **z* (Ringe and Taylor 2014, 86, §3.3.1, n. 24). The change precedes rhotacism (SC003 EAFRhotacism), which Ringe and Taylor place last in this sequence of northern developments (Ringe and Taylor 2014, 87, §3.3.1).

The corpus now witnesses this change directly. The interrogative pronoun 'who' is selected in its nominative singular masculine cell, PGmc **hwaz* (Gothic *hwas*), precisely the form Ringe and

Taylor cite for this loss (Ringe and Taylor 2014, 86, §3.3.1): the rule lengthens the short nucleus and deletes the sibilant, giving **hwā*, whence OE *hwā* ‘who’. The resulting back vowel never undergoes Anglo-Frisian brightening — “**hwǣ* does not exist”, as Campbell puts it (Campbell 1959, § 125, p. 49; Brunner 1965, § 137 Anm. 1, p. 129) — so the derivation ends with the attested form. Other members of Ringe and Taylor’s witness set remain outside the corpus because it selects oblique or plural cells for them — ‘cow’ and ‘meed’, for instance, enter the cascade in inflected forms whose **z*, where present, is word-internal.

SC097. Northern monosyllabic final **z*-loss (MonosyllabicFinalZLoss)

```
define MonosyllabicFinalZLoss [
    {*a} -> {*ā}, {*á} -> {*ā},
    {*e} -> {*ē}, {*é} -> {*ḗ},
    {*i} -> {*ī}, {*í} -> {*ī́},
    {*o} -> {*ō}, {*ó} -> {*ō},
    {*u} -> {*ū}, {*ú} -> {*ū}
        || .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
            _ {*z} .#.
] .o. [
    {*z} -> 0 ||
        .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
            EnglishStarVocalic+ _ .#.
];
```

The rule first lengthens a short nucleus standing immediately before word-final **z* in a monosyllable, then deletes the **z* after any vowel in a monosyllable. The principal synthetic controls run the change on Ringe and Taylor’s own witnesses, fed to the rule in their chronologically correct intermediate shapes. Short-nucleus inputs show both halves of the change at once: **hwaz* yields **hwā* and **hiz* yields **hī*, each with loss of the sibilant and compensatory lengthening (Ringe and Taylor 2014, 86, §3.3.1). A form whose nucleus is already bimoric skips the lengthening step and simply loses the sibilant: **maiz* yields **mai* at this stage, with its diphthong intact; the attested OE *mā* ‘more’ arises only later, when the stressed monophthongization (SC004 EAFAiMonophthongization) takes **ai* to **ā*. The word ‘cow’, whose history Ringe and Taylor themselves print with question marks and whose analysis remains disputed, is deliberately not used as a principal control; a long-vowel input of that shape (**kūz* yielding **kū*) merely repeats what **maiz* already demonstrates, and the word’s evidentiary weight is discussed in the historical dossier rather than leaned on here.

The corpus derivation of *hwā* ‘who’ now fixes this rule’s position empirically as well as philologically. It stands after SC020 EAFFinalZDeletion, since the two losses are historically distinct changes with the unstressed loss earlier (Ringe and Taylor 2014, 44–45, §3.1.1). Rhotacism (SC003 EAFRhotacism) follows in the executable cascade as in the historical account: Ringe and Taylor place rhotacism after this loss, at the end of the sequence of **z*-losses (Ringe and Taylor 2014, 87, §3.3.1), and the cascade composes it immediately after this rule, so a sibilant removed here can never surface as *-r* — were the order reversed, **hwaz* would rhotacize to **hwar* and ‘who’ could never be derived; the negative controls below confirm this. Consonant-final monosyllables are untouched: their nominative **-z*, where it ever existed, was eliminated before Proto-West Germanic under SC096 RootNounNomZLoss. Word-internal **z*, as in PGmc **déuzą* ‘deer’ on its way to OE *dēor* ‘deer’, does not meet the environment of this rule and duly rhotacizes.

3.14 Monosyllabic lengthening before final *r*

The model admits Old English *þēr* ‘there’ and *hwēr* ‘where’ from source-supported short-*a* antecedents. Their quantity and the date of the lengthening remain qualified: the forms do not establish a Proto-Germanic date, and they do not determine the order of SC015 PNWGmcILowering.

SC112. Stressed monosyllabic lengthening before inherited final *r* (PNWGmcMonosyllabicFinalRLengthening)

```
define PNWGmcMonosyllabicFinalRLengthening [
    [
        {*á} -> {*ā},
        {*é} -> {*ē},
        {*í} -> {*ī},
        {*ó} -> {*ō},
        {*ú} -> {*ū}
        || .#. EnglishStarConsonant* _ {*r} .#.
    ]
];
```

The early working placement preserves inherited final *r* before rhotacism, so secondary *r* does not trigger the rule. The guard also excludes disyllabic forms. This is a general formalization of the two admitted witnesses, not a lexical exception list or a claim of strict relative chronology against SC015 PNWGmcILowering or SC024 PNWGmcLongELowering.

3.15 West Germanic rhotacism

Historical discussion

Bammesberger states that Germanic *z yielded **r* in intervocalic position in Old English, while final *z was generally lost (Bammesberger 1992, 39). Ringe and Taylor argue that this merger of *z with **r* was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic *z-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

The internal identifier SC003 EAFRhotacism places the change in CAPR's Early Anglo-Frisian corridor, the operational post-Proto-West-Germanic stage on the English line; historically the change is a West Germanic rhotacism, later than Proto-Germanic. It is also distinct from SC020 EAFFinalZDeletion, which removes final *z before the surviving medial consonant becomes **r*.

SC003. West Germanic rhotacism (`EAFRhotacism`)

```
define EAFRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc ***mízdai*** 'meed' yields †***merde*** rather than expected OE ***meorde*** 'meed'. Moving rhotacism earlier within the tested range changes no output.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. The lower boundary rests on the historical analyses cited above: Ringe and Taylor put rhotacism at the end of the sequence of *z-losses — "first *z was lost in a variety of environments …, then all surviving *z became **r*" (Ringe and Taylor 2014, 87, §3.3.1) — and Crist observes that the deletions distinguish *z from **r* and so must precede the merger (Crist 2002, 2–3). The rule is accordingly ordered after the three final-*z losses (SC096 RootNounNomZLoss, SC020 EAFFinalZDeletion, and SC097 MonosyllabicFinalZLoss) and before breaking, so that the derivation follows the reconstructed chronology.

3.16 *mn*-dissimilation

Historical discussion

In the inherited *n*-stem paradigm the zero-grade oblique cells brought *m* and *n* into direct contact, and in that adjacent cluster the labial nasal dissimilated to a labial spirant: *mn* > *βn* (surfacing as *fn*). Old Norse preserves the older paradigmatic distribution, with the labial confined to the oblique cluster (*himinn* 'heaven' beside dative *hifni*); Old English and Old Saxon generalized it. Fulk treats the cluster change among developments common to Germanic, while warning that its surface results are irregular and that reverse *bn* > *mn* is later well attested in Northwest Germanic (Fulk 2018, 121, §6.11). The relevant *heofon* 'heaven' and *mōnaþ* 'month' material is discussed by Campbell (Campbell 1959, 189, 195, §§470, 484).

The underlying cluster change is therefore late Proto-Germanic / Common Germanic, not securely a pan-Northwest-Germanic innovation; `PNWGmc` remains a stable executable identifier only. The lexical evidence does not constrain a positive local cascade position.

SC022. Dissimilation of adjacent *mn* (`PNWGmcMnDissimilation`)

```
define PNWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _ {*n}
];
```

The rule fires only where *m* stands directly before *n*. It supplies the labial of *stefn* 'stem, trunk' from the *mn*-cluster of the *stamn*-family, and it is the historical change behind the labial of *heofon* 'heaven', which was generalized from the oblique cluster into the vowel-bearing stem before the Old English vocalic changes. (An earlier cross-syllable formulation that labialized an intervocalic *m* before a later nasal has been retired: it simulated paradigm levelling rather than a sound law.)

Moving SC022 PNWGmcMnDissimilation earlier or later leaves every output unchanged. Its executable place in this holding zone is therefore editorial/computational, while its historical classification rests on the handbook account of *mn*-dissimilation.

3.17 Word-final *n*-loss

Historical discussion

The change isolated here is far older than its position in the cascade suggests: it is the general (pre-)Proto-Germanic loss of word-final **n*, with nasalization of the preceding vowel, in polysyllables. Ringe's proof set for the law spans the whole grammar — nouns such as **yugón* > **jukǫ* 'yoke', pronouns such as **tón* > **þanǭ*, and even the verb form **dedǭ* 'I did' — so it is general phonology, not a fact about any one declension (Ringe 2017, 101–3). Gothic *tuggo* shares the weak nominative-singular outcome, and the nasalized reflex **-ǭ* remained contrastive into Proto-West Germanic before yielding OE *-e* (Ringe and Taylor 2014, 54–55, 58–59).

Within the present corpus the change surfaces in exactly one shape: the weak nouns are cited in the stem form **-ōn-*, and this rule carries them to the Proto-Germanic nominative singular in **-ǭ*, as in **túngōn* > **túngǭ* > *tunge* 'tongue', alongside *eorþe* 'earth', *heorte* 'heart', *nǣdre* 'adder', and thirteen further weak nouns. The masculine weak nominative singular in trimoric **-ô* never had a final **-n* to lose, and Proto-Germanic **sebun* 'seven', **nigun* 'nine', and **tehun* 'ten' kept their *-n* by lexical analogy among the numerals (Ringe 2017, 103); the rule's narrow **-ōn* environment leaves all of these correctly untouched.

SC023. Loss of word-final **n* after **ō* (PNWGmcNStemNLoss)

```
define PNWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

The verb *dōn* ‘do’ supplies the negative, counterfeeding witness for the chronology. PGmc **dōną* ‘do’ passes this rule untouched; only SC047 OEHeavySyllableNasalApocope later strips the final **ą* and creates a new word-final **-ōn*. That secondary **-n* survives into *dōn* precisely because the old loss was no longer active: if SC023 PNWGmcNStemNLoss is displaced after the apocope, it consumes the new nasal and the derivation collapses entirely (+*?*).

The retained *-n* of *dōn* ‘do’ therefore supplies a terminus ante quem for the loss — it must be dead before the apocope — while the seventeen weak nouns above are its positive witnesses; the lower boundary remains unattested within the cascade, as befits a change already complete in Proto-Germanic.

3.18 Nasal-spirant lengthening

Historical discussion

Germanic lost nasal consonants before voiceless fricatives twice, in two changes that are easily confused because their outcomes look alike. Both replace a sequence of vowel, nasal and fricative with a long nasalized vowel and the fricative. They differ in date, in geography, and in which fricatives they affect.

The earlier change is common Germanic and is treated in the opening chapter of this book, on the Proto-Germanic loss of a nasal before **x*. Its results are shared by every daughter language, and only **a*, **i* and **u* occur in its input (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7).

The later change belongs to the dialects bordering the North Sea, that is to Old English, Old Frisian and Old Saxon, the group traditionally called Ingvaeonic. Here the groups **mf*, **ns* and **nþ* likewise reject the nasal with compensatory lengthening and nasalization (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11; Brunner 1965, 176, §186.1; Luick 1914–-1940, 276, §301.1). Ringe and Taylor call it the most obvious phonological innovation of the northern dialects and list some thirty examples, among them **gans* ‘goose’ and **jugunþi* ‘youth’ (Ringe and Taylor 2014, 139–41). Campbell describes it as a later change similar to the common Germanic one, and Fulk as comparable to it; neither treats them as the same law.

The two are told apart by what the languages outside the North Sea area show. The common Germanic change left no nasal anywhere, so Old High German has *fūht* ‘damp’ and *fūst* ‘fist’ exactly as Old English does. The later change was confined to the north, so the cognates of its witnesses keep the nasal: Old High German *fimf*, *gans*, *ander*, *jugund* answer Old English *fīf* ‘five’, *gōs* ‘goose’, *ōþer* ‘other’, *ġeoguþ* ‘youth’. On this test **funhsti-* ‘fist’, whose long vowel is shared by Old High German *fūst*, Dutch *vuist* and German *Faust*, belongs to the earlier change and not to the North Sea law at all (Kroonen 2013, 160).

Where the vowel was **a*, the long nasalized vowel that this law produced was afterwards rounded to *ō* in Anglo-Frisian, which is why Old English has *gōs* ‘goose’, *tōþ* ‘tooth’ and *ōþer* ‘other’. That rounding is a third change again, and it is the same rounding that gives *fōn* ‘seize’ and *þōhte* ‘thought’ from the common Germanic law and *mōna* ‘moon’ and *spōn* ‘chip’ from inherited long *ā* before a surviving nasal; Campbell states that it reached all three sources at one and the same time (Campbell 1959, 50, §128 n. 1; Brunner 1965, 33, §26; Fulk 2018, 72, §4.11). It is treated in the chapter on the rounding of the long nasalized low vowel. That the vowel was rounded and did not simply merge shows that its nasality survived the law that created it: as Fulk observes, it did not fall together with the *ā* that came from **ai* (Fulk 2018, 55, §4.1). Ringe and Taylor take the nasalization to have remained subphonemic until it was lost separately in each daughter (Ringe and Taylor 2014, 141). Old Saxon shares the loss of the nasal and the nasalization, and rounds

only variably, which is why the law itself is described as North Sea Germanic and the rounding as Anglo-Frisian (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11).

The North Sea Germanic law is a single connected sound change: in every handbook account the nasal is lost *with* compensatory lengthening, and the lengthening is the compensation for the loss. It is stated here as two rules only because the vowel must be adjusted while the conditioning nasal is still present, before the nasal can be removed. The order of SC026 EAFNasalSpirantLengthening before SC027 EAFNasalSpirantLoss is a requirement of the statement, not evidence for two successive historical stages.

SC026. North Sea Germanic nasal-spirant law, first step (EAFNasalSpirantLengthening)

```
define EAFNasalSpirantLengthening [
    {*a} -> {*ã} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    # Stressed variants (*á -> *ã: stress absorbed into length, no stressed
    # long low tier exists):
    {*á} -> {*ã} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarNSGmcSpirant
];
```

The environment is nasal plus **f*, **þ* or **s*. The fricative **x* is excluded: nasal loss before **x* is the earlier change stated as SC103 PGmcNasalLossBeforeX. Campbell names the groups **mf*, **ns* and **nþ*, Fulk says the change affects **mf*, **ns* and **nþ*, Sievers and Brunner name the fricatives **f*, **þ* and **s*, and no example in Ringe and Taylor's list contains **x* (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11; Brunner 1965, 176, §186.1; Ringe and Taylor 2014, 139–41). As in the earlier change, only **a*, **i* and **u* occur. The outcome of **a* is the long nasalized **ą̄*, which SC104 EAFNasalizedLowRounding later rounds; the outcomes of **i* and **u* are written **ī* and **ū*, their nasality having no further consequence (Campbell 1959, 47, §121).

Two witnesses apply in the present corpus. PGmc **gánsz* 'goose' becomes **gą̄ns* 'goose', and PGmc **júgunθ* 'youth' becomes **júgūnθ* 'youth'. In *ġeoguþ* 'youth' the syllable carrying the lengthened vowel is unstressed, and the length is given up again by the later shortening of unstressed syllables; Sievers and Brunner note the same course in *beraþ* 'they carry' from **beranþi* 'they carry' through **berōþ* 'they carry' (Brunner 1965, 176, §186.1 Anm. 3; Luick 1914--1940, 276, §301.1).

If lengthening is stated after the loss of the nasal, it can no longer read its conditioning environment. The vowel must therefore be adjusted before that nasal is removed. This is an executable dependency within the statement of one historical change, not a date for either component. The earlier experiment's youth output also depended on the retired early glide-insertion architecture; it is not a current historical boundary.

3.19 Nasal-spirant loss

SC027. North Sea Germanic nasal-spirant law, second step (EAFNasalSpirantLoss)

```
define EAFNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarNSGmcSpirant
];
```

The nasal is removed in the environment that conditioned the lengthening, giving **gą̄s* 'goose' and **júgūθ* 'youth'. The rule completes the statement of the single change begun in SC026 EAFNasalSpirantLengthening; the two are not independent sound laws. The converse test,

stating the loss first, merely reproduces the same two wrong forms. Nothing in the material fixes a later boundary for the loss.

3.20 The hiatus-breaking *w* of the verba pura

Historical discussion

The small class of Germanic strong verbs whose roots ended in a vowel, the *verba pura*, reached Northwest Germanic with a morphologically expected hiatus. Ringe and Taylor reconstruct Proto-Germanic **sēanǫ* ‘to sow’ (Gothic *saian*), which the Northwest Germanic lowering of **ē₁* carried to **sāanǫ*; the West Germanic languages then “exhibit innovative consonants that eliminated” the hiatus (Ringe and Taylor 2014, 12). The repair differs by branch, and the difference dates and localizes the change. Old English and Old Frisian inserted *w*, as in Old English *sāwan* ‘to sow’ and Old Frisian *sāwinge* ‘sowing’ beside *grōwinge* ‘growth’, while Old Saxon and Old High German used *j* (*sāian* ‘to sow’, *sāen* ‘to sow’, *sājen* ‘to sow’) (Ringe and Taylor 2014, 12, 151). The insertion of *w* is therefore an Anglo-Frisian development and the consonant of *sāwan* is not inherited; it must not be projected back into the protoform.

The origin of that *w* is neither inherited nor morphological. Þórhallsdóttir, in the standard treatment of the problem, rejects in turn the derivation of the West Germanic *w* from a Proto-Indo-European *u*-perfect (Þórhallsdóttir 1993, 115–17), from a **ue/o*-present (Þórhallsdóttir 1993, 117–19), and from the transfer of a preterite *w* into the present (Þórhallsdóttir 1993, 121–22). The *w* “rather has an inner-Germanic phonological explanation” (Þórhallsdóttir 1993, 136). It began as an ordinary phonetic glide filling the hiatus, and it began in one narrow environment, before **u*. The cells in question, as Bremer had already seen, are the first singular and first plural of the present indicative together with the plural of the reduplicated preterite in **-um*, **-uþ*, **-un* (Þórhallsdóttir 1993, 120). Þórhallsdóttir sets out the pre-Old English present indicative with the hiatus still intact, first singular **sā’u*, second **sā’is*, third **sā’iþ*, first plural **sā’um*, second plural **sā’iþ*, third plural **sā’anþ*, and observes that “the glide *w* was a natural hiatus filler before the *u* of the endings of the 1st person singular and plural” (Þórhallsdóttir 1993, 127). The preterite stem **seuwun* arose by the same insertion before the **u* of the plural endings (Þórhallsdóttir 1993, 126).

From those few cells the *w* was carried through the whole paradigm by analogy. Three forces converge. Levelling proceeded out of the **u*-cells themselves; the preterite stem **seuw*, **seuwun* “would have assisted in extending the new *w* to all forms of the present paradigm” and to the preterite participle; and verbs with an etymological **w*, above all *flōwan* ‘to flow’ from the root **pleu-*, exerted their own attraction (Þórhallsdóttir 1993, 127). The result, complete already in pre-Old English, is the uniform pattern **sāwan*, **seuw*, **seuwun*, **sāwan-* (Þórhallsdóttir 1993, 136). Old Frisian inherits the same generalized type, so that its *w* and the Old English *w* are one Anglo-Frisian event (Þórhallsdóttir 1993, 130, 134). The scattered *w*-forms of Old High German are a separate and later inner-German development and are not to be identified with it (Þórhallsdóttir 1993, 119, 123).

This layering matters for how the rule below should be read. Old English citation forms are infinitives, and the infinitive **sā’an* is precisely one of the cells that never received *w* by the phonological rule, so the *w* of *sāwan* ‘to sow’ is analogical. The rule therefore models the outcome of the generalization, and its environment, any vowel after *ā*, is a citation-form proxy for a paradigm-wide result. It is not a reconstruction of the conditioning of the original insertion.

The chronology is fixed on both sides by Ringe and Taylor. The *w* must postdate the West Germanic loss of intervocalic **w*, or the new glide would itself have been swept away (Ringe and Taylor 2014, 151, n. 9); and it “must have occurred early enough to prevent fronting of **ā*” in pre-Old English (Ringe and Taylor 2014, 151). The whole reason *sāwan* ‘to sow’, *cnāwan* ‘to know’, *blāwan* ‘to blow’, and *māwan* ‘to mow’ keep their back vowel is that the *w* was already in place when the North Sea Germanic fronting applied. Since it is the generalized *w* that stands in the infinitive, this later boundary constrains the generalization itself. The rule accordingly stands before the fronting in the cascade, even though on the present corpus the two orders happen to produce the same outputs, since the fronting rule as implemented does not touch a prevocalic

ā in any case; the ordering encodes the historical chronology, and no corpus-internal contrast demonstrates it. That chronology rests on Ringe and Taylor alone, for Þórhallsdóttir's chapter nowhere discusses the fronting of **ā* and supplies no independent evidence for the order.

SC102. Generalized hiatus-breaking *w* after long *ā* (EAFHiatusWInsertion)

```
define EAFHiatusWInsertion [
    [..] -> {*w} || {*ā} _ EnglishStarVocalic
];
```

The rule is fed by SC024 PNWGmcLongELowering, which creates the *ā*-initial hiatus it repairs; **sḗanǫ* 'to sow' passes through **sāanǫ* to **sāwanǫ* on its way to *sāwan* 'to sow'. Displaced before the lowering, the rule can never apply, since the root vowel is still **ē* and no hiatus after *ā* exists, and the derivation loses its consonant altogether.

Its output in turn feeds the blocking environment of SC101 EAFLongAFronting: the inserted *w* is precisely what shields the *ā* of *sāwan* 'to sow' from fronting, as treated in that chapter. The corpus carries *sāwan* as the diagnostic witness of the class; *cnāwan* 'to know', *blāwan* 'to blow', *māwan* 'to mow', *wāwan* 'to blow (of wind)', and *þrāwan* 'to turn' instantiate the same derivation (Ringe and Taylor 2014, 151).

3.21 North Sea Germanic nasalization of long *ā* before a nasal

Historical discussion

In the dialects along the North Sea coast the long low vowel **ā* — centrally the vowel produced from **ē₁* by the Northwest Germanic lowering — was nasalized when a nasal consonant followed and survived. The nasalized vowel was afterwards rounded in Anglo-Frisian, which is why Old English has *mōna* 'moon', *mōnaþ* 'month' and *spōn* 'spoon' against Old High German *māno* 'moon', *mānōd* 'month', *spān* 'spoon' and Old Norse *máni* 'moon', *mánaðr* 'month', *spánn* 'spoon'. Campbell describes the split of Germanic *ǣ¹* before nasals in these terms and identifies the vowel that the rounding operated on as a nasalized and unrounded **ą̄* [Campbell (1959), p. 50, §127; p. 50, §128 n. 1]. Fulk places the rounded outcome of **ǣ* before nasals among the Anglo-Frisian changes and derives it from an earlier nasalized vowel (Fulk 2018, 72–73, §4.12).

The nasalization and the rounding have different geographies, and separating them resolves an apparent disagreement in the handbooks. Ringe and Taylor state that stressed low vowels were nasalized in the northern West Germanic dialects, Old Saxon among them (Ringe and Taylor 2014, 142, §5.1.2); Old Saxon accordingly shows *ōdar* 'other' and *sōd* 'true' beside unrounded *quān* 'wife' and *sāno* 'immediately' (Ringe and Taylor 2014, 150–51). What Old Saxon shares is the nasalization; what it shares only in part is the rounding, and for the vowel inherited from Proto-Germanic it does not share the rounding at all (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). The nasalization stated here is therefore North Sea Germanic, and the rounding treated in the chapter on the long nasalized low vowel is Anglo-Frisian.

The nasalization is the conditioned counterpart of the fronting of oral **ā* treated in the fronting chapter: a following nasal gives nasalization and eventual rounding, and its absence gives fronting. Together the two exhaust the fate of the old low vowel in this area. The comparative material does not order the two branches against each other, and no such ordering is claimed here.

SC025. Nasalization of long *ā* before nasals (EAFLongANasalRounding)

```
define EAFLongANasalRounding [
    {*ā} -> {*ā} || _ EnglishStarNasal
];
```

The rule consumes the **ā* created by SC024 PNWGmcLongELowering: displacing the lowering after this rule leaves the nasalization without an input, and **mḗnōθz* ‘month’ surfaces as †*mānaþ* in place of OE *mōnaþ* ‘month’, **spḗnuz* ‘spoon’ as †*spān* in place of *spōn* ‘spoon’. Its output is consumed in turn by SC104 EAFNasalizedLowRounding, which supplies the rounded vowel that the two words actually show.

Equally important is what the rule must precede. The monophthongization of **ai* in SC004 EAFAi-Monophthongization creates a new long *ā*, and that vowel was never nasalized and never rounded before nasals: *stān* ‘stone’ and *hām* ‘home’ keep *ā*. Stating the monophthongization before this rule makes the new vowel eligible for nasalization and hence for rounding, and the cascade then wrongly yields †*stōn* and †*hōm*. This is the same chronological inference Campbell draws for the fronting: the treatments of the old low vowel were complete, or at least under way, before *ai*-monophthongization supplied a new one (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70).

3.22 Anglo-Frisian rounding of the long nasalized low vowel

Historical discussion

Three separate developments described in earlier chapters end in the same sound: a long, nasalized, low vowel. The oldest is the Proto-Germanic loss of a nasal before **x*; the second is the North Sea Germanic nasal-spirant law; the third is the nasalization of the inherited long *ā* before a nasal that survived. In Old English and Old Frisian all three surface as *ō*, and Campbell states the unification without qualification: the nasalized vowel became identical with the *ō* inherited from Proto-Germanic already in prehistoric Old English, and the same change affected the nasalized vowel of the Proto-Germanic law and the nasalized vowel of the Ingvaeonic law at one and the same time (Campbell 1959, 50, §128 n. 1). One later sound change therefore accounts for all three, and the appearance of three independent roads to *ō* is an illusion created by looking only at the Old English surface.

That the intermediate vowel was nasalized and unrounded is shown by a form in which it was shortened very early. Campbell’s example is *samcucu* ‘half alive’, where the shortened reflex is *a* and never *o*; the vowel that the earlier laws produced was therefore a nasalized **ą̄*, and the rounding is a distinct and later event (Campbell 1959, 50, §128 n. 1). Sievers and Brunner describe the same nasalized long vowel and its rounded Old English outcome [Brunner (1965), p. 33, §26; p. 58, §64].

The geography is Anglo-Frisian. Old Saxon nasalized its stressed low vowels along with the rest of the northern West Germanic area (Ringe and Taylor 2014, 142, §5.1.2), and it shares the loss of the nasal in both of the earlier laws; the categorical, systematic rounding, however, it does not share. For the Proto-Germanic nasalized low vowel Old Saxon retains *ā* consistently, and for the vowel created by the nasal-spirant law it has *ā* or *ō* according to word and dialect, while Old English and Old Frisian have *ō* throughout (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). That partial and lexically variable Old Saxon rounding is comparative evidence bearing on the innovation rather than participation in it. Ringe treats the rounding as a parallel development of the diverging northern dialects and locates it in the northernmost of them, that is in Anglo-Frisian (Ringe 2017, 149–50, §3.2.7; Ringe and Taylor 2014, 142, §5.1.2). Luick groups the whole set of changes among those peculiar to the Anglo-Frisian dialect group and notes that the nasalized vowel later gave up its nasality (Luick 1914--1940, 276, §301.1). The rounding is accordingly Anglo-Frisian, and the nasalization that feeds it is the wider North Sea Germanic property.

Ringe and Taylor confirm that a single rounding covers all the sources: the rounding affected

the nasalized low vowels of the nasal-spirant law, the nasalized low vowels of the accompanying list, and the reflexes of Proto-Germanic **/anh/* alike (Ringe and Taylor 2014, 142, §5.1.2). The condition on the input is nasality and nothing else, which is why the long *ā* that Old English later won from **ai* escapes: that vowel was oral, and it came into being after the rounding had run its course. Fulk makes this the central argument for the long survival of the nasality, since the nasalized vowel developed to *ō* and did not fall together with Old English *ā* from **ai* (Fulk 2018, 55, §4.1).

SC104. Rounding of the long nasalized low vowel (EAFNasalizedLowRounding)

```
define EAFNasalizedLowRounding [
    {*ã} -> {*õ}
];
```

The rule is unconditioned, since the vowel it operates on exists only where one of the three earlier changes created it. Its inputs arrive from SC103 PGmcNasalLossBeforeX, from SC026 EAFNasalSpirantLengthening and from SC025 EAFLongANasalRounding, and each of those three stands in a feeding relation to it: stated before any one of them, the rule leaves that source's nasalized vowel untouched and the cascade returns no form at all for its witnesses.

Four lexemes in the present corpus reach Old English through this rule, and between them they witness all three sources. **gánsz* 'goose' arrives as **gą̄s* 'goose' from the North Sea Germanic law and gives *gōs* 'goose'; **mḗnōθz* 'month' arrives as **mą̄nōþ* 'month' and gives *mōnaþ* 'month'; **spḗnuz* 'spoon' arrives as **spą̄nu* 'spoon' and gives *spōn* 'spoon'. The Proto-Germanic law contributes **θánxtē* 'thought', the preterite of the verb 'to think', which arrives as **θą̄xtē* 'thought' and gives *þōhte* 'thought'. That is the very form cited for this rounding, beside Old Frisian *thochte*, and it is the evidence that the vowel did not fall together with the *ā* of *stān* 'stone' (Fulk 2018, 55, §4.1; Campbell 1959, 44, §119). The other firing of the Proto-Germanic law, the high vowel of *fȳst* 'fist', does not reach this rule at all.

The counterpart is what the rule leaves alone. **stáinaz* 'stone' and **xáimaz* 'home' acquire their long *ā* from SC004 EAFAiMonophthongization, which follows this rule, and their vowel was never nasalized; they surface as *stān* 'stone' and *hām* 'home'. Placing the monophthongization before the nasalization and the rounding makes that vowel eligible and the cascade then yields †*stōn* and †*hōm*, which is the chronological inference Campbell draws for the treatments of the old low vowel generally (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70).

3.23 North Sea Germanic fronting of long *ā*

Historical discussion

Long after the Northwest Germanic lowering of **ē₁* to **ā*, the dialects of the North Sea coast fronted the surviving oral **ā* to a low front vowel: West Saxon *ǣ* in *dǣd* 'deed', *slǣpan* 'to sleep', *lǣtan* 'to let', *rǣdan* 'to read', and *mǣl* 'meal', against Anglian, Kentish, and Old Frisian *ē* (*dēd* 'deed', *slēpa* 'to sleep', *jēr* 'year') (Ringe and Taylor 2014, 146–50; Campbell 1959, 50–51, §128). Old Saxon and Old High German keep the back vowel (Old Saxon *dād* 'deed', Old High German *tāt* 'deed', *slāfan* 'to sleep', *lāzan* 'to let'), but sporadic Old Saxon spellings in ⟨e⟩ show that the fronting lapped unevenly into Old Saxon territory (Ringe and Taylor 2014, 150). The change is therefore a North Sea Germanic development rather than an exclusively Anglo-Frisian one, though only Old English and Old Frisian carry it through systematically.

The fronting affected stressed, non-nasalized **ā*; nasalized **ą̄* was instead rounded, as treated in the rounding chapter. Ringe and Taylor further establish a conditioning by a following **w*. Before **w* plus a back or non-high vowel the **ā* was retained: the clearest witnesses are the *verba pura* — *sāwan* 'to sow', *cnāwan* 'to know', *blāwan* 'to blow', *māwan* 'to mow', *þrāwan* 'to turn' — together with *clāwu* 'claw' (Ringe and Taylor 2014, 151); the simplest hypothesis, which Ringe and Taylor adopt from Hogg, is that fronting never occurred in that environment (Ringe and Taylor 2014,

151; Hogg 1992a, 81). Before **w* plus a high front vocalic, by contrast, the fronting did apply: **lēwijaną* ‘to betray’ (Gothic *lēwjan* ‘to betray’, Old High German *gilāen* ‘to betray’) yields West Saxon *lǣwan* ‘to betray’, and the same environment appears in *eltǣwe* ‘entire’ and *brǣw* ‘eyelid’ (Ringe and Taylor 2014, 150). Both sides of the condition are encoded in the rule below and witnessed in the corpus. Bennett states the retaining environment independently from the West Saxon evidence alone, and slightly more broadly: West Saxon “shows *ǣ* as a regular isolative development of IE *ē* but has *ā* before *w* or *g* plus a back vowel”, with the paradigmatic alternation *wǣg* ‘wave’ beside plural *wāgas* ‘waves’ (Bennett 1950, 235, n. 6). The rule below encodes the *w* environment, for which the corpus supplies witnesses on both sides; the parallel *g* environment is left for separate treatment.

Whether this fronting restored a front vowel that had earlier been backed, or whether — as Fulk argues — the North Sea dialects simply retained an old front **ǣ* that was never backed at all, is the same dispute recorded in the lowering chapter (Fulk 2018, 60–61, §4.6; Campbell 1959, 50–51, §§128–129). On the retention analysis this chapter’s change dissolves into the non-event of staying put; the present model follows Ringe and Taylor’s two-step reconstruction, on the strength of the runic evidence for an early [aː] and the place-adverbs *þǣr* ‘there’ and *hwǣr* ‘where’ (Ringe and Taylor 2014, 13–14).

SC101. Fronting of long *ā* outside nasal and *w* environments (EAFLongAFronting)

```
define EAFLongAFronting [
    {*ā} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal - {*w}],
    {*ā} -> {*ǣ} || _ {*w} EnglishIUmlautTrigger
];
```

The first clause fronts **ā* before any oral consonant other than **w*; the second admits the fronting before **w* exactly when a high front vocalic follows. The two corpus witnesses of the *w* condition form a minimal contrast. **sḗaną* ‘to sow’, whose hiatus-filling *w* is supplied by SC102 EAFHiatusWInsertion, reaches this rule as **sāwaną* and is left unfronted, surfacing as *sāwan* ‘to sow’; were the *w*-block removed, the cascade would deliver †*sǣwan* instead. **lḗwijaną* ‘to betray’, whose inherited *w* is followed by **i*, is fronted by the second clause to **lǣwijaną* at this rule’s own stage — before, and independently of, the much later i-umlaut — and surfaces as *lǣwan* ‘to betray’. The rule consumes the **ā* created by SC024 PNWGmcLongELowering; displaced before the lowering, it has nothing to front, and **skḗpą* ‘sheep’ surfaces as †*sċāp* rather than OE *sċēap* ‘sheep’, **jḗrą* ‘year’ as †*ġār* rather than *ġēar* ‘year’, **slḗpaną* ‘to sleep’ as †*slāpan* rather than *slǣpan* ‘to sleep’.

Two later boundaries carry real historical content. First, the fronting must precede the completion of SC004 EAFAiMonophthongization: the *ā* that arose from **ai* was never fronted — *stān* ‘stone’, *hām* ‘home’, *lāþ* ‘hostile’, *rāp* ‘rope’, *tācn* ‘token’, *gāst* ‘spirit’ all keep the back vowel. Campbell draws exactly this chronological inference (Campbell 1959, 52–53, §132), and Ringe and Taylor endorse it as cogent (Ringe and Taylor 2014, 169–70); in the present cascade the inference is enforced by rule order, and displacing the fronting after the monophthongization wrongly yields †*lǣþ*, †*rǣp*, †*tǣcn*, †*sǣwol*, and †*ġēast*. Historically the two changes may well have overlapped in time; the discrete ordering is the grammar’s way of stating that inherited *ā* had been fronted before the new *ā* arose.

Second, the fronting must precede SC056 OEWsPalatalDiphthongization, which operated on the already-fronted vowel: *ǣ* > *ēa* after the palatals, as in *sċēap* ‘sheep’ and *ġēar* ‘year’ (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1). Displaced after the diphthongization, the cascade yields †*sċǣp* and †*ġǣr* instead.

Chapter 4

From Anglo-Frisian to Old English

4.1 Historical interval

This chapter follows the English daughter from the required Anglo-Frisian ancestor through prehistoric English to attested Old English. It includes changes earlier than the manuscript period, not merely changes within it. West Saxon is the principal target; each comparison still requires its actual dialect, paradigm cell and evidential status.

4.2 Scope and dialect variation

Not every change in this chapter has pan-Old-English scope. Some changes — most notably ordinary West Saxon palatal diphthongization and parts of the back-mutation history — are specifically West Saxon or more broadly southern Old English phenomena. The diphthong corridor also represents earlier West Germanic feeders; a West Saxon final outcome does not date every feeder to West Saxon (Ringe and Taylor 2014, 41–42, 65–66, 171–73, 215–17). (The West Saxon palatal-glide spellings, SC016, belong to the written surface of Old English and are treated in Chapter 5.)

The CAPR derivations target West Saxon Old English citation forms as the default comparator. Changes that belong to other dialects, or that are absent from West Saxon, may appear in lexical entries as comparanda rather than as derivational steps.

The distinction between West Saxon and Anglian comparanda requires the actual text and dialect, not merely a regional label. Hogg discusses the limits of the traditional divisions and the evidence for the principal textual varieties (Hogg 1992a, 3–8, §§1.5–1.12).

4.3 Chapter structure

The changes in this chapter fall into several natural historical subgroups, though the boundaries between them are not always sharp:

Prehistoric English contraction and fronting: Completed stressed **ai* > **ā* is now adopted on the English daughter in SC004 EAFAiMonophthongization. The new long vowel is distinguished from inherited oral long vowels; the source milestones allow overlap with their earlier restructuring (Campbell 1939, 90–91; Ringe and Taylor 2014, 170–71). This placement leaves possible conditioned ancestral onset and the runic interpretation qualifications explicit (Versloot 2017, 295–97, 318).

Early Old English changes linked to the Anglo-Frisian inheritance: Changes that feed directly on, or are closely related to, Anglo-Frisian brightening (SC043), whose section follows the earlier vowel developments at its adopted executable position in this chapter. The **awj* resolution, **au*

fronting and diphthong operations currently execute before ordinary-a brightening. They cannot be described as consuming its output. Their histories combine earlier glide reanalysis with English diphthong realization (Campbell 1959, 44–47; Ringe and Taylor 2014, 65–66, 171–73). The conventional breaking/restoration account instead presupposes fronted **æ*. English plain-a/au process identity and inherited stem identity remain separate questions.

A coordinated formalization of ordinary **a* fronting and **au* fronting with its completion has now been tested over the selected lexical material. Its twenty au paths wait until the ordinary fronting corridor and then converge with the current derivations before breaking. This establishes computational compatibility, not historical event identity. The production serialization remains unchanged, and the unstressed and surviving-long-final fronting components are not dated from this test. The completed ordinary stressed component is now independently characterized on the English daughter. Its English-episode interpretation remains explicitly dependent on its contraction and nucleus premises (Campbell 1939, 90–91; 1959, 52; Ringe and Taylor 2014, 170–75).

The earlier unstressed contribution is not the complete unstressed law: SC070 OEUnstressedFrontingEarly separately implements fronting before heterosyllabic nasals, whereas the coda-nasal history protects other endings (Campbell 1959, 140–41). The retained final-vowel helper connects unrounding's carried quantity to later shortening and merger; it does not establish an independent historical long-a fronting (Ringe and Taylor 2014, 58–59, 299–300).

Old English consonantal changes: Velar palatalization (SC052), palatalization of `*sk` (SC051), j-cluster coalescence (SC057), and related changes produce the characteristically Old English consonant phonemes. Hogg discusses these as OE consonant changes with class-specific conditions and chronology (Hogg 1979, 90–111).

Old English i-umlaut and its context: The i-umlaut changes vowels under a following high front vocoid; paradigm alternations are evidence for that phonological conditioning, not grammatical conditions on the law (Campbell 1959, 69–72, §§190–197; Ringe and Taylor 2014, 222). Ordinary West Saxon palatal diphthongization precedes mutation in the handbook account, while the later treatment of some mutation products after *sċ* follows it (Fulk 2018, 74; Campbell 1959, 68–69; Ringe and Taylor 2014, 215–17, 235). The adopted cascade now places the ordinary component before mutation and retains the late approximation afterward. Gift enters with the earlier-raised vowel, following Ringe's explicitly PGmc reconstruction (Ringe 2017, 135, pp. 151–153). Its lexical section discusses the published e-alternatives and the confidence in this choice.

Late Old English syllabic reduction and apocope: High-vowel apocope (SC063), medial syncope (SC065), and the cluster of late unstressed-vowel changes (SC069–SC078) represent the later stage of Old English phonological history, when the syllabic structure of the language began to shift toward the more reduced profile of Middle English.

4.4 The English partial chains

The working historical scaffold has several connected chains, not one uniquely demonstrated total order. Their evidential strength differs. An explicit handbook derivation, a contrast between original and secondary vowels, and the survival of a modeled output are not interchangeable arguments (Ringe and Taylor 2014, 170–73, 215–37; Hogg 1979, 100–110).

Historical relation	Evidence and interpretation	Limitation
Coronal assimilation before inherited ww reanalysis	Four and the second-person pronoun have assimilated inputs to vocalization	Does not date every j-created geminate
Inherited-long restructuring before completed ai contraction	New ai-derived long a escapes inherited-long fronting	Temporal overlap is possible
Completed ai contraction before ordinary short fronting	Conventional Campbell chain	Depends on the diphthong-nucleus premise

Historical relation	Evidence and interpretation	Limitation
Ordinary fronting before breaking/restoration	Conventional English reconstruction	Never-fronted and restored outputs can coincide
Breaking before ordinary palatal diphthongization before mutation	Broken inputs and worked handbook vowel histories	Does not date every consonantal palatal layer
Mutation before the later sc treatment	Ai-derived mutation products such as sheath	Exact later conditioner remains unresolved
Initial productive palatal cutoff before mutation-created fronts	Unrounded mutation vowels are the stronger controls	Medial fricative merger is a separate problem

The first three relations are supported and qualified by the earlier vowel accounts (Ringe and Taylor 2014, 41–42, 65–66, 170–73; Campbell 1939, 90–91). The palatal and mutation relations require their own evidence (Luick 1914--1940, 162–63; Fulk 2018, 74; Ringe and Taylor 2014, 215–17, 235; Hogg 1979, 100–110). The table states the defended historical targets; it does not claim that every present executable proxy already realizes them faithfully.

4.5 Early glide history and the English outcome

The diphthong corridor includes inherited ww, assimilated ww and geminates created later before j. These inputs must be distinguished before assigning one date to their long-vowel outputs. Ringe and Taylor describe inherited glide reanalysis separately from the English realization of its products (Ringe and Taylor 2014, 41–42, 65–66, 171–73). The model now separates the earlier reanalysis from English realization.

Witness	Adopted earlier checkpoint	Later English outcome	What it tests
Four	Assimilated ww is reanalyzed as eu plus w	fēower ‘four’	Assimilation feeder and retained glide
Hew	Inherited a plus ww gives au plus w	hēawan ‘hew’	Reanalysis distinct from English realization
You	Vocalized iu plus w with a surviving final i before apocope	ēow ‘you’	Prosody/apocope compatibility, not merely quantity
Hue	Later j-gemination creates a distinct glide input	hīew ‘hue’	Negative control for the earlier inherited-ww subset
Hay	Later awj history remains separate	hīeġ ‘hay’	Gemination and secondary glide resolution

These are existing lexical witnesses, not new corpus admissions. The current spelling differences also require their actual dialectal interpretation (Campbell 1959, 44–47; Ringe and Taylor 2014, 41–42, 57–58, 171–73). The reconstructed West Saxon strew target is a computational control, not an additional attested form.

The completed decomposition replaces the old consonantal-ww apocope proxy with the historical condition: a short final high vowel after a heavy syllable in a sentence-unstressed, phonologically final word. You selects that weak-final context independently of its unchanged PGmc reconstruction. Thus **iuwi* loses the final vowel to give **iuw* before mutation. A proclitic context retains the vowel, as in *ymbe* ‘around’; weak-final *and* ‘and’ and stressed *ġiest* ‘guest’ / *fȳr* ‘fire’

check the positive and negative conditions (Ringe and Taylor 2014, 41–42, 55, 57–58; Campbell 1959, 283).

The computational retained-i counterfactual for you yields a predicted †*īei*, not a source-backed strong Old English spelling: the existing w-loss before i intervenes before mutation. It tests the represented alternative context, not an additional attestation. Unmarked evaluation explicitly selects strong-final citation context; absence of an acute is never used to infer sentence stress. All selected final outputs remain unchanged.

The later realization paths are now explicit. Chew, dew, four, hew and you complete their earlier products through SC032 OEDiphthongLeveling. SC033 OEEwLongDiphthong supplies the separate j-created offglide reanalysis for *hīew* 'hue'. The selected input is inherited PGmc **xíwją*; its path is **xíwwją* to **xíuwją*, then long **xīowją* through SC110 OEJGlideIO, before mutation gives the West Saxon long *īe* reflex. Ringe and Taylor explicitly give the inherited i and long-io chain while leaving its phonemic analysis unclear (Ringe and Taylor 2014, 53, 250). Orel's and Kroonen's e reconstructions are real alternatives, and Fulk disputes traditional geminate dismantling; these qualifications are discussed in the lexical account (Orel 2003, 171–72; Kroonen 2013, 224; Fulk 2018, 59, 71–72). Knee instead selects the regular dative *cneowe* 'knee (dat.sg.)': its short pre-ending diphthong comes from SC044 OEBreaking, not singleton lengthening. The long endingless *cnēo* 'knee' and analogically restored final *w* in *cnēow* 'knee' remain a separate paradigm comparison; later long obliques must not be confused with the selected short cell (Campbell 1959, 232–33; Hogg 2011, 21–22, 86; Ringe and Taylor 2014, 187–88, 387). SC111 PWGmcEndinglessAw now supplies straw's regular endingless **strau*, feeding ordinary English realization to reconstructed **strēa*. The comparison remains attested *strēaw* 'straw': its restored final *w* belongs to paradigm levelling, so this regular derivation explicitly mismatches the target. The sources mark the free regular simplex and short obliques as unattested; the compound *strēaberige* 'strawberry' is not substituted for the selected noun (Campbell 1959, 232–33; Brunner 1965, 49, 110; Ringe and Taylor 2014, 172–73, 387). SC034 OEAwLongDiphthong retains only the three show cells as a disclosed proxy. Dictionary singleton stems and the inherited-geminate handbook analysis remain distinct, unresolved premises (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49). The unchanged SC106 OEJWWSimplification remains a technical compatibility operation, but hue now reaches it with singleton w and no longer feeds it. The before-j placement of early reanalysis is a representative serialization, not a new strict historical chronology claim (Campbell 1959, 45–47; Ringe and Taylor 2014, 65–66, 171–75).

4.6 Ordinary fronting, breaking and restoration

The preferred conventional account identifies ordinary non-nasal short-a fronting and fronting of au's first element as one English episode. The later offglide development is distinct. Early spellings record the fronted element, but do not measure the simultaneity of two laws (Campbell 1939, 91; 1959, 52–53; Luick 1914--1940, 130–31). English episode identity is not inherited Anglo-Frisian event identity. Nor does it identify the stressed-short process with every unstressed or long-final clause in the present model.

Breaking then has vowel, quantity, consonantal and dialectal conditions. Restoration in a following back-vowel environment can remove a previously fronted vowel. A final restored vowel alone cannot distinguish that history from original blocking of fronting (Hogg 1979, 90–97; Versloot 2025, 104, 123). The conventional account is retained as the working baseline because of the combined argument, not because the more elaborate intermediate path is inherently preferable.

Versloot's late Anglian alternatives must be compared in their exact domains. His mutation-created e argument and the unbroken neighboring environments are relevant to a restricted e-breaking component, not to a wholesale postmutation relocation of every West Saxon breaking law (Versloot 2025, 126–28, 131–35). A different dialectal target cannot be silently substituted for a West Saxon lexical target. Similarly, Frisian closed-syllable breaking is not English rC breaking merely because both are called breaking (Bremmer 2009, 33–35, 37).

4.7 Palatal consonants and the key–day problem

The useful chronological contrast is between original conditioning fronts and front vowels created by mutation. Rounded mutation products alone may support a weaker conclusion, a cutoff before later unrounding. Unrounded mutation-created fronts more directly test whether initial velar palatalization was still productive (Hogg 1979, 100–103; Laker 2007, 167–68). This cutoff does not by itself date assibilation or every earlier articulatory tendency.

Hogg's key and day paradigms identify the disputed merger premise. The initial key consonant survives a mutation-created unrounded front vowel; the day paradigm does not show the mutation that would follow if its new medial palatal element were already equivalent to inherited j. These statements concern particular historical cells, not arbitrary exceptions to a sound law (Hogg 1979, 102–10).

Diagnostic	Required distinction	What is not established
Key oblique vowel	Original fronts versus mutation-created fronts	Date of all palatal articulation
Day oblique vowel	Inherited j versus a palatal fricative at mutation	Unanimous agreement on merger timing
Geminate/postnasal g	Stop-class history versus singleton fricative	One universal g-to-j change
h	Breaking trigger versus subsequent palatal/weakening history	Identity with voiced-fricative chronology
sk	Own consonantal law versus later sc vowel treatment	A single combined palatalization event

Hogg considers several resolutions without endorsing one as demonstrated (Hogg 1979, 103–11). Ringe and Taylor, however, explicitly adopt merger after mutation (Ringe and Taylor 2014, 204). CAPR now implements that regular working account, preserving Hogg's phonetic objection rather than claiming consensus. Singleton g is a fricative, including initially; gg/ng are stops (Fulk 2018, 130–32).

At Hogg's pre-palatal, pre-OE checkpoint, key has **kājæ* and day **dæɣæ*. The inherited j in the former causes mutation, producing **kæjæ* without reactivating initial-k palatalization. In the latter, the newly palatal fricative remains ɟ and the vowel stays æ: **dæɟæ*. Native realization then gives *cǣġe* 'key' and *dæġe* 'day' (Hogg 1979, 105; Ringe and Taylor 2014, 204). Premature merger instead supplies a j trigger and wrongly raises the day vowel. These oblique cells are non-corpus diagnostics, not new selected PGmc reconstructions; the existing day nominative cannot substitute for them.

The complete key suffix exposed a second defect: unrestricted inherited-j normalization would erase its retained glide. The repaired SC082 OEIntervocalicJVocalization excludes non-high long front monophthongs. Its residual weak-suffix normalization is a telescoped representation, not a universal historical VjV vocalization law (Hogg 2011, 283–86; Ringe and Taylor 2014, 228).

The fricative merger has its own visible SC109 OEPalatalFricativeMerger. Its late execution after suffix raising is a tested computational holding zone, not a precisely proved historical date. Likewise stop ʧ/ʤ are eventual-reflex proxies, not early affricate claims. Cluster coalescence and a dotted written reflex cannot identify all these classes as one historical sound (Ringe and Taylor 2014, 203–4; Laker 2007, 167–68; Fulk 2018, 131–32).

4.8 Three different mutation and diphthongization arguments

Guest, sheath and gift do not establish the same chronological relation. The guest derivation passes through ordinary palatal diphthongization before mutation in both the handbook account

and the adopted model. The former model raised a simple vowel before diphthongizing it. Both paths reach *ġiest* ‘guest’, so their final agreement does not select the earlier path (Ringe and Taylor 2014, 216).

The following worked comparison abstracts from consonant notation and retains the vocalic trigger where it matters. For guest, the source starts with PGmc **gastiz* ‘guest’, continued as PWGmc **gasti* ‘guest’. The vowel history is **a* > **æ* > **ea* > **ie*; final loss of the trigger follows the mutation it caused (Ringe and Taylor 2014, 216, 287).

Checkpoint	Handbook and adopted guest path	Superseded modeled guest path
Ordinary fronting	Short a becomes æ	Short a becomes æ
After palatal conditioning is established	Original front vowel is available to ordinary PD	Original front vowel is available, but PD remains later
First disputed vowel operation	æ becomes ea by ordinary PD	æ becomes e at mutation
Second disputed vowel operation	ea becomes ie at mutation	e becomes ie in the bundled PD proxy
Written result	ġiest ‘guest’	ġiest ‘guest’

The agreement in the last row is precisely why the intermediate rows must be tested. This table does not treat the implementation’s palatal-consonant symbol as proof of a uniquely dated phonetic merger (Hogg 1979, 103–10).

Witness	Source-supported distinction	Present model consequence
Guest	Ordinary palatal diphthongization feeds mutation	Adopted æ > ea > ie replaces the output-equivalent old path
Sheath	Ai-derived ā mutates; a later sc layer can affect its product	Its unchanged late clause is separately retained as an approximation
Gift	Earlier inherited e-to-i raising is distinct from English mutation	Selected PGmc i is outside ordinary diphthongization
Sheep/year	Ordinary palatal diphthongization without a mutation trigger	Controls for the ordinary component
Cow/lung	Mutation changes the productive consonantal environment	Controls for cutoff, not diphthongal mutation

For sheath, *sċēaþ* ‘sheath’ beside *sċēþ* ‘sheath’ belongs to the separately described later group. Reporting both outcomes does not authorize lexical optionality in the law; the exact phonological layer and conditioner still require a defended specification (Campbell 1959, 68–69; Ringe and Taylor 2014, 235). The incremental repair preserves the literal late clause as a visible approximation, not a lexical exception or a newly established conditioner.

The sheath comparison instead follows the ai-derived long vowel: **ai* > **ā* > **ǣ*, with the final step supplied by mutation. The separately discussed later sc treatment can then give **ēa*. The vowel available to the earlier ordinary process and the vowel created later by mutation are therefore different historical inputs, which the adopted model now separates (Campbell 1959, 68–69; Ringe and Taylor 2014, 235).

For gift, Ringe reconstructs PGmc **giftiz* ‘gift’, with the Old English plural *ġifta* ‘wedding’, and describes the earlier raising separately (Ringe 2017, 135, pp. 151–153). Ringe and Taylor explicitly

distinguish that early change from the much later English vowel history (Ringe and Taylor 2014, 220). The selected input now follows that earlier-raised reconstruction. The choice is lexical and source-led, not a new raising rule justified by output fit. Orel and Kluge–Seebold print e-vowel alternatives (Orel 2003, 130; Kluge 2011, 359); the lexical section compares their evidence with the e-grade presentations and Ringe's qualified PGmc dating rather than claiming unanimous agreement.

4.9 Quantity, reduction and the written surface

Later apocope and syncope depend on prosody and syllable structure; earlier input conventions and surviving triggers must be inspected before their effects are used as dates for other changes. Morphological cells provide evidence for such environments, not grammatical conditions on phonological rules (Ringe and Taylor 2014, 57–59, 299–300). Inherited quantity, compensatory lengthening, diphthong quantity and later shortening must likewise remain separate dimensions.

The transition to Chapter 5 is a transition to the written surface, not proof that an orthographic change is a newly dated sound change. A single spelling can represent different historical consonant inputs, and a changed spelling need not change the phonetic derivation. The source discussions and worked lexical entries distinguish implemented laws from working reconstructions and localized unresolved components.

4.10 Executable serialization and unresolved interfaces

Sections follow executable order to make derivations inspectable, not because that serialization proves a unique total history. Final agreement can conceal different intermediate paths. Some proxies execute here with earlier historical classifications; renaming identifiers would not resolve that substantive distinction:

- SC041 (PWGmc Final Bare-*a Loss) and SC042 (Surviving Bimoric *ō Unrounding) carry Proto-West Germanic labels but execute in this stretch of the cascade.
- SC064 (NWGmc *-n Stem *n Loss) carries a Northwest Germanic label but executes after OE High-Vowel Apocope (SC063).
- SC049 (PGmc B Allophony) carries a Proto-Germanic label but executes here.

4.11 Sources

Campbell's *Old English Grammar* is the primary source for the dating and scope of individual changes in this chapter (Campbell 1959, 52–72). Hogg provides critical reassessments of palatal conditioning and its relative chronology (Hogg 1979, 90–112). Ringe and Taylor supply the most detailed relative-chronology analysis for the earlier portion of the chapter, through backmutation (Ringe and Taylor 2014, 169–73, 215–37). Fulk's *Comparative Grammar* provides additional coverage for phonological conditioning (Fulk 2018, 58–61, 72–74). For individual changes, source-specific citations appear in the relevant sound-change sections.

4.12 Awj resolution and the English brightening of au

Two changes stand between the Proto-Germanic diphthong **au* and its Old English reflex *ēa*. The earlier one repairs a West Germanic gemination and restores a diphthong that the gemination had obscured. The later one fronts the first element of every **au*, whether inherited or newly created, and belongs with Anglo-Frisian brightening.

The two are separate developments with separate domains. The first concerns a handful of words in which **w* stood before **j*. The second concerns the whole Old English history of **au*, for which *lēaf* 'leaf', *strēam* 'stream' and *brēad* 'bread' are ordinary witnesses, and to which the first change merely adds two more inputs.

4.13 Post-velar w-loss and loss of w before final i

The first rule is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor derive PGmc **singwan* ‘sing’ to Old English *singan* ‘sing’ (Ringe and Taylor 2014, 214, §6.4.2). This comparative evidence establishes the change, although no lexical evidence fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* ‘sea’ from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The first rule is restricted to the **ngw* sequence; the second has a specific lexical witness and defined earlier and later limits.

4.14 I-umlaut and West Saxon palatal diphthongization

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that “the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE” (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* ‘guest’, *giefan* ‘give’, *hierde* ‘shepherd’, and *ieldra* ‘older’ (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: “we come now to a change which is almost as uncontroversial as it is important” (Hogg 1992b, 113). His examples, such as *bryd* ‘bride’, *trymman* ‘strengthen’, *bedd* ‘bed’, *ciest* ‘chest’, and *wiersa* ‘worse’, likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992b, 113–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it “diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)” (Fulk 2018, 74, §4.13). Ringe and Taylor’s examples such as *gieldan* ‘pay’, *scield* ‘shield’, and *scieppan* ‘create’ show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

Two further distinctions are essential to their relative chronology. First, the raising of inherited Germanic **e* to **i* before a following high front vocoid is much earlier than Old English i-umlaut. Ringe reconstructs PGmc **giftiz* ‘gift’, with OE plural *ġifta* ‘wedding’, and discusses that earlier raising separately (Ringe 2017, 135, pp. 151–153). Ringe and Taylor likewise caution that raising of inherited **e* occurred hundreds of years before the Old English changes; a later repetition is rare and doubtful (Ringe and Taylor 2014, 220). The selected gift input now has the earlier-raised vowel. Its lack of ordinary diphthongization therefore does not date the general West Saxon process. Published e-reconstructions remain genuine alternatives (Orel 2003, 130; Kluge 2011, 359); the lexical discussion compares their evidence and explains the attributed working choice.

Second, the ordinary diphthongization of inherited/fronted non-high vowels must be distinguished from the later treatment of some mutation products after *sċ*. Campbell separates that small group from his general front-vowel diphthongization: the group includes *sċēaþ* ‘sheath’

beside *sċēþ* ‘sheath’ (Campbell 1959, 68–69, §§184–185). Ringe and Taylor explicitly place the treatment of the ai-derived mutation product after mutation, giving the same two sheath outcomes (Ringe and Taylor 2014, 235). Their ordinary West Saxon derivations, including *ġiest* ‘guest’ and *ċytel* ‘kettle’, instead pass through palatal diphthongization before mutation (Ringe and Taylor 2014, 215–17, 222).

The historically preferred working account therefore has ordinary palatal diphthongization before Old English mutation, with the separately evidenced later *sċ* treatment after it. The adopted cascade separates these operations: ordinary diphthongization precedes mutation, and the unchanged late approximation follows it. The sources’ alternate spellings do not by themselves supply an exceptionless conditioning law for the later group; its full conditioner remains unresolved.

4.15 English stressed ai contraction

Historical discussion

Inherited stressed **ái* yields **ā* in English. Ringe and Taylor discuss this development separately from the earlier unstressed contraction. The distinction between inherited long vowels and new **ā* requires inherited-long fronting to have been well under way before contraction completed; it does not prove that the two developments could not overlap (Ringe and Taylor 2014, 170–71). Campbell gives the stricter conventional sequence and places contraction before ordinary short-vowel fronting (Campbell 1959, 52–53). Later i-mutation of the new long vowel is a separate conditioned development, not that ordinary fronting (Campbell 1959, 69).

Versloot proposes a wave account of the regional outcomes, but CAPR does not adopt diffusion as a solution to the comparative tree problem. His readings of early English and Frisian inscriptions are relevant evidence independently of that mechanism, conditional on their provenance, etymology and phonetic interpretation (Versloot 2017, 295–97, 318). The required Anglo-Frisian ancestral node is retained. CAPR places completed English contraction on the daughter branch, while leaving a possible earlier conditioned onset on the common stem distinct. This is the defended working placement under the cited runic and comparative premises, not proof against every ancestral onset (Campbell 1939, 90–91; Versloot 2017, 295–97, 318). The inherited-long relation permits temporal overlap; the retained executable order does not assert a strict order between entire events.

The live selected-corpus census has twenty-three applications, all carrying stressed **ái*. Loam’s selected **láimą* ‘loam’ is explicitly a pre-Old-English model input, not an independent Proto-Germanic witness. The raw corpus’s additional roe reconstruction has no attested target and is excluded from that census. The unstressed development **ai* > **ē* is the separate earlier change SC014 PNWGmcUnstressedAiMonophthongization.

SC004. English stressed ai contraction (`EAFAiMonophthongization`)

```
define EAFAiMonophthongization Ctx([
    {*ái} -> {*ā}
]);
```

The soul form fixes the relation to interstress raising. If the monophthongization is delayed until after that change, PGmc **sáiwalō* ‘soul’ yields †*sāwel* rather than expected OE *sāwol* ‘soul’. An earlier placement changes no output. This shows that SC004 EAFAiMonophthongization must come before SC036 OEInterStressRaising in the modeled sequence.

The unstressed development **ai* > **ē* in final and nonfinal syllables is a separate and earlier change; see SC014 PNWGmcUnstressedAiMonophthongization.

4.16 Resolution of awj

Historical discussion of the resolution of **awj*

Old English *hīeġ* ‘hay’ and *strīeġan* ‘strew’ go back to forms in which **w* preceded **j*. West Germanic doubled every consonant except **r* before **j* after a short syllable, and **w* was an ordinary member of that law (Campbell 1959, 167, §407). Proto-Germanic **hawja-* therefore appears as West Germanic **hauuj* (Campbell 1959, 46, §120.2). Campbell writes the sequence as *auj* > *auuj* > *auj*, with the diphthong restored before the Old English developments begin.

Ringe and Taylor reach the same result and explain why it is possible (Ringe and Taylor 2014, 53, §3.1.3). They find the gemination of **wj* clearest where the preceding vowel was **i*, as in Proto-Germanic **niwjaz* ‘new’ and **siwjaną* ‘sew’, which give Old Saxon and Old High German *niuwi* and *siuwen*. Gemination was reversible, since it merged nothing and altered no underlying form, so the sequence Northwest Germanic **awj* to West Germanic **[aw’w’]* to pre-Old English **[auj]* can have run its course and then undone itself. Their derivations give Proto-Germanic **hawja* through **hauj-* to *hīeġ* ‘hay’, and Proto-Germanic **strawjaną* through **straujan* to Anglian *strēgan* ‘strew’ (Ringe and Taylor 2014, 173).

The strongest comparative argument for the doubled stage comes from paradigms in which some cells had **j* in the ending and others had **i*. Only the first group could double, and the two outcomes then sat side by side. Old High German has *hewi* ‘hay’ beside *houwi*, and Old English itself preserves both in one word, *glīg* ‘mirth’ from the undoubled nominative beside *glīowes* ‘of mirth’ in the genitive (Ringe and Taylor 2014, 53, §3.1.3). The continental forms *houwi* and *gistrouwen* ‘bestrew’, and Old Saxon *hoi* ‘hay’, point to the same West Germanic stage, which English alone went on to resolve (Ringe and Taylor 2014, 173).

The resolution did not treat every doubled **w* alike, and the difference is what allows the doubling to be seen apart from it. Campbell sets the two vowel types side by side: *auj* becomes *auuj* and then *auj*, while *iuj* becomes *iuuj* and then *iuj*. Thereafter they part company, since the *u* of *auuj* is generally lost while the *j* of *iuuj* is lost (Campbell 1959, 46, §120.2). Old English *hīeġ* ‘hay’ and *hīew* ‘form, hue’ begin from shapes that differ in a single vowel and end with different survivors. *Hīeġ* keeps its **j*, written *ġ*, while *hīew* keeps its **w*. Campbell’s other examples of the second type are *nīowe* ‘new’, *nīewe* ‘new’ and *glīow* ‘mirth’, *glīw* ‘mirth’.

One qualification belongs in the record. Fulk holds that **w* was never consonantal here, so that Proto-Germanic already had **straujaną* with its diphthong in place and there is no gemination to undo (Fulk 2018, 73, §4.10 n. 1). The account followed here is the handbook one, which the comparative material supports: the doubled stage that the continental and paradigm-internal forms point to is precisely the one Fulk denies ever existed. Both accounts agree that **auj* is what enters Old English, and the disagreement concerns whether a discrete change took place.

SC029. Resolution of **awj* to **auj* (`OEAwwjResolution`)

```
define OEAwwjResolution Ctx([
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j}
]);
```

The change is confined to the **a* type. PGmc **xáwją* ‘hay’ is doubled to **xáwwją* by the West Germanic law and then resolved to **xáują*, and PGmc **stráwjaną* ‘strew’ is doubled to **stráwwjaną* and resolved to **stráujaną*, yielding *hīeġ* ‘hay’ and *strīeġan* ‘strew’ once the later diphthong changes and *i*-umlaut have applied. PGmc **xíwją* ‘form, hue’ is doubled by the same law, and there the doubling holds: it passes through this change untouched and surfaces as *hīew* ‘form, hue’.

The resolution has to precede SC030 OEAuBrightening, since the fronting needs a diphthong to work on. Ringe and Taylor state the same dependence when they observe that these new instances of **au* went on to share the ordinary development (Ringe and Taylor 2014, 173).

4.17 English brightening of au

Historical discussion of the brightening of **au*

The fronting of **au* is the application to a diphthong of the same process that fronts plain **a*, the process usually called Anglo-Frisian brightening. Campbell arrives at the point while establishing the order of the early vowel changes, remarking that the normal development of Proto-Germanic **au* to Old English *ēa* shows that the change of *a* to *æ* would affect the first element of a diphthong (Campbell 1959, 52, §132). His ordered list accordingly places West Germanic *a* > Old English *æ* and West Germanic **au* > Old English **æu* in one and the same step. Fulk puts it in a single sentence, saying that this fronting of *a* applied also to the diphthong *au* in Old English (Fulk 2018, 73, §4.12).

The intermediate stage is directly attested. Ringe and Taylor describe **au* as first tensed and fronted to **æu*, a spelling still found occasionally in eighth-century documents, with the offglide unrounded and lowered only later (Ringe and Taylor 2014, 172). Early spellings such as *Eadbald* with an initial *aeo* preserve the rounded offglide, and the rounding survives in late Northumbrian (Fulk 2018, 73, §4.12). The outcome of the whole sequence is the *ēa* of *dēaþ* ‘death’, *ēage* ‘eye’ and *lēaf* ‘leaf’ (Campbell 1959, 53, §135).

The geographical reach of the diphthongal fronting is narrower than that of the plain one. Old Frisian shows no such fronting and has *ā*, so that Old English *ēac* ‘also’, *ēage* ‘eye’ and *bēam* ‘tree’ stand against Old Frisian *āk* ‘also’, *āge* ‘eye’ and *bām* ‘tree’ (Fulk 2018, 73, §4.12; Ringe and Taylor 2014, 172). A later change supplies independent confirmation. Old English *gēac* ‘cuckoo’ has a palatalized initial, which requires a front vowel to have followed it, while Old Frisian *gāk* ‘cuckoo’ has none (Fulk 2018, 73, §4.12). Brightening of plain **a* is shared with Frisian; brightening of the first element of **au* is English.

SC030. Brightening of **au* to **æu* (`OEAuBrightening`)

```
define OEAuBrightening Ctx([
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
]);
```

Most of the words that pass through the change carry **au* inherited straight from Proto-Germanic. PGmc **láubą* ‘leaf’ gives *lēaf* ‘leaf’, PGmc **stráumaz* ‘stream’ gives *strēam* ‘stream’, PGmc **bráudą* ‘bread’ gives *brēad* ‘bread’, and PGmc **dráugmaz* ‘dream’ gives *drēam* ‘dream’. Where a following **j* or **i* survives long enough to cause *i*-umlaut, the *ēa* appears in West Saxon as *īe*, as in *ġelīefan* ‘believe’ from PGmc **galáubijaną* ‘believe’ and *nīed* ‘need’ from PGmc **náudiz* ‘need’ (Campbell 1959, 46, §120.2). The two words supplied by SC029 OEAwwjResolution, *hīeġ* ‘hay’ and *strīeġan* ‘strew’, join this second group.

The proportions matter for what the rule is. This is the general Old English treatment of **au*, to which the resolution of **awj* contributes two further inputs; the history of *hīeġ* ‘hay’ and *strīeġan* ‘strew’ does not define it.

The fronted **æu* has no independent life. It is taken up at once by the SC032 OEDiphthongLeveling, which lowers the offglide and delivers *ēa*. Ringe and Taylor give that order explicitly when they place the tensing and fronting first and the unrounding and lowering later (Ringe and Taylor 2014, 172).

4.18 Short pre-w diphthongs and the j-created glide chain

Historical discussion

Ordinary diphthongization before singleton *w* must be distinguished from long diphthong formation in endingless forms and from the independently created *wwj* sequences. Their similar Old

English spellings do not establish one sound law assigning long quantity before every retained vowel ending.

For *cneowe* ‘knee (dat.sg.)’, the regular pre-ending stem has short *eo*. Hogg explicitly identifies the dative as a short-diphthong example, and Campbell and Luick distinguish its genitive and dative from the long endingless form (Hogg 2011, 21–22, 86, §§2.33, 5.22; Campbell 1959, 232–33, §584; Luick 1914--1940, 139, §134). Ringe and Taylor likewise distinguish short *cneow-* before syllabic endings from long endingless *cnēo* ‘knee’. The final *w* of *cnēow* ‘knee’ is generalized from the obliques, not retained by an unrestricted prevocalic lengthening law (Ringe and Taylor 2014, 187–88, 387, §§6.2.4, 7.2.4).

The selected comparison is therefore the dative *cneowe*, from the constructed PGmc dative **knéwai*, while citation **knéwą* remains the lexeme-level reconstruction. Fulk prefers the **-ai* analysis for the West Germanic dative line but discusses alternatives; the complete selected input is a paradigm construction, not a quotation of a whole-word reconstruction (Fulk 2018, 147, §7.8).

Its short diphthong is supplied by SC044 OEBreaking, with the source’s exclusion before a following high front vocalic. Long quantity could later spread from endingless forms into obliques, but that leveling is not the selected short dative’s sound-law history. Campbell describes the extension; Ringe and Taylor qualify the knee quantity inference because decisive verse evidence is elusive (Campbell 1959, 233; Ringe and Taylor 2014, 387).

Earlier inherited-glide reanalysis is independently represented by SC031 OEWWSimplification. Chew, dew, four, hew and you complete its products through SC032 OEDiphthongLeveling. The operation below neither repeats that inherited event nor implements the regular singleton history (Ringe and Taylor 2014, 41–42, 65–66, 171–75).

SC033. J-created palatalized-glide reanalysis (`OEEwLongDiphthong`)

```
define OEEwLongDiphthong Ctx([
    {*i} {*w} {*w} -> {*iu} {*w} || _ {*j},
    {*í} {*w} {*w} -> {*íu} {*w} || _ {*j}
]);
```

Hīew ‘hue’ enters with inherited PGmc **xíwją*, following Ringe and Taylor’s explicit **hiwją*. Their palatalized geminate develops through **iuw* before the separately realized long diphthong. The conditioner is palatalized *w*, carried here as *ww+j* and subsequently *w+j*; this does not assert a second independent phoneme where the authors leave the phonemic analysis unclear (Ringe and Taylor 2014, 53, 250).

Orel’s **xewjan* and Kroonen’s **heuja-* remain competing e reconstructions, not earlier stages invented to reconcile the authors. Their derivational explanations differ. The realized-i choice uses the inherited raising account, not a later hue-specific law (Orel 2003, 171–72; Kroonen 2013, 224; Ringe 2017, 151–53; Fulk 2018, 59). Campbell supports the contrasting *iwj* outcome, but Fulk questions traditional geminate dismantling and its consonantal premises. The detailed lexical account preserves those objections (Campbell 1959, 46; Fulk 2018, 71–72, 126).

The stable component now supplies offglide reanalysis, not long quantity. Singleton *iwj*, nonpalatal *iww*, unraised *ewwj* and the separate low-vowel *awj* history are excluded. Knee remains a short singleton-w comparison. No exact historical date or independent confidence is fabricated for this support encoding.

4.19 Technical support: j-created glide compatibility

SC106. Retained j-created glide compatibility (OEJWWSimplification)

```
define OEJWWSimplification Ctx([
    {*w} {*w} -> {*w} || _ {*j}
]);
```

The unchanged technical operation remains composed for compatibility. Hue's reanalysis already leaves a singleton glide, so it no longer feeds this simplification. Absence of a current application does not establish universal redundancy or a newly dated historical event. It neither restores nominative *w* in knee nor duplicates earlier inherited short-Vww reanalysis.

4.20 Technical support: palatalized-glide long-io realization

SC110. Palatalized-glide long-io realization (`OEJGlideIO`)

```
define OEJGlideIO Ctx([
    [{*iu}|{*íu}] -> {*īo} || _ {*w} {*j}
]);
```

The palatalized-glide product realizes long *īo*, not the ordinary long *ēo* supplied by SC032 OEDiphthongLeveling. Ringe and Taylor explicitly supply pre-OE long *īow* and early West Saxon long *īew*; the retained semivowel still conditions mutation (Ringe and Taylor 2014, 250). Existing SC055 OEIUmlaut supplies the latter change. Ordinary inherited *iu* without this palatal conditioner remains outside the component, as does short *io*. This distinct realization does not adjudicate all the inherited atomic and split-symbol clauses in SC032 OEDiphthongLeveling.

4.21 Quantity-distinct English diphthong realization

Historical discussion

The English histories of inherited diphthongs and their secondary equivalents must be distinguished from both smoothing and paradigm leveling. Ringe and Taylor identify a common tensing or lengthening of the syllabic nuclei, followed by further developments of the offglides; they place that tensing before breaking. This does not make all the eventual English spellings the products of a single simultaneous change (Ringe and Taylor 2014, 170–75).

The fronted continuation of **au* gives long *ēa*, including in *brēad* 'bread', *dēaw* 'dew' and *hēawan* 'hew'. First-element fronting is distinct from subsequent offglide unrounding and lowering. The latter was not universal: some Northumbrian varieties retained a rounded offglide. Thus this outcome is not exclusively West Saxon, but neither is it an exceptionless description of every English dialect (Ringe and Taylor 2014, 172–73; Campbell 1959, 53–54, 116–18).

The long *ēo* outcomes in *ċēowan* 'chew' and *fēower* 'four' belong to the **eu* history. Earlier inherited or assimilation-created geminate-glide reanalysis supplies the diphthong while retaining the following consonantal glide. Dew and hew similarly acquire **au* through SC031 OEWWSimplification, then reach the fronted-au path. Their early reanalysis is not the later English realization (Ringe and Taylor 2014, 41–42, 65–66, 172–74).

The **iu* history is different again. Early *īo* is not simply another spelling for the initial **eu* development. Subsequent io/eo convergence has dialect-specific histories: Northumbrian retains io, whereas West Saxon and parts of Mercian show convergence. The long *ēo* of *frēond* 'friend' and *ēow* 'you' is the selected endpoint here, not evidence that the merger preceded every intervening English change. Friend's contraction-created input and you's earlier glide reanalysis are also distinct origins (Ringe and Taylor 2014, 41–42, 174–75; Campbell 1959, 124–26).

SC032. Quantity-distinct diphthong realization (OEDiphthongLeveling)

```
define OEDiphthongLeveling Ctx([
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    # Split symbols from palatal glide (stress preserved where present):
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
]);
```

The retained definition is a compound realization proxy. Its atomic **aeu*, **eu* and **iu* symbols belong to the heavy, long-output tier; an acute mark does not itself indicate length. Atomic **aeu* records a fronted-au intermediate, not a claim that a three-vowel phonetic sequence is demonstrated. The direct **iu* to *ēo* mapping packages the selected later endpoint. The split-symbol e+u, é+u and i+u clauses instead produce short *eo*, retaining stress only in the é+u clause. Campbell distinguishes short breaking and back-umlaut diphthongs from their long equivalents, but his account does not specify these computational token boundaries (Campbell 1959, 116–18).

All 32 current applications use atomic long-tier inputs. The unaccented atomic alternatives and all three split-symbol clauses remain formal compatibility paths without a selected-word application. Their retention does not establish an independent palatal-glide origin, and no additional stressed-i clause is inferred.

The short dative *cneowe* ‘knee’ remains outside this long-output operation; its pre-w breaking is supplied by SC044 OEBreaking. The separately palatalized-glide history of *hīew* ‘hue’ reaches long *īo* through SC110 OEJGlideIO before the ordinary iu matcher. It then receives long *īe* through mutation, not ordinary *ēo* here. These contrasts preserve quantity and conditioning within the adopted selected cells (Campbell 1959, 232–33; Ringe and Taylor 2014, 187–88, 250, 387).

Fronting must supply its product before its offglide continuation: SC030 OEAuBrightening feeds this component, not every clause in the definition. Ringe and Taylor explicitly distinguish the two developments (Ringe and Taylor 2014, 172). Earlier displacement tests that left **aeu* unrealized produced *+?*, a computational rejection with no attested linguistic status. They therefore cannot independently establish the historical interval.

The input of *hēafod* ‘head’ must be realized into a recognized vowel before SC040 OEMedUnstressedULowering can lower its remaining medial vowel in the model. That internal recognition requirement is not proof that historical offglide change conditioned the lowering. Campbell records both *hēafud* and *hēafod*, with varying preservation of protected unstressed u (Campbell 1959, 155–56). No historical order for the entire nine-clause bundle follows from that one computational dependency.

4.22 Endingless *aw* and the retained show proxy

Historical discussion

The attested long nucleus does not establish one general law changing singleton *aw* to *ēaw* before every following vowel. For straw, the inherited nominative/accusative **stráwą* loses its ending and vocalizes final *w*, giving **strau*; the ordinary English development then gives reconstructed **strēa*. Attested *strēaw* ‘straw’ combines that long diphthong with *w* restored from inflected forms. Conversely, the long diphthong spreads into those forms. The regular free simplex and short obliques are explicitly unattested, although the endingless pattern survives in the compound *strēaberige* ‘strawberry’ (Campbell 1959, 46–47, 232–33; Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 172–73, 387).

The model retains attested *strēaw* ‘straw’ as its comparison target and produces reconstructed **strēa* through the regular cascade. This is an explicit mismatch, not an attested reconstructed endpoint or a sound law for analogical restoration. SC111 PWGmcEndinglessAw supplies the narrow endingless fragment; it does not imply that general final-vowel loss was conditioned by an adjacent *w*.

The inherited geminates of *dēaw* ‘dew’ and *hēawan* ‘hew’ instead undergo early SC031 OEWWSimplification, producing **au* plus retained **w*, followed by English fronting and SC032 OEDiphthongLeveling. Campbell distinguishes the glide histories; Ringe and Taylor distinguish earlier reanalysis from later English realization (Campbell 1959, 45–47, 53–54; Ringe and Taylor 2014, 65–66, 171–75). These retained-glide histories differ from endingless straw.

For *sċēawian* ‘show’, Orel and Kroonen print singleton dictionary inputs, while Campbell and Brunner include the verb among inherited geminate examples. The class-II suffix places a vowel between root *w* and *j*; this excludes direct root *aw+j* gemination but does not resolve the different root histories (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49). SC034 OEAwLongDiphthong therefore remains a disclosed support proxy for the three selected show cells, not an established general singleton sound law or a newly dated West Saxon event.

SC034. Retained show long-diphthong proxy (`OEAwLongDiphthong`)

```
define OEAwLongDiphthong Ctx([
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    # Stressed variant (prosodic tier):
    {*á} {*w} -> {*éa} {*w} || _ [EnglishStarVocalic | {*ô}]
]);
```

The current operation changes the show infinitive, imperative singular and present third singular. Straw already has its regular long nucleus without *w* and is unchanged here. Dew and hew no longer demonstrate a local simplification-then-lengthening chain here: their earlier reanalysis has already removed that input shape. The older displacement results for those geminates describe the superseded representation, not independent dates for the adopted early event. The two proxy clauses remain unchanged; their successful show finals do not decide the unresolved singleton-versus-geminate source premise. The old straw displacement observation measured dependence on a following vowel in the superseded proxy, not independent historical order (Campbell 1959, 45–47; Ringe and Taylor 2014, 171–75).

4.23 Inter-stress raising

Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell’s discussion of *weorold* ‘world’ / *weoruld* ‘world’ and Ringe and Taylor’s derivation of **weraldu* ‘world’ > **weruldu* ‘world’ > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

SC036. Raising of medial **a* between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising Ctx([
    {*a} -> {*u} || PGmcStarVowel EnglishStarConsonant* _ [EnglishStarConsonant -
    ↪  {*j}]+ [{*u}|{*ū}],
    # DEV_NOTES §16.6: Nebenton-conditioned raising of compound
    # second-element root *a. Luick §306, Brunner §402, R/T vol.2
    # §6.3.3 p.200-202: short *a in the trough between primary
```

```
    # stress and a weak/lost secondary stress raises to *u. The
    # grave-a itself is the prosodic conditioning marker; no
    # right-context is required — the compound-internal prosodic
    # position is encoded directly in the PROTOFORM.
    # Example: *wír-àldu → *wír-uldu → (u-lowering) → weorold.
    {*à} -> {*u}
]);
```

The two boundaries have unequal force. Before SC019 PNWGmcFinalLongORaising, PGmc ***sái-walō*** ‘soul’ yields †***sāwel*** rather than expected OE ***sāwol*** ‘soul’; after SC040 OEMedUnstressedU-Lowering, it yields †***sāwul*** rather than ***sāwol***, while PGmc ***wír-àldu*** ‘world’ yields †***weoruld*** rather than ***weorold*** ‘world’. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In handbook terms, medial ***a* > ***u*** belongs to the *world-* and *soul*-type low-stress vocalism that followed the earlier final-vowel changes.

4.24 Compound linking syncope

Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as ***reġnboga*** ‘rainbow’. The only boundary the lexical evidence supplies is the immediately following technical stress-stripping stage, which is not a sound change.

SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope Ctx([
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _ OEAnyConsonant+ PGmcStarGraveVowel
]);
```

The ***reġnboga*** ‘rainbow’ test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc ***régna-bùgô*** ‘rainbow’ yields †***reġnefoga*** rather than expected OE ***reġnboga*** ‘rainbow’, because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

4.25 Combinative u-umlaut in wi-forms

Historical discussion

The history of ***wuduwe*** ‘widow’ orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later ***weorold*** ‘world’ / ***weoruld*** ‘world’ alternation, while Ringe and Taylor give the same connection comparatively in ****widuwon-***, ****weraldu*** ‘world’, and ****jugunþi*** ‘youth’ (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

```
define OEWICombinativeUUmlaut Ctx([
    {*í} -> {*ú} || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
]);
```

The *wuduwe* ‘widow’ derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut is delayed until after SC040 OEMedUnstressedULowering, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe*. This one-sided witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

4.26 Final bare-*a* loss

Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

```
define PWGmcFinalBareALoss Ctx([
    {*a} -> 0 || _ .#.
]);
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final *z*-deletion, the change gives the wrong outputs: PGmc **bárdaz* ‘beard’ yields †*bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* ‘comb’ yields †*camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* ‘craft’ yields †*craft* rather than expected OE *crœft* ‘craft’, and PGmc **dágaz* ‘day’ yields †*dag* rather than expected *dæġ* ‘day’. The distant lower limit follows final *z*-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

4.27 Surviving bimoric **ō* unrounding

Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* ‘rest’ nevertheless undergoes unrounding before SC043 EAFBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992b, 102, 105; Ringe and Taylor 2014, 157–58, 189–90).

The sole witness establishes a local relation to brightening but supports no broader generalization.

SC042. Unrounding of the surviving bimoric **ō* (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding Ctx([
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ .#.
]);
```

The single *ræste* ‘rest’ derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 EAFFinalZDeletion or after SC043 EAFBrightening, PGmc **rástōz* ‘rest’ yields †*rasta* rather than expected OE *ræste*. Unrounding must therefore follow final *z*-loss and precede brightening, although only the relation to brightening is local.

4.28 Technical support: retained unstressed fronting

SC107. Retained unstressed contribution (EAFBrighteningUnstressed)

```
define EAFBrighteningUnstressed Ctx([
    {*a} -> {*æ} || _ [EnglishStarConsonant - EnglishStarNasal]
]);
```

This earlier contribution is preserved without identifying it with the entire unstressed history. The later syllabic rule also fronts surviving unstressed vowels before heterosyllabic nasals; an exception before every nasal would therefore be too broad (Campbell 1959, 140–41).

4.29 Ordinary English fronting

Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (Campbell 1959, 52, §131; Hogg 1992b, 102, 105; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

The traditional name does not itself establish one inherited event. Under Campbell’s premise that the daughter contractions precede ordinary fronting, comparable English and Frisian outcomes require separate daughter frontings in the strict tree. Chapter 3 develops that conditional argument without removing the ancestral node (Campbell 1939, 90–91). The model adopts the completed ordinary English stressed component on the English daughter under this conventional working account. This is not a claim that every author proves contraction before ordinary fronting: Ringe and Taylor explicitly question whether diphthong nuclei must behave like the plain short vowel (Ringe and Taylor 2014, 170–75). Earlier restricted ancestral fronting remains possible.

The separately retained unstressed and final-vowel components are not dated by that argument. The first supplies an earlier nonnasal contribution; SC070 OEUnstressedFrontingEarly separately implements the broader unstressed domain, including the contrast between coda and heterosyllabic nasals (Campbell 1959, 140–41, §§333–334). The second carries the model’s preserved final-vowel quantity into later shortening and merger. Its long intermediate is a representation choice, not independent evidence for a historical long-vowel fronting (Ringe and Taylor 2014, 58–59, 299–300).

SC043. Fronting of low **a* outside nasal environments (EAFBrightening)

```
define EAFBrightening Ctx([
    EAFBrighteningStressed
]);
```

Stressed component

```
define EAFBrighteningStressed [
    {*á} -> {*æ} || _ [EnglishStarConsonant - EnglishStarNasal | .#.]
];
```

Slay requires breaking to receive the fronted root vowel: delaying the stressed member until after SC044 OEBreaking gives *sleaan* | *slēaan*, rather than OE *slēan* ‘slay’. Rest involves two different changes, root fronting and the separately retained final-vowel representation. Its final-vowel dependency must not be used to date this stressed rule.

4.30 Technical support: retained final-vowel representation

SC108. Retained final-vowel representation (EAFBrighteningLongFinal)

```
define EAFBrighteningLongFinal Ctx([
    {*ā} -> {*ǣ} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
]);
```

In *ræste* ‘rest’, this helper receives the length-preserved outcome of SC042 PWGmcSurvivingBimoricOUnrounding. The preceding-nucleus guard excludes stressed monosyllabic *hwā* ‘who’. Later shortening and merger complete the ending; the carried long vowel does not establish a separate dated long-a law (Ringe and Taylor 2014, 58–59, 299–300; Campbell 1959, 49, §125).

4.31 Breaking

Historical discussion

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

Breaking has the fuller handbook treatment. The *feoh* ‘cattle’ and *feohtan* ‘fight’ type derivations preserve its velar-fricative trigger; they do not subsequently undergo velar-fricative palatalization.

SC044. Breaking before *h*, *rC*, *lC*, and conditioned *w* (`OEBreaking`)

```
define OEBreaking FootCtx(OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI);
```

Short *e* and *i* also develop short *eo* and *io* before singleton *w*, except when a high front vowel or *j* follows it. This is the regular pre-ending history of *cneowe* ‘knee (dat.sg.)’, not long-diphthong promotion (Luick 1914--1940, 139, §134; Hogg 2011, 86, §5.22; Ringe and Taylor 2014, 187–88, §6.2.4). The shared w conditioner makes that exclusion explicit:

```
define EnglishBreakingWContext [
    {*w} [[EnglishStarVocalic | EnglishStarConsonant]
        - [{*i} | {*í} | {*ī} | {*ǐ} | {*j}]] |
    {*w} .#.
];
```

Root stress notation does not determine vowel length. The short knee dative is the selected comparison; long endingless *cnēo* ‘knee’ and the restored *w* of *cnēow* ‘knee’ are separate paradigm histories, discussed with SC033 OEEwLongDiphthong.

Breaking must encounter the vowel created by brightening and must precede the rule that would otherwise palatalize its velar trigger in *feoh* ‘cattle’ and *feohtan* ‘fight’. Before SC043 EAFBrightening, PGmc **sláxaną* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. After SC045 OEVelarFricativePalatalization, PGmc **féxu* ‘cattle’ yields †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* ‘fight’ yields †*fehtan* rather than expected *feohtan*. The fronting relation feeds breaking. The fee/fight relation instead protects breaking’s velar trigger from premature palatalization: those forms are displacement negatives, not live SC045 OEVelarFricativePalatalization applications.

4.32 Velar-fricative palatalization

SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization Ctx([
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j}
]
    .o. EnglishStarAlphabet*);
```

The live population concerns voiceless x, as in *hēafod* ‘head’ and *heofon* ‘heaven’, not voiced-fricative merger. In *feoh* ‘cattle’ and *feohtan* ‘fight’, breaking removes the original front-vowel context: they do not change at this rule in the live derivation. Moving the rule before SC044 OEBreaking instead consumes its velar trigger, yielding †*fehu* and †*fehtan*. This is counterbleeding protection, not direct feeding.

The distant *six* ‘six’ displacement test supplies only a broader constraint: after SC060 OEWsPalatalUmlaut, PGmc **séxs* ‘six’ yields †*sihs* rather than expected OE *six*. Neither test dates voiced g. Its articulation and disputed merger are separately treated under SC052 OEVelarPalatalization and SC109 OEPalatalFricativeMerger (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32).

4.33 A-restoration

Historical discussion of A-restoration

Campbell’s restoration of *a* before following back vowels and Ringe and Taylor’s later retraction describe the same post-brightening development (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration FootCtx((
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
));
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 EAFBrightening, PGmc **bákaną* 'bake' yields †*bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* 'fare' yields †*færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákaną* again yields †*bæcan* instead of *bacan*, while PGmc **wádaną* 'wade' yields †*wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

4.34 Heavy-syllable nasal apocope

Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

SC047. Heavy-syllable nasal apocope of final **ą* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope Ctx([
    {*ą} -> 0 || OEAnyConsonant _ .#.
]);
```

The old lower-bound test depended on the superseded straw proxy: moving apocope before SC034 OEAwLongDiphthong made PGmc **stráwą* 'straw' yield †*stræw* rather than attested OE *strēaw* 'straw'. The adopted regular endingless branch now reaches reconstructed **strēa* through SC111 PWGmcEndinglessAw, before this apocope. The attested target retains analogically restored *w* and explicitly mismatches the regular output. The old failure therefore does not independently place nasal apocope after a universal singleton long-diphthong law (Campbell 1959, 46–47, 232–33; Ringe and Taylor 2014, 172–73, 387).

The separate recorded upper-bound test is unchanged: after SC048 OESecondaryNasalization, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan*, and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan*, alongside a broad *-en* failure set. That secondary-nasalization dependency is not newly adjudicated here.

4.35 Secondary nasalization

SC048. Secondary nasalization before final **n* (OESecondaryNasalization)

```
define OESecondaryNasalization Ctx([
    {*a} -> {*ą} || _ {*n} .#.
]);
```

The broad *-an*/*-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* ‘bake’ yields †*bacen* rather than expected OE *bacan* ‘bake’, and PGmc **bíndaną* ‘bind’ yields †*binden* rather than expected *bindan* ‘bind’. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* ‘steal’ yields †*steolan* rather than expected OE *stelan* ‘steal’, and PGmc **wébaną* ‘weave’ yields †*weofan* rather than expected *wefan* ‘weave’. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

4.36 B allophony

Historical discussion

The positional alternation of Germanic **b* is a Proto-Germanic distributional feature. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992b, 108). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick’s spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

The distribution is narrow, but later changes presuppose the stop-fricative alternation. CAPR implements the rule at a late cascade position for computational reasons: the alternation must interact with consonant environments shaped by intermediate rule applications. Its historical stage is Proto-Germanic.

SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony Ctx([
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
]);
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reġnboga* ‘rainbow’ develops as †*reġnfoga* rather than expected OE *reġnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

4.37 Sievers-law syncope

Historical discussion

Sievers' Law concerns a prosodic and morphological adjustment in heavy stems. It is a distributional rule distinct from b-allophony (SC049 PGmcBAllophony). Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

SC050. Sievers-law syncope (`SieversLawSyncope`)

```
define SieversLawSyncope Ctx([
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
]);
```

The Sievers-law reduction **-CijV-* > *-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* 'stretch' yields †*strecċan* rather than expected OE *streċċan* 'stretch'; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

4.38 Palatalization of **sk* to **sc*

Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992b, 106–8, 113). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

```
define OESkPalatalization Ctx([
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
```

```
]);
```

The non-fronted vowels of *flasce* ‘flask’ and *wascan* ‘wash’ fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* ‘flask’ and *wæscan* ‘wash’ rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sċeaft* ‘shaft’, *sċēar* ‘shear’, *sċēaþ* ‘sheath’, *sċēap* ‘sheep’, and *sċield* ‘shield’ before SC056 OEWsPalatalDiphthongization. The **sċea-* ’sea’/*sċie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

4.39 Class-distinct velar palatalization and fricative merger

Historical discussion

Singleton g, including initial g, was a fricative when palatalization occurred; gg and postnasal g were stops. Palatal articulation, productive cutoff, merger and later affrication are therefore separate questions (Fulk 2018, 130–32; Laker 2007, 166–68, 175–84). The dotted spelling in *dæġ* ‘day’ cannot establish an early affricate or identify every palatal consonant with inherited j.

The adopted working account keeps a palatal fricative ȷ distinct from j during mutation, then merges it with j. Ringe and Taylor explicitly propose this chronology (Ringe and Taylor 2014, 203–4). It is not unanimous: Hogg objects to the phonetic plausibility of delayed merger and leaves the key/day contradiction unresolved (Hogg 1979, 102–11). We prefer the regular class-distinct account because it explains the contrasting mutation triggers without lexical or grammatical conditions. The objection remains substantive, not silently superseded by a matching final spelling.

Eligibility also requires attribution. Ringe and Taylor’s medial-g table is narrower than Fulk’s before-front-vowel description (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32). The present fricative implementation retains its broader domain; this merger decision is not a covert adoption of the narrower table.

SC052. Singleton-fricative articulation (`OEGFricativePalatal`)

```
define OEGFricativePalatal [
    [{*g}|{*ɣ}] -> {*ȷ} ||
        [.#.|[EnglishStarAlphabet - [{*g}|{*n}|{*ŋ}]]]
        _ EnglishStarFrontVowel,
    [{*g}|{*ɣ}] -> {*ȷ} || EnglishStarFrontVowel _ .#.,
    [{*g}|{*ɣ}] -> {*ȷ} || EnglishStarFrontVowel
        _ [EnglishStarConsonant - [{*j}|{*g}]]
] .o. [
    [{*g}|{*ɣ}] -> {*ȷ} ||
        [.#.|[EnglishStarAlphabet - [{*g}|{*n}|{*ŋ}]]] _ {*j}
];
```

The guards keep gg/ng in their stop paths. The value ȷ is a real phonetic class, not a tag recording a consonant’s ancestry. It is transparent to a following mutation trigger, but is not itself j at that date. The source comparison normalizes older fricative notation to modern ɣ/ȷ, never to the digit 3 (Hogg 1979, 105; Ringe and Taylor 2014, 204).

SC052. K articulation and eventual reflex (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*í},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*í} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*í} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
];
```

Here ʧ is a telescoped eventual-reflex proxy, not a claim that early palatal stops were already affricates. Ringe and Taylor explicitly separate the initial palatal stop from subsequent affrication and its syncope dependencies (Ringe and Taylor 2014, 203–4). The *weccan* ‘wake’, *licgan* ‘lie’ and *lecgan* ‘lay’ examples concern inherited j-clusters, not independent proof of every plain-velar conditioner (Ringe and Taylor 2014, 213–14).

The deterministic kk-before-j path protects *streċċan* ‘stretch’. The original-front versus secondary-front distinction is more directly tested by unrounded key than by rounded *cȳ* ‘cow’; rounding alone can give only a cutoff before later unrounding (Hogg 1979, 100–105; Laker 2007, 167–68).

SC052. Retained stop paths (`OEVelarPalatalizationStops`)

```
define OEVelarPalatalizationStops [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

Singleton fricatives have already left this domain. The remaining gg/ng paths use ʤ as an eventual-reflex proxy; the initial singleton fricative is no longer represented as an early affricate. *Wicg* ‘horse’ is a geminate-class illustration, *sengan* ‘singe’ a postnasal control, and *lungen* ‘lung’ the unchanged back-vowel negative (Ringe and Taylor 2014, 203–4, 213–14; Fulk 2018, 131–32).

SC052. Combined articulation (`OEVelarPalatalization`)

```
define OEVelarPalatalization Ctx([
    OEGFricativePalatal .o. OEVelarPalatalizationStops
]);
```

The combined operation does not give every class one sharply dated palatalization/affrication/merger event. It supplies the class distinctions needed by ordinary palatal diphthongization and mutation. The common-stem onset remains a separate question (Luick 1914--1940, 835–41; Fulk 2018, 130–32).

Key and day at the mutation checkpoint

Hogg's oblique comparison starts here from pre-palatal, pre-OE **kājœ* and **dœɣœ*; neither is silently fed through a PGmc prefix (Hogg 1979, 105). The source target is *cǣġe* 'key', with velar initial k, versus *dœġe* 'day'.

Checkpoint	Key	Day under the adopted account
Before palatalization	**kājœ*	**dœɣœ*
Before mutation	Inherited j remains	Palatal ɟ is distinct from j
After mutation	**kǣjœ*, without new initial palatalization	**dœɟœ*, without æ-raising
After merger and native realization	*cǣġe*	*dœġe*

Premature merger instead predicts pre-OE †*dejœ* under these premises. The experiment tests that source-based contrast; the existing day nominative is not the oblique discriminator (Hogg 1979, 105–10; Ringe and Taylor 2014, 204).

The complete key suffix additionally requires SC082 OEIntervocalicJVocalization to retain j after its non-high long front vowel. A correct mutation checkpoint followed by model-only †*cǣie* would not constitute a correct native derivation (Hogg 1979, 105).

4.40 Pretonic front-vowel raising

Historical discussion of prefixal **a*-reduction

Old English *ġelīefan* 'believe' continues a formation with inherited **ga-*. Gothic *ga-* preserves the older vowel, while Old Saxon and Old High German *gi-* and Old English *ge-* exhibit reduced reflexes; Ringe and Taylor reconstruct **ga-* for both Proto-Germanic and Proto-West-Germanic. Comparable reduced prefixes do not by themselves establish a single shared West Germanic reduction event (Ringe and Taylor 2014, 206).

The English development comprises several changes, not an immediate replacement of **a* by **e*. Ordinary unstressed fronting supplies **œ*; early *gi-* and later *ge-* attest subsequent changes in the weak vowel. Campbell distinguishes fronting from the later merger, and Ringe and Taylor explicitly consider regular raising of pretonic **œ* after a palatal fricative. CAPR adopts that regular account as its working reconstruction: fronting, palatalization, raising, and later lowering. It does not formulate a sound law conditioned by the identity of a prefix (Campbell 1959, 140–41, §§333–334; Campbell 1959, 153–55, §§369–370; Ringe and Taylor 2014, 334–35).

Stress supplies the decisive distinction. The ordinarily weak vowel must be distinguished from the exceptional accented form in *ġeatwe* 'armour'. Metrical evidence independently supports the weakness of *ge-* and *be-* alongside stress on the following root; grammatical structure can help recover that stress, but is not itself the condition of the vowel change (Campbell 1959, 31, §74 and n. 5; Hogg 1992b, 99–100; Ringe and Taylor 2014, 335; Minkova and Zhou 2022, 535–36).

Morphological and prosodic boundaries need not coincide. Manuscript grouping in Old English, Old Saxon and Old High German sometimes joins weak *ge-* to preceding material rather than to the following root. This supports distinguishing the weak syllable from the headed root foot, not a universal prosodic boundary at every morpheme boundary. The accompanying modern German experiment concerns a different period and does not date English vowel raising. Here the derivations represent citation words with one weak pretonic syllable, not every possible sentence-level regrouping (Riegger et al. 2024, 2065–68).

SC035. Raising of pretonic *æ after a palatal fricative (OEPrefixAReduction)

```
define OEPrefixAReduction SentenceCtx([
    {*æ} -> {*i} || .#. {*ᶠ} OEAnyConsonant* {*j̨} _
]);
```

The vowel raises because it is short, pretonic and immediately follows the palatal fricative, not because it belongs to inherited *ga-*. The superscript `*ᶠ` is a nonsegmental annotation of one weak syllable before a positively identified headed foot. It is not a reconstructed sound. The context operations preserve that information after vowel changes consume the original accent marks; they do not introduce a phonetic intermediate vowel. The reconstruction itself has ordinary *æ and *i, not a protected breve-e.

Ringe and Taylor's regular-raising proposal provides the phonological account adopted here. They also mention possible influence from *bi-*. Berg prefers analogy because he finds the direct development unattractive, but does not supply positive evidence of substitution in that discussion. His Pforzen *ga-* is important comparative evidence against projecting *gi-* back indiscriminately into common West Germanic; it does not refute a later English development after palatalization. Regular phonology therefore has explanatory priority in the present account. The precise date of raising and its independently attested breadth remain less secure than the early *gi-* and later *ge-* sequence: no independent non-*ga-* word sharing the complete early environment has yet supplied a positive comparison (Ringe and Taylor 2014, 335; Berg 2014, 60–61).

The later evidence is wider than one derivation. Early Northumbrian *gistīga* 'climb into/onto' becomes *gestīga* 'climb into/onto', beside West Saxon *ġestīgan* 'climb into/onto'. Early Mercian *gidyrstig* 'bold' and *gibēaten* 'beaten' likewise have later *ge-* forms. These are historical spellings, not records of every reconstructed intermediate. The English *gi-* evidence is not exclusively Northumbrian. Hogg places the later lowering of unstressed *i*, including *be-* and *ge-*, marginally after the earlier front-vowel merger. Accordingly SC073 OEUnstressedAEMerger merges weak *æ with *e*, while SC076 OEPrefixIReduction subsequently lowers pretonic *i*. Neither process changes a stressed root vowel (Ringe and Taylor 2014, 334–35; Hogg 1992a, §§ 6.50–6.54 (held 2011 reprint, pp. 237–239)).

The absence of diphthongization in a weak vowel follows from the conditioning of SC056 OEWsPalatalDiphthongization, not from an exceptional ban on particular prefixes. Hogg requires a palatal immediately before a stressed vowel at the beginning of a stress foot, not necessarily at the beginning of a word. Thus *sċeaft* 'shaft' and *ġesċeaft* 'creation' both diphthongize the root vowel after the sh-sound, but the weak first vowel of the latter does not diphthongize after its j-sound. The adopted model derives both this comparison and *ġestīgan* 'climb into/onto'; the earlier compressed rule did not (Hogg 1992a, § 5.48 n. 1 (held 2011 reprint, p. 104); Ringe and Taylor 2014, 335).

In *ġelīefan* 'believe', ordinary weak fronting gives a pretonic *æ, palatalization supplies the fricative trigger, raising gives *i*, and late lowering gives *ge-*. The independently supported root-vowel history continues separately. This derivation establishes that the phonological account is coherent, not the unique date of raising. A fronting-and-merger account without raising reaches the same late vowel but leaves early *gi-* unexplained; the earlier protected-vowel compression obscures the distinction and excludes genuine clustered forms. These are the reasons for preferring the staged account, not merely its agreement with one final form (Ringe and Taylor 2014, 245, 267, 335).

Preserving stress-foot structure also preserves the real stressed developments: *ġiest* 'guest', *ġiefan* 'give' and *sċēap* 'sheep' retain ordinary palatal diphthongization, whereas *ġealla* 'gall' already has a diphthong from breaking. A plain vowel symbol is not therefore assumed to be weak: earlier fronting or diphthong leveling may have consumed its accent. SC107 EAFBrighteningUnstressed supplies the pretonic front vowel; the old requirement that a protective operation precede it was computational, not a historical date before SC043 EAFBrightening (Campbell 1959, 69–70, §§185–188; Ringe and Taylor 2014, 215–17, 335).

4.41 Post-velar w-loss

SC053. Loss of **w* after velars (`OEPostVelarWLoss`)

```
define OEPostVelarWLoss Ctx([
    {*w} -> 0 || {*n} {*g} _
]);
```

The comparative development `*singwan > singan` establishes narrow post-velar **w*-loss in the **ngw* sequence, yielding *singan* ‘sing’. Moving SC053 OEPostVelarWLoss earlier or later leaves every output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

4.42 W-loss before final i

SC054. Loss of **w* before final **i* (`OEWLossBeforeI`)

```
define OEWLossBeforeI Ctx([
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
]);
```

The history of *sǣ* ‘sea’ explains why non-initial **w* disappeared before final unstressed **i*. Campbell describes the loss, Ringe and Taylor derive the form from **saiwi-*/**sawi-*, and Luick gives the parallel trajectory (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 EAFFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields †*sǣw* rather than expected OE *sǣ* ‘sea’. The loss must therefore follow final *z*-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

4.43 West Saxon palatal diphthongization

SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization FootCtx([
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ǽ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ȷ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.]
]);
```

West Saxon *gieldan* ‘pay’, *scield* ‘shield’, and *scieppan* ‘create’ show diphthongization after an already palatal consonant (Ringe and Taylor 2014, 215–16, §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg's *giefan* 'give' and *sceap* 'sheep' belong to the same palatal-consonant environment (Hogg 1992b, 112). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (Fulk 2018, 74, §4.13).

The relevant beginning is that of a headed stress foot, not necessarily the word. Hogg's *sċeaft* 'shaft' and *ġesċeaft* 'creation' both have diphthongization in the stressed root after the sh-sound. The weak first vowel of *ġesċeaft* 'creation' does not diphthongize after the initial j-sound. The context operation in the printed rule preserves an explicitly identified pretonic syllable and applies the ordinary rule to the following headed domain. It does not suppress diphthongization throughout a prefixed word or infer weakness from a missing late accent (Hogg 1992a, § 5.48 n. 1 (held 2011 reprint, p. 104)).

This foot-based reformulation retains ordinary diphthongization before mutation; it does not reopen the separate later treatment of mutation products. Its complementary pretonic vocalism is discussed under SC035 OEPrefixAReduction.

The ordinary live applications comprise *ġiefan* 'give', *ġiest* 'guest', *sċeaft* 'shaft', *sċieran* 'shear', *sċēap* 'sheep', *sċield* 'shield' and *ġēar* 'year'. Gift is unchanged at this rule because its selected vowel already reflects earlier raising. Sheath changes from the mutation product **ǣ* to **ēa*; it supplies evidence for the later *sċ* layer, not the date of the ordinary treatment represented by the other examples.

The adopted ordinary process precedes mutation and preserves the split- diphthong and high-i negative controls. The reported absence of a later displacement failure supplies no positive historical terminus.

4.44 Old English i-umlaut

SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* 'follow', *gylden* 'golden', *wyrm* 'worm', and *giest* 'guest' exemplify the same `i`- or `j`-conditioned fronting across different vowels (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The selected cow and lung inputs are negative controls on the present palatalization rule. Moving the composite umlaut rule before that rule yields †*ċȳ* and †*lungen* instead of *cȳ* 'cow' and *lungen* 'lung'. This constrains the current rule's productive domain; it does not assign every consonantal layer the same date. In particular the selected *cȳ* is a dative-singular input, not evidence from an assumed generic plural.

The stored gift and sheath displacement failures constrain the current serialization, not a universal historical upper boundary. The former gift input left an earlier raising to this late rule; sheath represents the later *sċ* treatment of a mutation product. Moving the entire palatal-diphthongization bundle earlier strands sheath and diphthongizes the model's still-unraised gift vowel. Neither failure overturns the independently supported ordinary palatal-diphthongization-before-mutation account.

SC055. Raising under i-umlaut (`OEIUmlautRaising`)

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted æ to e continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The four displacement controls concern the composite rule. They are not independent tests of this raising component: for example, the cow vowel changes in the fronting component, not in this **æ*-raising clause. Component chronology must follow the source-supported assimilatory event, with ordinary palatal diphthongization and the later *sċ* extension kept distinct.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before m and n in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992b, 113).

SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon ie forms cited in the handbooks, including *giest* ‘guest’, *giefan* ‘give’, and *hierde* ‘shepherd’ in Campbell and *ciest* ‘chest’ in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992b, 113–14). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

Those composite displacement controls do not isolate diphthongal mutation: neither the cow input nor the gift input has a diphthong here. In the conventional history, diphthongs produced by ordinary palatal diphthongization can be mutated, as in the source derivation of *ġiest* ‘guest’ (Ringe and Taylor 2014, 216). The adopted derivation now follows that ea-to-ie source path. The former simple-vowel route converged on the same final form, showing why final agreement alone could not establish the relative chronology.

SC055. The composite i-umlaut rule (`OEIUmlaut`)

```
define OEIUmlaut FootCtx(OEIUmlautFronting
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong);
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The cow/lung displacement results require the present productive palatalization rule not to consume those mutation-created environments. The historical interpretation still depends on the relevant consonantal layer and inherited versus secondary front-vowel conditions.

There is no single source-supported upper boundary at the entire West Saxon bundle. Ordinary palatal diphthongization precedes mutation in the handbook account; the later *sċ* treatment of some mutation products follows it. The former gift input and sheath's later layer explained the old bundled serialization, not a date for every historical palatal process.

4.45 Technical support: late palatal diphthongization

SC105. Retained late palatal-diphthongization approximation (OELatePalatalDiphthong)

```
define OELatePalatalDiphthong Ctx([
    {*ǣ} -> {*ēa} || .#. [{*ɡ} | {*ɣ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.]
]);
```

This support operation retains exactly the former postmutation long-vowel clause. Its sole current application is *sċēaþ* 'sheath'. The four initial triggers are an unchanged approximation, not a source-proven complete historical conditioner. Campbell and Ringe–Taylor distinguish the later class (Campbell 1959, 68–69; Ringe and Taylor 2014, 235); the incremental implementation does not claim to have resolved it.

4.46 J-cluster coalescence

Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīeġan* 'bend' and *sēċan* 'seek' without assigning this later cluster adjustment the status of a major sound law (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

```
define OEJClusterCoalescence Ctx((
    # *bj coalescence removed: BAllophony (*b→*β) + JLossAfterHeavy handle
    # non-geminated *bj; for geminated *bbj, *j is consumed by JLossAfterHeavy.
    [{*g} {*j} -> {*ɣ}]
    .o. [{*k} {*j} -> {*ɡ}]
));
```

The forms *bīeġan* 'bend' and *sēċan* 'seek' determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc **báugijaną* 'bow' yields †*bēaġan* rather than expected OE *bīeġan*, and PGmc **sōkijaną* 'seek' yields †*sōċan* rather than expected *sēċan*. This constrains this computational cluster consumer. It does not independently date all singleton-fricative articulation or merger: the latter is distinct from this rule and is now explicit under SC109 OEPalatalFricativeMerger (Ringe and Taylor 2014, 204; Laker 2007, 167–68). Nothing in the present lexicon supplies a terminus ante quem for the cluster consumer.

4.47 Back mutation

Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* ‘gave’ and *weofan* ‘weave’. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (Ringe and Taylor 2014, 319, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (Campbell 1959, 86, §207), while Hogg draws the instructive comparison with breaking (Hogg 1992b, 113). Fulk accordingly separates back mutation from the earlier umlautal changes (Fulk 2018, 69, §4.8).

SC059. Back mutation before labials and liquids (`OEBackMutation`)

```
define OEBackMutation Ctx([
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    # Stressed variant (stress preserved on first element):
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
]);
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébaną* ‘give’ produce *ġeofan* ‘give’; the expected form is *ġiefan* ‘give’. **stélaną* ‘steal’ likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébaną* ‘weave’ yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

4.48 West Saxon palatal umlaut

Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut Ctx([
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    # Stressed variants (stress absorbed: result is short i):
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
]);
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage †*mieht* and †*nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

4.49 Weak-tail nasal loss

Historical discussion

The pathway from **dōną* ‘do’ to *dōn* ‘do’ supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (Campbell 1959, 144–45, §§345–349; Hogg 1992b, 121; Fulk 2018, 91, §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss Ctx([
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
]);
```

Final weak-tail **-ną* and **-mą* accordingly yield plain **-n* and **-m*.

Only *dōn* ‘do’ constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* ‘do’. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

4.50 High-vowel apocope

Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (Campbell 1959, 144–45, §§345–349; Hogg 1992b, 121; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

This later loss is distinct from the West Germanic weak-word loss of SC098 PWGmcUnstressedWordFinalIApocope. A selected proclitic context does not become word-final merely because its citation spelling ends there. Its temporary boundary annotation blocks the final-position conditions below and is removed only after their application; it is not a segment of the linguistic reconstruction. The vowel-replacement clauses themselves are unchanged (Ringe and Taylor 2014, 57–58, 284–303).

SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope Ctx([
    # Disyllabic: long syllable (long vowel + consonant(s)) + final vowel
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    # Disyllabic: diphthong + consonant(s) + final vowel
```

```
    # LONG diphthongs are heavy (any C count). SHORT diphthongs are heavy only
    # with 2+ consonants; a short diphthong + single C is a LIGHT syllable (from
    # back-umlaut of a light stem, Campbell §345; DEV_NOTES §17.17).
    # E.g. *spéru → speoru (LIGHT, retain -u); *xérdō → heord (HEAVY, lose -u).
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
   # Disyllabic: heavy closed syllable (short vowel + 2+ consonants) + final vowel
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    # Trisyllabic: long/diphthong + unstressed syllable + final vowel (Campbell
     ↪  §345)
    # Pattern: LongV/Diph + C+ + shortV + C+ + final_vowel
    # Example: heofon < *xemonų: *e*o + *β + *u + *n + *ų
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    # Trisyllabic with short-diphthong first syllable (from back-umlaut of a light
    # stem): trisyllabic apocope still fires regardless of stress-syllable weight
    # (Campbell §345). E.g. *xémonų → *xéomonų → heofon.
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    # Trisyllabic: short syllable + another syllable + final vowel (Campbell §345)
    # Pattern: shortV + C + shortV + C+ + final_vowel
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    # Trisyllabic with heavy-by-position first syllable (short V + 2+C) +
    # light second syllable + final high V (Campbell §345; DEV_NOTES §17.45.4).
    # E.g. *spénnilō → *spinnilu → *spinnil (then OELAdjacentSyncope + nn-degem
     ↪  → spinl).
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    # R/T §6.6.1: vowel-hiatus contraction — word-final *i after long vowel
```

```
    # with no intervening consonant (e.g. root noun dat.sg. *kūi → cȳ)
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    # Campbell §238/§346: final -u lost after /h/ (*x) regardless of weight.
    # Must fire before intervocalic h-loss would remove the conditioning.
    # E.g. *féxu → *féoxu → feoh (not feou).
    {*u} -> 0 || {*x} _ .#.,
    {*ų} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
] .o. [{*ᶜ} -> 0]);
```

Final **i*, **u*, and **ų* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* 'cow' yields †*cū* rather than expected OE *cȳ* 'cow', and PGmc **brūdiz* 'bride' yields †*brūd* rather than expected OE *brȳd* 'bride'. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* 'fright' yields †*fyrht* rather than expected OE *fyrhte* 'fright'. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

4.51 Stem-final n-loss after long i

Historical discussion

Evidence for post-apocope reduction is strikingly uneven. The inherited feminine *in*-stem represented by Gothic *faurhtei*, OE *fyrhtu*, and oblique OE *fyrhte* supplies the relevant evidence (Orel 2003, 120; Ringe and Taylor 2014, 380–81; Campbell 1959, 236, §589.7). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (Hogg 1992b, 121; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

```
define NWGmcInStemNLoss Ctx([{*n} -> 0 || {*ī} _ .#.]);
```

Only final **n* after long **ī* is at issue, as in the inherited *in*-stem behind OE *fyrhte* 'fright'.

CAPR models the oblique OE form through the Proto-Germanic genitive singular **fúrxtīnaz* 'fright', following the project convention of using an appropriate non-nominative paradigm cell when the nominative does not supply the required derivation. Within this selected genitive derivation, the same input fixes both ordering boundaries. Before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* 'fright' yields †*fyrhten* rather than expected OE *fyrhte* 'fright'. After SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* again yields †*fyrhten* rather than expected *fyrhte* 'fright'. I therefore order final bare-a loss, stem-final n-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the selected genitive derivation and depend on one inherited lexeme/paradigm.

4.52 Medial syncope before dentals

SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (Hogg 1992b, 121; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

```
define OEMedialSyncope FootCtx([
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
```

```
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
]);
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

4.53 L-adjacent syncope

Historical discussion

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *netele* 'nettle' supplies a concrete lexical type (Hogg 1992b, 121; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Brunner 1965, 144–45, §§158–159). Fulk places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope Ctx([
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
]);
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátilōn* 'nettle' yields †*nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* 'spindle' yields †*spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields †*spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

4.54 Dental assimilation

SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation Ctx([
    {*θ} -> 0 || {*t} _
]);
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (Hogg 1992b, 121; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

4.55 Preconsonantal degemination

SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination Ctx(OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN);
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* ‘spindle’ (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* ‘spindle’ yields †*spinnl* rather than expected OE *spinl* ‘spindle’. Syncope must therefore create the cluster before degemination simplifies it. Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

4.56 Early o-shortening

Historical discussion

After the principal palatal and umlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening Ctx([
    # EARLY: *ō → *a before nasals (runs BEFORE fronting)
    # This *a will undergo fronting: *a → *æ → e (Campbell §333, §369)
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
]);
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 PNWGmcNStemNLoss, PGmc **nédrōn* ‘adder’ yields †*nædran* rather than expected OE *nædre* ‘adder’, PGmc **érθōn* ‘earth’ yields †*eorþan* rather than expected *eorþe* ‘earth’, and PGmc **fláskōn* ‘flask’ yields †*flascan* rather than expected *flasce* ‘flask’. The same earlier shift also disrupts forms such as *heorte* ‘heart’ and *līne* ‘line’. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 PNWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

4.57 Early unstressed fronting

Historical discussion

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no lexical evidence fixes its later boundary.

SC070. Early fronting of unstressed **a* (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly FootCtx(OEUnstressedAFronting);
```

The rule fronts unstressed **a* to **æ* after the earlier shortening has created a frontable vowel but before the later shortening of unstressed **ō*. It produces endings such as OE *-en* in *lungen* ‘lungs’.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* ‘lungs’ yields †*lungen* rather than expected OE *lungen* ‘lungs’. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* ‘bears’ yields †*boreþ* rather than expected OE *boraþ* ‘bears’, and PGmc **mḗnōθz* ‘month’ yields †*mōneþ* rather than expected *mōnaþ* ‘month’. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

4.58 Late o-shortening

SC071. Later shortening of unstressed **ō* (OELateOShortening)

The following rule handles the later shortening stage.

```
define OELateOShortening Ctx([
    # LATE: *ō → *o elsewhere (runs AFTER fronting; Campbell §355: the late
    # product is "of too late origin to become æ").
    # Stausland Johnsen 2015:28-30: unstressed long vowels shorten regularly in
    # pre-OE AFTER the loss of short unstressed vowels (the *bursti-/*burstīz
    # argument). The shortened vowel is a plain short *o — the ONLY source of
    # unstressed *o at this stage (the earliest OE texts have just two
    # unstressed back vowels, u and a: Stausland Johnsen 2015:21; classical-OE
    # unstressed o arises only from the later 9th-c. lowering, SC040). Its
    # split — u in medial syllables, a in final syllables — is carried out by
    # OEMedUnstressedORaising (SC099) and OEFinalUnstressedOLowering (SC100)
    # immediately below, replacing the old one-step *ō → *a compression that
    # hard-coded the final-syllable outcome for all positions (van Helten's
    # refuted conditioning aside, medial *ō would have surfaced wrongly as a:
    # **wundade).
    {*ō} -> {*o} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
]);
```

The rule shortens the remaining unstressed long **ō* after fronting. The shortened vowel is then resolved by the following medial/final distribution, not directly as *a* (Stausland Johnsen 2015,

28–31).

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* ‘bears’ yield †*boreþ* rather than expected OE *boraþ* ‘bears’. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly.

4.59 Medial raising of shortened unstressed o

SC099. Medial raising of shortened unstressed **o* (OEMedUnstressedORaising)

```
define OEMedUnstressedORaising Ctx([
    {*o} -> {*u} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]* EnglishStarVocalic
]);
```

After SC071 OELateOShortening, the shortened vowel gives *u* in an unstressed medial syllable. The rule encodes Stausland Johnsen’s statistically supported account of West Saxon ō-verb pasts, not a general rule for inherited short **o* or for nominal morphology (Stausland Johnsen 2015, 28–31, 36). His diagnostic derivation is PGmc **wúndōdē* ‘wounded’ > †*wundode* > OE *wundude* ‘wounded’ (Stausland Johnsen 2015, 28–29).

4.60 Final lowering of shortened unstressed o

SC100. Final lowering of shortened unstressed **o* (OEFinalUnstressedOLowering)

```
define OEFinalUnstressedOLowering Ctx([
    {*o} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]* .#.
]);
```

In a final syllable the same shortened vowel gives *a*. Thus the existing month control continues PGmc **mḗnōθz* ‘month’ through shortened **o* to OE *mōnaþ* ‘month’, while **wúndōdē* ‘wounded’ takes SC099 OEMedUnstressedORaising instead. The medial/final contrast and its chronology after long-vowel shortening are Stausland Johnsen’s analysis (Stausland Johnsen 2015, 28–31).

4.61 Unstressed long-vowel shortening

Historical discussion

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **œ* with **e* (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening FootCtx(OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8);
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. If the rule is delayed until after SC073 OEUnstressedAE-Merger, PGmc **nédrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

4.62 Unstressed ae-merger

SC073. Merger of unstressed **æ* with **e* (OEUnstressedAEMerger)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger SentenceCtx([
    [{*æ} -> {*e} || .#. {*ᶠ} OEAnyConsonant* _]
    .o. FootCtx(OEWeakTailReduction3)
]);
```

The rule merges unstressed **æ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Pretonic vowels also participate. A positively identified weak initial syllable is eligible without the preceding-vowel condition appropriate to post-tonic endings. The superscript `*ᶠ` records this prosodic information, not a segment or a grammatical prefix. Stressed roots are not thereby included in the merger. Later lowering of pretonic **i*, including the early *gi-* of *gistīga* ‘climb into/onto’, belongs to SC076 OEPrefixIReduction; Hogg places it marginally after the earlier merger (Campbell 1959, 153–55, §§369–370; Hogg 1992a, §§ 6.50–6.54 (held 2011 reprint, pp. 237–239); Ringe and Taylor 2014, 334–35).

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressed-LongVowelShortening, PGmc **nédrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The lexical evidence fixes the local order after SC072 OEUnstressedLongVowelShortening and places the merger before the later h-loss and contraction.

4.63 Medial unstressed-i lowering

Historical discussion

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (Hogg 1992b, 121; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *sċilling* ‘shilling’ type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 FootCtx([
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _
]);
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhti* rather than expected OE *fyrhte* ‘fright’. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* ‘shilling’ yields †*sċilleng* rather than expected *sċilling* ‘shilling’. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

4.64 Preservation of i before ng

SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering Ctx([
    {*e} -> {*i} || _ {*n} {*g}
]);
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* ‘shilling’.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skíllingaz* ‘shilling’ yield †*sċilleng* rather than expected OE *sċilling* ‘shilling’. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

4.65 Medial unstressed-u lowering

SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering FootCtx([
```

```
    {*u} -> {*o} || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]] [EnglishStarConsonant
     ↪  | EnglishPalatalConsonant]+ _ [[EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] - {*m}]
]);
```

Medial unstressed **u* tends to lower to **o* in the later Old English period, while accented **u* preserves the following unstressed vowel (Campbell 1959, 155–56, §373). This lowering follows the earlier shortening and resolution of unstressed long vowels, rather than preceding that history. The eighth-century unstressed back-vowel system and its later development distinguish the two periods (Stausland Johnsen 2015, 21–22, 28–31).

The inherited ordering evidence from *wuduwe* ‘widow’ remains relevant: delaying SC039 OEWICombinativeUUmlaut until after lowering yields †*wudowe* rather than the expected form. In the adopted sequence, *wuduwe* retains its medial vowel after accented **u*, whereas *weorold* ‘world’ undergoes lowering. The reciprocal early-lowering experiment is no longer equivalent: the intervening vowel-resolution history restores medial **u*, so that movement does not now produce the same *widow* failure.

The formerly reported *ġeoguþ* ‘youth’ boundary does not establish that lowering preceded SC072 OEUnstressedLongVowelShortening. That experiment inserted a front glide vowel early in the derivation; delayed lowering then encountered an altered preceding nucleus and produced †*ġeogoþ*. With the adopted late SC016 OEWsPalatalGlide spelling, the root retains phonological **u* through lowering and preserves the medial vowel. Moving lowering across shortening in this current architecture leaves *ġeoguþ* unchanged. The spelling evidence is therefore not an independent date for the lowering; the later placement rests on the historical vowel chronology, not a reversal of that obsolete experiment.

4.66 Pretonic i-lowering

Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk’s discussion of prefix vowels accounts for OE **be-* and **ne-* (Fulk 2018, 97, §5.7). Hogg explicitly includes *be-* and *ge-* in the lowering of unstressed **i*, marginally later than the earlier front-vowel merger. Ringe and Taylor supply early *gistīga* ‘climb into/onto’ beside later *gestīga* ‘climb into/onto’, and place *bi-* and *ni* alongside this development (Hogg 1992a, §§ 6.50–6.54 (held 2011 reprint, pp. 237–239); Ringe and Taylor 2014, 334–35).

The manuscript evidence, rather than the older proxy’s displacement results, supports this limited relative chronology.

SC076. Pretonic i-lowering (`OEPrefixIReduction`)

```
define OEPrefixIReduction SentenceCtx([
    [{*i} -> {*e} || .#. {*ᶠ} OEAnyConsonant* _]
    .o. [{*ᶠ} -> 0]
]);
```

The rule lowers short **i* in a positively identified weak pretonic syllable, not specifically in morphemes beginning with a particular consonant. Thus the same conditioning treats inherited weak **bi-* and the raised vowel in **gi-*; stressed roots remain outside its domain. The superscript `*ᶠ` is a nonsegmental prosodic annotation, removed once it is no longer needed. The rule therefore produces **e* directly, not a protected intermediate breve-e.

This bounded implementation requires a following headed syllable. The separately unstressed negative particle *ni* ‘not’ > *ne* ‘not’ supports the wider history of weak-vowel lowering, but is not made into a newly encoded selected input or a claim that the present pretonic condition covers every weak particle (Ringe and Taylor 2014, 335).

The earlier displacement experiment found no final-output boundary for the old consonant-identity proxy. That negative result is not a test of the reformulated prosodic rule and does not date it. The relative placement after the earlier merger follows the handbook account; its precise interval remains approximate.

4.67 Weak-tail reduction

Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc ***bákaną*** ‘bake’ yields †***bacen*** rather than expected OE ***bacan*** ‘bake’, and PGmc ***bíndaną*** ‘bind’ yields †***binden*** rather than expected ***bindan*** ‘bind’, alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc ***fléuxaną*** ‘flee’ yields †***flēoan*** rather than expected OE ***flēon*** ‘flee’, and PGmc ***sláxaną*** ‘slay’ yields †***sleaan*** rather than expected ***slēan*** ‘slay’.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

4.68 J-loss after heavy syllables

Historical discussion

After SC079 OEJLossAfterHeavy removes ***j*** in heavy environments, forms such as ***lungen*** ‘lungs’ acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final ***nn*** outcome in the following derivation.

SC079. Loss of ***j*** after heavy syllables (`OEJLossAfterHeavy`)

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

The rule removes ***j*** after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is ***j***.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* ‘believe’ yields †*ġelēafan* rather than expected OE *ġelīefan* ‘believe’, PGmc **báugijaną* ‘bow’ yields †*bēaġan* rather than expected *bīeġan* ‘bow’, and PGmc **fúlgijaną* ‘follow’ yields †*fulġan* rather than expected *fylġan* ‘follow’. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* ‘lungs’ yields †*lungenn* rather than expected OE *lungen* ‘lungs’. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

4.69 Final geminate simplification

SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* ‘lungs’ yield †*lungenn* rather than expected OE *lungen* ‘lungs’. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

4.70 J-strengthening after front diphthongs

Historical discussion

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then normalizes part of the inherited intervocalic **j* domain, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ǽa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjaną* ‘strew’ yields †*strēaġan* rather than expected OE *strīeġan* ‘strew’. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields †*strīeian* rather than *strīeġan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīeġan* ‘strew’ derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

4.71 Intervocalic j-vocalization

SC082. Bounded inherited-j normalization (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} ||
        [EnglishStarVocalic - [{*æ}|{*ē}|{*é}]] _ EnglishStarVocalic
];
```

The former unrestricted VjV matcher wrongly gave model-only †*cæie* instead of *cæġe* ‘key’. Hogg’s pre-OE **kæjæ* retains the glide (Hogg 1979, 105). The bounded rule therefore excludes preceding non-high long front monophthongs.

This is not a generic physical vocalization law. Hogg distinguishes high-vowel coalescence from spellings that merely represent a consonantal glide and from unstressed alternations (Hogg 2011, 283–86, §§7.69–7.76). The remaining weak-suffix e+j to e+i to i path telescopes later raising/contraction, rather than asserting that its e+i intermediate is a source reconstruction (Ringe and Taylor 2014, 228). SC083 OEUnstressedEIContraction consumes that modeled sequence. The full residual proxy domain is not claimed to be one independently established historical event.

The separately represented fricative ȷ passes through this zone without being treated as inherited j. Its later merger belongs to SC109 OEPalatalFricativeMerger.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjaną* ‘strew’ yield †*strīeian* rather than expected OE *strīeġan* ‘strew’. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjaną* ‘bore’ yield †*boreian* rather than expected OE *borian* ‘bore’, PGmc **xándlōjaną* ‘handle’ yield †*handleian* rather than expected *handlian* ‘handle’, and PGmc **mákōjaną* ‘make’ yield †*maceian* rather than expected *macian* ‘make’. The witness forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

4.72 Unstressed ei-contraction

SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ {*i}
];
```

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* ‘bore’ and *liccian* ‘lick’.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **búrōjaną* ‘bore’ yield †*boreian* rather than expected OE *borian* ‘bore’, and PGmc **líkkōjaną* ‘lick’ yield †*licceian* rather than expected *liccian* ‘lick’. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

4.73 Intervocalic h-loss

Historical discussion

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (Fulk 2018, 270, §12.21), and Luick describes the corresponding West Germanic contractions (Luick 1914--1940, 165).

SC085. Loss of intervocalic **h* (`OEHLoss`)

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* ‘flee’ yields †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxaną* ‘slay’ yields †*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yields †*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* yields †*tāe* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

4.74 Contraction after h-loss

SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*éo},
    {*éo} {*o} -> {*éo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*é} {*a} -> {*é},
    {*é} {*e} -> {*é},
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*í} {*a} -> {*í},
    {*í} {*e} -> {*í},
```

```
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
    {*ū} {*e} -> {*ū}
];
```

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* ‘flee’, *slēan* ‘slay’, and *tēon* ‘draw’.

Moving contraction before SC085 OEHLoss makes PGmc **fléuxaną* ‘flee’ yield †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxaną* ‘slay’ yield †*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yield †*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* ‘toe’ yield †*tāe* rather than expected *tā* ‘toe’. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

4.75 R-metathesis

Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (Brunner 1965, 159, §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (Luick 1914--1940, 201).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstaną* ‘burst’ yield †*beorstan* rather than expected OE *berstan* ‘burst’. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters no output.

The lexical evidence fixes the earlier relation but does not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

4.76 Late unstressed suffix raising

Historical discussion

The late history of unstressed **-ag-* differs from prehistoric i-mutation. Ringe and Taylor give the sequence through fronting and raising to *-ig-*, explicitly placing its final stage long after mutation. Their examples include *hunigʒ* 'honey'; a velar intermediate can survive before a back vowel (Ringe and Taylor 2014, 334–35, §6.9.6). These are the printed pages, not the PDF sheet labels.

SC089. Late suffix consumer (`OELateUnstressedAgSuffix`)

```
define OELateUnstressedAgSuffix (
    [{*a} -> {*e} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ _
         ↪  {*g}]
    .o.
    [{*g} -> {*ʝ} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ {*e}
         ↪  _ .#.]
    .o.
    [{*e} -> {*i} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ _
         ↪  [{*ʝ}|{*ʠ}]]
);
```

The singleton consonant is a palatal fricative, not an early affricate (Fulk 2018, 130–32). The raising consumer therefore recognizes ʝ alongside retained stop-reflex compatibility. This does not make ʝ a prehistoric mutation trigger: later raising and earlier i-mutation have different conditioning histories.

The present late applications are *hunigʒ* 'honey' and *wīþigʒ* 'withy'. The word-final guard remains narrow; this implementation is not an exhaustive model of every inflected suffix form. SC109 OEPalatalFricativeMerger follows here as a disclosed serialization, not proof of a uniquely dated historical merger immediately after this raising (Ringe and Taylor 2014, 204, pp. 334–335).

4.77 Technical support: postmutation fricative merger

SC109. Postmutation merger serialization (OEPalatalFricativeMerger)

```
define OEPalatalFricativeMerger [
    {*ʝ} -> {*j}
];
```

The historical commitment is merger after mutation (Ringe and Taylor 2014, 204). Its execution after SC089 OELateUnstressedAgSuffix is a computational holding zone, not proof of an exact historical date. It keeps the fricative out of inherited-j normalization while preserving late palatal raising, syllable weight, reduction and native rendering. SC057 OEJClusterCoalescence remains a different cluster process. Stop affrication dates and the precise disputed eligibility domain are not settled by this merger.

Chapter 5

Old English orthography and the written surface

5.1 Historical interval

This short chapter stands apart from the derivational chapters that precede it. The earlier chapters primarily discuss changes to the spoken language, alongside explicitly identified technical support operations. The material treated here belongs instead to the written surface of Old English — scribal conventions that determined how the results of the completed phonological history were committed to parchment.

In the executable model these conventions apply after every phonological rule, in the final written-surface block. That serialization renders the results of the phonological reconstruction; it does not independently date the origin of each spelling practice relative to every phonological change.

5.2 Scope

The one rule treated here is the West Saxon palatal-glide spelling (SC016), by which back vowels following word-initial [j] — spelled *g* — came to be written with a preceding front glide letter, as in *geoc* ‘yoke’ for spoken [jok] and *geogu̯þ* ‘youth’ for a form whose root vowel remained [u]. Ringe and Taylor state the modern assessment directly: the *eo* of *geoc* is a spelling convention, and the word was pronounced [jok] (Ringe and Taylor 2014, 5). Hogg reaches the same verdict for the back-vowel cases generally (Hogg 1992b, 113). The older handbooks — Campbell, Brunner, Bülbring, Luick — analysed the same spellings as rising diphthongs; the section below presents both views (Campbell 1959, 17, § 44; Brunner 1965, 64–65, § 92).

5.3 Sources

Ringe and Taylor provide the modern phonological interpretation (Ringe and Taylor 2014, 5). Campbell (Campbell 1959, 17, 64–67, §§ 44, 170–176), Brunner (Brunner 1965, 64–65, § 92), and Bülbring (Bülbring 1902, 120, §§ 298–299) document the distribution of the spellings; Hogg supplies the critical reassessment (Hogg 1992b, 113).

5.4 West Saxon palatal-glide spelling before back vowels

Historical discussion

West Saxon spellings such as *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeoguþ* ‘youth’ write a front glide letter between a word-initial palatal and a following back vowel. Campbell describes the phenomenon as the development of rising diphthongs when “palatal glides developed before back vowels” and cites *ġeoc* directly (Campbell 1959, 17, §44); Brunner separates the *u*-cases (*ġeong*, *ġeoguþ*) from the *o*-cases (*ġioc* ‘yoke’, *ġeoc*) (Brunner 1965, 64–65, §92.1); Bülbring likewise treats *iuguð* ‘youth’ and *iuc* under *ju* but derives *ġioc*, *ġeoc* from West Germanic **jok* (Bülbring 1902, 120, §§298–299); and Luick groups all of these under his “schwebende Diphthonge” after palatal onsets (Luick 1914–1940, 158–59, §169).

The phonological interpretation of these spellings is disputed. The older handbook tradition — Campbell, Brunner, Bülbring, Luick — reads them as genuine rising diphthongs. The modern assessment is orthographic: Ringe and Taylor state flatly that *ġeoc* “is /jok/”, the digraph being a spelling convention that became universal after word-initial /j/ (Ringe and Taylor 2014, 5), and Hogg concludes that the back-vowel cases were “never anything more than an orthographic variation”, judging Campbell’s arguments to the contrary “insubstantial” (Hogg 1992b, 113; Campbell 1959, 66–67, §176). This model follows Ringe and Taylor and Hogg: the rule is a spelling convention applied to the finished phonology, and it therefore stands at the end of the derivation, in the written-surface stage of the cascade.

Its position also settles a relative chronology. The *o* of *ġeoc* ‘yoke’ is itself the product of Northwest Germanic u-lowering (SC017 PNWGmcULowering): Fulk lists *ġeoc* as a regular lowering example beside OIcel *ok* and OHG *joh* (Fulk 2018, 56, §4.3), and Campbell gives *ġeoc* among the regular *u* > *o* words (Campbell 1959, 43, §115). The lowering therefore feeds the spelling: first **juk-* became **jok-* in Northwest Germanic, and only much later did West Saxon scribes write the result as *ġeoc*. Where lowering did not apply, as in *ġeoguþ* ‘youth’, whose root *u* was protected by the high vowel of the following syllable, the same convention wrote the retained *u* with the same digraph (Brunner 1965, 64–65, §92.1).

SC016. West Saxon palatal-glide spelling before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
    {*ó} -> {*éo} || .#. ġ _ ,
    {*ú} -> {*éo} || .#. ġ _ ,
    {*o} -> {*eo} || .#. ġ _ ,
    {*u} -> {*eo} || .#. ġ _
];
```

The rule rewrites a back vowel after word-initial *ġ* as the digraph spelling, covering both the lowered *o*-cases (*ġeoc* ‘yoke’) and the retained *u*-cases (*ġeoguþ* ‘youth’). Because it is a convention of the written language, it applies after every phonological change; in particular it follows SC017 PNWGmcULowering, which supplies the *o* of *ġeoc*. If the spelling rule were placed before u-lowering, the derivation would have to treat an Old English scribal practice as a Northwest Germanic sound change, an ordering that no source supports. The witnesses *ġeoc* and *ġeoguþ* between them fix both faces of the rule: one shows the convention applied to lowered *o*, the other to unlowered *u*. The handbook domain is broader (it also includes *a*/*ā*/*ō* contexts after word-initial palatals), but this executable rule is intentionally complete for the currently selected corpus witnesses rather than a maximal dialectal enumeration.

Part II

Lexical derivations

Word-by-word derivations

Introduction

The catalogue groups words by the kind of historical explanation they require. Regular derivations establish the reach of the sound laws. The remaining classes mark, without concealment, the places where attestation, morphology, analogy, or an unresolved mismatch intervenes.

The catalogue proceeds from words; the sound-change account proceeds from rules. Together they allow each proposed sound law to be tested against its lexical witnesses and each etymology against the complete ordered history. SC-numbered rules cited in these entries are discussed in the companion sound-change volume, or in the sound-change chapters of the combined book.

Data and sources

This volume assembles the lexical corpus from the aligned Germanic dataset and the compact derivation traces that accompany each entry. Comparative dictionaries, Old English dictionaries, and historical grammars are cited in the prose where they bear on particular lexical arguments.

The result is a lexical catalogue rather than a separate report on citation method or trace machinery.

Transducer and derivation method

Four objects must be distinguished in every derivation: the citation reconstruction, the selected input, the transducer outcome, and the Old English target. The summary identifies them where they differ; the boxed trace then divides the changes into Earlier Germanic and Old English stages.

Derivation classes

The lexical catalogue is ordered by seven derivation classes and, within each class, alphabetically by headword. Counts in this alpha are:

- Regular derivations: 88
- Attested variants: 4
- Early analogy: 36
- Late analogy: 29
- Reconstructed Old English comparators: 3
- Known but unmodelled developments: 2
- Unexplained or deliberately unmodelled exceptions: 5

Chapter 6

Regular derivations

In these entries the selected Germanic input yields the Old English reflex by regular sound change. They establish the standard against which the following analogical and exceptional histories must be judged.

6.1 adder — OE *nǣdre*

Derivation: **nǽdrōn* > *nǣdre* (regular).

Derivation trace

Proto input: **nǽdrōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**nǣdræ*
PNWGmc Long E Lowering	**nādrōn*	OE Unstressed AE Merger	**nǣdre*
PNWGmc N Stem N Loss	**nādrǭ*		
EAF Long A Fronting	**nǣdrǭ*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdre*

Reconstruction and comparative evidence

Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nǣdre* ‘adder’, *næddre* ‘adder’ under the latter (Kroonen 2013, 381, 386). Orel’s feminine entry gives **nēdrōn* ‘adder’ ~ **nadrōn* ‘adder’ and relates them to the masculine formation. These comparison spellings use plain *d* for Orel’s printed barred *đ*; the quantity and the two alternatives are retained (Orel 2003, 286).

The derivational input retains the long-*e* feminine alternative. The lexical comparison supports that formation without erasing Orel’s alternative or the distinction between his notation and CAPR’s derivational input.

Reconstruction alternatives and their limits

Orel’s short-*a* feminine is an admitted alternative within his own account, not proof that he and Kroonen advocate exclusive histories. The unresolved premise is the reflex distribution and formation history licensing that alternative beside the selected long-*ē* feminine (Orel 2003, 286; Kroonen 2013, 381, 386). Kroonen’s uncertain sewing or ‘winder’ connection and masculine

comparanda cannot substitute for that feminine argument. CAPR's acute marks its input stress; it is not independently quoted in these headwords.

Fulk's probable shortening accompanying pre-*r* gemination concerns the later Old English variant. He calls the motivation obscure and considers analogical extension; this does not explain inherited *ē* versus *a* or supply another complete ancestral adder word (Fulk 2018, 132). Regular production of *nǣdre* therefore supports the selected path without settling the alternative formation.

Old English evidence

The Old English word is securely represented by *nǣdre* 'adder', with *næddre* 'adder' as a secondary variant. Clark Hall cross-references *næddre* to *nædre* 'adder', and Fulk treats *næddre* as the later geminated form beside the older base (Clark Hall 1960, 225; Fulk 2018, 132).

Development to Old English

From **nédrōn* 'adder', the stressed long mid vowel develops to Old English *nǣdre* 'adder', and the weak feminine ending remains as final *-e*, giving *nǣdre*. The doubled consonant of *næddre* 'adder' is secondary and does not alter the inherited base form.

6.2 bake — OE *bacan*

Derivation: **bákaną* > *bacan* (regular).

Derivation trace

Proto input: **bákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**bækaną*
[no change]	OE A Restoration	**bakaną*
	OE Heavy Syllable Nasal Apocope	**bakan*
Early Anglo-Frisian	OE Secondary Nasalization	**bakąn*
[no change]	OE Weak Tail Reduction	**bakan*

Old English form: *bacan*

Reconstruction and comparative evidence

Orel reconstructs the verb as **bakanan* 'bake' and cites Old English *bacan* 'bake' beside Old High German *backan, bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* 'bake' to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

Singleton and geminate present alternatives

The selected infinitive keeps singleton *k*. Kroonen explicitly distinguishes **bakan-* beside **bakkan-*, considering contamination with, or back-formation from, the iterative **bak(k)ōn-*; a present *n*-suffix explanation is less likely (Kroonen 2013, 49). Orel prints singleton **bakanan* and separately the long-*ō* preterite (Orel 2003, 33). The inspected passages do not establish Orel's reason for excluding the geminate from this citation. This is a qualified present-formation question, not proof of a mutually exclusive reconstruction debate.

Later finite-cell front vowels and the inherited preterite grade are not rival infinitive vowels. Ringe and Taylor distinguish West Saxon infinitive *bacan* 'bake' from second singular *becst* 'you bake', preserving the umlauted vowel in syncopated forms while *a* spread elsewhere

(Ringe and Taylor 2014, 232–33). Regular infinitive production does not resolve the earlier singleton/geminate distribution.

Old English evidence

Bosworth-Toller and Clark Hall both record *bacan* ‘bake’ as the ordinary Old English verb (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English

From **bákaną* ‘bake’, Anglo-Frisian brightening first gives **bækaną* ‘bake’. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* ‘bake’ (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: **bákaną* > *bacan*.

6.3 beech — OE *bōc*

Derivation: **bōkō* > *bōc* (regular).

Derivation trace

Proto input: **bōkō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bōk*
PNWGmc Final Long O Raising	**bōku*		
Early Anglo-Frisian			
[no change]			

Old English form: *bōc*

Reconstruction and comparative evidence

Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc* ‘beech’, *bēce* ‘beech’ among its reflexes (Kroonen 2013). The derivational input **bōkō* ‘beech’ is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence

Kroonen’s Old English evidence already separates the paradigm material: *boc* ‘beech’ as the nominative form and *bēce* ‘beech’ as an oblique form (Kroonen 2013). The relevant comparator is therefore *bōc* ‘beech’; *bēċe* ‘beech’ remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English

With nominative input **bōkō* ‘beech’, the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc* ‘beech’. The regular comparison is therefore **bōkō* > *bōc*.

6.4 begin — OE *beġinnan*

Derivation: **bigínnaną* > *beġinnan* (regular).

Derivation trace

Proto input: **bigínnaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**bigínnan*
[no change]		OE Secondary Nasalization	**bigínnąn*
Early Anglo-Frisian		OE Velar Palatalization	**bi̯ínnąn*
[no change]		OE Prefix I Reduction	**bei̯ínnąn*
		OE Weak Tail Reduction	**bei̯ínnan*
		OE Palatal Fricative Merger	**bei̯ínnan*

Old English form: *beġinnan*

Reconstruction and comparative evidence

The verb is modeled here as inherited **bigínnaną* ‘begin’. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (Ringe and Taylor 2014), and Campbell lists *ginnan* ‘begin’ among familiar examples of palatal *g* in this verb family (Campbell 1959, 174).

Old English evidence

Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* ‘begin’ / *beginnan* ‘begin’ (Bosworth and Toller 1898, 84; Clark Hall 1960). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan* ‘begin’.

Development note

The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (Ringe and Taylor 2014).

Development to Old English

From **bigínnaną* ‘begin’, heavy-syllable nasal apocope yields **bigínnan* ‘begin’. Intervocalic **g* between front vowels then palatalizes to *ġ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan* ‘begin’.

6.5 bier — OE *bǣr*

Derivation: **bḗrō* > *bǣr* (regular).

Derivation trace

Proto input: **bḗrō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bǣr*
PNWGmc Long E Lowering	**bārō*		
PNWGmc Final Long O Raising	**bāru*		
EAF Long A Fronting	**bǣru*		
Early Anglo-Frisian			
[no change]			

Old English form: *bǣr*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **bērō-* f. ‘bier’ and cites Old English *bar* ‘bier’, *bær* ‘bier’ among the reflexes (Kroonen 2013, 717). The derivational input **bḗrō* ‘bier’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller lemmatize the noun as *bær* ‘bier’, and Kroonen also records *bar* ‘bier’ beside it (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717). The target *bǣr* ‘bier’ is therefore a normalized long-vowel spelling of the same noun.

Source note

Lexicographic spellings vary between *bær* ‘bier’ and *bar* ‘bier’. The normalized target *bǣr* ‘bier’ represents the same long vowel (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717).

Development to Old English

From **bḗrō* ‘bier’, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *ǣ*, and high-vowel apocope yields *bǣr* ‘bier’. The resulting noun matches the normalized Old English target.

6.6 birth — OE *byrd*

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace

Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**byrdi*
EAF Final Z Deletion	**búrdi*	OE High Vowel Apocope	**byrd*
Early Anglo-Frisian			
[no change]			

Old English form: *byrd*

Reconstruction and comparative evidence

Kroonen cites the noun under stem-level **burdi-* and gives Old English *(ge-)byrd* among the reflexes (Kroonen 2013). The derivational input **búrdiz* ‘birth’ is the nominative-style form that stands behind that stem label.

Old English evidence

Clark Hall and Bosworth-Toller both attest simplex *byrd* ‘birth’ as an Old English noun meaning ‘birth’ (Clark Hall 1960; Bosworth and Toller 1898, 125). The prefixed form *gebyrd* ‘birth’ is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* ‘birth’ entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (Kroonen 2013; Bosworth and Toller 1898, 125; Campbell 1959).

Form note

The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Kastovsky's discussion of deverbal feminines provides the broader derivational setting (Kastovsky 1992, 358).

Development to Old English

From **búrdiz* 'birth', loss of final *z* gives **búrdi* 'birth'. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

6.7 bone — OE *bān*

Derivation: **báiną* > *bān* (regular).

Derivation trace

Proto input: **báiną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**bān*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**bāną*		

Old English form: *bān*

Reconstruction and comparative evidence

Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* 'bone' (Kroonen 2013; Orel 2003). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān* 'bone'.

Old English evidence

Clark Hall and Bosworth-Toller record *bān* 'bone' as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898). Bright's glossary also distinguishes citation-form *bān* 'bone' from oblique *bāne* 'bone' (Bright 1971).

Source note

The comparative headwords **baina-* and **bainan* 'bone' provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną* 'bone'.

Development to Old English

West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną* 'bone'; heavy-syllable nasal apocope then yields *bān* 'bone'. The resulting form matches the attested Old English citation noun.

6.8 book — OE *bōc*

Derivation: **bōkz* > *bōc* (regular).

Derivation trace

Proto input: **bōkz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *bōc*

Reconstruction and comparative evidence

The noun is an athematic consonant stem (root noun), and the sources print its nominative singular in three different notations that encode the same morphology. Orel's citation form is **bōkz* 'book', with the explicit nominative marker **-z* that he writes across the whole root-noun class (Orel 2003, 52). Kroonen cites the bare stem **bōk-* 'book, beech' and treats Gothic *bōka* 'letter' as a separate ō-stem formation (Kroonen 2013, 71–72). Kluge/Seebold print a third variant with voiceless sibilant, **bōks* 'book' (Kluge 2011, 158). These are citation conventions, not competing facts: Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306). No scholar is simply "wrong" here; the notations differ in whether they make the inflectional marker visible.

The derivational input **bōkz* 'book' follows Orel's morphologically explicit convention: it records the nominative marker whose historical elimination the derivation then models. The word is distinct from the ō-stem 'beech' word, whose derivational input here is the separate nominative **bōkō* 'beech'.

Old English evidence

Old English *bōc* 'book' is the endingless nominative-accusative singular of a root noun, with the mutated plural *bēċ* 'books' preserving the class's characteristic i-umlaut alternation (Fulk 2018, 165–66). Comparative root-noun evidence shows the branch-by-branch fate of the ending: Gothic keeps its sibilant in this class (*baúrgs* 'city', *nahts* 'night'), while Old Norse *bók* 'book' keeps a reflex of **-z* as *-r* only in the masculine root nouns and drops it in feminines (Fulk 2018, 165–67). West Germanic alone is uniform: none of its daughters shows any ending in the root-noun nominative (Ringe and Taylor 2014, 118).

Development to Old English

The nominative marker of the consonant stems was eliminated before Proto-West Germanic: endingless nominatives arose by Szemerényi's law after sonorant-final stems and spread analogically through the class (Fulk 2018, 143), so that no West Germanic daughter inherits an ending here (Ringe and Taylor 2014, 118). From the derivational input **bōkz* 'book' the loss of the marker gives **bōk* 'book', which is already the Old English form: *bōc* 'book' differs only in spelling. This early morphological loss is a different development from the later West Germanic loss of final **z* in unstressed syllables (as in **bárdaz* 'beard') and from the still later northern loss in stressed monosyllables (as in the pronouns); the root-noun ending was gone before either of those changes applied.

6.9 both — OE *bū*

Derivation: **bō* > *bū* (regular).

Derivation trace

Proto input: **bō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Stressed Monosyllable O Raising	**bū*	
Early Anglo-Frisian		
[no change]		

Old English form: *bū*

Reconstruction and comparative evidence

Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai* 'both', **bans* 'both (pl.)', **bōz* 'both'/ **bōns* 'both', **bō* 'both', with Old English *bēġen* 'both', *bā* 'both', and neuter *bū* 'both' (Kroonen 2013, 47). For the present entry, the relevant inherited form is the unextended neuter dual **bō* 'both'.

The older explanation of *bēġen* 'both' derives it from **bō-jen-*, and Orel still gives OE *bezen* 'both' (< **bō-jenō* 'both') beside ON *báðir* 'both', OFris *bēthe* 'both', OS *be-thia* 'both', and OHG *bēde* 'both' (Orel 2003, 65). Fulk reports that explanation cautiously and notes Seebold's preference for a **bō-þ-* analysis instead (Fulk 2018, § 10.1). That debate concerns *bēġen* and the extended forms behind Modern English *both* 'both', German *beide* 'both', and Dutch *beide* 'both'; it does not displace the inherited neuter **bō* 'both' > *bū* 'both' treated here.

Old English evidence

The Old English dual paradigm is well established. Brunner gives masculine *bēġen* 'both', feminine *bā* 'both', and neuter *bū* 'both' beside *bā*, with compounds such as *bā twā* 'both (two)', *bū tū* 'both (two, neut.)', and *bām* 'both' *twām* 'two' (Brunner 1965, § 324 Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēġen* 'both', feminine *bā*, neuter *bā*, *bū*, genitive *bēġra* 'both', *bēġ(e)a*, and dative *bǣm* 'both' (Campbell 1959, § 683; Fulk 2018, § 10.1).

bū 'both' is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēġen* 'both' belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* 'both' remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English

**bō* 'both' is a stressed monosyllabic form. Campbell cites *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (Campbell 1959, § 122). Brunner states the same development more directly: Auslautendes *ō* erscheint als û in *bū* … cu … *hū*, *tū* (Brunner 1965, § 69).

The development is therefore straightforward: **bō* 'both' > *bū* 'both'.

Form comparison

The comparison below sets the relevant forms side by side. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
**bō* > *bū*	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry

Form	Source / stage	Status	Relevance to this entry
bēġen	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the Old English form here
bā	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator
báðir, German *beide*, Dutch *beide*, Modern English *both*	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE *bū*

6.10 bow — OE *bīeġan*

Derivation: **báugijaną* > *bīeġan* (regular).

Derivation trace

Proto input: **báugijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**báeugijaną*
[no change]	OE Diphthong Leveling	**bēagijaną*
	OE Heavy Syllable Nasal Apocope	**bēagijan*
Early Anglo-Frisian	OE Secondary Nasalization	**bēagijąn*
[no change]	Sievers Law Syncope	**bēagjąn*
	OE Velar Palatalization	**bēaǧjąn*
	OE I Umlaut	**bīejjąn*
	OE Weak Tail Reduction	**bīejjan*
	OE J Loss After Heavy	**bīejan*
	OE Palatal Fricative Merger	**bīejan*

Old English form: *bīeġan*

Reconstruction and comparative evidence

Kroonen reconstructs the weak verb as **baugjan-* ‘to (make) bend’ and cites Old English *biegan* ‘bow, bend’ among its reflexes (Kroonen 2013). Ringe and Taylor give the northwest Germanic preform **baugijana* ‘bow’, with later **béagjan* ‘bow’ behind West Saxon Old English *biegan* ‘bow, bend’ (Ringe and Taylor 2014). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence

Clark Hall lemmatizes *biegan* ‘bow, bend’ as an Old English weak verb, and Bosworth-Toller records *bigan* ‘bow, bend’ as an attested form (Clark Hall 1960; Bosworth and Toller 1898, 102). The form *bīeġan* ‘bow, bend’ used here is a normalized spelling of that attested Old English weak verb.

Development to Old English

From **báugijaną* ‘bow’, the stem reaches pre-Old-English **bēagjan* ‘bow’, after which palatalization of **gj* and i-umlaut yield West Saxon *biegan* ‘bow’; Campbell lists *biegan* ‘bow’ among the

regular *ie* outcomes of **éa* under i-umlaut (Ringe and Taylor 2014; Campbell 1959, 80). The development is therefore straightforward: **báugijaną* > *bīeġan* ‘bow’.

6.11 breeches — OE *brēċ*

Derivation: **brōkiz* > *brēċ* (regular).

Derivation trace

Proto input: **brōkiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Palatalization	**brōʧi*
EAF Final Z Deletion	**brōki*	OE I Umlaut	**brēʧi*
Early Anglo-Frisian		OE High Vowel Apocope	**brēʧ*
[no change]			

Old English form: *brēċ*

Reconstruction and comparative evidence

Kroonen cites the noun under **brōk-*, with Old English *brōc* ‘breeches’ and plural ‘breeches’ among its reflexes (Kroonen 2013). Ringe and Taylor give the plural development directly as PNWGmc **brōkiz* ‘breeches’ > **breeci* ‘breeches’ > OE *bréc* ‘breeches’ (Ringe and Taylor 2014). The deeper verbal base belongs to the noun’s etymological background, while the derivational input here is the plural noun form **brōkiz* ‘breeches’.

Old English evidence

Bright notes *brōc* ‘breeches’ with plural *brēc* ‘breeches’, and Clark Hall gives *brēc* fp. breeches while also listing *broc* ‘breeches’ as a feminine noun probably represented chiefly in the plural (Bright 1971; Clark Hall 1960, 64). I write *brēċ* ‘breeches’ for the long vowel and palatal consonant; the attested plural is *brēc*.

Development to Old English

After loss of final *-z*, the stem ends in *-ki*, so the velar palatalizes and *ō* ‘breeches’ undergoes i-umlaut to *ē* ‘breeches’; final high-vowel apocope then yields *brēċ* ‘breeches’ (Ringe and Taylor 2014). The development is therefore regular: **brōkiz* ‘breeches’ > *brēċ* ‘breeches’.

6.12 calf — OE *ċealf*

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace

Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kálb*
EAF Final Z Deletion	**kálba*	EAF Brightening	**kœlb*
Early Anglo-Frisian		OE Breaking	**kealb*
		PGmc B Allophony	**kealβ*
[no change]		OE Velar Palatalization	**ʧealβ*

Old English form: *ċealf*

Reconstruction and comparative evidence

Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz* ‘calf’ with plural **kalbizō* ‘calves’, while Orel cites **kalbaz* ‘calf’ as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz*, **kalbiz-* (Kroonen 2013; Orel 2003, 248; Ringe and Taylor 2014, 220). The derivational input here is the singular **kálbaz* ‘calf’, since the entry concerns the citation-form noun.

Old English evidence

Clark Hall gives *cealf* ‘calf’ I. (æ, e) nm. (nap. *cealfru* ‘calves’), and Bosworth-Toller likewise records *Caelf* / *Cealf* ‘calf’ beside plural forms such as *calfur* and *cealfru* ‘calves’ (Clark Hall 1960; Bosworth and Toller 1898, 131). Campbell and Fulk show the same singular-plus-*-r-* plural pattern (Campbell 1959; Fulk 2018, 193). I write *ċealf* ‘calf’ for the palatalized initial; the attested dictionary headword is *cealf* ‘calf’.

Development to Old English

After loss of final *-z* and bare *-a*, Anglo-Frisian brightening gives **kœlb* ‘calf’, and breaking before *l* plus consonant yields **kealb* ‘calf’. Ringe and Taylor’s account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *ċ-*, so **kálbaz* ‘calf’ develops regularly to *ċealf* ‘calf’ (Ringe and Taylor 2014, 220).

6.13 coat — OE *rocc*

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace

Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**rókk*
PNWGmc U Lowering	**rókkaz*		
EAF Final Z Deletion	**rókka*		
Early Anglo-Frisian			
[no change]			

Old English form: *rocc*

Reconstruction and comparative evidence

Orel cites a masculine **rukkaz* ‘coat’ for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (Orel 2003; Kroonen 2013, 290).

Old English evidence

Clark Hall and Bosworth-Toller record *rocc* ‘coat’ as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* ‘breast-coat’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

With **rúkkaz* ‘coat’ as the derivational input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note

This entry concerns the garment noun only. The stone word seen in *stānrocc* ‘stone-coat’ belongs to a different lexical history (Clark Hall 1960; Bosworth and Toller 1898).

6.14 corn — OE *corn*

Derivation: **kúrną* > *corn* (regular).

Derivation trace

Proto input: **kúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**kórn*
PNWGmc U Lowering	**kórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *corn*

Reconstruction and comparative evidence

Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan* ‘corn’, both with Old English *corn* ‘corn, grain’ among the reflexes (Kroonen 2013; Orel 2003, 264). The singular form **kúrną* ‘corn’ is the nominative-accusative singular appropriate to the citation noun.

Old English evidence

Clark Hall gives *corn n. ‘corn,’ grain*, Bright’s glossary lists *corn, n.* with genitive singular *cornes*, and Bosworth-Toller treats *corn* ‘corn, grain’ as an ordinary noun headword (Clark Hall 1960; Bright 1971, 347; Bosworth and Toller 1898, 144). The target is therefore an attested citation form, while forms such as *cornes* simply provide paradigm background.

Development to Old English

With northwest Germanic lowering, **kúrną* ‘corn’ becomes **kórną* ‘corn’, and later loss of final nasal after a heavy syllable yields **kórn* ‘corn’, whence *corn* ‘corn, grain’. The oblique form **kurnăn* ‘corn’ belongs to comparative background rather than to the derivational input of this entry.

6.15 deed — OE *dǣd*

Derivation: **dǽdiz* > *dǣd* (regular).

Derivation trace

Proto input: **dḗdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**dǣd*
PNWGmc Long E Lowering	**dādiz*		
EAF Final Z Deletion	**dādi*		
EAF Long A Fronting	**dǣdi*		
Early Anglo-Frisian			
[no change]			

Old English form: *dǣd*

Reconstruction and comparative evidence

Orel reconstructs the noun as **dēdiz* ‘deed’, and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* ‘deed’ through northwest Germanic **dadiz* ‘deed’ (Orel 2003; Ringe and Taylor 2014). The acute accent in **dḗdiz* ‘deed’ marks stress on the same reconstructed long vowel.

Old English evidence

Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* ‘deed’ beside non-West-Saxon *dēd* ‘deed’ (Campbell 1959; Brunner 1965). Clark Hall likewise lists *dæd* ‘deed’ and cross-refers Anglian *dēd* to it (Clark Hall 1960). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English

From inherited **dēdiz* ‘deed’, loss of final *-z* and the West Saxon lowering of stressed long *ē* yield *dǣd* ‘deed’; Anglian *dēd* ‘deed’ preserves the non-West-Saxon outcome (Campbell 1959; Brunner 1965). The development treated here is therefore the regular West Saxon line.

6.16 dew — OE *dēaw*

Derivation: **dáwwō* > *dēaw* (regular).

Derivation trace

Proto input: **dáwwō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**dáeuwu*
OE WW Simplification	**dáuwō*	OE Diphthong Leveling	**dēawu*
PNWGmc Final Long O Raising	**dáuwu*	OE High Vowel Apocope	**dēaw*
Early Anglo-Frisian			
[no change]			

Old English form: *dēaw*

Reconstruction and development

The selected PGmc **dáwwō* gives OE *dēaw* ‘dew’. Its inherited geminate is reanalyzed early: short **a* plus **ww* becomes **au* with retained consonantal **w*. The nucleus subsequently undergoes

English fronting and offglide realization. Early reanalysis must therefore be distinguished from the later long English surface diphthong (Ringe and Taylor 2014, 65–66, 171–75; Campbell 1959, 45–47).

The current path is SC031 OEWWSimplification, then SC030 OEAuBrightening, then SC032 OEDiphthongLeveling. The unchanged reconstruction and final form no longer depend on a late unconditional ww deletion followed by singleton-aw lengthening. The selected evaluation uses the explicitly declared strong-final citation context; lexical accent does not infer sentence stress (Ringe and Taylor 2014, 55, 57–58).

6.17 door — OE *dor*

Derivation: **dúrą* > *dor* (regular).

Derivation trace

Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**dór*
PNWGmc U Lowering	**dórą*		
Early Anglo-Frisian			
[no change]			

Old English form: *dor*

Reconstruction and comparative evidence

Kroonen reconstructs a neuter **dura-* ‘gate, (single) door’ and cites Old English *dor* ‘door’ among its reflexes. In the same entry he separates Old English *duru* ‘door’, Old Frisian *dore*, and Old High German *tura* ‘door’ as reflexes of **durō-* instead (Kroonen 2013).

Old English evidence

Clark Hall records *dor* ‘door’ as a neuter noun and separately records feminine *duru* ‘door’ with its own inflection (Clark Hall 1960). Ringe and Taylor likewise treat *duru* ‘door’ as an early Old English u-stem, originally a root noun shifted into that class (Ringe and Taylor 2014). The Old English form here is therefore the attested neuter *dor* ‘door’, while *duru* ‘door’ remains a parallel Old English reflex from another stem history.

Development to Old English

From **dúrą* ‘door’, Northwest Germanic u-lowering gives **dórą* ‘door’, and heavy-syllable nasal apocope then yields *dor* ‘door’. The regular development treated in this entry is therefore **dúrą* > *dor*; the feminine *duru* ‘door’ belongs to the separate line identified by Kroonen and Ringe-Taylor (Kroonen 2013; Ringe and Taylor 2014).

6.18 fare — OE *faran*

Derivation: **fáraną* > *faran* (regular).

Derivation trace

Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**færaną*
[no change]	OE A Restoration	**faraną*
	OE Heavy Syllable Nasal Apocope	**faran*
Early Anglo-Frisian	OE Secondary Nasalization	**farąn*
[no change]	OE Weak Tail Reduction	**faran*

Old English form: *faran*

Reconstruction and comparative evidence

Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan* ‘fare’, both with Old English *faran* ‘fare’ among the reflexes (Kroonen 2013; Orel 2003, 132). Campbell also uses *faran* ‘fare’ as a standard example of Old English A-restoration (Campbell 1959, 61).

Old English evidence

Clark Hall lemmatizes the strong verb as *faran* ‘fare’ and separately records weak *færan* ‘to frighten’; *fære* ‘fare’, *færst* ‘fare’, and *færð* ‘fare’ belong to present-tense forms of *faran* rather than to the infinitive itself (Clark Hall 1960). Bosworth-Toller preserves the same distinction (Bosworth and Toller 1898, 108). The Old English form here is therefore the attested citation infinitive *faran* ‘fare’.

Development to Old English

From **fáraną* ‘fare’, Anglo-Frisian brightening first gives **færaną* ‘fare’, but A-restoration before single *r* returns **faraną* ‘fare’; later apocope and weak-tail reduction yield *faran* ‘fare’ (Campbell 1959, 61). Fulk’s contrast with participial faren- < **farœn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (Fulk 2018).

6.19 fell — OE *fell*

Derivation: **féllą* > *fell* (regular).

Derivation trace

Proto input: **féllą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**féll*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *fell*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **fella-* ‘membrane, skin, hide’ and cites Old English *fell* ‘skin, fell’ beside Dutch *vel* ‘skin, fell’ and German *Fell* (Kroonen 2013). The derivational input **féllą* ‘fell’ is the derivable singular form of that same inherited noun.

Old English evidence

Clark Hall records *fell* ‘skin, fell’ as the noun ‘fell, skin, hide’, and Bright’s glossary likewise gives *fell* ‘skin, fell’ with inflected forms such as accusative singular *fel* and dative plural *fellum* (Clark Hall 1960; Bright 1971). The target is therefore the attested noun *fell* ‘skin, fell’, not the verb *fellan* or the preterite *feoll*.

Development to Old English

With **féllą* ‘fell’, no special earlier reshaping is needed: heavy-syllable nasal apocope yields **féll* ‘fell’, surfacing as *fell* ‘skin, fell’. The regular development treated here is therefore **féllą* > *fell*.

6.20 fern — OE *fearn*

Derivation: **fárnaz* > *fearn* (regular).

Derivation trace

Proto input: **fárnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fárn*
EAF Final Z Deletion	**fárna*	EAF Brightening	**fœrn*
Early Anglo-Frisian		OE Breaking	**fearn*
[no change]			

Old English form: *fearn*

Reconstruction and comparative evidence

Kroonen cites the noun as masculine **farna-* and gives Old English *fearn, fern*, while Orel gives the same lexeme as neuter **farnan* ‘fern’ with Old English *fearn* ‘fern’ (Kroonen 2013; Orel 2003, 133). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style **fárnaz* ‘fern’.

Old English evidence

Clark Hall gives *fearn* ‘fern’ as an Old English noun, and Bosworth-Toller records *fearn* ‘fern’ with inflected forms such as *fearnes*, *fearna*, and *fearne* (Clark Hall 1960; Bosworth and Toller 1898, 219). Kroonen also records *fern* ‘fern’, but the local lexical sources give stronger support to *fearn* ‘fern’ as the citation target (Kroonen 2013).

Development to Old English

From **fárnaz* ‘fern’, loss of final *-z* and final *-a* gives **fárn* ‘fern’; Anglo-Frisian brightening then yields **fœrn* ‘fern’, and breaking before *r* plus consonant gives *fearn* ‘fern’ (Campbell 1959; Ringe and Taylor 2014). The development treated here is therefore the regular *rC*-breaking line.

6.21 field — OE *feld*

Derivation: **félθuz* > *feld* (regular).

Derivation trace

Proto input: **félθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**féld*
EAFL Th Voicing	**félduz*		
EAF Final Z Deletion	**féldu*		
Early Anglo-Frisian			
[no change]			

Old English form: *feld*

Reconstruction and comparative evidence

Ringe and Taylor treat Old English *feld* ‘field’ as one of the cases where earlier **felþu- ~ *feldu-* may reflect either inherited *þ ~ *d* alternation or the regular West Germanic development **lþ > ld* (Ringe and Taylor 2014, 170). The broader proto label **félθuz* ‘field’ can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence

Clark Hall records *feld* ‘field’ with oblique forms such as *felda* and *felde*, and Campbell notes early place-name spellings in *-felth* beside the later standard form (Clark Hall 1960, 114; Campbell 1959, 169). The Old English form here is therefore the attested citation noun *feld* ‘field’, with the older *-felth* spellings as historical support rather than as rival targets.

Development to Old English

In the modeled pathway, medial **lþ* becomes *ld*, final *-z* is lost, and high-vowel apocope then yields *feld* ‘field’. Whether the voiced dental ultimately reflects inherited alternation or the regular **lþ > ld* development, both accounts converge on the same Old English form (Ringe and Taylor 2014, 170).

6.22 flea — OE *flēah*

Derivation: **fláuxz > flēah* (regular).

Derivation trace

Proto input: **fláuxz*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**fláeux*
[no change]	OE Diphthong Leveling	**flēax*
Early Anglo-Frisian		
[no change]		

Old English form: *flēah*

Reconstruction and comparative evidence

This word carries a genuine stem-class dispute, not merely a notational one. Orel reconstructs a root noun with his class-wide explicit nominative marker, **flauxz* ‘flea’ (Orel 2003, 105). Kroonen instead reconstructs an ō-stem **flauhō-* ‘flea’, with no *-z* at any layer of the analysis (Kroonen

2013, 145). Kluge and Seebold offer yet a third analysis, a masculine a-stem **flauha-* ‘flea’ (Kluge 2011, 304), whose nominative singular **flauhaz* would have carried its *-z* in an unstressed syllable and lost it under the West Germanic unstressed loss rather than with the root nouns. The attested forms do not decide the question: Old English *flēah* ‘flea’ and Old Norse *fló* ‘flea’ are compatible with all three reconstructions. This is a case where reputable sources disagree about the morphological class itself, and the disagreement is documented here rather than silently normalized away.

The derivational input **fláuxz* ‘flea’ adopts Orel’s root-noun analysis with its morphologically explicit nominative marker. Under Kroonen’s ō-stem analysis the nominative would instead have ended in **-ō* and developed like the other ō-stems; the choice of Orel’s form keeps the word in the root-noun class alongside ‘book’, ‘goose’, and ‘louse’, whose endingless West Germanic nominatives it shares (Ringe and Taylor 2014, 118).

Old English evidence

Old English *flēah* ‘flea’ is the endingless nominative-accusative singular. The root-noun class to which Orel assigns it shows no nominative ending anywhere in West Germanic (Ringe and Taylor 2014, 118); on Kroonen’s ō-stem analysis the endingless form would instead reflect the regular fate of the ō-stem nominative in this shape. Either way the attested Old English form is the expected one, which is precisely why the class question remains open.

Development to Old English

On the selected root-noun analysis, the nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **fláux* ‘flea’ from the derivational input **fláuxz* ‘flea’. The diphthong is then fronted and levelled in Old English, yielding **flēax* ‘flea’, spelled *flēah* ‘flea’ with final *h* for the fricative. The early loss of the root-noun marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the derivational input is a stressed monosyllable, but its **-z* stands after a consonant and was gone long before the northern change, whose domain is vowel-final monosyllables.

6.23 fly — OE *flēogan*

Derivation: **fléuganą* > *flēogan* (regular).

Derivation trace

Proto input: **fléuganą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**flēoganą*
[no change]	OE Heavy Syllable Nasal Apocope	**flēogan*
	OE Secondary Nasalization	**flēogą̄n*
Early Anglo-Frisian	OE Weak Tail Reduction	**flēogan*
[no change]		

Old English form: *flēogan*

Reconstruction and comparative evidence

Ringe and Taylor derive the verb as **fleugana* ‘fly’ > OE *fléogan* ‘fly’ and elsewhere contrast West Saxon *fléogan* ‘fly’ with Anglian *flégan* ‘fly’, alongside related forms noted in the form note below (Ringe and Taylor 2014). The derivational input **fléuganą* ‘fly’ represents that inherited strong verb in the notation used here.

Old English evidence

Clark Hall and Bosworth-Toller record *flēogan* ‘fly’ as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag* ‘flew’, *flugon* ‘flew’, *flogen* ‘flown’ with present *fleogeð* ‘flies’ (Clark Hall 1960; Bosworth and Toller 1898; Bright 1971). The target in this entry is therefore the attested verbal infinitive itself.

Form note

Ringe and Taylor also list related *fléoge* ‘fly’ / *flége* ‘fly’ and Anglian *flégan* ‘fly’, which belong to the same family but do not replace the infinitive *flēogan* ‘fly’ treated here (Ringe and Taylor 2014).

Development to Old English

From **fléuganą* ‘fly’, Old English diphthong leveling gives **flēoganą* ‘fly’; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* ‘fly’ (Ringe and Taylor 2014). The development is therefore regular: **fléuganą* > *flēogan* ‘fly’.

6.24 forlorn — OE *lēosan*

Derivation: **léusaną* > *lēosan* (regular).

Derivation trace

Proto input: **léusaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**lēosaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēosan*
	OE Secondary Nasalization	**lēosąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**lēosan*
[no change]		

Old English form: *lēosan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fraliusan* ‘lose’ and Old English *for-lēosan* ‘lose, forlorn’; Orel likewise gives Old English *for-leósan* ‘lose’ (Kroonen 2013; Orel 2003). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence

The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* ‘lose, forlorn’ and especially in the participle *forloren* ‘forlorn’, recorded by Ringe and Taylor and in the dictionaries (Ringe and Taylor 2014; Clark Hall 1960; Bosworth and Toller 1898). The simplex infinitive *lēosan* ‘lose’ represents the verbal base itself.

Form note

As a base-form comparison, the simplex infinitive is *lēosan* ‘lose’, while the English adjective continues the prefixed Old English family *forlēosan* ‘lose’ / *forloren* ‘forlorn’ (Ringe and Taylor 2014).

Development to Old English

From **léusaną* ‘lose’, Old English diphthong leveling gives **lēosaną* ‘lose’, and later nasal apocope and weak-tail reduction yield *lēosan* ‘lose’ (Ringe and Taylor 2014). The prefixed forms follow the same verbal base with added *for-*.

6.25 four — OE *fēower*

Derivation: **fédwōr* > *fēower* (regular).

Derivation trace

Proto input: **fédwōr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Diphthong Leveling	**fēowar*
PWGmc Final Or Lowering	**fédwar*	EAF Brightening Unstressed	**fēowær*
PWGmc Coronal W Assimilation	**féwwar*	OE Unstressed AE Merger	**fēower*
OE WW Simplification	**féuwar*		
Early Anglo-Frisian			
[no change]			

Old English form: *fēower*

Reconstruction and development

The selected PGmc **fédwōr* gives OE *fēower* ‘four’. Coronal-w assimilation creates the geminate; short **e* plus **ww* is then reanalyzed as an **eu* nucleus with retained consonantal **w*. Ringe and Taylor explicitly use this history alongside the oblique plural pronoun *ēow* ‘you’ (Ringe and Taylor 2014, 41–42).

Thus SC008 PWGmcCoronalWAssimilation feeds SC031 OEWWSimplification. The earlier product reaches English long-diphthong realization through SC032 OEDiphthongLeveling, rather than receiving an English surface vowel in the West Germanic reanalysis (Ringe and Taylor 2014, 65–66, 171–75). The selected context is the ordinary strong-final citation convention, not the weak-final selection of the pronoun. No reconstruction or final target has changed.

6.26 gang — OE *gang*

Derivation: **gángaz* > *gang* (regular).

Derivation trace

Proto input: **gángaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**gáng*
EAF Final Z Deletion	**gánga*		
Early Anglo-Frisian			
[no change]			

Old English form: *gang*

Reconstruction and comparative evidence

Orel reconstructs the noun as **gangaz* ‘gang’ and cites Old English *gang* ‘going, way’ beside Old Norse *gangr*, Old Frisian *gang* / *gong*, Old Saxon *gang* ‘going, way’, and Old High German *gang* ‘going, way’ (Orel 2003). The derivational input **gángaz* ‘gang’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller both record *gang* ‘going, way’ as the noun ‘going, journey, way’, and Bright’s glossary gives *gong (gang), m., path, course* (Clark Hall 1960; Bosworth and Toller 1898, 159; Bright 1971, 392). The target is therefore the attested noun headword itself.

Form note

This entry concerns the noun *gang* ‘going, way’, not the separate verb *gangan* (Clark Hall 1960; Bosworth and Toller 1898, 159).

Development to Old English

From **gángaz* ‘gang’, loss of final *-z* gives **gánga* ‘gang’, and later loss of final bare *-a* yields *gang* ‘going, way’. The development is therefore regular: **gángaz* > *gang*.

6.27 gift — OE *ġift*

Derivation: **gíftiz* > *ġift* (regular).

Derivation trace

Proto input: **gíftiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Palatalization	**jífti*
EAF Final Z Deletion	**gífti*	OE High Vowel Apocope	**jíft*
Early Anglo-Frisian		OE Palatal Fricative Merger	**jíft*
[no change]			

Old English form: *ġift*

A. The selected reconstruction and the question it answers

Old English *ġift* ‘gift’ is a small word with unusually important chronological consequences. A simple high vowel follows an initial palatal consonant, whereas *ġiefan* ‘give’ shows the diphthongal treatment associated with that consonant. To explain the difference we must ask which vowel was present when ordinary palatal diphthongization operated, not simply which order of rules can reproduce the final spellings. Campbell lists the short-i gift headword among his initial-palatal examples (Campbell 1959, 173–74, §427).

We select PGmc **giftiz* ‘gift’ following Ringe’s explicit reconstruction (Ringe 2017, 135). Our acute root-stress convention does not add a different vowel or claim that the source prints an accent. Both the comparative reconstruction and the selected computational input use the raised vowel. This is a regular inherited derivation, not a late analogical substitute input. The decision is nevertheless not a claim that every etymological source prints the same reconstruction. A substantial e-form tradition must be explained and preserved rather than suppressed.

B. Orel: the full e-vowel noun reconstruction

Orel's noun entry gives an e-vowel reconstruction, normalized here as **geftiz* 'gift', and connects it with the verb 'give' (Orel 2003, 130). It has the full iz ending, not merely a bare root or a stem ending in a hyphen. His evidence includes the Old English gift headword and the Old Frisian counterpart *jeft* 'gift, privilege', alongside the wider inherited formation. The e is plainly printed in the original page: it is not an OCR accident or a vowel introduced by our transcription. Only the initial velar sign is normalized to g here; no argument about stop versus fricative history is drawn from that typographic normalization.

Orel thus provides genuine published support for the older e-input used in our model. However, the entry does not explain why e is reconstructed despite the high-vowel reflex, identify a specific later raising date, or argue against the early raising account. Its derivation from 'give' supplies an etymological relationship; that alone does not demonstrate that the noun's realized root vowel remained identical to the verb's at every historical stage. The cited bibliography points to morphology and verbal etymology treatments, including Bammesberger and Seebold, considered below (Orel 2003, 130). Those references are evidence of the scholarly tradition behind the entry, not a license to attribute an unstated phonological argument to Orel.

C. Kluge–Seebold: an e-vowel feminine stem

Kluge–Seebold gives **gefti-* 'gift', explicitly a Germanic feminine stem, and relates the noun to 'give' (Kluge 2011, 359). This is independent confirmation that an e-vowel etymological presentation is established in major reference works. It is not the same printed object as Orel's full nominative reconstruction. We must not silently append z and then present the expanded form as a quotation from Kluge–Seebold.

The entry cites the Old English gift headword and the Old Frisian form *jeft(e)* 'gift' within the inherited family (Kluge 2011, 359). Its principal discussion concerns the word's semantic history; it does not supply a worked chronology of root-vowel raising versus palatal diphthongization. Consequently the entry establishes the stem and its etymological affiliation more directly than it establishes the vowel quality at a narrowly dated phonological node. This does not prove that the stem notation is purely abstract. It means that such an interpretation, if adopted, would require an argument beyond the entry.

D. Bammesberger: the e-grade derivational analysis

Bammesberger places the formation among ti-stems built on an e-grade verbal root and prints a segmented e-vowel form (Bammesberger 1990, 142, §5.4.2). His discussion relates the noun to the Germanic root of 'give' and treats the consonantal sequence ft as regular in the ti-abstract formed from a labial-final root. On the same page the formation also participates in his explanation of the structural pattern of other abstracts. The continuation lists further e-grade formations with Old English i-vowel reflexes (Bammesberger 1990, 142–43).

This is a substantive reason for retaining e in the derivational analysis: the noun belongs to an e-grade formation, not an arbitrarily invented high-vowel root. But derivational grade and realized surface quality after a regular conditioned sound change are different questions. An e-grade formation can have a later i-reflex without becoming a zero-grade formation. The passage supports the morphological analysis; it does not itself decide that the inherited e survived until Old English mutation or state a competing date for raising. We therefore retain its e-grade evidence in the deeper explanation without using it alone to override a separately argued phonological reconstruction (Bammesberger 1990, 142–43).

E. Seebold: segmented forms and the limits of notation

Seebold prints a segmented e-vowel formation with the full nominal ending and supplies more differentiated Old English evidence than a single normalized citation (Seebold 1970, 218). He distinguishes *giftu* 'wedding', explicitly plural, *gyft* 'marriage', and *feoh-gift* 'gift of money'. A separate homonymous noun associated with 'food' belongs to a different formation and must not

be recruited as attestation for the noun treated here. These distinctions matter: the formation's identity is well supported, but a plural or compound citation is not automatically evidence for a particular singular manuscript token.

His methodological introduction also makes clear that the reconstructed notation is not uniformly a direct transcription of one phonological moment. Earlier states are reconstructed for selected developments, including the development of ei to long i (Seebold 1970, 35). The symbols for stem classes are modelled on common-Germanic forms without claiming exact phonetic reproduction (Seebold 1970, 35–36). These explicit cautions make it particularly important not to read every segment mechanically as an input to a later sound-change cascade. They do not, however, specifically explain the root e in this noun: the warning about stem-class endings cannot be generalized into proof that all reconstructed root vowels are abstract morphological labels. We leave that interpretive limit visible rather than declaring the e/i discrepancy merely notational.

F. Ringe: the realized PGmc i-form and its phonological rationale

Ringe explicitly gives PGmc **giftiz* 'gift' as the nominal formation associated with the verb 'give' (Ringe 2017, 135). His discussion places it in the broader treatment of productive nominal formation and consonantal developments. The Old English comparison is plural *ġifta* 'wedding', with a change of stem class. We do not turn that plural into a claim that Ringe attests our normalized singular target.

The vowel has a separate phonological explanation. Ringe states that e was raised to i when a high front vocalic followed in the same or immediately succeeding syllable (Ringe 2017, 151–52). The noun's following i supplies precisely such an environment. This explains how an etymological e-grade formation can be represented with i in a realized phonological reconstruction. It is not a lexical exception, a grammatical condition on nouns, or a repair licensed only because this one word otherwise produces an unwanted diphthong.

Ringe is more cautious about the exact ancestral dating than a bare reconstruction might suggest. The merger of stressed e and i in Gothic prevents direct demonstration of the raising there. His indirect argument depends on raising preceding j-loss and subsequent contraction, particularly the history of the numeral 'three'. He calls a PGmc date the most probable hypothesis, not completely certain (Ringe 2017, 152–53). That qualification belongs in our account. It concerns the placement of the change in the deeper ancestry; it must not be misrepresented as doubt that his noun entry actually prints an i-vowel reconstruction.

Ringe and Taylor independently insist on separating this inherited raising from Old English front mutation: the older change occurred hundreds of years earlier, and alleged later repetitions are rare and doubtful (Ringe and Taylor 2014, 220). Their discussion therefore supplies an important chronological check against explaining this noun's inherited high vowel by the general Old English mutation rule.

G. Why we choose i, and what changes in the derivation

For the input to a PGmc-to-OE cascade we prefer the explicitly staged phonological reconstruction, rather than carrying an earlier etymological e-grade representation forward until a convenient later rule can correct it. Ringe's noun and regular raising account together supply that reconstruction (Ringe 2017, 135, pp. 151–153). The competing reference forms remain part of the evidence: we neither count dictionary spellings as votes nor conclude that their authors were wrong merely because they did not give a worked chronology.

With earlier i already present, the noun undergoes the inherited consonantal and ending developments, then initial palatalization. The singleton initial is a palatal fricative, distinct from gg/ng stops; the adopted model keeps it distinct from inherited j during mutation (Fulk 2018, 130–32; Ringe and Taylor 2014, 204). Ordinary palatal diphthongization does not change its high vowel, and Old English mutation need not raise an e that is no longer there. Loss of the final high vowel and surface spelling yield *ġift* 'gift'. Our earlier implementation instead allowed a selected e-input to reach late mutation and used that correction to justify putting all palatal diphthongization afterward. The right final spelling concealed the wrongly conflated histories.

The independent ordinary chronology is breaking, then palatal diphthongization, then mutation (Fulk 2018, 74; Ringe and Taylor 2014, 215–17, 222). The worked contrast is *ġiest* 'guest': its low-front vowel first becomes ea after the palatal, and ea then becomes ie under mutation. Gift's earlier i is a negative control on ordinary diphthongization, not an independent proof of that relative order. The later treatment of *sċēaþ* 'sheath' is separately retained after mutation; its mutation-derived long vowel does not require ordinary diphthongization as a whole to be late (Campbell 1959, 68–69; Ringe and Taylor 2014, 235). The complete conditioning of that later treatment remains unresolved.

H. Confidence and the remaining alternatives

Our confidence is strongest in the distinction between inherited raising and later OE mutation, and in the independently supported ordinary diphthongization chronology. The choice of Ringe's i-input is well motivated as an explicit working phonological reconstruction, but is not advertised as unanimous: Orel's e-form, Kluge–Seebold's e-stem and the e-grade derivational presentations remain genuine alternatives (Orel 2003, 130; Kluge 2011, 359; Bammesberger 1990, 142–43; Seebold 1970, 218). We have not established that every one of those forms denotes exactly the same historical phonological checkpoint as Ringe's input.

Confidence in the precise PGmc date is qualified by Ringe's own indirect evidence (Ringe 2017, 152–53). Confidence in our normalized target as a headword is distinct from certification of a diplomatic paradigm cell (Campbell 1959, 173–74). Variant spellings alone do not establish a different exceptionless sound law, lexical diffusion, or an analogical history; those explanations would need their own evidence. We make no such new claim here.

The computational comparison is deliberately secondary. The adopted input and ordinary chronology preserve every previously generated final form, while realizing the source sequence for guest and leaving sheath's derivation unchanged. That demonstrates compatibility of the selected account with the present corpus, not its historical truth by output score. If stronger evidence eventually requires an e-vowel at the input node, the necessary earlier regular raising must be specified and tested as such; it must not be silently delegated again to late OE mutation.

6.28 give — OE *ġiefan*

Derivation: **gébanᾱ* > *ġiefan* (regular).

Derivation trace

Proto input: **gébanᾱ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**géban*
[no change]	OE Secondary Nasalization	**gébą̄n*
	PGmc B Allophony	**géβą̄n*
Early Anglo-Frisian	OE Velar Palatalization	**ǰéβą̄n*
[no change]	OE Ws Palatal Diphthongization	**ǰíeβą̄n*
	OE Weak Tail Reduction	**ǰíeβan*
	OE Palatal Fricative Merger	**ǰíeβan*

Old English form: *ġiefan*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* 'give' among its reflexes (Kroonen 2013). Ringe and Taylor contrast West Saxon *giefan* 'give' with Mercian *forgeofan* 'give' and Northumbrian *geafa* 'give', showing that the inherited verb takes different later dialectal shapes (Ringe and Taylor 2014).

Old English evidence

Campbell gives *gefan* (W-S *giefan* ‘give’) among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* ‘give’ with forms such as *geaf* and *giefen* (Campbell 1959; Clark Hall 1960). I write *ġiefan* ‘give’ for the palatal initial.

Dialect note

West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* ‘give’ or *for-geofan* ‘give’ continue the same verb without the West Saxon vocalism (Ringe and Taylor 2014).

Development to Old English

The singleton initial belongs to the fricative class, not the gg/ng stop class. The adopted consonantal account distinguishes its palatal j̨ from inherited j during mutation and merges them afterwards (Fulk 2018, 130–32; Ringe and Taylor 2014, 204).

From **gébaną* ‘give’, initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* ‘give’ (Campbell 1959; Ringe and Taylor 2014). The result is therefore the regular West Saxon infinitive.

6.29 gold — OE *gold*

Derivation: **gúlθą* > *gold* (regular).

Derivation trace

Proto input: **gúlθą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**góld*
EAFL Th Voicing	**gúldą*		
PNWGmc U Lowering	**góldą*		
Early Anglo-Frisian			
[no change]			

Old English form: *gold*

Reconstruction and comparative evidence

Ringe and Taylor cite the noun as **gulþa-* / **gulda-*, and Kroonen gives the same pair (Ringe and Taylor 2014, 42; Kroonen 2013). The derivational input **gúlθą* ‘gold’ preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence

Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (Bosworth and Toller 1898, 121; Clark Hall 1960, 152). The target is therefore the attested citation form itself.

Development note

Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa-* / **gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (Ringe and Taylor 2014, 42).

Development to Old English

From **gúlθą* ‘gold’, the regular consonant development gives **gúldą* ‘gold’; Northwest Germanic / Old English lowering then yields **góldą* ‘gold’, and apocope gives *gold* (Campbell 1959; Ringe and Taylor 2014, 42).

6.30 goose — OE *gōs*

Derivation: **gánsz* > *gōs* (regular).

Derivation trace

Proto input: **gánsz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
EAF Nasal Spirant Lengthening	**gãns*	
EAF Nasal Spirant Loss	**gãs*	
EAF Nasalized Low Rounding	**gōs*	
Early Anglo-Frisian		
[no change]		

Old English form: *gōs*

Reconstruction and comparative evidence

The ‘goose’ word is a root noun whose nominative singular has an unusually layered history. Orel’s citation form is **gansz* ‘goose’, with his class-wide explicit nominative marker **-z* stacked on the root-final *s* (Orel 2003, 126). Kroonen reconstructs an endingless nominative **gans* ‘goose’ with Verner-voiced *z* only in the oblique cells (genitive **gunzaz* ‘goose (gen.)’) (Kroonen 2013, 168–69), and Kluge/Seebold treat the German cognate under the same root noun (Kluge 2011, 331). Bammesberger supplies the deeper layer: by Szemerényi’s law the Proto-Indo-European nominative **ǵʰanss* yielded **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196). A mainstream specialist thus reconstructs a Proto-Germanic ‘goose’ nominative with a final voiced sibilant; Orel’s notation is not idiosyncratic, and Kroonen’s endingless citation reflects a different analytical layer of the same paradigm rather than a contradictory fact.

The derivational input **gánsz* ‘goose’ follows Orel’s morphologically explicit convention, keeping the nominative marker visible so that its historical elimination is modelled rather than presupposed.

Old English evidence

Old English *gōs* ‘goose’ is the endingless nominative-accusative singular, with mutated plural *ġēs* ‘geese’ (Fulk 2018, 165–66). The comparative picture matches the rest of the root-noun class: Gothic replaced the word (its ‘goose’ is the u-stem remodeling behind Spanish *ganso* ‘goose’ (Orel 2003, 126)), and no West Germanic daughter shows any nominative ending (Ringe and Taylor 2014, 118).

Development to Old English

The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **gáns* ‘goose’ from the derivational input **gánsz* ‘goose’. The remaining development is the familiar Ingvaeonic nasal-spirant sequence: the vowel lengthens and rounds before the nasal + voiceless fricative cluster and the nasal is then lost, giving **gōs* ‘goose’, which is the attested Old English form. The early loss of the nominative marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; had the marker survived to the later monosyllabic change, a lengthened vowel before *s* would not be the expected outcome of this shape, since the root-noun ending was already gone.

6.31 grave — OE *grafan*

Derivation: **grábaną* > *grafan* (regular).

Derivation trace

Proto input: **grábaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**græbaną*
[no change]	OE A Restoration	**grabaną*
	OE Heavy Syllable Nasal Apocope	**graban*
Early Anglo-Frisian	OE Secondary Nasalization	**grabąn*
[no change]	PGmc B Allophony	**graβąn*
	OE Weak Tail Reduction	**graβan*

Old English form: *grafan*

Reconstruction and comparative evidence

Campbell gives *grafan* ‘dig, grave’ among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence

Clark Hall records *grafan* ‘dig, grave’ as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (Clark Hall 1960). The target here is the attested infinitive headword of the verb.

Development to Old English

From **grábaną* ‘grave’, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* ‘dig, grave’ (Campbell 1959, 61; Ringe and Taylor 2014).

Form note

Noun *græf* ‘grave’ and verbal forms such as *grǽfð* ‘graves’ or past participial *grǽfen* ‘graven’ belong to other lexical or paradigm positions and do not replace the infinitive *grafan* ‘grave’ as the target here (Clark Hall 1960).

6.32 guest — OE *ġiest*

Derivation: **gástiz* > *ġiest* (regular).

Derivation trace

Proto input: **gástiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**gæsti*
EAF Final Z Deletion	**gásti*	OE Velar Palatalization	**jæsti*
Early Anglo-Frisian		OE Ws Palatal Diphthongization	**jeasti*
		OE I Umlaut	**jiesti*
[no change]		OE High Vowel Apocope	**jiest*
		OE Palatal Fricative Merger	**jiest*

Old English form: *ġiest*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* ‘guest’ type (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record the word under forms such as *gist* ‘guest’, *gest* ‘guest’, *giest* ‘guest’, and *gyst* ‘guest’ (Bosworth and Toller 1898; Clark Hall 1960). The Old English form here, *ġiest*, ‘guest’ is the normalized West Saxon form within that attested family.

Development to Old English

From **gástiz* ‘guest’, ordinary English fronting supplies the low-front vowel. Initial singleton g palatalizes as a fricative, not an early affricate (Fulk 2018, 130–32). Ordinary West Saxon palatal diphthongization changes æ to ea before mutation changes ea to ie. The high-vocalic ending is lost afterwards; the regular outcome is *ġiest* ‘guest’ (Ringe and Taylor 2014, 215–17, 222, 287).

Dialect note

West Saxon *ġiest* ‘guest’ is the Old English form here. Anglian *gest* ‘guest’ and related spellings remain real Old English comparators rather than corrections to that choice (Ringe and Taylor 2014; Bosworth and Toller 1898).

6.33 hair — OE *hǣr*

Derivation: **xérą* > *hǣr* (regular).

Derivation trace

Proto input: **xérą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çǣrą*
PNWGmc Long E Lowering	**xārą*	OE Heavy Syllable Nasal Apocope	**çǣr*
EAF Long A Fronting	**xǣrą*		
Early Anglo-Frisian			
[no change]			

Old English form: *hǣr*

Reconstruction and comparative evidence

Kroonen cites the ordinary Proto-Germanic hair word as **hēra-* (Kroonen 2013). The derivational input **xérą* ‘hair’ represents that same long-ē stem in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hær* ‘hair’ / *hǣr* ‘hair’ as the ordinary Old English noun (Clark Hall 1960, 158; Bosworth and Toller 1898, 510). The target is therefore the attested headword itself.

Development to Old English

From **xérą* ‘hair’, Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hǣr* ‘hair’. The development treated here is straightforward and does not require any special paradigm choice.

Form note

Older references to **xazwăz* ‘hair’ belong to a different lexeme, and the separate *haddr* / *heordan* / *hād-* material does not displace the ordinary simplex *hǣr* ‘hair’ treated here (Kroonen 2013; Clark Hall 1960, 158).

6.34 harvest — OE *hierfest*

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace

Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xærbistu*
EAF Final Z Deletion	**xárbistu*	OE Breaking	**xearbistu*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çearbistu*
		PGmc B Allophony	**çearβistu*
[no change]		OE I Umlaut	**çierβistu*
		OE High Vowel Apocope	**çierβist*
		OE Med Unstressed I Lowering1	**çierβest*

Old English form: *hierfest*

Reconstruction and comparative evidence

Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest* / *hyrfest* type (Bammesberger 1997; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record *hærfest* ‘harvest’, with *herfest* ‘harvest’ as a variant in the lexical tradition (Bosworth and Toller 1898; Clark Hall 1960). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English

From **xárbistuz* ‘harvest’, Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest* ‘harvest’. That is the regular West Saxon development treated here (Ringe and Taylor 2014; Campbell 1959).

Source note

The Old English form here, *hierfest*, represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest* ‘harvest’ / *herfest* ‘harvest’, commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (Bammesberger 1997; Ringe and Taylor 2014).

6.35 hedge — OE *heġġ*

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace

Proto input: **xágjaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xággj*
PWGmc J Gemination	**xággjaz*	EAF Brightening	**xæggj*
EAF Final Z Deletion	**xággja*	OE Velar Fricative Palatalization	**çæggj*
Early Anglo-Frisian		OE Velar Palatalization	**çædʒdʒj*
[no change]		OE I Umlaut	**çedʒdʒj*
		OE J Loss After Heavy	**çedʒdʒ*

Old English form: *heġġ*

Reconstruction and comparative evidence

The derivational input models a palatal **-gj-** ‘hedge’ noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heġġ* ‘hedge’ type (Campbell 1959).

Old English evidence

Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* ‘hedge’ / *heċġ* ‘hedge’ (Bosworth and Toller 1898; Clark Hall 1960). The lexical item itself is therefore well attested even though the form compared here is normalized.

Development to Old English

This is the geminate-stop class, distinct from the singleton fricative in *dæġ* ‘day’. The model’s palatal stop-reflex proxy does not assert that affrication had already occurred at the articulation checkpoint (Ringe and Taylor 2014, 203–4; Fulk 2018, 131–32).

From **xágjaz* ‘hedge’, West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heġġ* ‘hedge’. The development is treated as regular rather than exceptional.

Form note

Standard dictionary spelling is *hecġ* ‘hedge’ or *hecg* ‘hedge’. Normalized *hegġ* ‘hedge’ is the Old English form here, while the ordinary lexicographic forms remain the main Old English citation evidence (Bosworth and Toller 1898; Clark Hall 1960).

6.36 helm — OE *helm*

Derivation: **xélmaz* > *helm* (regular).

Derivation trace

Proto input: **xélmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xélm*
EAF Final Z Deletion	**xélma*	OE Velar Fricative Palatalization	**çélm*
Early Anglo-Frisian			
[no change]			

Old English form: *helm*

Reconstruction and comparative evidence

Kroonen cites the helmet noun as **helma-* and separately distinguishes a different lexeme **helman-* ‘rudder’ (Kroonen 2013). The derivational input **xélmaz* ‘helm’ is the nominative-style form used for the helmet noun itself.

Old English evidence

Clark Hall and Bosworth-Toller record *helm* ‘helm, helmet’ as the ordinary Old English noun for ‘helmet’, while *helma* ‘helm’ belongs to a separate rudder lexeme (Clark Hall 1960; Bosworth and Toller 1898, 542).

Development to Old English

From **xélmaz* ‘helm’, loss of final *z* and later loss of the short final vowel yield *helm* ‘helm, helmet’. The development is therefore a straightforward citation-form match.

Form note

Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma* ‘helm’, which is a different noun meaning ‘helm, rudder’ (Kroonen 2013; Clark Hall 1960).

6.37 help — OE *helpan*

Derivation: **xélpaną* > *helpan* (regular).

Derivation trace

Proto input: **xélpaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Velar Fricative Palatalization	**çélpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**çélpan*
	OE Secondary Nasalization	**çélpąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**çélpan*
[no change]		

Old English form: *helpan*

Reconstruction and comparative evidence

The entry treats the strong verb itself rather than the separate noun *help* ‘help’. Bright’s principal parts *helpan, healp, hulpon, holpen* show the ordinary Old English strong-verb family continued by this input (Bright 1971).

Old English evidence

Clark Hall and Bosworth-Toller record *helpan* ‘help’ as the verbal headword ‘to help’ (Clark Hall 1960; Bosworth and Toller 1898, 542). The target is therefore the attested infinitive citation form.

Development to Old English

From **xélpaną* ‘help’, no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* ‘help’ directly.

Form note

Noun *help* ‘help’ belongs to a separate lexical line and should not replace verbal *helpan* ‘help’ as the target here (Clark Hall 1960; Bosworth and Toller 1898, 542).

6.38 hew — OE *hēawan*

Derivation: **xáwwaną* > *hēawan* (regular).

Derivation trace

Proto input: **xáwwaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**xáeuwaną*
OE WW Simplification	**xáuwaną*	OE Diphthong Leveling	**xēawaną*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çēawaną*
		OE Heavy Syllable Nasal Apocope	**çēawan*
[no change]		OE Secondary Nasalization	**çēawąn*
		OE Weak Tail Reduction	**çēawan*

Old English form: *hēawan*

Reconstruction and development

PGmc **xáwwaną* gives OE *hēawan* ‘hew’. Its inherited short-vowel-plus-geminate sequence yields **au* plus retained consonantal **w* in the earlier West Germanic reanalysis. English fronting and realization then produce the long diphthong. The geminate is not simply deleted at the later English checkpoint (Ringe and Taylor 2014, 65–66, 171–75; Campbell 1959, 45–47).

Together with dew, this word follows SC031 OEWWSimplification, SC030 OEAuBrightening, and SC032 OEDiphthongLeveling. It contrasts with the j-created geminates of hay and reconstructed

West Saxon strew, which retain their independently governed resolution. The earlier and later components are separate even though the final spelling is unchanged (Campbell 1959, 45–47; Ringe and Taylor 2014, 53, 173).

6.39 hind — OE *hind*

Derivation: **xéndjō* > *hind* (regular).

Derivation trace

Proto input: **xéndjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çéndju*
PNWGmc Final Long O Raising	**xéndju*	OE I Umlaut	**çindju*
Early Anglo-Frisian		OE High Vowel Apocope	**çindj*
		OE J Loss After Heavy	**çind*
[no change]			

Old English form: *hind*

Reconstruction and comparative evidence

Kroonen cites the animal name as **hindō-* f. ‘hind’ (Kroonen 2013). The derivational input **xéndjō* ‘hind’ represents that same noun in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hind* ‘hind, doe’ as the noun ‘hind, female deer’ (Clark Hall 1960; Bosworth and Toller 1898, 554). The target is therefore the attested lexical item itself.

Development to Old English

From **xéndjō* ‘hind’, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note

hindan ‘from behind, behind’ is a different Old English lexeme and does not belong to the noun history of *hind* (Clark Hall 1960; Bosworth and Toller 1898, 554).

6.40 hold — OE *healdan*

Derivation: **xáldaną* > *healdan* (regular).

Derivation trace

Proto input: **xáldaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xældaną*
[no change]	OE Breaking	**xealdaną*
	OE Velar Fricative Palatalization	**çealdaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**çealdan*
[no change]	OE Secondary Nasalization	**çealdą̊n*
	OE Weak Tail Reduction	**çealdan*

Old English form: *healdan*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the verb as a regular *a* + *lC* breaking case, with West Saxon *healdan* ‘hold’ opposed to Anglian and Mercian *haldan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bright gives the ordinary strong-verb citation form and principal parts *healdan, heold, heoldon, healden* (Bright 1971). The target is therefore the attested infinitive headword itself.

Development to Old English

From **xáldaną* ‘hold’, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Dialect note

West Saxon *healdan* ‘hold’ is the Old English form here. Anglian and Mercian *haldan* ‘hold’ are genuine non-West-Saxon doublets rather than corrections to that choice (Campbell 1959; Ringe and Taylor 2014).

6.41 horn — OE *horn*

Derivation: **xúrną* > *horn* (regular).

Derivation trace

Proto input: **xúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**xórn*
PNWGmc U Lowering	**xórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *horn*

Reconstruction and comparative evidence

Kroonen and Orel cite lemma-style Proto-Germanic headwords of the **hurna-* / **xurnan* ‘horn’ type for this noun (Kroonen 2013; Orel 2003, 234). The derivational input **xúrną* ‘horn’ is the nominative-style form used in the derivation here.

Old English evidence

Clark Hall, Bosworth-Toller, and Bright all record *horn* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898, 108; Bright 1971). The target is therefore the attested citation form.

Development to Old English

From **xúrną* ‘horn’, Northwest Germanic u-lowering gives **xórną* ‘horn’, and later loss of the final nasal leaves *horn*. The development treated here is fully regular.

Form note

The note’s oblique **xurnăn* ‘horn’ belongs to comparative stem background only. It does not replace the derivational input **xúrną* ‘horn’ as the derivational form used here (Kroonen 2013; Orel 2003, 234).

6.42 hue — OE *hīew*

Derivation: **xíwją* > *hīew* (regular).

Derivation trace

Proto input: **xíwją*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Ew Long Diphthong	**xíuwją*
PWGmc J Gemination	**xíwwją*	OE J Glide IO	**xīowją*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çīowją*
		OE Heavy Syllable Nasal Apocope	**çīowj*
[no change]		OE I Umlaut	**çīewj*
		OE J Loss After Heavy	**çīew*

Old English form: *hīew*

Reconstruction and comparative evidence

The selected reconstruction is realized Proto-Germanic **xíwją* ‘form, appearance’, a neuter *ja*-stem. It follows Ringe and Taylor’s explicit PGmc **hiwją* ‘form, appearance’, normalized with CAPR’s dorsal *x* and lexical acute (Ringe and Taylor 2014, 250). Both the citation reconstruction and the derivational input use i. This is an adopted working reconstruction, not a claim that all dictionaries print it or that their e forms have been shown to belong to a different historical stage.

The published e reconstructions

Orel gives **xewjan* ‘form, appearance’, neuter, with Gothic *hiwi* ‘surface appearance’, Old Norse *hý* ‘down on plants’ and Old English *hīw* ‘form’. He probably derives the noun from **xawwanan* and lists older etymological alternatives (Orel 2003, 171–72). The e vowel is therefore not an invented CAPR spelling. The dictionary entry supplies a real competing reconstruction; it does not license us to call its e an abstract underlying grade or an automatically pre-PGmc form. His final *-an* must also be distinguished from the other authors’ stem citation or nasal-vowel notation.

Kroonen gives **heuja-*, a neuter stem ‘visible layer, appearance’, on the comparative set Gothic hiwi ‘surface appearance, complexion’, Old Norse hý ‘down on plants’, Norwegian *hy*, and Old English *hīw*, *hēow* ‘shape, form, hue, colour’. He considers an early derivative of **kieh₁-u-*, with **heuja-* probably arising from **kēu-ió-* by Dybo’s law (Kroonen 2013, 224). This is not the same derivational argument as Orel’s. Their agreement on e cannot be treated as two identical histories, and their different endings cannot be silently collapsed into one quoted whole-word form. Writing prevocalic u as w and h as x explains a normalization; it does not by itself explain the disagreement between e and i.

The realized-i account and its early conditioning

Ringe and Taylor explicitly begin with PGmc **hiwją*. They distinguish a nongeminated **hiwi* from the stem with palatalized geminate w, and propose its development through **iuw* to pre-OE long **īow*, with a still-palatalizing semivowel, then early West Saxon long *īew*. They call the development surprising and its phonemic analysis unclear (Ringe and Taylor 2014, 250). The i choice is consequently not inferred merely from the final Old English spelling: an author explicitly assigns it to Proto-Germanic and offers a complete subsequent history for this noun.

Ringe separately describes early e-to-i raising before a high front vocalic in the same or following syllable. His example **néwios* to PGmc **niwjaz* 'new' supplies the relevant w+j class; he qualifies the evidence for the precise early date (Ringe 2017, 151–53). This passage explains the inherited-i preference but is not itself a quotation of a whole-word hue reconstruction. Fulk likewise treats PGmc e-to-i, and eu-to-iu, before following i/j separately from later front umlaut (Fulk 2018, 59). Under this working account, the raising precedes the supplied PGmc input. It must not be replaced by a later one-word e-to-i operation simply because that would conceal an input disagreement.

The glide objection and the strength of the choice

Campbell distinguishes the traditional awj and iwj developments and prints the hue reflex in the iwj class (Campbell 1959, 46, §120.2). Ringe and Taylor's independent gemination comparison uses new, sew and mirth, including nongeminated beside geminated paradigm stems (Ringe and Taylor 2014, 53). These support distinguishing the consonant classes and successive operations; they do not settle every phonemic detail of the offglide.

Fulk presents a substantive objection to the traditional dismantling account. He questions how w retained the presumed consonantal status in these inputs and how the resulting geminate was dismantled. His discussion considers a different strew input and paradigm regularization for new, rather than simply endorsing every conventional arrow (Fulk 2018, 71–72). His general statement of j-gemination after short vowels, not diphthongs, belongs to that issue (Fulk 2018, 126, §6.15). He does not supply a replacement complete hue history in these passages. Neither his objection nor Ringe and Taylor's uncertainty is removed by our choosing a computable working account.

We prefer realized PGmc i because the explicit reconstruction, independently motivated early raising and complete hue chain form a coherent account of the selected West Saxon reflex. This is not an author majority, an automatic preference for the most recent book, or an argument from successful final spelling. Confidence is stronger in the inherited high-front conditioning and the selected dialectal reflex than in a uniquely recoverable phonemic analysis of palatalized w. The e reconstructions remain alternatives with positive published support; they have not been disproved or silently reclassified.

These differences therefore warrant extended discussion despite a regular selected result: the authors' formation premises, citation endpoints and later phonemic uncertainties are distinct questions. No majority vote or normalization of final spellings resolves them. The existing source-led *i* choice and full glide analysis are retained, not replaced by a summary of dictionary vowels.

Old English evidence

Campbell prints *hīew* 'form' beside *hīow* 'form' in the section that derives this stem type, alongside *nīowe*, *nīewe* 'new' and *glīow*, *glīw* 'mirth' (Campbell 1959, § 120.2, p. 46). He distinguishes non-West-Saxon io from West Saxon ie. CAPR selects *hīew* as the source-supported West Saxon counterpart, not as the only attested spelling. Orel gives *hīw*, and Kroonen gives *hīw*, *hēow* (Orel 2003, 171–72; Kroonen 2013, 224).

Development to Old English

Campbell's decisive class contrast is that awj and iwj share gemination but thereafter lose different glides: the low-vowel type yields the palatal consonant, whereas the high-front type retains

w and yields io/ie (Campbell 1959, 46, §120.2). For hue, the adopted cascade now follows the independently specified Ringe-Taylor chain rather than using long ēo as an output-compatible proxy for their long īo (Ringe and Taylor 2014, 250).

Step	Selected reconstruction / represented checkpoint	Interpretation
PGmc input	**xíwją*	Inherited i, with early raising already reflected in the reconstruction
J-gemination	**xíwwją*	Palatalized geminate carried as ww+j
J-created reanalysis	**xíuwją*	Offglide formation, retaining palatalized w as w+j
English realization	**xīowją*	Long īo, distinct from ordinary ēo
Palatal articulation and ending loss	**çīowj*	Independent dorsal history and reduction in the existing cascade
Mutation	**çīewj*	Existing long-diphthong mutation conditioned by the retained palatal carrier
Later glide loss and spelling	*hīew*	Retained w and the selected native orthography

These are CAPR's checkpoints, not diplomatic quotations of all the authors' intermediate forms. In particular, superscript j on a source w denotes palatalization. Our separate j symbol carries that phonetic conditioning until the relevant consumer; it does not decide that a second j phoneme existed alongside every palatalized w. No new phonetic marker or lexical exception is needed.

Knee now selects the regular dative *cneowe* 'knee (dat.sg.)', whose short diphthong is produced by SC044 OEBreaking; it no longer shares this long-output component. The source distinguishes its short pre-ending stem from long endingless *cnēo* 'knee' and restored *w* in *cnēow* (Hogg 2011, 21–22, 86; Ringe and Taylor 2014, 387).

The j-created geminate is distinct from the earlier inherited-Vww operation. Its offglide checkpoint belongs to SC033 OEEwLongDiphthong; its long-io realization is separately represented by SC110 OEJGlideIO, before unchanged ordinary SC032 OEDiphthongLeveling. The composed SC106 OEJWWSimplification compatibility operation no longer acts on hue, which already has singleton w. Its absence of a live application is not universal proof of redundancy. The earlier inherited-reanalysis placement remains a representative serialization, not a newly proved strict chronology (Campbell 1959, 45–47; Ringe and Taylor 2014, 53, 65–66, 171–75, 250).

The contrast with *hīeġ* 'hay', from **xáwją*, remains exact: the selected inputs differ in one vowel and both geminate, but the low-vowel word loses w and retains the palatal consonant while hue retains w. The low-vowel resolution and subsequent fronting do not apply here (Campbell 1959, 46; Ringe and Taylor 2014, 53, 250).

Why this row is in the corpus

The corpus originally lacked a high-front witness separating gemination of w from the low-vowel resolution illustrated by hay and strew. Hue remains a positive witness for SC010 PWGmcJGemination and a negative control for SC029 OEAwwjResolution. The corrected input and distinct intermediate history now make that contrast scientifically informative rather than merely spelling-compatible. No additional lexeme or analogical mechanism is introduced.

6.43 lead — OE *lǣdan*

Derivation: **láidijaną* > *lǣdan* (regular).

Derivation trace

Proto input: **láidijaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lādijan*
[no change]		OE Secondary Nasalization	**lādijąn*
Early Anglo-Frisian		Sievers Law Syncope	**lādjąn*
EAF Ai Monophthongization	**lādijaną*	OE I Umlaut	**lædjąn*
		OE Weak Tail Reduction	**lædjan*
		OE J Loss After Heavy	**lædan*

Old English form: *lǣdan*

Reconstruction and comparative evidence

Ringe and Taylor derive Old English *lǣdan* ‘lead’ from Proto-Germanic **laidijaną* ‘lead’, and Kroonen likewise cites a weak verb of the **laidjan-* type for ‘lead’ (Ringe and Taylor 2014; Kroonen 2013, 363).

Old English evidence

Clark Hall and Bosworth-Toller both record *lædan* ‘lead’ / *lǣdan* ‘lead’ as the ordinary Old English verb ‘to lead, guide, conduct’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **láidijaną* ‘lead’, monophthongization of **ai* first gives a **lād-* stage. Later syncope, i-mutation, weak-tail reduction, and loss of *j* after a heavy stem yield *lǣdan* ‘lead’, so the development represented here is fully regular (Ringe and Taylor 2014).

6.44 learn — OE *leornian*

Derivation: **líznōjaną* > *leornian* (regular).

Derivation trace

Proto input: **líznōjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornōjaną*
PNWGmc I Lowering	**léznōjaną*	OE Heavy Syllable Nasal Apocope	**léornōjan*
EAF Rhotacism	**lérnōjaną*	OE Secondary Nasalization	**léornōjąn*
Early Anglo-Frisian		OE I Umlaut	**léornējąn*
		OE Unstressed Long Vowel Shortening	**léornejąn*
[no change]		OE Weak Tail Reduction	**léornejan*
		OE Intervocalic J Vocalization	**léorneian*
		OE Unstressed EI Contraction	**léornian*

Old English form: *leornian*

Comparative reconstruction and the inherited formation

Old English *leornian* ‘learn’ belongs with Old High German *lernēn* ‘learn’ beside *lirnēn* ‘learn’, and Old Saxon *līnon* ‘learn’. Kroonen reconstructs the weak verb **liznōn-*; Orel likewise preserves a high-vowel comparative formation. The reconstructed **z* is important both to the nasal formation and to the later rhotic of English. Neither the dictionary spelling of the English lemma nor its modern pronunciation establishes an inherited **e* (Kroonen 2013, 340; Orel 2003, 247).

The selected infinitive **líznōjaną* ‘learn’ retains that inherited root. Its extended weak-verb shape should not be confused with a verbatim quotation of Kroonen’s citation stem, or with the distinct imperative and third-singular cells. The family underwent conjugational remodelling; an English class II paradigm does not by itself recover every Proto-Germanic ending. Ringe and Taylor discuss the earlier formation separately from its later English continuation (Ringe and Taylor 2014, 38, 84).

Attested vowels and selected cells

The infinitive selected here is independently attested. In Ælfric’s Treatise the clause beginning *Gif þu nelt witan* continues *þu scealt leornian*: the modal requires an infinitive, so the grammatical identification does not depend on a dictionary headword. The edition prints this on p.71. The same volume gives the singular imperative *leorna* ‘learn!’ in Deuteronomy14:23, followed by a clause with the singular addressee *þu*. These are textual occurrences, not cells generated from an instructional paradigm (Crawford 1922, 71, 351).

For the companion present indicative third singular, Miller’s Bede prints *leornað* ‘learns’ after *swa hwylc swa* ‘whoever’, coordinated with *rædeð* ‘reads’. The singular subject disambiguates the ending, which in isolation could resemble a plural. CAPR represents final dental-fricative spelling with thorn and therefore selects *leornaþ* ‘learns’, without changing the root vowel, ending vowel or grammatical cell (Miller 1891, 380, ll. 26–27).

The alternative *liornian* ‘learn’ and finite *liorna* ‘learn!’ remain important. Campbell’s traditional account relates *io/eo* to breaking, later vowel developments and levelling. It must not be compressed into a universal equation of *liorn- with conservative Northumbrian and* leorn- with secondary West Saxon: early West Saxon also has the high-vowel spelling. A conservative spelling is not necessarily an unchanged inherited vowel (Campbell 1959, 48, 59, 125).

Lowering, mutation and the weak paradigm

The account adopted in SC015 PNWGmcILowering lowers **i* before preconsonantal **z*, independently of the following vowel. For the selected infinitive, early **lizn-* > *lezn-* is followed by rhotacism and English breaking before **rn*. The resulting *leorn- is not obtained by selecting an inherited mid-vowel reconstruction. Hill explains the competing high-vowel forms through later i-mutation in extended stems and subsequent paradigmatic history (Hill 2017, 153, n. 25).

That explanation is plausible but not a recovered complete paradigm. Jasanoff had already connected the family with the history of weak class III. He considers a lost third-singular **lizniþ* as a possible source of the high vowel of OHG *lirnēn*, rather than requiring a disputed first-singular reconstruction. The proposal matters because it supplies a particular morphological source for mutation or levelling, not a general appeal to analogy wherever the desired vowel is missing (Jasanoff 1973, 865, 868).

The original distribution of suffixes, their extension and the direction of later levelling remain uncertain. Later class II morphology cannot be projected backwards merely to exclude every earlier high-front trigger; conversely, one possible trigger does not establish that all three selected cells once contained it. The modern discussion of the class III family preserves these limitations (Ringe and Taylor 2014, 38, 84; Tarsi and Laker 2026, 42–43).

Interpretation of the selected derivation

The three entries represent one etymological family: an infinitive, an imperative singular and a present third singular, not three independent proofs of a sound law. Their inherited inputs remain distinct, while the selected *leorn- vocalism receives a regular consonant-conditioned history. Selection of these textually supported vowels does not disprove the* liorn- tradition or resolve the whole weak-verb paradigm (Crawford 1922, 71, 351; Miller 1891, 380; Jasanoff 1973, 865, 868).

The finite entries retain an analogical family classification for the conjugational remodelling, not because finite-cell selection is itself analogy. No additional analogical replacement of the selected root vowel is required. The distinction between regular phonology of the form compared here and uncertain earlier paradigm history is essential to interpreting the derivations (Jasanoff 1973, 865, 868; Tarsi and Laker 2026, 42–43).

6.45 lid — OE *hlid*

Derivation: **xlídą* > *hlid* (regular).

Derivation trace

Proto input: **xlídą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xlíd*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hlid*

Reconstruction and comparative evidence

Orel cites a neuter lexeme of the **xliđ-* type with Old English *hlid* ‘lid, cover’, and Lloyd includes OE *hlid* ‘lid’ beside ON *hliþó* ‘lid’ and OHG *(h)lit* ‘lid’ among forms that retain *i* (Orel 2003; Lloyd 1966).

Old English evidence

Clark Hall and Bosworth-Toller record *hlid* ‘lid, cover’ as the noun ‘lid, cover, door, gate’ (Clark Hall 1960; Bosworth and Toller 1898, 563).

Development to Old English

The derivational input already represents the later Germanic *hliđ-* stage used for the derivation here. From **xlídą* ‘lid’, heavy-syllable apocope yields *hlid* ‘lid’, and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (Lloyd 1966).

Form note

An earlier etymological stage **liþuz* ‘lid’ belongs to comparative background only. The form represented here is the later **xlídą* ‘lid’ > *hlid* ‘lid’ line that matches the attested Old English noun (Orel 2003; Lloyd 1966).

6.46 light — OE *līehtan*

Derivation: **léuxtijaną* > *līehtan* (regular).

Derivation trace

Proto input: **léuxtijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**lēoxtijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēoxtijan*
Early Anglo-Frisian	OE Secondary Nasalization	**lēoxtijąn*
[no change]	Sievers Law Syncope	**lēoxtjąn*
	OE I Umlaut	**līextjąn*
	OE Weak Tail Reduction	**līextjan*
	OE J Loss After Heavy	**līextan*

Old English form: *līehtan*

Reconstruction and comparative evidence

Fulk gives Proto-Germanic **liuxtijanan* ‘light’ with Old English *līehtan* ‘illuminate’, and Ringe and Taylor likewise derive West Saxon *liehtan* ‘light, illuminate’ from the same weak-verb formation (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan* ‘illuminate’, *lihtan* ‘light, illuminate’, and *līhtan* ‘illuminate’, distinct from the related noun *lēoht* ‘light’ and adjective *leoht* / *liht* ‘light’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **léuxtijaną* ‘light’, the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* ‘illuminate’ stage, and is represented here by normalized *līehtan* ‘illuminate’. The word treated in this entry is therefore the verb ‘to light, illuminate’, not the related noun from **leuxtą* ‘light’ (Fulk 2018; Ringe and Taylor 2014).

Dialect note

Ringe and Taylor and Campbell distinguish West Saxon *liehtan* ‘light, illuminate’ from Anglian *lihtan* ‘light, illuminate’, while later West Saxon also shows *lyhtan* ‘light, illuminate’ (Ringe and Taylor 2014; Campbell 1959).

6.47 linden — OE *lind*

Derivation: **líndō* > *lind* (regular).

Derivation trace

Proto input: **líndō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**línd*
PNWGmc Final Long O Raising	**líndu*		
Early Anglo-Frisian			
[no change]			

Old English form: *lind*

Reconstruction and comparative evidence

Kroonen cites **lindō-* ‘lime tree’ and gives Old English *lind* ‘linden, lime tree’ as the relevant reflex (Kroonen 2013).

Old English evidence

Clark Hall and Bosworth-Toller both record *lind* ‘linden, lime tree’ as the noun ‘lime-tree, linden’ (Clark Hall 1960; Bosworth and Toller 1898, 630).

Development to Old English

From **líndō* ‘linden’, Northwest Germanic final **ō* raising first gives a **líndu* ‘linden’ stage, and later high-vowel apocope yields *lind* ‘linden, lime tree’. The development is therefore straightforward and regular.

Form note

The Old English noun represented here is *lind* ‘linden, lime tree’. Clark Hall also has a separate adjectival *linden* ‘made of linden-wood’, but that is not the noun counterpart for this entry (Clark Hall 1960).

6.48 louse — OE *lūs*

Derivation: **lūsz* > *lūs* (regular).

Derivation trace

Proto input: **lūsz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *lūs*

Reconstruction and comparative evidence

Orel’s citation form is **lusz* ‘louse’, with the explicit nominative marker **-z* he writes across the root-noun class; his entry points to Bammesberger and Griepentrog for the morphology (Orel 2003, 252). Kroonen has no ‘louse’ entry, so his dictionary cannot be cited for this word (Kroonen 2013, 345). Kluge/Seebold carry the German cognate as the same root noun (Kluge 2011, 563). Bammesberger shows that the root-final *s* is itself secondary: the root **luw-* ‘louse’ was extended with *s* on the model of the ‘mouse’ word (Bammesberger 1990, 195). The ‘mouse’ parallel also

shows why no extra sibilant is audible in the nominative: Ringe reconstructs endingless **mūs* ‘mouse’, with the expected double sibilant of the nominative removed by degemination (Ringe 2017, 149).

The derivational input **lūsz* ‘louse’ follows Orel’s morphologically explicit convention. The final **-z* is an inflectional marker made visible in the citation form, not a claim that a phonetic cluster *[sz]* was ever pronounced; the notation records the morphology whose elimination the derivation models.

Old English evidence

Old English *lūs* ‘louse’ is the endingless nominative-accusative singular of a root noun, with mutated plural *lȳs* ‘lice’ (Fulk 2018, 165–66). Old Norse *lús* ‘louse’ is likewise endingless, in keeping with the North Germanic redistribution that drops the reflex of **-z* in feminine root nouns (Fulk 2018, 167). No West Germanic daughter shows any ending in this class (Ringe and Taylor 2014, 118).

Development to Old English

The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118). From the derivational input **lūsz* ‘louse’ the loss of the marker gives **lūs* ‘louse’, which is already the attested Old English form. Because the root itself ends in *s*, the marker’s elimination here is best read together with the degemination that Ringe describes for ‘mouse’: at no stage does the reconstruction require an audible *-sz* sequence. This early morphological development is distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the root-noun ending was gone before either applied, and the root-final *s* of *lūs* ‘louse’ was never eligible for either change.

6.49 milk — OE *meoloc*

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace

Proto input: **mélukz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Back Mutation	**méoluk*
EAF Final Z Deletion	**méluk*	OE Med Unstressed U Lowering	**méolok*
Early Anglo-Frisian			
[no change]			

Old English form: *meoloc*

Reconstruction and comparative evidence

Kroonen and Orel reconstruct the noun with stem notation **meluk-* and nominative **melukz* ‘milk’, and the nominative-style input used here is **mélukz* ‘milk’ (Kroonen 2013; Orel 2003, 306).

Old English evidence

Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* ‘milk’ < *meoluc* ‘milk’ < **mélukz* ‘milk’, Campbell likewise discusses *meoluc* ‘milk’ and *meoloc* ‘milk’, and Anglian shows *milc* ‘milk’ (Ringe and Taylor 2014; Campbell 1959).

Development to Old English

The unsyncopated line from **mélukz* ‘milk’ loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc* ‘milk’. That fuller unsyncopated outcome is the form represented here (Ringe and Taylor 2014).

Form comparison

Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc* / *meoluc* preserves the fuller nominal shape (Campbell 1959; Ringe and Taylor 2014).

6.50 mother — OE *mōder*

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace

Proto input: **mōdēr*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**mōder*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *mōder*

Reconstruction and comparative evidence

Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder-* with citation **mōdēr* ‘mother’ (Kroonen 2013; Orel 2003).

Old English evidence

The transmitted Old English headword tradition is *mōdor* ‘mother’ / *modor* ‘mother’, with oblique *mēder* ‘mother’ in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (Clark Hall 1960; Campbell 1959; Ringe and Taylor 2014).

Development to Old English

From **mōdēr* ‘mother’, the regular suffixal development yields *mōder* ‘mother’. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* ‘mother’ reflects later levelling within the r-stem paradigm (Campbell 1959; Ringe and Taylor 2014).

Form comparison

The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* ‘mother’ is the regularized nominative represented here, but dictionaries usually print *mōdor* / *modor* ‘mother’, and the oblique evidence survives in *mēder* ‘mother’ (Clark Hall 1960; Ringe and Taylor 2014).

6.51 nest — OE *nest*

Derivation: **nístą* > *nest* (regular).

Derivation trace

Proto input: **nístą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**nést*
PNWGmc I Lowering	**néstą*		
Early Anglo-Frisian			
[no change]			

Old English form: *nest*

Comparative formation and lowering

Old English *nest* ‘nest’ continues an inherited neuter formation. Kroonen reconstructs **nista-*, Orel the full-word **nistan*. Their comparanda include Latin *nīdus* ‘nest’ and Sanskrit *nīḍá-* ‘nest’. The traditional analysis combines an element meaning ‘down’ with a zero-grade formation of the root ‘sit’. The English word must not be confused with the separate food noun also spelled *nest* (Kroonen 2013, 388, 391; Orel 2003, 287).

The selected **nístą* ‘nest’ retains the inherited high vowel. SC015 PNWGmcILowering lowers it immediately before **st*, rather than because of the next vowel. This is the principal independent lexical support for Hill’s **st* condition. The contrast with **sk* in the fish family is useful, but it does not turn one inherited nest etymon into numerous independent witnesses (Hill 2017, 153–55).

Later mutation in a derivative with a high-front trigger can conceal an earlier mid vowel. Such derivatives test chronological compatibility; they do not establish every detail of a lost paradigm. The general productivity of lowering before **st* is less securely demonstrated than lowering before inherited **r*, and the adopted unified law retains that evidential qualification (Hill 2017, 153–56).

6.52 net — OE *nett*

Derivation: **nátją* > *nett* (regular).

Derivation trace

Proto input: **nátją*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**nættją*
PWGmc J Gemination	**náttją*	OE Heavy Syllable Nasal Apocope	**nættj*
Early Anglo-Frisian		OE I Umlaut	**nettj*
		OE J Loss After Heavy	**nett*
[no change]			

Old English form: *nett*

Reconstruction and comparative evidence

Orel gives **natjan* ‘net’ with Old English *nett* ‘net’, and Fulk’s account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (Orel 2003; Fulk 2018).

Old English evidence

Clark Hall and Bosworth-Toller record *nett* ‘net’ as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (Clark Hall 1960; Bosworth and Toller 1898, 29; Campbell 1959).

Development to Old English

From **nátją* ‘net’, West Germanic j-gemination first gives **náttją* ‘net’. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett* ‘net’, so the development represented here is regular (Fulk 2018).

Form note

Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* ‘net’ (Campbell 1959; Orel 2003).

6.53 nightmare — OE *mare*

Derivation: **márōn* > *mare* (regular).

Derivation trace

Proto input: **márōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mærǭ*
PNWGmc N Stem N Loss	**márǭ*	OE A Restoration	**marǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**maræ*
		OE Unstressed AE Merger	**mare*
[no change]			

Old English form: *mare*

Reconstruction and comparative evidence

Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **marōn-*, with Old English *mare, maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *mare* ‘nightmare, monster’ and also preserves related variant forms *mera* / *mere* (Clark Hall 1960, 213).

Development to Old English

The selected simplex input **márōn* ‘nightmare’ regularly gives *mare* ‘nightmare’ after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (Ringe and Taylor 2014).

Form note

The concept corresponds to an unattested compound **nihtmare* ‘nightmare’, but the Old English lexical evidence is for simplex *mare* ‘nightmare’, with oblique *maran* and variant *mere* / *mera* (Ringe and Taylor 2014; Clark Hall 1960, 213).

6.54 sheep — OE *sċēap*

Derivation: **skép̄ą* > *sċēap* (regular).

Derivation trace

Proto input: **skép̄ą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**skǣp*
PNWGmc Long E Lowering	**skāpą*	OE Sk Palatalization	**ʃǣp*
EAF Long A Fronting	**skǣpą*	OE Ws Palatal Diphthongization	**ʃēap*
Early Anglo-Frisian			
[no change]			

Old English form: *sċēap*

Reconstruction and comparative evidence

Ringe and Taylor cite a later West Germanic **skap* ‘sheep’ > WS *scéap* ‘sheep’, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *scēap* ‘sheep’ with spelling variation, and Campbell likewise lists West Saxon *scéap* ‘sheep’ among the palatal-diphthongized forms (Clark Hall 1960; Campbell 1959).

Development to Old English

From **skép̄ą* ‘sheep’, Northwest Germanic lowering gives **skǣpą* ‘sheep’; after apocope and palatalization the West Saxon branch diphthongizes to *sċēap* ‘sheep’. The development represented here is therefore fully regular.

Dialect note

Ringe and Taylor contrast West Saxon *scéap* ‘sheep’ with Mercian and Kentish *scép* ‘sheep’, and Campbell also notes Northumbrian *scip* ‘sheep’. The form represented here is the West Saxon headword (Ringe and Taylor 2014; Campbell 1959).

6.55 shilling — OE *sċilling*

Derivation: **skíllingaz* > *sċilling* (regular).

Derivation trace

Proto input: **skíllingaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**skílling*
EAF Final Z Deletion	**skíllinga*	OE Sk Palatalization	**fílling*
Early Anglo-Frisian		OE Med Unstressed I Lowering1	**fílleng*
[no change]		OE Med Unstressed I Lowering	**fílling*

Old English form: *sċilling*

Reconstruction and comparative evidence

Kroonen treats the cognate set under **skellinga- ~ *skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* ‘shilling’ among the reflexes (Kroonen 2013; Orel 2003). The derivational input **skíllingaz* ‘shilling’ is the nominative-style form used here to represent that inherited **-ing-* derivative.

Old English evidence

Clark Hall records *scilling* ‘shilling’, and Campbell cites it among nouns with unstressed *i* in derivational *-ing* (Clark Hall 1960; Campbell 1959). The target is the ordinary OE citation form, normalized as *sċilling* ‘shilling’.

Development to Old English

From **skíllingaz* ‘shilling’, loss of final *-az* yields **skílling* ‘shilling’. Old English palatalization of initial *sk* before front vocalism then gives *sċilling* ‘shilling’. The *i* of derivational *-ing-* remains, so the regular outcome is *sċilling* ‘shilling’, not **sċilleng* ‘shilling’ (Campbell 1959; Hogg 1992a).

Form note

Kroonen’s **skellinga- ~ *skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The derivational input **skíllingaz* ‘shilling’ is the specific form used for the derivation represented here (Kroonen 2013).

6.56 shovel — OE *sċofl*

Derivation: **skúflō* > *sċofl* (regular).

Derivation trace

Proto input: **skúflō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Sk Palatalization	**fóflu*
PNWGmc U Lowering	**skóflō*	OE High Vowel Apocope	**fófl*
PNWGmc Final Long O Raising	**skóflu*		
Early Anglo-Frisian			
[no change]			

Old English form: *sċofl*

Reconstruction and development

The selected short-root-vowel **skúflō* ‘shovel’ follows the feminine formation discussed by Orel (Orel 2003, 347). With lowering before final oral raising, its root becomes **o* before the ending becomes **u*; subsequent apocope and consonant development give *sċofl* ‘shovel’. No later root-vowel levelling is needed for the form compared here.

This is a conditional witness, not independent proof of the order. Earlier raising instead predicts a high-root nominative beside a lowered oblique. Luick’s inferred **scufel* beside *scofel* ‘shovel’ makes alternation credible but does not directly establish o-generalization in the early English paradigm (Luick 1914--1940, 105–6, §78). The assessment of that additional analogy, alongside wool and the other ō-stems, belongs to SC019 PNWGmcFinalLongORaising.

6.57 show — OE *sċēawian*

Derivation: **skáwōjaną* > *sċēawian* (regular).

Derivation trace

Proto input: **skáwōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skǽawōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**skǽawōjan*
Early Anglo-Frisian	OE Secondary Nasalization	**skǽawōjąn*
[no change]	OE Sk Palatalization	**ʃǽawōjąn*
	OE I Umlaut	**ʃǽawējąn*
	OE Unstressed Long Vowel Shortening	**ʃǽawejąn*
	OE Weak Tail Reduction	**ʃǽawejan*
	OE Intervocalic J Vocalization	**ʃǽaweian*
	OE Unstressed EI Contraction	**ʃǽawian*

Old English form: *sċēawian*

Reconstruction and comparative evidence

Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* ‘show’ among the reflexes (Orel 2003, 337; Kroonen 2013, 442). The dictionary singleton stem is not silently identified with Campbell and Brunner’s inherited-geminate analysis (Campbell 1959, 45; Brunner 1965, 49). Brunner also records the later West Saxon variant *scāwian* ‘show’ in his discussion of diphthong accent shift; it is not evidence for the same prehistoric singleton quantity (Brunner 1965, 106–7).

Old English evidence

The held fourth edition of Bright’s reader contains *scēawian* ‘show’ and the related imperative *scēawa* ‘show’ in the Blickling homily (Bright 1917, 71). This is not a newly verified passage in the separate Cassidy/Ringler edition. The sources use *scēawian* ‘show’, while *sċēawian* ‘show’ supplies CAPR’s normalized spelling.

Development to Old English

From **skáwōjaną* ‘show’, the transducer retains SC034 OEAwLongDiphthong as a computational proxy mapping to long *ea+w* before the class-II suffix. Its **ō* separates root **w* from **j*, excluding direct *aw+j* gemination. Neither that exclusion nor the successful *sċēawian* ‘show’ output establishes an exceptionless singleton lengthening law. The derivational input and *regular* computational classification remain unchanged, conditional on this retained proxy; their historical basis is provisional, not a resolution of the root-history premise. Campbell and Brunner’s inherited-geminate account differs from the dictionary singleton stem (Campbell 1959, 45–46; Brunner 1965, 49; Orel 2003, 337; Kroonen 2013, 442).

The finite companions select an imperative singular and a present third singular, not alternative dictionary infinitives. Their class-II endings are independently supported even though successful root outputs cannot decide the disputed prehistoric quantity (Ringe and Taylor 2014, 80, 314; Campbell 1959, 148).

Form note

The difference between *scēawian* ‘show’ and *sċēawian* ‘show’ is CAPR’s editorial normalization of initial <*sc*>, not a different lexeme or paradigm cell. It does not itself settle the consonant’s prehistoric articulation.

6.58 sleep — OE *slǣpan*

Derivation: **slépaną* > *slǣpan* (regular).

Derivation trace

Proto input: **slépaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**slǣpan*
PNWGmc Long E Lowering	**slāpaną*	OE Secondary Nasalization	**slǣpąn*
EAF Long A Fronting	**slǣpaną*	OE Weak Tail Reduction	**slǣpan*
Early Anglo-Frisian			
[no change]			

Old English form: *slǣpan*

Reconstruction and comparative evidence

Kroonen preserves the comparative verb as **slēpan-*, and Fulk cites the same family under root **slēb-* (Kroonen 2013; Fulk 2018, 120). The derivational input **slépaną* ‘sleep’ is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence

Clark Hall gives *slœpan* ‘sleep’ with preterite *slēp* ‘slept’, *slēap* ‘slept’, and Bright likewise lists *slœpan* ‘sleep’ (*slāpan* ‘sleep’), *slēp* ‘slept’, *slēpon* ‘slept’, *slēpen* ‘slept’ (Clark Hall 1960; Bright 1971, 435). The target represented here is therefore the normalized infinitive *slǣpan* ‘sleep’, not the preterite forms and not the separate noun *slǣp* ‘sleep’.

Development to Old English

From **slépaną* ‘sleep’, Northwest Germanic lowering gives **slǣpaną* ‘sleep’. The later OE tail developments then yield *slǣpan* ‘sleep’ regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan* ‘sleep’ / *slœpan* ‘sleep’ and Anglian or Kentish *slēpan* ‘sleep’, but those do not displace the infinitive chosen here (Brunner 1965; Bülbring 1902).

Form note

The note concerns lemma type rather than a special derivational problem: this row represents the verb *slǣpan* ‘sleep’, whereas *slǣp* ‘sleep’ belongs to noun or lookup background and *slēp* ‘slept’ / *slēap* ‘slept’ are preterite forms (Clark Hall 1960; Bright 1971, 435).

6.59 smear — OE *smierwan*

Derivation: **smérwijaną* > *smierwan* (regular).

Derivation trace

Proto input: **smérwijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**sméorwijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**sméorwijan*
	OE Secondary Nasalization	**sméorwijąn*
Early Anglo-Frisian	Sievers Law Syncope	**sméorwjąn*
[no change]	OE I Umlaut	**smíerwjąn*
	OE Weak Tail Reduction	**smíerwjan*
	OE J Loss After Heavy	**smíerwan*

Old English form: *smierwan*

Reconstruction and comparative evidence

Kroonen gives the comparative headword as **smerwjan-* (Kroonen 2013). Ringe and Taylor instead cite a later-stage **smirwijana* ‘smear’, from which they derive West Saxon *smierwan* ‘smear’, Mercian *smirwan* ‘smear’, and Northumbrian *smiriga* (Ringe and Taylor 2014). The derivational input **smérwijaną* ‘smear’ therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence

The target represented here is the West Saxon citation form *smierwan* ‘smear’. Campbell’s Anglian discussion explains the contrasting *smirwan* ‘smear’, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (Campbell 1959; Clark Hall 1960; Brunner 1965; Bright 1971).

Development to Old English

From **smérwijaną* ‘smear’, breaking before *r* + *consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan* ‘smear’. Anglian *smirwan* ‘smear’ reflects the well-known failure of breaking in this environment, not a different lexeme (Campbell 1959; Ringe and Taylor 2014).

Dialect note

The entry therefore represents the West Saxon member of a broader OE family: *smierwan* ‘smear’ in West Saxon, *smirwan* ‘smear’ in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (Ringe and Taylor 2014; Clark Hall 1960).

6.60 sorrow — OE *sorg*

Derivation: **súrgō* > *sorg* (regular).

Derivation trace

Proto input: **súrgō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**sórg*
PNWGmc U Lowering	**sórgō*		
PNWGmc Final Long O Raising	**sórgu*		
Early Anglo-Frisian			
[no change]			

Old English form: *sorg*

Reconstruction and development

The selected **súrgō* ‘sorrow’ belongs to the feminine formation discussed by Kroonen (Kroonen 2013, 493). Lowering before final oral raising gives a low root before the ending becomes **u*; later apocope yields *sorg* ‘sorrow’. This accounts for the selected form without an additional root-vowel generalization.

Earlier raising would instead protect the nominative root while leaving the nasal accusative’s non-high conditioner available for lowering. Generalization of that oblique root could recover the low-root paradigm, but has not been independently established for this particular early English noun. Generic evidence for such levelling is not specific attestation of this operation (Ringe and Taylor 2014, 32–34; Stiles 2012, 47–48). The comparative economy argument and its limits are developed in SC019 PNWGmcFinalLongORaising.

6.61 span — OE *spannan*

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace

Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**spánnan*
[no change]	OE Secondary Nasalization	**spánnąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**spánnan*
[no change]		

Old English form: *spannan*

Reconstruction and comparative evidence

Kroonen cites the inherited verb as **spannan-*, with OE *spannan* ‘span, fasten’ among the reflexes (Kroonen 2013). The derivational input **spánnaną* ‘span’ is the infinitive-style form used here for that same verbal lexeme.

Old English evidence

Clark Hall lists noun *spann* and verb *spannan* ‘span, fasten’ as separate headwords, and Brunner likewise records *sponnan, spannan stv.* (Clark Hall 1960; Brunner 1965). The target is the strong-verb infinitive, not the noun.

Development to Old English

From **spánnaną* ‘span’, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan* ‘span, fasten’. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note

English *span* can also reach the noun *spann* in local lookup material. The form represented here is the verb *spannan* ‘span, fasten’, with the noun treated elsewhere (Clark Hall 1960).

6.62 spar — OE *spearra*

Derivation: **spárrô* > *spearra* (regular).

Derivation trace

Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærrô*
[no change]	OE Breaking	**spearrô*
	OE Unstressed Long Vowel Shortening	**spearra*
Early Anglo-Frisian		
[no change]		

Old English form: *spearra*

Reconstruction and comparative evidence

Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* ‘spar, rafter’ (Kroonen 2013; Orel 2003). The derivational input **spárrô* ‘spar’ is the OE-facing nominal form used here for that same lexeme.

Old English evidence

The noun is *spearra* ‘spar, rafter’. English gloss overlap also reaches the unrelated verb *sperran* ‘to bar’, which does not belong to this row.

Development to Old English

From **spárrô* ‘spar’, Anglo-Frisian brightening gives **spærrô* ‘spar’, and OE breaking before geminate *rr* yields **spearrô* ‘spar’, later *spearra* ‘spar, rafter’. The development is therefore regular for a breaking-conditioned noun of this type (Luick 1914--1940).

Form note

This entry concerns the noun *spearra* ‘spar, rafter’ only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (Kroonen 2013; Orel 2003).

6.63 still — OE *stillan*

Derivation: **stéllijaną* > *stillan* (regular).

Derivation trace

Proto input: **stéllijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**stéllijan*
[no change]	OE Secondary Nasalization	**stéllijǫn*
	Sievers Law Syncope	**stélljǫn*
Early Anglo-Frisian	OE I Umlaut	**stilljǫn*
[no change]	OE Weak Tail Reduction	**stilljan*
	OE J Loss After Heavy	**stillan*

Old English form: *stillan*

Reconstruction and comparative evidence

The wider West Germanic family includes adjective *still* ‘still, quiet’ and verb *stillen* (Kluge 2011). The derivational input **stéllijaną* ‘still’ represents the verbal j-formation used for the OE row.

Old English evidence

Clark Hall gives *stillan* as the verb and separately *stille* ‘still, quiet’ as the adjective (Clark Hall 1960). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* ‘still, stop’ and related forms (Bosworth and Toller 1898, 724). The Old English form here is the verb *stillan*, not the adjective.

Development to Old English

As a heavy-stem Class I weak verb, **stéllijaną* ‘still’ undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note

The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* ‘still, quiet’ belongs to the related adjectival branch of the family (Clark Hall 1960; Kluge 2011).

6.64 summer — OE *sumer*

Derivation: **súmaraz* > *sumer* (regular).

Derivation trace

Proto input: **súmaraz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**súmar*
EAF Final Z Deletion	**súmara*	EAF Brightening Unstressed	**súmær*
Early Anglo-Frisian		OE Unstressed AE Merger	**súmer*
[no change]			

Old English form: *sumer*

Reconstruction and comparative evidence

Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz* ‘summer’, while Orel preserves an alternate **sumeraz* ‘summer’ (Kroonen 2013; Ringe and Taylor 2014; Orel 2003, 425). The derivational input **súmaraz* ‘summer’ follows the **a* vocalism that underlies the regular development represented here.

Old English evidence

Clark Hall gives *sumor* ‘summer’ m., gs. *sumeres* ‘summer’, ds. *sumera* ‘summer’ / *sumere* ‘summer’, and Bright likewise lists *sumor (sumer)* with genitive *sumeres* ‘summer (gen.)’ (Clark Hall 1960; Bright 1971, 440). The tradition therefore preserves both *sumor* ‘summer’ and *sumer* ‘summer’, with the oblique forms strongly supporting second-syllable *e*.

Development to Old English

From **súmaraz* ‘summer’, loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer* ‘summer’. The form compared here is the regularized *e*-form, while the common citation form *sumor* ‘summer’ remains part of the attested OE tradition (Ringe and Taylor 2014).

Form note

The entry does not deny *sumor* ‘summer’. It represents *sumer* ‘summer’ as the regular outcome chosen here, while *sumor* ‘summer’ remains a common headword spelling and *sumeres* ‘summer (gen.)’ / *sumere* ‘summer (dat.)’ show that the *e*-vocalism was also real in Old English (Clark Hall 1960; Bright 1971, 440).

6.65 sunder — OE *sundrian*

Derivation: **súndrōjaną* > *sundrian* (regular).

Derivation trace

Proto input: **súndrōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**súndrōjan*
[no change]	OE Secondary Nasalization	**súndrōjąn*
	OE I Umlaut	**súndrējąn*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**súndrejąn*
[no change]	OE Weak Tail Reduction	**súndrejan*
	OE Intervocalic J Vocalization	**súndreian*
	OE Unstressed EI Contraction	**súndrian*

Old English form: *sundrian*

Reconstruction and comparative evidence

Orel distinguishes three related formations: adverbial **sunþraz* ‘asunder’ > *sundor* ‘apart, asunder’, Class I verbal **sunþrjanan* ‘sunder’ > *syndrian* ‘separate, sunder’, and Class II verbal **sunþrōjanan* ‘sunder’ > *sundrian* ‘sunder’ (Orel 2003).

Kluge-Seebold aligns the cognate set with German *sondern* ‘separate’ and OE *gesundrian* ‘separate’, so this entry belongs with the Class II verb, not the adverb (Kluge 2011).

Old English evidence

Clark Hall and Bosworth-Toller list *sundrian* ‘sunder’ and *syndrian* ‘sunder’ separately from adverbial *sundor* ‘asunder’; both also record the prefixed verbal family *ā-sundrian* ‘sunder apart’ (Clark Hall 1960, 296; Bosworth and Toller 1898). The target is therefore the weak verb *sundrian* ‘sunder’.

Development to Old English

From **súndrōjaną* ‘sunder’, the Class II weak-verb suffix yields regular OE *-ian*, producing *sundrian* ‘sunder’. Because this is the **-ōjan-* verb and not the Class I **-jan-* formation, the form represented here does not belong to the umlauted *syndrian* ‘separate’ branch.

Form note

The earlier confusion was lexical, not phonological: *sundor* ‘apart’ is the separate adverb, and *syndrian* ‘separate’ is a related but different verb. The verb treated here is the Class II verb *sundrian* (Orel 2003; Clark Hall 1960, 296).

6.66 swallow — OE *swealwe*

Derivation: **swálwōn* > *swealwe* (regular).

Derivation trace

Proto input: **swálwōn*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**swælwǭ*
PNWGmc N Stem N Loss **swálwǭ*	OE Breaking	**swealwǭ*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**swealwæ*
	OE Unstressed AE Merger	**swealwe*
[no change]		

Old English form: *swealwe*

Reconstruction and comparative evidence

Kroonen gives the bird name as **swalwōn-*, and Ringe and Taylor cite the later West Germanic stage **swalwa* ‘swallow’, from which West Saxon *swealwe* ‘swallow’ and Mercian *swalwe* ‘swallow’ develop (Kroonen 2013, 535; Ringe and Taylor 2014, 200). The selected etymological comparison belongs to the swallow-bird family, not to the verb *swelgan*.

Old English evidence

Clark Hall records *swealwe (a, o)* as the noun headword (Clark Hall 1960). Campbell and Brunner also preserve later or oblique-family forms such as *swaluwe*, *swalewan*, and *swealuwe*, but those belong to wider variation around the noun rather than to the citation form represented here (Campbell 1959; Brunner 1965).

Development to Old English

From **swálwōn* ‘swallow’, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (Ringe and Taylor 2014, 200; Kroonen 2013, 535).

6.67 swine — OE *swīn*

Derivation: citation reconstruction **swíńą*; form followed here **swíńą* > *swīn* (regular).

Derivation trace

Proto input: **swíńą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**swī́n*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *swīn*

Reconstruction and comparative evidence

Kroonen cites the noun as **swina-* ‘pig’ (Kroonen 2013). The selected comparative form here is the bare neuter citation cell **swī́ną* ‘swine’, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence

Clark Hall records *swin (y)* as the ordinary noun headword (Clark Hall 1960). The target here is that singular citation form *swīn* ‘swine’, not a plural glossed in Modern English as *swine*.

Development to Old English

From derivational input **swī́ną* ‘swine’, loss of the final nasal vowel yields *swīn* ‘swine, pig’. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note

The derivational input writes stressed long *ī* as **ī́*, so comparative **swī́ną* ‘swine’ and derivational **swī́ną* ‘swine’ represent the same lexical form.

6.68 there — OE *þǣr*

Derivation: **θár* > *þǣr* (regular).

Derivation trace

Proto input: **θár*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Monosyllabic Final R Lengthening	**θār*	
EAF Long A Fronting	**θǣr*	
Early Anglo-Frisian		
[no change]		

Old English form: *þǣr*

Old English evidence

Old English *þǣr* ‘there’ is the normalized West Saxon adverb. Clark Hall lemmatizes *ðǣr* ‘there’; the substitution of thorn for eth does not change the vowel or the identity of the word (Clark Hall 1960, 305). Hill contrasts West Saxon *ðǣr* with Anglian *ðēr* ‘there’: both have long vowels, with different dialectal qualities (Hill 2017, 142–43). These are normalized lexical and dialect forms, rather than diplomatic transcriptions of individual manuscripts.

Reconstruction and comparative evidence

Kroonen reconstructs **þar* ‘there’, formed from the demonstrative stem **þa-* with locative **-r* (Kroonen 2013, 534). Orel likewise identifies the demonstrative formation, but prints the alternatives **þar* and **þēr* (Orel 2003, 416). His long-vowel citation must be kept distinct from the earlier short-vowel antecedent. The selected **θár* ‘there’ represents the short antecedent with an explicit root accent and a phonological dental symbol; neither addition is a literal quotation of the dictionaries.

The comparative distribution leaves the date of lengthening open. Gothic *þar* ‘there’ has a spelling that does not distinguish short *a* from long *ā*. More informative for conditioning is the contrast between Old High German *dār* ‘there’ and the short vowels of *dăra* ‘thither’ and *dărōt* ‘thither’. Hill uses this contrast to support lengthening in monosyllables ending in inherited *r* (Hill 2017, 142–48).

Development to Old English

The adopted derivation applies SC112, stressed monosyllabic lengthening before inherited final *r*, giving **þar* > **þār*. Subsequent SC101, long-*ā* fronting supplies **þǣr*, yielding West Saxon *þǣr* without a change of formation or an analogical replacement. The vowel becomes long while it is still back; this is distinct from lengthening an already fronted short vowel.

Hill argues for a Proto-Germanic date, whereas Ringe and Taylor place the lengthening in West Germanic and discuss deictic stress and reanalysis as possible explanations (Hill 2017, 146–48; Ringe and Taylor 2014, 65). The regular short-*a*-to-*ǣ* development is conditional on the chosen antecedent and early lengthening. Its successful outcome does not establish that absolute date, and a word without inherited **i* cannot independently order SC015 relative to lengthening.

6.69 think — OE *þenċan*

Derivation: **θánkijanǫ* > *þenċan* (regular).

Derivation trace

Proto input: **θánkijanǫ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**θánkijan*
[no change]	OE Secondary Nasalization	**θánkijǫn*
	Sievers Law Syncope	**θánkjǫn*
Early Anglo-Frisian	OE Velar Palatalization	**θántʃjǫn*
[no change]	OE I Umlaut	**θentʃjǫn*
	OE Weak Tail Reduction	**θentʃjan*
	OE J Loss After Heavy	**θentʃan*

Old English form: *þenċan*

Reconstruction and comparative evidence

Kroonen gives the verb as **þankjan-* ‘to think’, and Ringe and Taylor cite fully inflected **þankijanǫ* ‘think’ beside OE *þenċan* ‘think’ (Kroonen 2013; Ringe and Taylor 2014). The noun **þankaz* ‘think’ belongs only to the wider derivational background.

Old English evidence

Bosworth-Toller preserves the verb under *þencan* ‘think’ / *geþencan* ‘think’, and the citation form here is the ordinary infinitive *þenċan* ‘think’ (Bosworth and Toller 1898).

Development to Old English

From **θánkijanǫ* ‘think’, palatalization before **j* and i-umlaut produce *þenċan* ‘think’. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note

Campbell’s assibilation discussion uses the same verb *þencan* ‘think’; the class-III relic *hycgan* is a different lexeme (Campbell 1959; Hogg 1992a).

6.70 thorn — OE *þorn*

Derivation: **θúrnaz* > *þorn* (regular).

Derivation trace

Proto input: **θúrnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**θórn*
PNWGmc U Lowering	**θórnaz*		
EAF Final Z Deletion	**θórna*		
Early Anglo-Frisian			
[no change]			

Old English form: *þorn*

Reconstruction and comparative evidence

Kroonen gives **þurna-* ‘thorn, briar’, while Orel preserves the masculine pair **þurnuz* ‘thorn’ and **þurnaz* ‘thorn’ (Kroonen 2013; Orel 2003). The derivational input **θúrnaz* ‘thorn’ belongs to that same comparative family.

Old English evidence

Bright lists *þorn* ‘thorn’ (m.) and Clark Hall likewise treats *þorn* ‘thorn’ as the ordinary noun headword (Bright 1971; Clark Hall 1960).

Development to Old English

The inherited stem shows regular lowering of *u* to *o* before *r*, and final loss yields *þorn* ‘thorn’. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note

The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form *þorn* ‘thorn’.

6.71 thought — OE *þōhte*

Derivation: **θánxtē* > *þōhte* (regular).

Derivation trace

Proto input: **θánxtē*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**θōxte*
EAF Nasalized Low Rounding	**θōxtē*		
Early Anglo-Frisian			
[no change]			

Old English form: *þōhte*

Reconstruction and comparative evidence

The derivational input is the class I weak preterite third singular of the verb ‘think’, PGmc **θánxtē* ‘thought’ (traditionally *þanhtē). Ringe gives the principal parts in full as *þankijaną, *þanhtē, *þanhtaz, with the comparative set Gothic þagkjan ‘think’, þāhta ‘thought’, ON þekkja ‘think’, þátti ‘thought’, OE *þenċan*, *þōhte*, *þōht*, OHG denken ‘think’, dāhta ‘thought’ (Ringe 2017, 281). The preterite stem is reconstructed a second time, independently of the paradigm list, among the past participles that show pre-*t devoicing: post-PIE *tong- gives PGmc *þank- in *þankijaną beside *þanhtaz ‘thought’ (Ringe 2017, 136). In both passages the Proto-Germanic form carries the nasal before the dorsal fricative. The nasal is therefore present in the derivational input and its loss is derived by the cascade instead of being built into the reconstruction.

The corpus already holds the present stem of the same verb as *think*, PGmc **θánkijaną* ‘think’. The preterite is entered as a separate row because the two stems have entirely different vowel histories, and only the preterite passes through the nasal-loss complex.

The connectorless formation

Ringe’s full principal parts establish an inherited finite word, not a CAPR assembly from a present stem and a general ending. His set contains exactly five reconstructable Proto-Germanic class-I verbs with connectorless pasts; Ringe and Taylor distinguish that inherited set from later extension (Ringe 2017, 281; Ringe and Taylor 2014, 71–75, 97–99). The selected third singular is the verb ‘thought’, not the noun.

Its deeper formation is a different question. Fulk discusses **þaŋxta* and the cluster **þaŋk-t-*, reviewing retention of a present-only suffix, loss of an earlier connector, and analogical remodelling without finding a consensus (Fulk 2018, 299–301). Those morphological illustrations are not additional complete selected words. In particular, his counterfactual buy form **bu3-dē* predicts the wrong Gothic result; it is neither an endorsed input nor evidence for thought’s finite cell (Fulk 2018, 300). Successful nasal-loss and rounding phonology does not explain the connector’s absence. The inherited reconstruction is secure within this account while that formation-level premise remains qualified.

Old English evidence

West Saxon *þōhte* ‘thought’ is the standard preterite singular of *þenċan* (Clark Hall 1960; Bosworth and Toller 1898). Fulk names this exact form as the Anglo-Frisian witness for the rounding of the lengthened nasalized low vowel: the lengthened vowels “may have remained nasalized for a considerable time, well past the close of the NWGmc. period, since *ą produced this way developed to ō in Anglo-Frisian (as in OE pret. sg. þōhte, OFris. thochte ‘thought’) and did not fall together with OE ā < ai or OFris. ā < ai, au” (Fulk 2018, § 4.1, p. 55). The Old Frisian counterpart thochte ‘thought’ confirms that the development is Anglo-Frisian and not an English innovation.

Development to Old English

The diagnostic root-vowel history consists of nasal loss and rounding. Loss of the nasal before the dorsal fricative nasalizes and lengthens the preceding vowel, so *θánxtē becomes *θãxtē; the Anglo-Frisian rounding of the nasalized low vowel then gives *θōxtē, which surfaces as *þōhte* with the fricative written *h* (Ringe 2017, 281; Fulk 2018, § 4.1, p. 55). The trace also records ordinary final-vowel shortening; that endpoint adjustment is separate from the diagnostic root-vowel history.

Three non-firings matter as much as the firings. The Ingvaeonic nasal-spirant law does not apply, correctly, because the dorsal fricative is not one of the *f, *þ, *s spirants that law governs. Preconsonantal loss of the dorsal fricative does not apply either, so the fricative survives to the surface; this is what makes the preterite a cleaner diagnostic than *fist* (Ringe 2017, 281). Finally the rounding does not touch ordinary long low vowels of other origin, as *stone* and *home* show.

Why this row is in the corpus

The nasal-loss rule has a high-vowel branch and a low-vowel branch. Before this row only the high-vowel branch had a corpus witness, in *fist* *fúnxstiz, whose input vowel is *u and which never reaches the rounding rule at all. The low-vowel pathway *aNx > *ą̃x > *ōx rested on the handbooks alone. This row supplies an attested instance of it, and with it the feeding relation between the nasal loss and the rounding, which was thereby promoted from stage entailment to an independently demonstrated ordering. See SC103 PGmcNasalLossBeforeX and SC104 EAFNasalizedLowRounding.

6.72 tide — OE *tīd*

Derivation: citation reconstruction **tī́diz*; form followed here **tī́diz* > *tīd* (regular).

Derivation trace

Proto input: **tī́diz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**tī́d*
EAF Final Z Deletion	**tī́di*		
Early Anglo-Frisian			
[no change]			

Old English form: *tīd*

Reconstruction and comparative evidence

Kroonen gives **tīdi-* ‘time’, and Orel’s **tīđiz* ‘tide’ points to the same feminine noun (Kroonen 2013; Orel 2003). The related verb *tīdan* ‘happen’ is separate.

Old English evidence

Bright records *tīd* ‘tide’ with singular *tīde* ‘tide’ and plural *tīda* ‘tides’, and Clark Hall treats *tīd* ‘tide’ as the ordinary noun ‘time, period, season’ (Bright 1971; Clark Hall 1960, 309).

Development to Old English

From **tī́diz* ‘tide’, final *z* is lost and the high final vowel drops, leaving *tīd* ‘tide’. The development is straightforward for a feminine i-stem.

Lexical note

English *tide* can also lead to the separate weak verb *tīdan* ‘happen’; the noun is *tīd* ‘tide’.

6.73 token — OE *tācn*

Derivation: **táikną* > *tācn* (regular).

Derivation trace

Proto input: **táikną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**tākn*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**tākną*		

Old English form: *tācn*

Reconstruction and comparative evidence

Kroonen cites **taikna-* and Orel **taiknan* ‘token’ for the noun ‘sign, token’ (Kroonen 2013; Orel 2003, 438). The derivational input **táikną* ‘token’ is the simple citation-form noun used for the derivation.

Old English evidence

Campbell and Sievers-Brunner preserve both unbroken *tācn* ‘token’ and broken *tācen* ‘token’, with oblique *tācnes* ‘token’ remaining unbroken (Campbell 1959; Brunner 1965).

Development to Old English

Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn* ‘token’. The unbroken form is therefore a regular Old English outcome.

Form note

tācn ‘token’ is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen* ‘token’, but that does not displace the older unbroken form (Campbell 1959; Brunner 1965).

6.74 town — OE *tūn*

Derivation: **tūną* > *tūn* (regular).

Derivation trace

Proto input: **tūną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tūn*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *tūn*

Reconstruction and comparative evidence

Kroonen cites **tūna-* ‘fenced area’, while Orel gives **tūnan* ‘town’ and **tūnaz* ‘town’ (Kroonen 2013; Orel 2003, 452). The derivational input **tūną* ‘town’ is the simple citation-form noun used in the derivation.

Old English evidence

Clark Hall records *tūn* ‘town’ as the ordinary headword ‘enclosure, yard, village, town’ (Clark Hall 1960).

Development to Old English

The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* ‘town’ regularly (Brunner 1965).

Source note

The comparative headwords vary, but the Old English target here is the direct citation form *tūn* ‘town’, not an oblique **tūnăn* ‘town’.

6.75 wade — OE *wadan*

Derivation: **wádaną* > *wadan* (regular).

Derivation trace

Proto input: **wádaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wædaną*
[no change]	OE A Restoration	**wadaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**wadan*
[no change]	OE Secondary Nasalization	**wadą̃n*
	OE Weak Tail Reduction	**wadan*

Old English form: *wadan*

Reconstruction and comparative evidence

Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* ‘wade, go through’ among the standard open-syllable examples (Campbell 1959; Ringe and Taylor 2014; Luick 1914--1940, 239).

Old English evidence

Clark Hall gives *wadan* ‘wade, go through’ as the verb ‘to go, move, stride, advance’, and Bright lists the same infinitive in the strong-verb paradigm (Clark Hall 1960; Bright 1971).

Development to Old English

From **wádaną* ‘wade’, Anglo-Frisian brightening first gives **wædaną* ‘wade’. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* ‘wade, go through’ (Campbell 1959; Ringe and Taylor 2014).

Development note

The infinitive belongs to the A-restoration class. The citation form is therefore *wadan* ‘wade’, not a fronted *wæden* ‘waded’-type output.

6.76 warp — OE *weorpan*

Derivation: **wérpaną* > *weorpan* (regular).

Derivation trace

Proto input: **wérpaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**wéorpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**wéorpan*
	OE Secondary Nasalization	**wéorpąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**wéorpan*
[no change]		

Old English form: *weorpan*

Reconstruction and comparative evidence

Ringe and Taylor distinguish preterite **warp* ‘warp’ from infinitive **werpana* ‘warp’, and the derivational input here is the verbal form **wérpaną* ‘warp’ (Ringe and Taylor 2014).

Old English evidence

Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan, wearp, wurpon, worpen* (Clark Hall 1960; Bright 1971).

Development to Old English

Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (Campbell 1959; Hogg 1992a).

Lexical note

English *warp* also points to related *wearp* material. The target is the infinitive *weorpan*.

6.77 wash — OE *wascan*

Derivation: **wáskaną* > *wascan* (regular).

Derivation trace

Proto input: **wáskaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæskaną*
[no change]	OE A Restoration	**waskaną*
	OE Heavy Syllable Nasal Apocope	**waskan*
Early Anglo-Frisian	OE Secondary Nasalization	**waskąn*
[no change]	OE Weak Tail Reduction	**waskan*

Old English form: *wascan*

Reconstruction and comparative evidence

Kroonen cites **waskan-*, Orel **waskanan* ‘wash’, and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (Kroonen 2013; Orel 2003, 489; Ringe and Taylor 2014, 142).

Old English evidence

Clark Hall heads the verb as *wascan* ‘wash’, while Sievers-Brunner also notes the variant *wæscan* ‘wash’ (Clark Hall 1960; Brunner 1965).

Development to Old English

From **wáskaną* ‘wash’, brightening gives **wæskaną* ‘wash’. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (Campbell 1959; Ringe and Taylor 2014, 142).

Form note

The conservative citation form *wascan* ‘wash’ is selected here. Spellings such as *wæscan* ‘wash’ or *wasċan* ‘wash’ belong to variant or normalized background rather than to the target of this entry.

6.78 wax — OE *weaxan*

Derivation: **wáxsaną* > *weaxan* (regular).

Derivation trace

Proto input: **wáxsaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæxsaną*
[no change]	OE Breaking	**weaxsaną*
	OE Heavy Syllable Nasal Apocope	**weaxsan*
Early Anglo-Frisian	OE Secondary Nasalization	**weaxsąn*
[no change]	OE Weak Tail Reduction	**weaxsan*
	OE Xs Merge	**weaXSan*

Old English form: *weaxan*

Reconstruction and comparative evidence

Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanan* ‘wax’, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (Kroonen 2013; Orel 2003, 478; Ringe and Taylor 2014).

Old English evidence

Clark Hall gives *weaxan* ‘wax’ as the verb headword and separately records bare *wax* ‘wax’ as a preterite form; Bright likewise treats *weaxan* ‘wax’ as the infinitive (Clark Hall 1960; Bright 1971).

Development to Old English

From **wáxsanǫ* ‘wax’, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs* > *s* belongs to forms where another consonant follows, such as *wæstm* ‘growth’, not to the infinitive itself (Campbell 1959; Brunner 1965).

Lexical note

The target here is the infinitive *weaxan* ‘wax’. Noun *weax* ‘wax’ and preterite *wax* / *wēox* ‘waxed’ belong to different lexical or paradigm slots.

6.79 way — OE *weġ*

Derivation: **wégaz* > *weġ* (regular).

Derivation trace

Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wég*
EAF Final Z Deletion	**wéga*	OE Velar Palatalization	**wéj*
Early Anglo-Frisian		OE Palatal Fricative Merger	**wéj*
[no change]			

Old English form: *weġ*

Reconstruction and comparative evidence

Kroonen cites the noun as **wega-* ‘way, road’, while the selected derivational form here is nominative-singular **wégaz* ‘way’ (Kroonen 2013). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weġ* ‘way’ and inflected *wegas* / *wegum* (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014, 341).

Old English evidence

The Old English singular is the ordinary noun *weg* ‘way’, here normalized as *weġ* ‘way’ to show the palatal final. The contrasting plural and oblique forms *wegas* ‘ways’, *wegum* ‘ways’ keep a velar stop before the following back vowel (Campbell 1959; Ringe and Taylor 2014, 341).

Development to Old English

From **wégaz* ‘way’, final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *weġ* ‘way’, whereas *wegas* ‘ways’ remains velar because the following *a* blocks the same outcome (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014, 341).

Form note

Normalized *weġ* ‘way’ and dictionary *weg* ‘way’ represent the same noun. No Old English form *wē* ‘we’ is supported by the evidence for ‘way’.

6.80 weapon — OE *wǣpn*

Derivation: **wépnǫ* > *wǣpn* (regular).

Derivation trace

Proto input: **wépnǫ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**wǣpn*
PNWGmc Long E Lowering	**wāpnǫ*		
EAF Long A Fronting	**wǣpnǫ*		
Early Anglo-Frisian			
[no change]			

Old English form: *wǣpn*

Reconstruction and comparative evidence

Kroonen reconstructs a double-stem noun **wēbna-* ~ **wēpna-* and cites OE *wæpn* ‘weapon’ among its reflexes (Kroonen 2013, 617). The derivational input **wépnǫ* ‘weapon’ represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence

Campbell’s cluster-noun discussion preserves unbroken *wépn* ‘weapon’ beside broken *wépen* ‘weapon’-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen* ‘weapon’ / *wapen* ‘weapon’ with unbroken oblique *wǣpnes* ‘weapon’, while Clark Hall lemmatizes the noun under *wapen* ‘weapon’ and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

Development to Old English

Northwest Germanic lowering gives *wǣpn* ‘weapon’, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The Old English form here is the attested unbroken form *wǣpn* ‘weapon’.

Form note

The ordinary late West Saxon simplex headword is *wǣpen* ‘weapon’, but Campbell’s noun-class discussion also preserves unbroken *wépn* ‘weapon’ beside broken *wépen* ‘weapon’-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* ‘weapon’ remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

6.81 wether — OE *weþer*

Derivation: **wéθruz* > *weþer* (regular).

Derivation trace

Proto input: **wéθruz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**wéθr*
EAF Final Z Deletion	**wéθru*	OE Epenthetic Vowel	**wéθer*
Early Anglo-Frisian			
[no change]			

Old English form: *weþer*

Comparative reconstruction and development

Old English *weþer* ‘wether, ram’ belongs with Gothic *wiþrus* ‘lamb, young ram’, Old Norse *veðr* ‘ram’, Old Saxon *withar* ‘wether’ and Old High German *widar* ‘ram’. These are masculine animal names, not the neuter weather noun, despite the similar spellings. Kroonen reconstructs **weþru-*; Kluge–Seebold likewise gives a masculine Germanic **weþru-*, while allowing a similar stem formation (Kroonen 2013, 584; Kluge 2011, 985).

The inherited root has short **e*. Gothic *i* does not require Proto-Germanic **i*: inherited short **e* and **i* merged as *i* in Gothic outside its breaking environments (Fulk 2018, 59). The continental *i*-forms require equal care. Raising before a following *u* is independently documented in Old Saxon and Old High German, but it is not universal there; that general development is not a complete, independently dated history of every cell of this noun (Fulk 2018, 69). The comparison consequently supports inherited **e* without requiring every daughter-language vowel to preserve it unchanged.

The principal formation question concerns the inflection, not the root vowel or masculine gender. Orel records both **weþruz* and **weþraz*, respectively masculine *u*- and *a*-stem alternatives, and notes traces of *u*-inflection in Norse (Orel 2003, 459–60). Gothic nominative *wiþrus* supplies particularly useful positive evidence for the older *u*-stem. CAPR therefore adopts the masculine nominative singular **weþruz*, represented as **wéθruz*, while retaining **weþraz* as a legitimate alternative. The latter may represent early inflectional remodelling; the dictionaries do not establish its precise date.

The Old English masculine paradigm, including genitive *weþeres*, does not decide the original Germanic stem class. Bosworth–Toller records *weþer, es; m.* and the senses ‘a wether, a ram’ (Bosworth and Toller 1898, 1212). Masculine *u*-stems underwent reclassification towards other declensions in West Germanic, and heavy-stem nominatives lost their final short **u* (Fulk 2018, 163). In the selected nominative, loss of final **-z* and **-u* leaves the consonantal ending resolved as unstressed *-er* in *weþer*, consistent with the vocalization of final postconsonantal sonorants (Fulk 2018, 91–92). The root vowel is already **e*; no early lowering of **i* is needed for this derivation.

The further etymology is less secure. Orel connects the noun with the Indo-European word for ‘year’, comparing Hittite, Greek and Albanian year-words and names for young cattle (Orel 2003, 459–60). Kroonen makes that connection conditional: if correct, the original designation was ‘yearling, one-year-old animal’, with Sanskrit *vatsá-* ‘calf’ among the comparanda (Kroonen 2013, 584). Kluge–Seebold notes phonological difficulties with the traditional comparison involving Latin *vitulus* ‘calf’, and also considers derivation from a root meaning ‘sheep’ (Kluge 2011, 985–86). A young-animal designation could develop into ‘ram’ or ‘wether’, but this attractive semantic account is not as secure as the Germanic masculine *u*-stem reconstruction.

6.82 where — OE *hwǣr*

Derivation: **xwár* > *hwǣr* (regular).

Derivation trace

Proto input: **xwár*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Monosyllabic Final R Lengthening	**xwār*	
EAF Long A Fronting	**xwǣr*	
Early Anglo-Frisian		
[no change]		

Old English form: *hwǣr*

Old English evidence

Clark Hall records *hwǣr* ‘where’, the West Saxon comparison form used here (Clark Hall 1960, 170). Hill sets it beside Anglian *hwēr* ‘where’ (Hill 2017, 142–43). The contrast concerns the quality of a long vowel, not lengthening in West Saxon alone. The macrons belong to normalized scholarly forms; they do not imply that every manuscript token explicitly marks quantity.

Reconstruction and comparative evidence

Kroonen gives **hwar* ‘where’, an interrogative formation with locative **-r*. He explicitly derives the long-vowel West Germanic forms by monosyllabic lengthening, rejecting inherited **hwēr* as their source (Kroonen 2013, 263). Orel instead preserves both **xwar* ‘where’ and **xwēr* as citation alternatives, connecting the formation with pronominal **xwa-* (Orel 2003, 198). The selected **xwár* ‘where’ follows the short-vowel line. Its acute marks the accented root, while **xw-* represents the Germanic dorsal-plus-glide sequence corresponding to the dictionaries’ **hw-* or **xw-*.

The extended interrogative supplies an important morphological comparison. Gothic *ƕarjis* ‘which’ and Old Norse *hverr* ‘who, which’ support an inherited disyllabic **hwarja-* with short root vowel. Hill argues that such a formation could precede lengthening of the monosyllabic adverb; its short vowel therefore need not date lengthening to West Germanic. Gothic adverbial *ƕar* ‘where’ itself leaves quantity ambiguous in writing (Hill 2017, 142, 147–48).

Development to Old English

From short **hwar*, SC112, stressed monosyllabic lengthening before inherited final *r* gives **hwār*; then SC101, long-*ā* fronting gives **hwǣr*. The initial consonant sequence is represented orthographically as *hw-*. The derivational input with short **a* thus reaches *hwǣr* through lengthening followed by fronting, without inflectional retargeting. The parallel *þǣr* ‘there’ makes the same vowel development visible in a different pronominal formation.

The historical interpretation remains narrower than a uniquely dated reconstruction. Hill’s Proto-Germanic account depends in part on his explanation of the Scandinavian short-vowel forms, while Ringe and Taylor retain a later West Germanic lengthening (Hill 2017, 146–48; Ringe and Taylor 2014, 65). The early placement adopted here preserves inherited-*r* conditioning before rhotacism could obscure consonant ancestry. That analytical choice does not establish an independently dated ordering against SC015 or the lowering of inherited **ē₁* in SC024.

6.83 who — OE *hwā*

Derivation: **xwáz* > *hwā* (regular).

Derivation trace

Proto input: **xwáz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
Monosyllabic Final Z Loss	**xwā*	
Early Anglo-Frisian		
[no change]		

Old English form: *hwā*

Reconstruction and comparative evidence

The derivational input is the nominative singular masculine of the interrogative pronoun, PGmc **xwáz* ‘who’ (traditionally *hwaz; Gothic ƕas ‘who’). Ringe and Taylor cite exactly this form among the witnesses for the northern West Germanic loss of word-final *-z in stressed monosyllables: “*hwaz ‘who’ > OE hwā” (Ringe and Taylor 2014, 86). The nominative is the natural citation cell of the interrogative and the direct ancestor of the Old English headword form, so no paradigm-cell retargeting is involved.

Old English evidence

West Saxon *hwā* ‘who’ is the standard headword form (Campbell 1959, § 125, p. 49). The vowel is decisive for the derivation: Campbell states that the back vowel of *hwā* never underwent Anglo-Frisian brightening — “*hwǣ does not exist” — and Sievers-Brunner records the same restriction (Campbell 1959, § 125, p. 49; Brunner 1965, § 137 Anm. 1, p. 129).

Development to Old English

The derivation is two historical steps. First, the later northern West Germanic loss of word-final *-z in stressed monosyllables removed the sibilant with compensatory lengthening of the short nucleus: **xwáz* ‘who’ > **hwā* ‘who’ (Ringe and Taylor 2014, 86). Old High German instead rhotacized the sibilant (cf. OHG wer ‘who’ type inflection), which is why this loss counts among the diagnostic Ingvaeonic features (Fulk 2018, 18). Second, the resulting long back *ā stayed back: Anglo-Frisian brightening of long final *ā requires a preceding nucleus in the word, and in this monosyllable there is none, so the form surfaces directly as OE *hwā* (Campbell 1959, § 125, p. 49).

An alternative account derives the endingless pronoun from a generalized unaccented sentence variant *hwa with later lengthening (Campbell 1959, 166; Brunner 1965, § 182, p. 160). Ringe and Taylor reject the unaccented-variant analysis for this witness set because *mā* ‘more’ and *cū* ‘cow’ are not plausibly unaccented words (Ringe and Taylor 2014, 86). The project follows Ringe and Taylor; the alternative is recorded, not adopted.

Why this row is in the corpus

This row makes SC097, the monosyllabic final *-z loss, corpus-witnessed: before its addition the change was validated only by synthetic controls. It also witnesses the ordering SC097 before SC003 rhotacism — a rhotacized* †hwar *could never yield* hwā* — and exercises the narrowed long-final clause of Anglo-Frisian brightening. See `docs/sound_changes/audits/corpus-maturation-01-candidate-adjudication.md` §1.

6.84 will — OE *willa*

Derivation: **wéljô* > *willa* (regular).

Derivation trace

Proto input: **wéljô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**willjô*
PWGmc J Gemination	**wéllj̄ô*	OE Unstressed Long Vowel Shortening	**willja*
Early Anglo-Frisian		OE J Loss After Heavy	**willa*
[no change]			

Old English form: *willa*

Reconstruction and comparative evidence

Kroonen separates noun **weljan* ‘will, wish’ from verb **weljan* ‘to want’, while Orel and Kluge represent the noun as **weljōn* ‘will’ (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form **wéljô* ‘will’ is the noun-side input used for this row.

Old English evidence

Clark Hall lemmatizes noun *willa m.* separately from verb *willan* ‘will, want’ (Clark Hall 1960, 368). The Old English form here is the noun citation form, not the related verb.

Development to Old English

From **wéljô* ‘will’, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa* ‘will, desire’. The noun is therefore a regular weak masculine outcome.

Lexical note

The target here is the noun *willa* ‘will, wish’. Related verb *willan* ‘will, want’ belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

6.85 wind — OE *windan*

Derivation: **wíndaną* > *windan* (regular).

Derivation trace

Proto input: **wíndaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**wíndan*
[no change]	OE Secondary Nasalization	**wíndą̄n*
Early Anglo-Frisian	OE Weak Tail Reduction	**wíndan*
[no change]		

Old English form: *windan*

Reconstruction and comparative evidence

Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (Kroonen 2013). Fulk and Ringe and Taylor derive the dental directly from PIE **wendh-*, not from a Verner alternant (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller record *windan* ‘wind, coil’ as the verb headword (Clark Hall 1960; Bosworth and Toller 1898, 101). The Old English form here is the ordinary infinitive of the strong verb.

Development to Old English

The derivational input **wíndanǫ* ‘wind’ yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note

English *wind* also names the noun. This row represents the class-III verb, not the noun (Kroonen 2013; Clark Hall 1960).

6.86 wold — OE *weald*

Derivation: **wálθuz* > *weald* (regular).

Derivation trace

Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**wældu*
EAFL Th Voicing	**wálduz*	OE Breaking	**wealdu*
EAF Final Z Deletion	**wáldu*	OE High Vowel Apocope	**weald*
Early Anglo-Frisian			
[no change]			

Old English form: *weald*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **walþu-* and gives OE *weald* ‘forest, woodland’ beside other West Germanic *wald* ‘forest’ forms (Kroonen 2013). The derivational input **wálθuz* ‘wold’ is the nominative singular used for the derivation.

Old English evidence

Clark Hall makes *weald* ‘forest, woodland’ the main noun headword and cross-refers *wald* ‘forest’ and *wold* ‘plain, open country’ to it (Clark Hall 1960). Clark Hall glosses *wald* as *(N,VPs)* — Northumbrian and Vespasian Psalter (Mercian) — making this an explicitly dialect-labelled Anglian form.

Development to Old English

**lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald* ‘forest, woodland’. The noun is therefore a regular breaking outcome.

Dialect note

wald ‘forest’ survives as an Anglian variant in the same family (Clark Hall *wald (N,VPs)*: Northumbrian and Vespasian Psalter). The Old English form here is normalized *weald* ‘forest, woodland’, not the variant form (Clark Hall 1960; Ringe and Taylor 2014).

6.87 yarn — OE *ġearn*

Derivation: **gárną* > *ġearn* (regular).

Derivation trace

Proto input: **gárną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gærną*
[no change]	OE Breaking	**gearną*
	OE Heavy Syllable Nasal Apocope	**gearn*
Early Anglo-Frisian	OE Velar Palatalization	**ǰearn*
[no change]	OE Palatal Fricative Merger	**jearn*

Old English form: *ġearn*

Reconstruction and comparative evidence

Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* ‘yarn’ > **geern* ‘yarn’ > **gearn* ‘yarn’ > OE *gearn* ‘yarn’ (Kroonen 2013; Ringe and Taylor 2014, 220). The derivational input **gárną* ‘yarn’ is the nominal citation form used here, while oblique **garnăn* ‘yarn’ belongs only to comparative background.

Old English evidence

Clark Hall records *gearn (e) n.* ‘yarn, spun wool’, and Bosworth-Toller glosses *gearn* ‘yarn’ with Latin *filatum* ‘spun thread’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **gárną* ‘yarn’, brightening and breaking before *rn* yield *gearn* ‘yarn’; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn* ‘yarn’. The derivation is regular.

Form note

Dictionary *gearn* ‘yarn’ and normalized *ġearn* ‘yarn’ refer to the same noun. The comparative stem **garna-* and oblique **garnăn* ‘yarn’ do not replace the derivational input **gárną* ‘yarn’.

6.88 you — OE *ēow*

Derivation: **ízwiz* > *ēow* (regular).

Derivation trace

Proto input: **ízwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Diphthong Leveling	**ēow*
PWGmc Coronal W Assimilation	**íwwiz*		
OE WW Simplification	**íuwiz*		
EAF Final Z Deletion	**íuwi*		
PWGmc Unstressed Word Final I Apocope	**íuw*		
Early Anglo-Frisian			
[no change]			

Old English form: *ēow*

Reconstruction and comparative evidence

The derivational input is the dative(-accusative) plural of the second-person plural pronoun, PGmc **ízwiz* ‘you (dat. pl.)’ (Gothic izwis ‘you’). Ringe and Taylor print the development in full: “PGmc *izwiz ‘you (dat. pl.)’ (Goth. izwis) > *iwwi > PWGmc *iuwi ~ *iuw (see 3.1.4) > OE īow, OF iū, OS, OHG iu” (Ringe and Taylor 2014, 41–42), following Stiles’s demonstration of the coronal-w assimilation (Stiles 1985, 89–94). Fulk’s discussion of inherited geminate glides after short vowels supplies the wider reanalysis context, not a replacement citation for that assimilation (Fulk 2018, 117). The oblique plural is the cell that survives as the Old English pronoun *ēow*.

Competing origins of the inherited oblique

Kroonen derives **izwiz* by metathesis from **iw-siz*, with a suffix abstracted from the first-person plural, and separately distinguishes older **iwiz* and the dual formations (Kroonen 2013, 275–76). Fulk regards the reported Krahe–Meid reduplication **wiz-wiz* and initial-*w* dissimilation as probably most plausible, while explicitly stating that there is no consensus (Fulk 2018, 187–88). This is his assessment of a reported theory, not a newly verified original argument or a named rebuttal of Kroonen.

Ringe’s Katz-based account instead combines case formation and sandhi history from Indo-European obliques including **uswé*, with later prothetic *i*; he also records Weiss’s dissimilatory alternative (Ringe 2017, 233–35). The competing proposals concern the genealogy and segmentation of the same relevant plural oblique. Agreement on a later inherited input does not choose between metathesis and suffix abstraction, reduplication and dissimilation, or sandhi and case formation. The regular English path and its independently selected weak-final context cannot settle that earlier morphology.

Old English evidence

West Saxon *ēow* ‘you’ beside early West Saxon and Northumbrian *īow* ‘you’ shows the normal West Saxon treatment of the diphthong (Campbell 1959, § 702, p. 283). The corpus targets the West Saxon citation form.

Development to Old English

Coronal-w assimilation first supplies the geminate in **ízwiz* > **íwwiz*. Early inherited-glide reanalysis then gives an **iu* nucleus with retained consonantal **w*, rather than an already completed English long diphthong. Final-z loss exposes the last high vowel, and weak-final apocope selects the apocopated member of Ringe and Taylor’s PWGmc **iuwi* / **iuw* doublet. The later English realization gives *ēow* without an i-mutation trigger (Ringe and Taylor 2014, 41–42, 44–45, 57–58, 171–75).

The selected sentence context is weak and phonologically final at early apocope. It is recorded independently of the unchanged PGmc reconstruction and independently of the root accent: the acute is not a sentence-stress mark. Under a retained-i context the present model produces predicted †*īei*, because w-loss before i intervenes before mutation. That is a computational negative control, not an attested strong Old English spelling. Proclitic *ymbe* ‘around’, weak-final *and* ‘and’, and stressed *ġiest* ‘guest’ / *fȳr* ‘fire’ test the boundary and stress contrasts; no pronoun-specific or lexeme-ID sound-law condition is used (Ringe and Taylor 2014, 55, 57–58).

Why this row is in the corpus

The word supplies the assimilation input to SC031 OEWWSimplification and the selected weak-final context for SC098 PWGmcUnstressedWordFinalIApocope. It demonstrates loss of the mutation trigger, not a late English vocalization-before-degemination order. That older interpretation belonged to the superseded ww proxy. The independently justified assimilation feeder and the prosodic condition now remain separate (Ringe and Taylor 2014, 41–42, 55, 57–58).

Chapter 7

Attested variants and comparison forms

Here the Old English comparator belongs to a documented set of variants. The variation is part of the evidence and not an inconvenience to be normalized away.

7.1 cud — OE *cwedu*

Derivation: citation reconstruction **kwíθuz*; form followed here **kwéðuz* > *cwedu* (attested variant).

Derivation trace

Proto input: **kwéðuz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PWGmc Dental Hardening	**kwéduz*	
EAF Final Z Deletion	**kwédu*	
Early Anglo-Frisian		
[no change]		

Old English form: *cwedu*

Reconstruction and comparative evidence

Kroonen gives the resin word under homonym 2 of **kwedu-* ‘cud’ and supplies Old English variants *cwidu* ‘cud’, *cweodu* ‘cud’, and *c(w)udu* ‘cud’ (Kroonen 2013, 315). The homonym number is not part of the reconstructed phonological stem. Orel also lists *cwidu*, but his feminine headword is a different formation, not another quotation of Kroonen’s masculine u-stem (Orel 2003, 227). The derivational input **kwéðuz* ‘cud’ supplies an e-vowel and voiced dental for the chosen comparison *cwedu* ‘cud’. It is a CAPR whole-word selection, not a diplomatic quotation of either dictionary’s complete form. Its relationship to the different citation reconstruction above remains under comparative review.

Old English evidence

The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss *cwidu* ‘cud’ > *cwudu* ‘cud’ > *cudu* ‘cud’ and also note late West Saxon *cweodu* ‘cud’. Their starting reconstruction is explicitly West Germanic with i, not an explicitly

printed Proto-Germanic form in this passage (Ringe and Taylor 2014, 323). Clark Hall's headword *cwudu* 'cud' admits eo and i variants, but does not itself attest the selected bare-e spelling (Clark Hall 1960, 69). The claim that CAPR's comparison is the conservative attested variant needs further source verification; these passages do not establish it.

Development to Old English

The current modeled chain takes **kwéðuz* 'cud' to *cwedu* 'cud', with the voiced dental hardened along the way. This records the executable result, not independent proof that the derivational input is the best source-backed history. Ringe and Taylor support back umlaut and further reduction within the i-starting variant family; their passage does not establish the older account's claim that the i-form is leveled from the bare-e comparison (Ringe and Taylor 2014, 323). Corpus fields and classification remain unchanged pending the complete source audit.

Variant comparison

Variant type	Old English form	Comment
selected comparison	*cwedu*	current target; its attestation and conservative status require verification
i-grade comparison	*cwidu*	source-backed variant; leveling from the Old English form here is not established by the cited passage
back-umlauted forms	*cweodu*, *cwudu*	later developments within the same OE tradition
reduced form	*cudu*	further reduced member of the same variant set

7.2 ten — OE *tēon*

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace

Proto input: **téxun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**téoxun*
[no change]	OE Med Unstressed U Lowering	**téoxon*
	OE H Loss	**téoon*
Early Anglo-Frisian	OE Contraction	**téon*
[no change]		

Old English form: *tēon*

Reconstruction and comparative evidence

**tēon* 'ten' is the bare cardinal form. Fulk states that Old English *tien* 'ten' shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *tēon* 'ten' (Fulk 2018, § 10.2). Brunner gives the same contrast more broadly: **tēon* 'ten' develops from **tëhun* 'ten', while West Saxon *tien* 'ten', *tȳn* 'ten' belong to a different, umlauted branch of the numeral history (Brunner 1965, §§ 129.2, 129 Anm. 6, 234).

The comparison form *tēon* 'ten' therefore represents the bare cardinal's un-umlauted line, not the later umlauted simplex tradition.

Old English evidence

The attested simplex forms are varied. Campbell gives *tien* ‘ten’, north-western West Saxon *tēn* ‘ten’, and late Northumbrian *tēo* ‘ten’, *tēa* ‘ten’ (Campbell 1959, § 682). Brunner likewise lists West Saxon *tien* ‘ten’, *tȳn* ‘ten’ beside *tēn* ‘ten’, *tēo* ‘ten’, *tēa* ‘ten’ in other dialects (Brunner 1965, § 325).

Exact simplex *tēon* ‘ten’ is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* ‘tenth’ and *tēontig* ‘ten’ (Brunner 1965, § 129.2; Fulk 2018, § 10.2). The comparison form *tēon* ‘ten’ is therefore a normalized spelling of that un-umlauted base.

Development to Old English

From **téxun* ‘ten’, lowering of medial unstressed *u* gives **téxon* ‘ten’, breaking gives **téoxon* ‘ten’, loss of intervocalic *h*/*x* yields **téoon* ‘ten’, and contraction produces **tḗon* ‘ten’, written *tēon* ‘ten’. This is the regular bare-cardinal path.

The umlauted forms *tien* / *tīen* ‘ten’ belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (Fulk 2018, § 10.2; Brunner 1965, § 129 Anm. 6).

Variant comparison

The comparison below sets the relevant forms side by side. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
tēon	normalized un-umlauted comparison form; trace-supported	Old English form here
tien / *tīen*	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here
tēn / *tēo* / *tēa*	attested un-umlauted simplex variants in other dialects	support the same branch as the comparison form

7.3 three — OE *þrīe*

Derivation: **θréjez* > *þrīe* (attested variant).

Derivation trace

Proto input: **θréjez*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**θrije*
EAF Final Z Deletion	**θréje*	OE Intervocalic J Vocalization	**θriie*
Early Anglo-Frisian		OE Contraction	**θrīe*
[no change]			

Old English form: *þrīe*

Reconstruction and comparative evidence

Kroonen cites the numeral under a broader stem-style reconstruction rather than under one Old English-ready paradigm cell (Kroonen 2013, 586). The input **θréjez* ‘three’ is therefore best understood as the inherited masculine nominative-accusative singular.

The Old English numeral has no uniform citation form across the paradigm. The masculine singular line must be kept apart from feminine-neuter *þrēo* ‘three’ and from later reduced spellings of the masculine form.

Old English evidence

Campbell gives masculine nominative-accusative *þrīe* ‘three’, feminine and neuter nominative-accusative *þrēo* ‘three’, genitive *þrēora* ‘three’, and dative *þrim* ‘three’, adding that late West Saxon has *þry* ‘three’ / *þri* ‘three’ for masculine nominative-accusative beside *þrīe* ‘three’ (Campbell 1959, § 683). Fulk presents the same masculine *þrīe* ‘three’ beside the wider numeral paradigm (Fulk 2018, § 10.1).

The target is therefore an attested Old English paradigm cell. *þrī* ‘three’ / *þry* ‘three’ belongs to later reduction or headword-style citation, whereas *þrīe* ‘three’ is the conservative masculine nominative-accusative form.

Development to Old English

From **θréjez* ‘three’, loss of final *-z* leaves **θréje* ‘three’ as an intermediate form. The following **θrije* ‘three’ fronts the stem vowel, then vocalizes between vowels, and contraction yields **þrīe* ‘three’. The compact trace records the same sequence as θrije > θriie > *þrīe*.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the selected masculine cell from the later reduced form and from the rest of the numeral paradigm.

Form	Status	Relevance to this entry
þrīe	attested masculine nom./acc.; regular output	Old English form here
þrī / *þry*	later reduced masculine variant	genuine OE variant, but not the conservative comparison form
þrēo	attested feminine-neuter nom./acc.	same numeral, different paradigm cell
þrēora, *þrim*	attested genitive and dative forms	confirm the wider paradigm, not the cell compared here

7.4 wasp — OE *wæfs*

Derivation: **wábsaz* > *wæfs* (attested variant).

Derivation trace

Proto input: **wábsaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wábs*
EAF Final Z Deletion	**wábsa*	EAF Brightening	**wæbs*
Early Anglo-Frisian		PGmc B Allophony	**wæβs*
[no change]			

Old English form: *wæfs*

Reconstruction and comparative evidence

The Proto-Germanic form **wábsaz* ‘wasp’ reaches Old English without any special change of stem or paradigm cell. The question in this entry is instead which attested Old English member of the variant set should serve as the comparison form.

Fulk presents the Old English forms together as *wæfs* ‘wasp’ with variants *wæsp* ‘wasp’ and *wæps* ‘wasp’ (Fulk 2018, § 6.5). Bülbring and Brunner then make the chronology more explicit by deriving later *wæps* and late West Saxon *wasp* ‘wasp’ from earlier *waefs* / *wæfs* through restricted metatheses (Bülbring 1902, § 484 Anm. 3; Brunner 1965, §§ 193, 204).

Old English evidence

The earliest directly cited Old English form is *wæfs* ‘wasp’, written *waefs* ‘wasp’ in the Épinal-Corpus material discussed by Bülbring and Brunner (Bülbring 1902, § 484 Anm. 3; Brunner 1965, § 193). Later Old English also shows *wæps* ‘wasp’ and *wæsp* ‘wasp’ / *wasp* ‘wasp’, and dictionary practice often favors *wæps* or later spellings as headwords (Clark Hall 1960, 341).

This entry therefore distinguishes chronological priority from headword habit. *wæfs* ‘wasp’ is not a convenient reconstruction: it is an attested Old English form and also the one that matches the regular development most closely.

Development to Old English

From **wábsaz* ‘wasp’, the regular Old English path passes through loss of final *z*, Anglo-Frisian fronting, and the allophonic development of *b* to a fricative before *s*, yielding *wæfs* ‘wasp’.

The later forms *wæps* ‘wasp’ and *wæsp* ‘wasp’ / *wasp* ‘wasp’ belong to subsequent, lexically restricted metatheses. They are genuine Old English forms, but they are later within the variant history.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the earliest attested and regular form from the later metathesized doublets.

Form	Status	Relevance to this entry
wæfs	earliest attested OE form; regular output	Old English form here
wæps	later attested metathesized variant	genuine OE doublet, but secondary
wæsp / *wasp*	later West Saxon metathesized variant	genuine OE doublet, but not the form compared here

Chapter 8

Early analogy and pre-Old-English input selection

Here an analogical change already separates the transducer input from the lexeme's citation reconstruction before the specifically Old English changes apply. Each entry must therefore justify the remodeled input independently of the successful output.

8.1 bottom — OE *botm*

Derivation: citation reconstruction **búdmaz*; form followed here **búttmaz* > *botm* (early analogy).

Derivation trace

Proto input: **búttmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**bóttm*
PNWGmc U Lowering	**bóttmaz*	OE Preconsonantal Degemination	**bótm*
EAF Final Z Deletion	**bóttma*		
Early Anglo-Frisian			
[no change]			

Old English form: *botm*

Reconstruction and comparative evidence

Kroonen reconstructs the word as a stem complex **budmō* 'bottom' with genitive **buttaz* 'bottom', summarized as **budman-* ~ **buttman-*, and gives Old English *botm* 'bottom' as the reflex (Kroonen 2013, 120). The comparative label **búdmaz* 'bottom' names the lexeme-level stem complex, while the derivational input **búttmaz* 'bottom' represents the pre-Old-English form with oblique **butt-* generalized into the nominative formation.

Orel likewise preserves both sides of the comparison under **budmaz* 'bottom' and **butmaz* 'bottom' (Orel 2003, 100). The derivational input is thus a historical stem choice, not an arbitrary respelling.

Old English evidence

The Old English noun itself is secure. Clark Hall gives *botm* 'bottom' (Clark Hall 1960, 63). Bosworth-Toller cross-references *bodan* 'bottom, floor' to *botm* 'bottom', showing the wider

reflex family without weakening the attested lemma (Bosworth and Toller 1898, 112).

Development to Old English

Once the oblique **butt-* stem has been generalized, the derivational input **búttmaz* ‘bottom’ develops regularly to *botm* ‘bottom’. The analogical step is therefore early: it belongs to pre-Old-English stem formation rather than to a later choice among Old English paradigm cells.

8.2 brand — OE *brandes*

Derivation: citation reconstruction **brándaz*; form followed here **brándas* > *brandes* (early analogy).

Derivation trace

Proto input: **brándas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**brándæs*
[no change]	OE Unstressed AE Merger	**brándes*
Early Anglo-Frisian		
[no change]		

Old English form: *brandes*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **brándaz* ‘brand’, continued by Old English *brand* ‘brand’ and its continental cognates (Orel 2003, 53). The selected input **brándas* ‘brand’ is not a different lexeme but the genitive singular of that same a-stem noun.

Both forms belong to the same root and stem class but occupy different inherited inflectional cells. The derivational input is the oblique cell.

Old English evidence

Old English dictionaries lemmatize the noun as *brand* ‘brand’ (Clark Hall 1960, 49; Bosworth and Toller 1898, 116). Bosworth-Toller also records inflectional forms such as *brandas* ‘brand’, *branda* ‘brand’, and *brandum* ‘brand’ under the same entry (Bosworth and Toller 1898, 116).

The specific comparison form in this entry, *brandes* ‘brand’, is the expected genitive singular of that a-stem noun. It is therefore an inferred Old English paradigm form rather than the ordinary dictionary headword.

Development to Old English

From **brándas* ‘brand’, the regular Old English development passes through the usual unstressed-vowel weakening of the inflectional ending, yielding *brandes* ‘brand’. Nothing in the stem itself requires a special repair. The root consonants and the stressed vowel are the same as in the citation lemma *brand* ‘brand’.

The analytical weight of the entry lies in the ending. By choosing the oblique singular rather than the nominative citation form, the entry presents the same lexeme in a different inherited cell.

Form comparison

The comparison below sets the relevant forms side by side. It separates the citation lemma from the selected oblique singular.

Form / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**brándaz*	expected OE lemma *brand*	*brand*	regular headword-level outcome
genitive singular	**brándas*	regular output: *brandes*	*brandes*	exact match for the oblique cell

The noun itself is straightforwardly inherited. The main point of the entry is that *brandes* ‘brand (gen.)’ belongs to the same regular a-stem paradigm as *brand* ‘brand’, even though the citation lemma remains the nominative singular.

8.3 breast — OE *brēost*

Derivation: citation reconstruction **brústz*; form followed here **bréustą* > *brēost* (early analogy).

Derivation trace

Proto input: **bréustą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**brēostą*
[no change]	OE Heavy Syllable Nasal Apocope	**brēost*
Early Anglo-Frisian		
[no change]		

Old English form: *brēost*

Reconstruction and comparative evidence

The word family shows two related but distinct Proto-Germanic formations. The root noun **brust-* lies behind forms such as Gothic *brusts* ‘breast’, whereas Old English *brēost* ‘breast’ belongs to a thematic formation **breusta-*, alongside Old Norse *brjóst* ‘breast’ and Old Saxon *briost* ‘breast’ (Kroonen 2013, 114; Orel 2003, 95; Ringe and Taylor 2014, 43).

The derivational input **bréustą* ‘breast’ therefore differs from the citation label **brústz* ‘breast’ because Old English reflects the thematic branch rather than the root noun. The morphological choice comes before the Old English sound changes themselves.

Old English evidence

Clark Hall records the noun as *brēost* ‘breast’ / *breóst* ‘breast’ (Clark Hall 1960, 65). The form is an established Old English lexeme, not a reconstructed target assembled from comparative evidence alone.

What requires explanation is not the Old English attestation but the relation between that attested noun and the broader Germanic word family. The relevant comparison form is therefore the thematic Old English noun *brēost* ‘breast’.

Development to Old English

From **bréustą* ‘breast’, the regular Old English development gives *brēost* ‘breast’, with the expected *eu* > *ēo* vowel history (Campbell 1959, § 115). No special repair is needed once the correct thematic formation is chosen.

The earlier mismatch arose only if the word was forced into the root-noun line. The Old English noun itself continues the thematic branch cleanly and directly.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader root-noun family label from the thematic formation actually continued in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader root-noun family	**brústz*	root-noun type outcomes outside OE	non-OE comparanda	useful family label, but not the direct source of *brēost*
selected thematic formation	**bréustą*	regular output: *brēost*	*brēost*	exact match between formation and attested OE noun

The relevant point is the formation split. *brēost* ‘breast’ is the regular Old English outcome of the thematic **breusta-* branch, not of the root noun **brust-*.

8.4 craft — OE *crœft*

Derivation: citation reconstruction **kráftiz*; form followed here **kráftaz* > *crœft* (early analogy).

Derivation trace

Proto input: **kráftaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kráft*
EAF Final Z Deletion	**kráfta*	EAF Brightening	**krœft*
Early Anglo-Frisian			
[no change]			

Old English form: *crœft*

Reconstruction and comparative evidence

Comparative sources disagree about the older stem class. Kroonen gives a u-stem **kraftu-*, while Orel prints **kraftiz* ‘craft’ and **kraftuz* ‘craft’ (Kroonen 2013, 340; Orel 2003, 259). The comparative label **kráftiz* ‘craft’ remains in view as a lexeme-level shorthand, while **kráftaz* ‘craft’ is the pre-Old-English form used for the Old English derivation.

Old English evidence

The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *crœft* ‘craft’ as the headword (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

Development to Old English

The comparison is between possible pre-Old-English inputs. The i-stem comparator **kráftiz* ‘craft’ gives *creft*, while the u-stem comparator **kráftuz* ‘craft’ gives *craft*. The a-stem-shaped input **kráftaz* ‘craft’ yields *crœft* ‘craft’ and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of **-iz* ‘craft’ and without the back-vowel outcome associated with the u-stem comparator.

Form comparison

Candidate input	OE output	Result
*kráftiz	*creft*	non-match; i-stem comparator
*kráftuz	*craft*	non-match; u-stem comparator
*kráftaz	*crœft*	exact match; selected pre-OE input

8.5 dill — OE *dile*

Derivation: citation reconstruction **déljaz*; form followed here **déliz* > *dile* (early analogy).

Derivation trace

Proto input: **déliz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**dili*
EAF Final Z Deletion	**déli*	OE Med Unstressed I Lowering1	**dile*
Early Anglo-Frisian			
[no change]			

Old English form: *dile*

Reconstruction and comparative evidence

Comparative evidence preserves both an i-stem and a ja-stem formation, with Old English *dile* ‘dill’ on one side and continental forms such as Old Saxon *dilli* ‘dill’ and Old High German *tilli* ‘dill’ on the other (Fulk 2018, 170). The derivational input **déliz* ‘dill’ therefore represents the i-stem side of the paradigm, whereas the citation label **déljaz* ‘dill’ is a broader comparative headword.

The stem class determines the Old English consonant shape. A ja-stem with **-lj-* would be expected to produce gemination, but the Old English noun shows a single *l*. Fulk’s discussion of ja-stems transferred to the i-stems provides the relevant morphological background for the OE side (Fulk 2018, 170).

Old English evidence

Old English dictionaries record the plant name as *dile* ‘dill’, alongside the variant *dili* ‘dill’ (Bosworth and Toller 1898, 164; Clark Hall 1960, 95). The form discussed here is therefore an attested Old English noun with single *l*.

The Old English evidence is the relevant point. Whatever broader comparative headword is chosen for the family, the inherited form reflected in OE is the i-stem type *dile* ‘dill’, not a geminated *dill* ‘dill’ outcome.

Development to Old English

From **déliz* ‘dill’, regular loss of final *z* and the later lowering of unstressed *i* yield *dile* ‘dill’. The stem itself remains ungeminated throughout that path.

The contrast is morphological rather than phonological. If the word were forced through a ja-stem **-lj-* pathway, the expected result would show *ll*. The attested Old English noun instead matches the i-stem development.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the stem class actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative ja-stem label	**déljaz*	ja-stem type outcome with gemination	*dill*-type comparison	useful comparative label, but not the OE form
selected i-stem formation	**déliz*	regular output: *dile*	*dile*	exact match between formation and attested OE noun

The single *l* is the decisive diagnostic. It identifies *dile* ‘dill’ with the i-stem formation rather than with the continental ja-stem branch.

8.6 fast — OE *festan*

Derivation: citation reconstruction **fastēną*; form followed here **fástijaną* > *festan* (early analogy).

Derivation trace

Proto input: **fástijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**fæstijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**fæstijan*
	OE Secondary Nasalization	**fæstijąn*
Early Anglo-Frisian	Sievers Law Syncope	**fæstjąn*
[no change]	OE I Umlaut	**festjąn*
	OE Weak Tail Reduction	**festjan*
	OE J Loss After Heavy	**festan*

Old English form: *festan*

Reconstruction and comparative evidence

Kroonen places the verb within the wider **fastu-* adjective family and its derived **fasten-* verbal line, the comparative background behind Old English 'to fast' (Kroonen 2013, 171). Ringe and Taylor, however, distinguish the Old English verb more closely: they treat OE 'to fast' as originally a class-I weak verb that later acquired the stative meaning through lexical confusion (Ringe and Taylor 2014, 110).

The derivational input **fástijaną* 'fast' therefore represents the inherited class-I formation reflected in Old English, whereas the citation label **fastēną* 'fast' belongs to the broader comparative presentation of the lexeme.

Old English evidence

Old English dictionaries record forms such as *festan* 'fast', alongside related *fæstan* 'fast' / *fǣstan* 'fast' spellings and meanings (Bosworth and Toller 1898, 213). The form selected here is *festan*, which fits the regular class-I phonological development.

The *æ*-forms remain relevant, but they do not control the entry. In the present analysis they belong to a later analogical reshaping under the adjective *fæst* 'fast', whereas *festan* 'fast' is the regular inherited class-I comparison form.

Development to Old English

From **fástijaną* 'fast', Anglo-Frisian brightening and subsequent i-umlaut produce the fronted vowel seen in *festan* 'fast'. The later weak-tail reductions and loss of *j* after a heavy syllable complete the regular Old English outcome.

What makes the entry non-regular is not the phonology of *festan* 'fast' itself, but the choice of formation. Old English continues the class-I verb, even though the comparative headword is often given under the parallel **fastēn-* family.

Class comparison

The comparison below sets the relevant forms side by side. It distinguishes the comparative class-III headword from the class-I formation actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative class-III headword	**fastēną*	class-III type outcome, not *festan*	wider family context	useful family label, but not the direct source of the target
selected class-I weak verb	**fástijaną*	regular output: *festan*	*festan*	exact match between formation and attested OE verb
later analogical reshaping	adjective-driven *fæst* influence	*fæstan* / *fǣstan* type spellings	fæstan-type evidence	genuine later OE reshaping, but secondary to the Old English form here

The relevant point is the class split. *festan* 'fast' is the regular Old English outcome of the class-I formation, while the better-known *æ*-forms belong to a later analogical layer.

8.7 flask — OE *flasce*

Derivation: citation reconstruction **flaskō*; form followed here **fláskōn* > *flasce* (early analogy).

Derivation trace

Proto input: **fláskōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**flæskǭ*
PNWGmc N Stem N Loss	**fláskǭ*	OE A Restoration	**flaskǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**flaskæ*
		OE Unstressed AE Merger	**flaske*
[no change]			

Old English form: *flasce*

Reconstruction and comparative evidence

The wider Germanic family is often cited under a form such as **flaskō* ‘flask’, but the evidence relevant for Old English points instead to a weak feminine formation **fláskōn* ‘flask’ / **flaskǭ* ‘flask’ (Orel 2003, 104). That distinction is crucial for the suffixal history of the noun.

The derivational input therefore differs from the citation label in stem class. Old English *flasce* ‘flask’ belongs with the weak feminine line, and the plural or oblique forms *flascan* ‘flask’ support that analysis (Ringe and Taylor 2014, 192).

Old English evidence

Old English dictionaries record the noun as *flasce* ‘flask’, with inflectional support from forms such as *flascan* ‘flask’; a later West Saxon *flaxe* ‘flask’ is also noted as a secondary variant (Bosworth and Toller 1898, 235; Clark Hall 1960, 121).

The relevant comparison form is therefore the weak feminine noun *flasce* ‘flask’. The plural and oblique evidence helps explain why the vowel and ending are preserved as they are in the singular.

Development to Old English

From **fláskōn* ‘flask’, the weak feminine passes through the expected loss of *n* and the later Old English development of the unstressed ending, reaching *flasce* ‘flask’. Campbell cites restored *a* in exactly this environment, including *flasce* ‘flask’ after inflected *flascan* ‘flask’ (Campbell 1959, § 158). Once the weak feminine formation is chosen, the noun follows a regular path to its Old English shape.

The decisive issue is morphological rather than phonological. A simple strong feminine citation form does not capture the OE weak noun as cleanly as the selected **fláskōn* ‘flask’ does.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak feminine formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader comparative headword	**flaskō*	broader family label	wider family context	useful lexeme label, but not the cleanest OE-facing derivation
selected weak feminine formation	**fláskōn*	regular output: *flasce*	*flasce*	exact match between formation and attested OE noun

The weak feminine suffix is the relevant point. It aligns the inherited form with attested *flasce* ‘flask’ and its supporting paradigm forms.

8.8 follow — OE *fylġan*

Derivation: citation reconstruction **fulgēną*; form followed here **fúlgijaną* > *fylġan* (early analogy).

Derivation trace

Proto input: **fúlgijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**fúlgijan*
[no change]	OE Secondary Nasalization	**fúlgijąn*
	Sievers Law Syncope	**fúlgjąn*
Early Anglo-Frisian	OE Velar Palatalization	**fúljjąn*
[no change]	OE I Umlaut	**fyljjąn*
	OE Weak Tail Reduction	**fyljjan*
	OE J Loss After Heavy	**fyljan*
	OE Palatal Fricative Merger	**fyljan*

Old English form: *fylġan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb as **fulgen-* and gives Old English *fylgan* ‘follow’, *folgian* ‘follow’, adding that Old Norse *fylgja* ‘follow’ and Old English *fylg(e)an* continue a formation **fulgjan-* (Kroonen 2013, 159). The comparative headword and the class-I formation are therefore related but not identical.

Ringe and Taylor distinguish PNWGmc **fulgija-* ‘follow’ ~ **fulgai-* ‘follow’ > OE *fylgan* ‘follow’ ~ *folgian* ‘follow’ and describe it as a dual formation that probably reflects an older alternation between j-present and e-stative (Ringe and Taylor 2014, 293–94). This is a stem-class choice, not a spelling choice. The derivational input **fúlgijaną* ‘follow’ belongs to the class-I **fulgija-* / **fulgjan-* branch; the citation form **fulgēną* ‘follow’ belongs to the parallel class-II history behind *folgian* ‘follow’.

Old English evidence

The Old English evidence preserves both formations. Clark Hall lists *fylgan* ‘follow’ with variant spellings *fylgian* ‘follow’ and *fyligan* ‘follow’ (Clark Hall 1960, 125). Bosworth-Toller likewise has a separate *fylgean* ‘follow’ entry (Bosworth and Toller 1898, 275).

Bright notes traces of the older conjugation in *fylg(e)an* ‘follow’ (Bright 1971, 77) and lists *folgian* ‘follow’ (*fylgean* ‘follow’) in the glossary (Bright 1971, 364). The relevant comparison form in this entry is therefore the class-I verb *fylgan* ‘follow’ / *fylgean*, here normalized as *fylġan* ‘follow’. The spelling with ġ represents the palatalized velar before a front-vocalic environment.

Development to Old English

**fúlgijaną* ‘follow’ is a class-I weak-verb formation. In the class-I branch the **j* blocks NWGmc lowering of *u* to *o*, since Ringe and Taylor formulate that lowering for environments in which no **j* intervened (Ringe and Taylor 2014, 96). The same front-vocalic environment then triggers i-umlaut, so *u* becomes *y* (Ringe and Taylor 2014, § 6.6.2).

The subsequent Old English developments are palatalization of the velar, weak-tail reduction, and loss of *j* after a heavy syllable, yielding *fylġan* ‘follow’. This is the regular outcome of the class-I formation. The class-II form *folgian* ‘follow’ belongs to the parallel **-ē-* / **-ai-* branch and is not the form modeled here.

Class comparison

A class comparison identifies which inherited formation corresponds to the established Old English form under discussion. The comparison below is manual; no full automatic class probe is presented here.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation class-II formation	**fulgēną*	probe output: *folgon*	*folgian* ‘follow’	mismatch: the regular output is not the remodeled infinitive *folgian*
parallel class-II branch	PNWGmc **fulgai-*	Ringe-Taylor: OE *folgian* ‘follow’	*folgian* ‘follow’	documents the separate class-II branch, but not the target of this entry
selected class-I formation	**fúlgijaną*	regular output: *fylġan* ‘follow’	*fylġan* ‘follow’ / *fylgan* ‘follow’	exact match between input, output, and class

The relevant point is the class split. *fylġan* ‘follow’ is the regular Old English outcome of the class-I **fulgija-* / **fulgjan-* formation, whereas *folgian* ‘follow’ belongs to the parallel class-II branch.

8.9 gall — OE *ġealla*

Derivation: citation reconstruction **gállą*; form followed here **gállô* > *ġealla* (early analogy).

Derivation trace

Proto input: **gállô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gœllô*
[no change]	OE Breaking	**geallô*
	OE Velar Palatalization	**jeallô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**jealla*
[no change]	OE Palatal Fricative Merger	**jealla*

Old English form: *ġealla*

Reconstruction and comparative evidence

The wider cognate family can be presented under a form such as **gállą* ‘gall’, but the Old English noun itself belongs with a weak noun **gallōn-* ‘gall’, cited here as **gállô* ‘gall’ (Kroonen 2013, 165). The derivational input therefore differs from the broader comparative headword in stem class.

The stem class determines the Old English shape. The weak masculine pathway preserves the ending needed for *ġealla* ‘gall’, whereas a simple strong-noun headword does not align as closely with the attested OE noun.

Old English evidence

Old English dictionaries record the noun as *gealla* ‘gall’, and Bright also gives the dative *geallan* ‘gall’, confirming a weak-noun paradigm (Bosworth and Toller 1898, 297; Clark Hall 1960, 145; Bright 1971, 372). The normalized spelling *ġealla* ‘gall’ uses ġ for the palatal consonant.

Campbell also notes dialectal variation, contrasting West Saxon or Kentish *gealla* ‘gall’ with Anglian *galla* ‘gall’ (Campbell 1959, § 486). The target of this entry is the West Saxon type *ġealla* ‘gall’.

Development to Old English

From **gállô* ‘gall’, the weak noun develops through the expected Old English history of the suffix and the regular breaking environment before *ll*, yielding *ġealla* ‘gall’ (Campbell 1959, § 486). Once the weak masculine input is chosen, the noun follows a regular path to its attested Old English form.

The decisive issue is therefore morphological. Old English reflects the weak noun, while the broader family label belongs to a different way of presenting the cognate set.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak noun formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader family label	**gállą*	broader cognate-set headword	wider family context	useful lexeme label, but not the direct source of *ġealla*
selected weak noun	**gállô*	regular output: *ġealla* ‘gall’	*ġealla* ‘gall’	exact match between formation and attested OE noun

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
dialectal Anglian continuation	weak noun branch	Anglian *galla* ‘gall’ type	*galla* ‘gall’	genuine OE variant, but not the West Saxon form used here

The weak-noun stem class is the relevant point. It gives a direct route to attested *ġealla* ‘gall’, while the broader comparative label serves only as a family heading.

8.10 heaven — OE *heofon*

Derivation: citation reconstruction **xémenaz*; form followed here **xébun* > *heofon* (early analogy).

Derivation trace

Proto input: **xébun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Velar Fricative Palatalization	**çébun*
[no change]	PGmc B Allophony	**çéβun*
	OE Back Mutation	**çéoβun*
Early Anglo-Frisian	OE Med Unstressed U Lowering	**çéoβon*
[no change]		

Old English form: *heofon*

A. Selected CAPR derivation

Item	Value
lexical item	heaven
citation reconstruction / lexeme label	*xémenaz
form followed here	*xébun (northern WGmc / pre-OE* hebun*)*
Old English target	heofon
classification	early_analogy
derivational result	*xébun > heofon (multiplicity 1, exact match)

Compact live trace (corrected cascade, literal adjacent-*mn* SC022):

step	form
proto input	**xébun*
OE Med Unstressed U Lowering	**xébon*
OE Velar Fricative Palatalization	**çébon*
PGmc B Allophony	**çéβon*
OE Back Mutation	**çéoβon*
Old English Orthography	*heofon*

No rule in this span creates the labial: it enters with the input *b* ($\rightarrow \beta \rightarrow f$). Only vocalic and surface changes are modelled.

B. Why *xébun is not arbitrary

Ringe and Taylor reconstruct northern West Germanic **hebun* 'sky, heaven', gen. **hebunas* 'heaven', whence West Saxon and Northumbrian *heofon* 'heaven', Mercian *heofen* 'heaven' (Ringe and Taylor 2014, 272, 324). R&T's *hebun* is the lexeme CAPR writes **xébun* (canonical *x* for source *h*), dated to that northern West Germanic stage, not Proto-Germanic; §D explains its labial.

The word is an inherited mn-stem (Kroonen 2013, 220): PIE *$h_2$ék-mōn, gen.* h_2k-mn-ós, remodelled in Germanic to nom. **hemō* 'heaven', gen. **hemnaz* 'heaven'. In the zero-grade oblique cluster cells the sequence *-mn- is adjacent, and the change* mn > βn/bn/fn* is regular there (Fulk §6.14 (Fulk 2018, 121)). Old Norse preserves exactly this paradigmatic distribution — *himinn* 'heaven' (nominative, with *m*) beside dative *hifni* 'heaven' (oblique, with *f* from the syncopated cluster) — while Old English and Old Saxon (OS *heban* 'heaven', *hebenes* 'heaven' (Fulk 2018)) generalized the labial into the vowel-bearing stem. The *b* of *xébun is therefore analogical* within this nominative input*, but its historical source is regular and recoverable — back to the inherited genitive (see §E).

C. The unresolved *u*

The medial *u* is the less secure element and must not be treated as certain merely because the cascade needs a back vowel to trigger back mutation. It is the n-stem suffixal vowel, the genuine conditioning environment for West Saxon back umlaut of *e* (parallel to *seven*, §D), and it is supported by Ringe–Taylor's reconstruction and by the Old English back-mutation pattern (Ringe and Taylor 2014, 324). But its exact deeper ablaut grade — *u* here, against the *i* of the cognate vowel-stem **hemina-* (*himinn* 'heaven', Go. *himins* 'heaven') — is not fully recoverable. It is the irreducible residue of the reconstruction; the *b* is not.

D. Regular modelled span

The northern-WGmc / pre-OE → Old English development **xébun* > *heofon* is regular and is the exact structural parallel of the independently regular numeral *seven*, **sébun* > *seofon* (row 2174, *DERIVATION_CLASS* regular). Both have a medial *u* and a labial, and both undergo West Saxon back umlaut of *e* before *u* across that labial (Ringe and Taylor 2014, 324; Hogg 1992a). The live traces run in lock-step (medial *u*-lowering, labial spirantization, back mutation, orthography). Ringe and Taylor explicitly treat *heaven* and *seven* as parallel back-umlaut examples, which is why *seven* is retained here as the control.

E. Deep oblique controls

The obliques go deeper than the selected nominative and derive the labial themselves. Live outputs under the corrected literal adjacent-*mn* cascade, with evidential status marked:

cell / form	type	corrected-cascade output	status	note
nom. citation **xémenaz*	reconstructed (citation)	*hemen*	counterfactual (no labial)	citation a-stem; no cluster, so no regular *f*
gen.sg cluster **xémnas*	reconstructed (PIE *h_2k-mn-ós)	*hefnes*	reconstructed, **not attested in OE**	labial **regular** here (mn > βn)
dat.sg cluster **xémni*	reconstructed	*hifn*	attested only as ON *hifni* 'heaven'	labial + i-umlaut regular
dat.pl cluster **xémnum*	reconstructed	*hefnum*	reconstructed, **not attested in OE**	labial regular

cell / form	type	corrected-cascade output	status	note
remodelled stem **xébun*	reconstructed (R&T *hebun)	*heofon*	**attested** (selected row)	labial pre-supplied
R&T genitive **hebunas*	reconstructed	*heofones*	**attested**	labial pre-supplied
OE oblique *heofnes* / *heofnum*	**attested**	—	actual OE forms	carry *eo*; from the re-vowelled stem + late syncope
ON *himinn* : *hifni* 'heaven'	**attested** (North Gmc)	—	attested cluster line	keeps the *m* : *f* alternation
OS *heban* / *hebenes*	**attested** (Old Saxon)	—	labial generalized	vowel-bearing throughout

Distinctions kept explicit: attested OE forms (*heofon*, *heofones*, *heofnes*, *heofnum*) descend from the re-vowelled stem; the cluster outputs *hefnes*/*hifn*/*hefnum* are reconstructed regular outputs, and for Old English they are counterfactual — the actual attested obliques carry *eo* from back mutation on the retained/restored medial vowel, then late syncope (*heofonum* > *heofnum*), not from a bare *-fn-* cluster. The genuine attested witness of the pure cluster line is Old Norse *hifni* 'heaven'.

F. Why the oblique is not the selected row

The oblique lines are historically essential — they show that the labial has a regular origin, all the way back to the inherited genitive *h_2k-mn-ós — but they are inferior as the single CAPR input→target datum. In Old English every oblique cluster form encounters morphological re-vowelling (restoration of a medial vowel into the labialized* -βn- *stem) before it reaches an attested Old English shape. That re-vowelling is a non-regular event; it interrupts any single deterministic path from a cluster oblique to an attested OE form. The nominative input* xébun begins after that restructuring and then proceeds by regular sound change to a secure attested target — the longest clean modellable span.

G. Attestation warning

Bare Old English *hefn* is not securely attested and must not be reintroduced as an attested comparator. The earliest witnesses (Cædmon's Hymn) are *hefun* 'heaven', *hefen* 'heaven', and (Moore Bede) *heben* 'heaven' — all with a medial vowel. The reconstructed cluster output *hefn* (§E) is a reconstruction, never a manuscript form.

H. Classification rationale

The row is *early_analogy*, not *late_analogy* and not *known_unmodelled*.

- The selected row models exactly one form, **xébun → heofon*. The necessary paradigm restructuring (labial levelling + suffix-vowel generalization) precedes the derivational input; once that input is chosen, the modelled development is regular. This is the definition of *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply).
- It is not *late_analogy*: the analogy is in the input, not a later paradigm-cell selection; the later history of *other* paradigm cells (oblique syncope, *heo-* levelling) does not touch the modelled nominative derivation.
- It is not *known_unmodelled*: that label (a legitimate, conservative alternative discussed in `dossier-heaven-paradigm-history-2026.md`) would discard a real, defensible regular span. **xébun* > *heofon is* a clean regular derivation to a secure attested target, so CAPR models it.

The old cross-syllable *mV…n* proxy in SC022 — which formerly fabricated the labial from an intervocalic *m* so that the previous input **xémonų* could reach *heofon* — has been retired in favour of the literal historical adjacent *mn* > *βn*. See `dossier-heaven-paradigm-history-2026.md` §§13–15 and the implementation audit `audits/heaven-sc022-implementation-2026.md`.

8.11 knight — OE *cniht*

Derivation: citation reconstruction **knixtaz*; form followed here **knéxtaz* > *cniht* (early analogy).

Derivation trace

Proto input: **knéxtaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**knéxt*
EAF Final Z Deletion	**knéxta*	OE Breaking	**knéoxt*
Early Anglo-Frisian		OE Ws Palatal Umlaut	**knixt*
[no change]			

Old English form: *cniht*

Reconstruction and comparative evidence

The comparative sources align on an *e*-grade reconstruction for this noun. Ringe and Taylor cite **kneht* ‘knight’, and Orel gives **knextaz* ‘knight’ (Ringe and Taylor 2014, 142; Orel 2003, 256). Kluge-Seebold likewise points to **knehta-* (Kluge 2011, 506). The derivational input **knéxtaz* ‘knight’ follows that comparative evidence.

A competing citation reconstruction **knixtaz* ‘knight’ remains possible as a label for the word family, but it is not the reconstruction followed here. The Old English development discussed below is based on **knéxtaz* ‘knight’.

Old English evidence

Old English dictionaries record the noun as *cniht* ‘knight’ (Clark Hall 1960, 63; Bosworth and Toller 1898, 71). Campbell cites plural *cneohtas* ‘knights’ among the broken forms, showing the same vowel environment from another point in the paradigm (Campbell 1959, § 146).

The target is therefore an ordinary attested Old English noun. No reconstructed OE comparator is needed here.

Development to Old English

From **knéxtaz* ‘knight’, the relevant Old English changes include breaking before the velar cluster and then the later reduction that yields *cniht* ‘knight’. Campbell later notes the early West-Saxon alternation *cniht* ‘knight’ beside plural *cneohtas* ‘knights’ (Campbell 1959, § 305). Sievers-Brunner gives the same contrast as *cniht … cneohtas* (Brunner 1965, § 122). With that corrected input, the derivation is straightforward.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the handbook-supported *e*-grade input from a competing citation reconstruction.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing citation reconstruction	**knixtaz*	not the reconstruction followed here	broader citation tradition	useful as a competing label, but not the source-based choice used for the OE derivation
handbook-supported reconstruction	**knéxtaz*	regular output: *cniht* ‘knight’	*cniht* ‘knight’	exact match between comparative reconstruction and attested OE noun
related plural evidence	same stem family	plural *cneohtas* ‘knights’ type background	*cneohtas* ‘knights’	supports the vowel environment, but not the Old English form here cell

8.12 lade — OE *hladan*

Derivation: citation reconstruction **laθōjanǫ*; form followed here **xláðanǫ* > *hladan* (early analogy).

Derivation trace

Proto input: **xláðanǫ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xlædanǫ*
PWGmc Dental Hardening **xládanǫ*	OE A Restoration	**xladanǫ*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**xladan*
	OE Secondary Nasalization	**xladǫn*
[no change]	OE Weak Tail Reduction	**xladan*

Old English form: *hladan*

Reconstruction and comparative evidence

Ringe and Taylor cite the strong verb *hladan* ‘lade, load’ directly (Ringe and Taylor 2014, 248). The citation label **laθōjanǫ* ‘lade’ is used here only as a broader family heading, not as the direct source of the OE derivation.

The derivational input therefore marks an early stem choice. The entry follows the strong Verner-grade form that reaches Old English *hladan* directly.

Old English evidence

Bosworth-Toller records the verb as *hladan* and preserves the expected strong-verb paradigm material around it (Bosworth and Toller 1898, 559). Clark Hall likewise records *hladan* (Clark Hall 1960, 159). The target is an attested infinitive rather than a reconstructed paradigm cell.

For this entry the relevant comparison form is the infinitive *hladan* itself. The question is how that attested strong verb relates to the broader comparative family.

Development to Old English

From **xláðaną* ‘lade’, the verb passes through the expected early voiced stop stage, Anglo-Frisian brightening, and the later A-restoration that returns the root vowel to *a* before the full infinitival ending. Campbell treats verbs such as *hladan* as showing restored *a* in the present system (Campbell 1959, § 744). Ringe and Taylor likewise derive *hladan* from the voiced strong-verb line (Ringe and Taylor 2014, 248). The resulting Old English infinitive is *hladan*.

Class comparison

The comparison below sets the relevant forms side by side. It separates the wider weak-verb family label from the strong verb actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative weak-verb label	*laθōjaną	wider family background	broader family context	useful family label, but not the direct source of *hladan*
selected strong Verner-grade input	*xláðaną	regular output: *hladan*	*hladan*	exact match between formation and attested OE infinitive

8.13 lap — OE *lappa*

Derivation: citation reconstruction **lábbaz*; form followed here **láppô* > *lappa* (early analogy).

Derivation trace

Proto input: **láppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**læppô*
[no change]	OE A Restoration	**lappô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**lappa*
[no change]		

Old English form: *lappa*

Reconstruction and comparative evidence

The comparative sources point to a weak noun with *pp*: Orel gives **lappōn* ‘lap’, and the Old English dictionary tradition preserves variant *læppa* ‘lap’ (Orel 2003, 236; Clark Hall 1960, 180). The derivational input **láppô* ‘lap’ follows that evidence.

A competing comparative label **lábbaz* ‘lap’ has also circulated for the word family, but the cited handbooks do not make it the direct source of the Old English weak noun. The form relevant to the OE development is the weak masculine input **láppô* ‘lap’.

Old English evidence

Campbell cites *lappa* ‘lap’ as a case of restored *a* (Campbell 1959, § 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ (Brunner 1965, § 10). The dictionary tradition also preserves *læppa* ‘lap’ (Clark Hall 1960, 180; Bosworth and Toller 1898, 613).

The target of this entry is the restored singular *lappa* ‘lap’. The variant *læppa* ‘lap’ and the oblique or plural *leappan* remain part of the Old English record and help frame the noun’s vowel history.

Development to Old English

Campbell explicitly lists *lappa* ‘lap’ among the forms with restored *a* (Campbell 1959, § 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ at the same Old English stage (Brunner 1965, § 10). With the weak masculine input chosen, the selected *lappa* ‘lap’ outcome is therefore the regular Old English comparison.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the weak masculine formation from a competing voiced comparative label.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing voiced comparative label	*lábbaz	not the form followed for the OE weak-noun derivation	broader comparative background	useful as a competing label, but not the source-based choice used here
selected weak masculine noun	*láppô	regular output: *lappa*	*lappa*	exact match between formation and attested OE noun
attested OE variant line	same noun family	*læppa*, *leappan*	*læppa* / *leappan*	useful control forms within the same OE tradition

8.14 laugh — OE *hliehhan*

Derivation: citation reconstruction **lákaną*; form followed here **xláxjaną* > *hliehhan* (early analogy).

Derivation trace

Proto input: **xláxjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xlæxxjaną*
PWGmc J Gemination	**xláxxjaną*	OE Breaking	**xleaxxjaną*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**xleaxçjaną*
		OE Heavy Syllable Nasal Apocope	**xleaxçjan*
[no change]		OE Secondary Nasalization	**xleaxçjąn*
		OE I Umlaut	**xliexçjąn*
		OE Weak Tail Reduction	**xliexçjan*
		OE J Loss After Heavy	**xliexçan*

Old English form: *hliehhan*

Reconstruction and comparative evidence

The wider Germanic family includes a non-j branch represented here by the citation label **lákaną* ‘laugh’, while the derivational input **xláxjaną* ‘laugh’ reflects the j-present line behind Old English *hliehhan* ‘laugh’.

This branch supplies the geminate fricative and the vowel development characteristic of the Old English verb. The comparative family label and the OE-facing input are therefore related but not identical.

Old English evidence

Bosworth-Toller records *hlihhan* ‘laugh’ as the verb ‘to laugh’ (Bosworth and Toller 1898, 551). Clark Hall cross-references *hlæhan* ‘laugh’, *hlehhan* ‘laugh’, and *hlihhan* ‘laugh’ to *hliehhan* ‘laugh’ (Clark Hall 1960, 160–61). Bright’s glossary likewise gives *hlihhan (hliehhan, hlyhhan)* (Bright 1971, 315). The target of this entry is the West Saxon *hliehhan* ‘laugh’.

The variants belong to the same background, but the attested lemma *hliehhan* ‘laugh’ supplies the evidence followed here.

Development to Old English

From **xláxjaną* ‘laugh’, West Germanic j-gemination yields the doubled consonant (Campbell 1959, § 407). Ringe and Taylor derive Old English *hliehhan* ‘laugh’ from the j-present branch via breaking before the palatalized geminate (Ringe and Taylor 2014, 240). They separately compare the related noun *hleahtor* as the outcome of **hlahtraz* ‘laugh’ (Ringe and Taylor 2014, 328).

Branch comparison

The comparison below sets the relevant forms side by side. It separates the wider non-j family label from the j-present branch actually reflected in Old English.

Formation / branch	Candidate input	Expected or documented OE outcome	OE comparison form	Result
wider non-j family	*lákaną	comparative background outside the selected OE line	wider family context	useful family label, but not the direct source of *hliehhan*
selected j-present branch	*xláxjaną	regular output: *hliehhan*	*hliehhan*	exact match between branch and attested OE lemma
attested OE variants	same OE verb line	*hlæhhan*, *hlehhan*	*hlæhhan* / *hlehhan*	genuine variant evidence, but secondary to the form compared here

8.15 loam — OE *lām*

Derivation: citation reconstruction **laimōn*; form followed here **láimą* > *lām* (early analogy).

Derivation trace

Proto input: **láimą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lām*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**lāmą*		

Old English form: *lām*

Reconstruction and comparative evidence

The inherited comparative noun is given as **laimōn* ‘loam’ or **laiman-*, and both Orel and Kroonen identify Old English *lām* ‘loam’ as a neuter reflex of that family (Orel 2003, 272; Kroonen 2013, 363). The form followed here, **láimą* ‘loam’, differs from the comparative headword because it represents the stem class that matches the Old English noun most directly.

This is therefore a class shift within the history of the English branch rather than a dispute about the OE target itself.

Old English evidence

Old English dictionaries record the noun as *lām* ‘loam’, a neuter word for ‘loam, clay, mud’ (Bosworth and Toller 1898, 604; Clark Hall 1960, 196). The target is an attested citation form rather than a reconstructed comparator.

The relevant question is not whether *lām* ‘loam’ is Old English, but which inherited formation best accounts for that attested neuter noun.

Development to Old English

From **láimą* ‘loam’, regular monophthongization of *ai* and the later loss of the final nasal syllable yield *lām* ‘loam’. With that OE-facing input, the phonological development is straightforward.

Class comparison

The comparison below sets the inherited comparative n-stem label beside the OE-facing stem class used to derive the attested noun.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited comparative noun	*laimōn	comparative family background	wider family context	useful headword, but not the direct OE-facing input
OE-facing stem class followed here	*láimą	regular output: *lām*	*lām*	exact match between the form followed here and the attested OE noun

8.16 lung — OE *lungen*

Derivation: citation reconstruction **lungō*; form followed here **lúnganjō* > *lungen* (early analogy).

Derivation trace

Proto input: **lúnganjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**lúngennju*
PWGmc J Gemination	**lúngannjō*	OE High Vowel Apocope	**lúngennj*
PNWGmc Final Long O Raising	**lúngannju*	OE J Loss After Heavy	**lúngenn*
Early Anglo-Frisian		OE Final Geminate Simplification	**lúngen*
[no change]			

Old English form: *lungen*

Reconstruction and comparative evidence

Kroonen treats the basic noun as **lungōn-* and also cites an OE-facing derivative **lungunjō-*, continued by Old English *lungen* ‘lung’ and close West Germanic cognates (Kroonen 2013, 384). The derivational input **lúnganjō* ‘lung’ models that derived feminine formation rather than the base noun. The notation differs slightly from Kroonen’s **lungunjō-*, but both point to the same derived feminine line.

The difference between the citation label and the derivational input is therefore derivational. Old English *lungen* ‘lung’ is not a direct reflex of the bare base noun **lungō* ‘lung’; it belongs to an expanded feminine formation.

Old English evidence

Old English dictionaries record the noun as *lungen* ‘lung’, with inflected forms such as *lungenne* and *lungena* (Bosworth and Toller 1898, 634). Clark Hall also preserves a small family of compounds such as *lungenādl* ‘lung-disease’, *lungensealf*, and *lungenwyrt* (Clark Hall 1960, 191).

The target is an attested Old English lexeme with its own paradigm, not a rescued inflectional cell.

Development to Old English

From the selected derived input, the expected derivational consonant and vowel adjustments lead to *lungen* ‘lung’. Once the expanded feminine formation is chosen, the Old English outcome is regular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the base noun from the derived feminine formation reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
base noun	*lungō	base-noun outcome without the OE derivative suffix	broader family context	useful headword, but not the direct source of *lungen*

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
derived OE-facing formation	*lúnganjō	regular output: *lungen*	*lungen*	exact match between the derived formation and the attested OE noun
Kroonen's cited derivative	**lungunjō-*	comparative support for the same OE-facing formation	*lungen* and cognate set	supports the derived feminine formation, with notation differing from the normalized input form used here

8.17 navel — OE *nafola*

Derivation: citation reconstruction **nablô*; form followed here **nábulô* > *nafola* (early analogy).

Derivation trace

Proto input: **nábulô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**nœbulô*
[no change]	OE A Restoration	**nabulô*
	PGmc B Allophony	**naβulô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**naβula*
[no change]	OE Med Unstressed U Lowering	**naβola*

Old English form: *nafola*

Reconstruction and comparative evidence

Kroonen instead gives a nasal-suffix navel formation with Old English *nafela* 'navel' among its reflexes (Kroonen 2013, 420), while Ringe and Taylor give the derivational pathway **nabulō* 'navel' > **nœbula* 'navel' > *nafola* 'navel' (Ringe and Taylor 2014, 270). The difference is one of stage and notation rather than of lexeme identity: the derivational input **nábulô* 'navel' is the pre-syncope form needed for the Old English development.

For the Old English comparison, the crucial point is simply that the pre-OE form still contains a medial vowel.

Old English evidence

Ringe and Taylor note the early West Saxon shift *nafola* 'navel' > *nafela* 'navel' (Ringe and Taylor 2014, 336). Campbell likewise records *nafela* 'navel' beside Corpus *nabula* 'navel' (Campbell 1959, § 159). The target of this entry is the nominative singular *nafola* 'navel', the form that matches the selected derivational pathway most directly.

nafela 'navel' is the better-known later West Saxon spelling, while *nabula* 'navel' preserves a less reduced medial vowel. These forms belong to the same lexical history, but this entry is centered

on *nafola* ‘navel’.

Development to Old English

Ringe and Taylor give the pre-OE line **nabulō* ‘navel’ > **næbula* ‘navel’ > OE *nafola* ‘navel’ (Ringe and Taylor 2014, 270). The trace represents that same development with stress-marked notation and explicit intermediate weakening. Intervocalic *b* then surfaces as *f*, and final weak-tail shortening gives *nafola* ‘navel’.

The medial vowel is still present when A-restoration applies. That is why the selected pre-syncope input differs from the syncopated comparative headword.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative citation form from the pre-syncope input and from the later OE spellings.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
syncopated comparative headword	**nablô*	reduced *næfla*-type outcome rather than *nafola* ‘navel’	not the Old English form here	useful citation form, but too reduced for the pathway modeled here
selected pre-syncope input	**nábulô*	regular output: *nafola* ‘navel’	*nafola* ‘navel’	exact match between derivational input and target
later OE reduction stages	same lexical history	attested *nafela* ‘navel’; Corpus *nabula* ‘navel’	*nafela* ‘navel’ / *nabula* ‘navel’	related OE spellings, but not the chosen comparator

8.18 neck — OE *hnecca*

Derivation: citation reconstruction **xnákkaz*; form followed here **xnékkô* > *hnecca* (early analogy).

Derivation trace

Proto input: **xnékkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**xnékka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hnecca*

Reconstruction and comparative evidence

The noun belongs to an ablauting n-stem family. Kroonen reconstructs a paradigm with nominative **hnekkō* ‘neck’, genitive **hnukkaz* ‘neck’, and accusative plural **hnakkuns* ‘neck’, and

he places Old English *hnecca* ‘neck, nape’ among the e-grade descendants (Kroonen 2011, 167). Kluge-Seebold likewise identifies *ae. hnecca* as an ablaut partner of the a-grade *Nacken* family (Kluge 2011, 347).

A competing comparative label **xnákkaz* ‘neck’ belongs to the wider family, and Orel also gives an a-grade headword line (Orel 2003, 218). The derivational input **xnékkô* ‘neck’, however, is the form that matches the Old English branch.

Old English evidence

Clark Hall records the weak masculine noun *hnecca* ‘neck, nape’ (Clark Hall 1960, 162). Bosworth-Toller likewise records *hnecca* ‘neck, nape’ (Bosworth and Toller 1898, 567). The target is therefore an attested citation form, not an oblique cell or a reconstructed lemma.

The phonological question is upstream of the Old English evidence. The attested noun already shows that the branch continued an e-grade form rather than the a-grade seen in much of the continental material.

Development to Old English

From **xnékkô* ‘neck’, the derivation is straightforward. The trace shortens the final long vowel to **xnékka* ‘neck’, and Old English orthography gives *hnecca* ‘neck’.

The derivation depends on the earlier selection of the e-grade weak-noun form continued by Old English.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the wider a-grade family from the selected e-grade Old English branch.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing comparative label	**xnákkaz*	broader a-grade family rather than the selected OE source	continental *Nacken* line	useful family label, but not the input followed for the Old English derivation
weak noun with a-grade	**xnakkô*	expected *hnacca* type outcome	*hnacca*	fixes the class, but not the vowel grade
selected e-grade nominative	**xnékkô*	regular output: *hnecca*	*hnecca*	exact match between derivational input and attested OE noun
oblique paradigm background	**hnukkaz*, **hnakkuns*	ON/OHG/German a-grade continuation	*hnakki* / *Nacken*	shows the wider ablaut family, but not the chosen OE branch

8.19 needle — OE *nǣdl*

Derivation: citation reconstruction **néθlō*; form followed here **néðlō* > *nǣdl* (early analogy).

Derivation trace

Proto input: **néðlō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**nǣdl*
PNWGmc Long E Lowering	**nāðlō*		
PWGmc Dental Hardening	**nādlō*		
PNWGmc Final Long O Raising	**nādlu*		
EAF Long A Fronting	**nǣdlu*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdl*

Reconstruction and comparative evidence

Ringe and Taylor treat the word as a voiced/voiceless alternant, citing **nēþlō* ‘needle’ ~ **nēdlō* ‘needle’ > OE *nédl* ‘needle’ (Ringe and Taylor 2014, 329). The form followed here, **néðlō* ‘needle’, is the voiced Verner-grade form used for the Old English comparison, while the citation form **néθlō* ‘needle’ remains the broader lexeme label.

The development discussed here follows the Ringe-Taylor alternant framework.

Old English evidence

Clark Hall records the attested citation form *nǣdl* ‘needle’ (Clark Hall 1960, 210). Campbell lists *nédl* ‘needle’ among the expected unbroken forms after *t* and *d* (Campbell 1959, § 367). Hogg also includes *nidi* / *nǣdl* ‘needle’ in the same broader cluster history (Hogg 1992b).

The target is therefore an attested citation form. No oblique-cell substitution is involved in this entry.

Development to Old English

Ringe and Taylor give the historical line **nēþlō* ‘needle’ ~ **nēdlō* ‘needle’ > OE *nédl* ‘needle’ (Ringe and Taylor 2014, 329). Campbell likewise lists *nédl* ‘needle’ among the expected unbroken forms after *t* and *d* (Campbell 1959, § 367). The trace expresses the same pathway with the voiced alternant followed here for the Old English comparison.

The essential choice lies in which Proto-Germanic alternant is taken as the starting point. Once the voiced form is chosen, the rest of the pathway is regular.

Alternant comparison

The comparison below sets the relevant forms side by side. It separates the broader citation headword from the voiced alternant used for Old English.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative voiceless headword	**néθlō*	broader word-family label rather than the OE-facing alternant	**nēþlō* line	useful citation form, but not the derivational input for the Old English comparison

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
voiced Verner alternant followed here	**néðlō*	regular output: *nǣdl*	*nǣdl*	exact match between the form followed here and the attested OE noun
later hardening stage	*nédlō	intermediate pre-OE stage in the same derivation	*nǣdl*	genuine stage in the pathway, but not the Proto-Germanic form followed here

8.20 nose — OE *nosu*

Derivation: citation reconstruction **nasō*; form followed here **núsō* > *nosu* (early analogy).

Derivation trace

Proto input: **núsō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc U Lowering	**nósō*	
PNWGmc Final Long O Raising	**nósu*	
Early Anglo-Frisian		
[no change]		

Old English form: *nosu*

Reconstruction and comparative evidence

Kroonen reconstructs a Germanic ablaut pair **nasō-* ~ **nusō-* and adds that the root **nus-* is likely to have arisen as a secondary zero grade after a remodeling of the older paradigm (Kroonen 2013, 383). Campbell is more specific for Old English, citing *nosu* ‘nose’ < **nusō* ‘nose’ (Campbell 1959, 43).

The citation reconstruction **nasō* ‘nose’ is therefore best treated as the full-grade comparative headword, while the derivational input **núsō* ‘nose’ represents the remodeled zero-grade line continued by the Old English form discussed here. Orel’s **nasō* ‘nose’ alongside OE *nasu* ‘nose’ preserves the competing full-grade notation and shows that the two lines should not be collapsed without comment (Orel 2003, 320).

Old English evidence

Nosu ‘nose’ is an attested Old English noun. Ringe and Taylor list it among the few surviving early Old English feminine u-stems (Ringe and Taylor 2014, 385). Clark Hall likewise gives *nosu f.* ‘nose’, with genitive-dative singular *nosa*, and cross-refers *nasu* ‘nose’ to *nosu* ‘nose’ (Clark Hall 1960, 211, 217).

The selected OE target is therefore attested *nosu* ‘nose’, not a reconstructed placeholder. The lexicographical record also gives *nasu* ‘nose’ for the full-grade side of the tradition.

Development to Old English

From **núsō* ‘nose’, the regular path is the one documented by the current trace: **núsō* ‘nose’ > **nósō* ‘nose’ > **nósu* ‘nose’ > *nosu* ‘nose’. The early special step lies in the choice of the zero-grade input, not in any late Old English repair.

With that input chosen, the OE development is straightforward. The full-grade line behind **nasō* ‘nose’ instead points toward *nasu* ‘nose’, not to the form treated here.

The low root of *nosu* ‘nose’ supports the economy of lowering before final oral raising, but only conditional on that selected remodeled stem. Reversing the changes yields a high-root nominative, whose replacement by a lowered oblique root would be an additional, individually unproved levelling. The word is therefore not a pristine inherited nominative proving the chronology. The comparative argument and its limits are developed in SC019 PNWGmcFinalLongORaising (Fulk 2018, 56; Kroonen 2013, 383).

Stem comparison

The comparison below sets the relevant forms side by side. It separates the full-grade comparative line from the remodeled zero-grade input that yields the Old English form.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
full-grade comparative line	*nasō	expected full-grade continuation *nasu*	*nasu*	useful comparative background, but not the Old English-facing input
remodeled zero-grade line	*núsō	regular output: *nosu*	*nosu*	exact match between derivational input and attested OE noun

8.21 sap — OE *sæp*

Derivation: citation reconstruction **sapōn*; form followed here **sápą* > *sæp* (early analogy).

Derivation trace

Proto input: **sápą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**sæpą*
[no change]	OE Heavy Syllable Nasal Apocope	**sæp*
Early Anglo-Frisian		
[no change]		

Old English form: *sæp*

Reconstruction and comparative evidence

The comparative sources do not give one uniform inherited stem. Kroonen preserves the word family as **saf* ‘sap’ / **ppan-*, with Old English *sæp* ‘sap’ (Kroonen 2013, 420). Orel preserves the

comparative notation **sapōn* ‘sap’ and **sapan* ‘sap’ (Orel 2003, 319).

The derivational input **sápą* ‘sap’ therefore does not replace those comparative labels. It identifies the OE-facing stem shape that yields the attested noun treated here.

Old English evidence

Clark Hall records *sæp* ‘sap’ (e) n. (Clark Hall 1960, 247). The target is therefore an attested neuter Old English noun. Orel’s plain *sap* ‘sap, juice’ notation belongs to comparative normalization, not to the spelling adopted here for the Old English form (Orel 2003, 319).

Development to Old English

From **sápą* ‘sap’, Anglo-Frisian brightening yields *sæ* ‘sap’, and heavy-syllable nasal apocope then produces *sæp* ‘sap’. That is the regular path documented by the current trace.

The competing comparative lines do not give the same result. The inherited n-stem notation **sapōn* ‘sap’ yields *sape* ‘sap’, while an i-stem continuation from the **sapi-* line leads to *sep* ‘sap’ / *sepe* ‘sap’ rather than to *sæp* ‘sap’. The special step in this entry is therefore the early stem choice, not a late OE paradigm-cell selection.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative stem lines from the Old English-facing input.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative n-stem line	**sapōn*	local comparator output: *sape* ‘sap’	*sape* ‘sap’	useful comparative background, but not the source of attested *sæp*
inferred i-stem comparator from **sapi-*	**sapiz*	local comparator output: *sepe* ‘sap’	*sepe* ‘sap’	confirms that an i-triggering stem does not reach the target
selected a-stem input	**sápą*	regular output: *sæp* ‘sap’	*sæp* ‘sap’	exact match between derivational input and attested OE noun

8.22 sea — OE *sǣ*

Derivation: citation reconstruction **sái*; form followed here **sáiwiz* > *sǣ* (early analogy).

Derivation trace

Proto input: **sáiwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE W Loss Before I	**sāi*
EAF Final Z Deletion	**sáiwi*	OE I Umlaut	**sǣi*
Early Anglo-Frisian		OE High Vowel Apocope	**sǣ*
EAF Ai Monophthongization	**sāwi*		

Old English form: *sǣ*

Reconstruction and comparative evidence

Kroonen gives the noun in stem notation as **saiwi-*, an i-stem whose English reflex is cited as OE *sæ* ‘sea’ (Kroonen 2013, 423). Ringe and Taylor write the fuller form **saiwiz* ‘sea’ and derive it through **sawi* ‘sea’ > **sei* ‘sea’ > OE *sǣ* ‘sea’ (Ringe and Taylor 2014, § 6.7.1). The comparative headword is therefore shorter than the form required for the English history: **sái* ‘sea’ names the lexeme, but **sáiwiz* ‘sea’ preserves the medial **w* and the final high vowel that control the later development.

Old English evidence

The Old English noun is the ordinary word for ‘sea’. Kroonen cites it as *sæ* ‘sea’; the normalized form here is *sǣ* ‘sea’ (Kroonen 2013, 423). Campbell likewise treats *sea* as continuing the same **saiui* ‘sea’ > **sǣi* ‘sea’ history, with loss of *u*/*w* before *i* (Campbell 1959, § 406).

Development to Old English

Once the fuller i-stem input is chosen, the development is regular. After Proto-West-Germanic monophthongization **sáiwiz* ‘sea’ > **sāwiz* ‘sea’ and final **-z* loss **sāwiz* ‘sea’ > **sāwi* ‘sea’, the non-initial **w* disappears before unstressed **i*, and the following high vowel fronts the root vowel before final apocope. The documented chain is **sáiwiz* ‘sea’ > **sāwiz* ‘sea’ > **sāwi* ‘sea’ > **sāi* ‘sea’ > **sǣi* ‘sea’ > *sǣ* ‘sea’ (Ringe and Taylor 2014, § 6.7.1; Campbell 1959, § 406).

Stem and stage comparison

The comparison below sets the relevant forms side by side. It separates the abbreviated comparative headword from the fuller i-stem input that yields the Old English form.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
abbreviated comparative headword	**sái*	too short to preserve the **w … *i* environment needed for the documented chronology	*sǣ*	useful comparative label, but not the Old English-facing input
selected i-stem input	**sáiwiz*	documented regular output: *sǣ*	*sǣ*	exact match between derivational input and Old English target

8.23 sieve — OE *sife*

Derivation: citation reconstruction **síbaz*; form followed here **síbi* > *sife* (early analogy).

Derivation trace

Proto input: **síbi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	PGmc B Allophony	**síβi*
[no change]	OE Med Unstressed I Lowering1	**síβe*
Early Anglo-Frisian		
[no change]		

Old English form: *sife*

Reconstruction and comparative evidence

Kluge-Seebold gives wg. **sibi-* n. … ae. *sife*, and Campbell groups *sife* ‘sieve’ with short neuter i-stems such as *spere* (Kluge 2011, 847; Campbell 1959, § 609). The older morphological background is the s-stem **sib-iz* ‘sieve’, but the derivational input is the normalized i-stem form **síbi* ‘sieve’.

Kroonen’s nearby **sebjō-* entry belongs to the separate kinship lexeme that yields Old English *sibb* ‘sieve’, not to the sieve word. Orel’s **sibaz* ‘sieve’ … OE *sife* ‘sieve’ preserves a broader handbook notation, but that a-stem shape does not fit the Old English form treated here (Orel 2003, 328).

Old English evidence

Clark Hall gives *sibi* ‘sieve’ (GL) … = *sife* ‘sieve’ and also *sife* ‘sieve’ n. ‘sieve’ (Clark Hall 1960, 263). Campbell likewise cites Corpus Glossary *sibi* ‘sieve’ and treats *sife* ‘sieve’ as a short neuter i-stem (Campbell 1959, §§ 444, 609). The normalized Old English target is therefore *sife* ‘sieve’, while *sibi* ‘sieve’ is an attested earlier spelling rather than a separate lexeme.

Development to Old English

From **síbi* ‘sieve’, the regular derivation gives **síβi* ‘sieve’ > **síβe* ‘sieve’ > *sife* ‘sieve’. Medial *b* is realized as a spirant and later written *f*, while the final unstressed *i* lowers to *e*. The older s-stem background **sib-iz* ‘sieve’ explains the morphology, but the derivational input **síbi* is the immediate pre-Old-English form.

Stem comparison

The comparison below sets the relevant forms side by side. It distinguishes the accepted i-stem line from its rejected competitors.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
ja-stem kinship line	Kroonen **sebjō-* / comparator **sibja*	OE *sibb* ‘sieve’	*sibb* ‘sieve’	separate lexeme, not the target treated here
a-stem handbook line	**síbaz*	expected *sif*	*sif*	wrong ending for the attested noun
selected i-stem line from older **sib-iz*	**síbi*	documented regular output: *sife* ‘sieve’	*sife* ‘sieve’; early spelling *sibi* ‘sieve’	exact match between derivational input and Old English evidence

8.24 spare — OE *sparian*

Derivation: citation reconstruction **sparēną*; form followed here **spárōjaną* > *sparian* (early analogy).

Derivation trace

Proto input: **spárōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærōjaną*
[no change]	OE A Restoration	**sparōjaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**sparōjan*
[no change]	OE Secondary Nasalization	**sparōjąn*
	OE I Umlaut	**sparējąn*
	OE Unstressed Long Vowel Shortening	**sparejąn*
	OE Weak Tail Reduction	**sparejan*
	OE Intervocalic J Vocalization	**spareian*
	OE Unstressed EI Contraction	**sparian*

Old English form: *sparian*

Reconstruction and comparative evidence

Kroonen keeps the inherited verb under class-III **sparēn-* (Kroonen 2013, 465). Orel similarly preserves **sparēnan* ‘spare’ (Orel 2003, 362). Ringe and Taylor, however, reconstruct **sparai-* ~ **sparja-* for the English branch and derive the citation verb from a class-II line (Ringe and Taylor 2014, 162, 191). The derivational input **spárōjaną* ‘spare’ therefore represents the refashioned class-II formation behind Old English *sparian* ‘spare’, while the citation reconstruction **sparēną* ‘spare’ remains the inherited comparative headword.

Old English evidence

Campbell says that *sparian* ‘spare’ does not show the ordinary class-III characteristics, but the Ritual forms, normalized here as *spæria* ‘spare’, *spær* ‘spare’, and *spærede* ‘spare’, together with Vespasian Psalter *spearad* ‘spare’, point to primitive Old English forms both with and without back vowels (Campbell 1959, § 764). Brunner likewise records Northumbrian *spæria* ‘spare’, *spærede* ‘spare’ beside common Old English *sparian* ‘spare’ and Vespasian Psalter *spearad* ‘spare’ (Brunner 1965, § 364 Anm. 11). The citation form treated here is *sparian* ‘spare’; the Anglian forms are relics of the older formation, not alternative headwords of equal status.

Development to Old English

Once the class-II formation **spárōjaną* ‘spare’ is chosen, the remaining development is regular. The regular derivation shows brightening, restoration of *a* before the back vocalism of the suffix, later i-mutation within the weak ending, weak-tail reduction, and contraction to *sparian* ‘spare’. By contrast, Brunner’s rule against further apocope of final *-e* explains why Ritual *spær* ‘spare’ cannot be the regular continuation of inherited **spárē* ‘spare’ (Brunner 1965, § 150).

Formation comparison

The comparison below sets the relevant forms side by side. It contrasts the inherited class-III formation with the refashioned class-II one that yields the citation verb.

Formation / comparison	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited class-III infinitive	**spárēną*	paradigm comparison / probe output: *sparen*	*sparian* ‘spare’	wrong class and wrong ending for the citation verb
inherited class-III imperative singular	**spárē*	paradigm comparison / probe output: *spære*	Ritual *spær* ‘spare’	loss of final *-e* is not regular, so the relic form cannot control the entry
inherited class-III finite present	**spárēθi*	paradigm comparison / probe output: *spæreþ*	*spearad* ‘spare’	attested form is mixed, not a direct continuation of the inherited cell
selected class-II formation	**spárōjaną*	documented regular output: *sparian* ‘spare’	*sparian* ‘spare’	exact match between derivational input and Old English citation form

8.25 staff — OE *stæf*

Derivation: citation reconstruction **stábiz*; form followed here **stábaz* > *stæf* (early analogy).

Derivation trace

Proto input: **stábaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**stáb*
EAF Final Z Deletion	**stába*	EAF Brightening	**stæb*
Early Anglo-Frisian		PGmc B Allophony	**stæβ*
[no change]			

Old English form: *stæf*

Reconstruction and comparative evidence

The comparative dictionaries do not give one uniform stem class. Kroonen reconstructs an a-stem **staba-* (Kroonen 2013, 471). Orel writes **stábiz* ‘staff’ ~ **stábaz* ‘staff’ (Orel 2003, 368). A direct i-stem input in **-iz* ‘staff’ would predict i-mutation in Old English, whereas the attested noun keeps *æ*.

Old English evidence

The Old English noun itself is the ordinary citation form *stæf* ‘staff’. Luick lists *stæf* ‘staff’ among closed monosyllables with *æ* (Luick 1914--1940, 176). Ringe and Taylor pair singular *stæf* ‘staff’ with plural *stafas* ‘staffs’ (Ringe and Taylor 2014, 193). The normalized form here is therefore *stæf*; later English *staff* with *a* belongs to a later stage of the word’s history.

Development to Old English

With the selected a-stem input, the development is regular. Final **-z* disappears, bare final *-a* is lost, Anglo-Frisian brightening gives *æ* in the closed monosyllable, and medial *b* surfaces as a fricative written *f*. The documented chain is **stábaz* ‘staff’ > **stába* ‘staff’ > **stáb* ‘staff’ > *stæb* ‘staff’ > *stæf* ‘staff’. A direct continuation of **stábiz* ‘staff’, by contrast, would produce i-mutated *stefe* ‘stave, staff’ rather than the attested singular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the rejected i-stem line from the selected a-stem input.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
comparative i-stem line	**stábiz*	expected *stefe* after i-mutation	*stæf*	wrong vowel for the attested singular
mixed comparative notation	Orel **stabiz* ~ **stabaz*; Kluge **stabi-*/a-	source-level stem-class uncertainty	*stæf*	useful comparative background, but not a single OE-facing input
selected a-stem input	**stábaz*	documented regular output: *stæf*	*stæf*	exact match between derivational input and Old English target

8.26 stem — OE *stefn*

Derivation: citation reconstruction **stámnaz*; form followed here **stámniz* > *stefn* (early analogy).

Derivation trace

Proto input: **stámniz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**stæβni*
EAF Final Z Deletion	**stámni*	OE I Umlaut	**steβni*
PNWGmc Mn Dissimilation	**stáβni*	OE High Vowel Apocope	**steβn*
Early Anglo-Frisian			
[no change]			

Old English form: *stefn*

Reconstruction and comparative evidence

Orel reconstructs the Proto-Germanic source as **stámnaz* ‘stem, trunk’ (also variant **stamniz* ‘stem, trunk’), citing Old Norse *stafn* ‘stem of a ship’, Old Frisian *stevene* ‘stem of a ship’ (fem.), and Old Saxon *stamn* ‘stem’ as its main continuants (Orel 2003, 371).

This word is etymologically unrelated to the Old English homonym *stefn* ‘voice, sound’ / *stemn* ‘voice, sound’, which descends from a distinct Proto-Germanic **stebnō* ‘voice, sound’ / **stemnō* ‘voice, sound’ etymon. The two OE lexemes are distinguished in Clark Hall, Bosworth-Toller,

Brunner, and Luick; they collide on the surface forms *stefn* and *stemn* through historically independent developments.

Kroonen relates the 'stem' family to the *stam(m)* 'stem, trunk' group attested in Old High German, Dutch, and German, without reconstructing a separate 'prow' proto-form (Kroonen 2013, 479–80).

Old English evidence

Clark Hall records *stefn* 'stem, trunk, root, prow, foundation', the weak n-stem *stefna* 'prow or stern of a ship', and the related *stofn* 'trunk, stem, branch, shoot' (Clark Hall 1960, 276, 341).

Luick §211 explicitly distinguishes OE *stefn* 'stem, trunk' / *stemn* 'stem, trunk' — noting cognates with Old Saxon *stamn* 'stem' and Middle English *stam* 'stem' — from the separate OE voice/sound word (*stefn* 'voice, sound' / *stemn* 'voice, sound') treated in §75 (Luick 1914--1940, § 211). Brunner §205 lists *stefn, stemn Stamm* among words showing the *fn*/*mn* alternation (Brunner 1965, § 205).

The primary comparison form used here is *stefn* 'stem, trunk, root, prow', the form attested most directly in the semantic range relevant to English *stem*. The n-stem *stefna* 'prow/stern' is a closely related but more narrowly attested nautical specialization.

Development to Old English

The derivation of OE *stefn* 'stem, trunk' from the i-stem input **stámniz* 'stem, trunk' (Orel's attested i-stem variant; the citation reconstruction remains **stámnaz* 'stem, trunk') is now regular and modelled, with multiplicity 1. Live trace under the corrected literal adjacent-*mn* SC022:

step	form
proto input	**stámniz*
EAF Final Z Deletion	**stámni*
PNWGmc Mn Dissimilation (adjacent mn > βn)	**stáβni*
EAF Brightening (á > æ)	**stæβni*
OE i-Umlaut (æ > e, from the i-stem ending)	**steβni*
OE High Vowel Apocope	**steβn*
Old English Orthography (β > f)	*stefn*

The consonantal change mn → *βn*/*fn* is attested comparatively: Old Norse *stafn* 'stem of a ship' and Old Saxon *stamn* 'stem' preserve the *fn*/*mn* variants expected from this family. The handbooks treat the change as a pre-Old-English development (Luick §211; Brunner §205), and the FST now encodes it as the historical adjacent *mn* > *βn* (the rule *PNWGmcMnDissimilation* / SC022). The *e* of *stefn* is regular from *á* via brightening (*á* > *æ*) plus i-umlaut triggered by the i-stem ending (*æ* > *e*), so both the vowel and the consonant follow by regular sound change once the i-stem input is selected.

Note the contrast with *heaven*. Here SC022 (*PNWGmcMnDissimilation*) fires directly inside the selected derivation, deriving the labial of *stefn* from the adjacent *-mn-* cluster of **stámniz*. In *heaven*, by contrast, the same historical change *mn* > *βn*/*fn* operated only in the deeper cluster-bearing oblique prehistory; the resulting labial was then generalized into the vowel-bearing **hebun-* stem, so the selected CAPR path **xébun* > *heofon* begins *after* that analogy and does not itself contain SC022 (see `dossier-heaven-paradigm-history-2026.md` §§13–15 and `audits/heaven-sc022-implementation-2026.md`). The earlier cross-syllable *mV...n* proxy — which had formerly fabricated a labial from an intervocalic *m* — has been retired in favour of this literal adjacent *mn* > *βn*. Selecting the i-stem input **stámniz* over the a-stem citation **stámnaz* is a pre-OE input selection; once selected, the Old English development is regular. Hence the classification *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply), not *known_unmodelled*.

Homonym note

Old English *stefn*/*stemn* is a surface homonym for two etymologically unrelated words:

1. *stefn*/*stemn* 'voice, sound': from PGmc **stebnō* (Ringe-Taylor, Orel); via *bn* → *fn* (b-allophony), giving *stefn* 'voice, sound'. This is a different row from 2216.
2. *stefn*/*stefna*/*stofn*/*stemn* 'stem, trunk, prow': from PGmc **stámnaz* 'stem/trunk' (Orel); via the pre-OE cluster change *mn* → *fn*. This is row 2216.

The form followed here, **stébnō*, used in earlier versions of this entry was the wrong homonym's transponent and must not be used here.

Source comparison

Form or label	Status	OE relation	Result
**stámnaz* 'stem, trunk'	Orel's PGmc a-stem citation for the stem/trunk/prow family	lexeme-level citation	retained as PROTO (citation)
**stámniz* 'stem, trunk'	Orel's attested i-stem variant	**selected derivational input**	regular OE output *stefn* (mult 1)
stefn III 'stem, trunk, root, prow'	primary OE target (strong masc.)	stem/trunk sense per Clark Hall	target form
stefna 'prow/stern'	OE n-stem specialization	nautical sense per Clark Hall	comparison form
stofn 'trunk'	OE *o*-grade variant or earlier stage	trunk/stem sense	comparison form
stemn 'trunk'	late West Saxon *fn* → *mn* doublet	secondary form	comparison form
**stébnō* (voice word)	**wrong homonym** — belongs to voice/sound dossier	no relation to stem/trunk sense	**must not be used as row 2216's derivational input**

8.27 swan — OE *swanes*

Derivation: citation reconstruction **swánaz*; form followed here **swánas* > *swanes* (early analogy).

Derivation trace

Proto input: **swánas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**swánæs*
[no change]	OE Unstressed AE Merger	**swánes*
Early Anglo-Frisian		
[no change]		

Old English form: *swanes*

Reconstruction and comparative evidence

The Germanic noun is ordinarily cited as the masculine a-stem **swánaz* 'swan' (Orel 2003, 367). The derivational input **swánas* 'swan' is not a competing lexeme reconstruction. It is the genitive singular **swánas* of the same paradigm.

The question here is therefore one of paradigm cell rather than stem history. The citation form remains **swánaz* ‘swan’ > *swan* ‘swan’; the comparison form is the genitive singular **swánas* ‘swan’ > *swanes* ‘swan’.

Old English evidence

Bright’s glossary records the ordinary noun as *swan* ‘swan’, m. and also gives the exact inflected form *swanes* ‘swan’, citing the phrase *swanes feðre* (Bright 1971, 441).

The target is therefore an attested Old English genitive singular, not a reconstruction or the ordinary citation lemma.

Development to Old English

From **swánas* ‘swan’, Anglo-Frisian brightening gives **swánæs* ‘swan’, and subsequent merger of unstressed *æ* with *e* yields *swanes* ‘swan’. The comparison is straightforward once the genitive singular is chosen as the relevant cell.

Paradigm-cell comparison

The comparison below sets the relevant forms side by side. It separates the ordinary citation form from the selected inflected cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**swánaz*	OE headword *swan* ‘swan’	*swan* ‘swan’	ordinary lexeme line
genitive singular	**swánas*	regular output: *swanes* ‘swan’	*swanes* ‘swan’	attested cell

8.28 thousand — OE *þūsend*

Derivation: citation reconstruction **θūs-undī*; form followed here **θūs-èndi* > *þūsend* (early analogy).

Derivation trace

Proto input: **θūsèndi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Strip Secondary Stress	**θūsend*
PWGmc Early I Apocope	**θūsènd*		
Early Anglo-Frisian			
[no change]			

Old English form: *þūsend*

Reconstruction and comparative evidence

Kroonen reconstructs the Germanic numeral as **þūsundī-* and cites Old English *þūsend* ‘thousand’ among its continuations (Kroonen 2013, 554). The derivational input **θūs-èndi* ‘thousand’ is not the same claim. It is an OE-oriented transponent with the second-member vowel already resolved to *e* and the final high vowel already shortened for apocope.

The chronology must explain why Old English shows *þūsend* ‘thousand’ while related languages such as Old Saxon and Old High German keep *u* in the second syllable? (Kroonen 2013, 554).

Old English evidence

Old English *þūsend* ‘thousand’ is an ordinary citation form, not a selected oblique or paradigm cell. Campbell treats it as a neuter noun with normal case forms (Campbell 1959, § 689). The problem lies in the internal history of the word, not in its lexical status.

Development to Old English

If the old final *-ī* had remained long enough to trigger ordinary double umlaut, Campbell’s rule would point toward a form of the **þȳsend* ‘thousand’ type rather than attested *þūsend* ‘thousand’ (Campbell 1959, § 203). Preserved root *ū* therefore argues that the umlaut-triggering vowel was lost or neutralized before the ordinary OE umlaut could develop.

That early loss, however, does not by itself explain the medial *e*. Luick compares the word with *ærende* ‘errand/message’ and later groups *thousand* with forms reshaped on that pattern (Luick 1914--1940, §§ 198, 492). Viredaz is more cautious, arguing that Old English *e* in this weak position may simply write schwa and so need not prove a unique *ærende* ‘message’-type analogy (Viredaz 2025, § 2.1.4).

The selected transponent **θūs-èndi* ‘thousand’ captures the OE-side state from which the regular derivation reaches *þūsend* ‘thousand’.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the secure chronology from the more interpretive account of the second-syllable vowel.

Stage / interpretation	Candidate form	OE relation	Result
surviving *-ī* with ordinary double umlaut	**þūsundī-* treated as still umlaut-active in OE	would point toward **þȳsend*	excluded by preserved *ū*
early loss of the trigger without further reshaping	**þūsund-* type	explains *ū*, but not why OE alone has medial *e*	incomplete account
selected OE-oriented transponent	**θūs-èndi*	regular output: *þūsend* ‘thousand’	selected modeling input

8.29 timber — OE *timber*

Derivation: citation reconstruction **tímrą*; form followed here **tímbrą* > *timber* (early analogy).

Derivation trace

Proto input: **tímbrą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tímbr*
[no change]	OE Epenthetic Vowel	**tímber*
Early Anglo-Frisian		
[no change]		

Old English form: *timber*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **timbra-* and cites Old English *timber* ‘timber’ among its continuations (Kroonen 2013, 517). Ringe and Taylor instead state the history from **timra* ‘timber’ through West Germanic **timbr* ‘timber’ to Old English *timber* ‘timber’ (Ringe and Taylor 2014, 327).

The difference is therefore not over the Old English noun itself. It concerns whether medial *b* belongs in the comparative citation form or appears in an early pre-Old-English stage of the cluster.

Old English evidence

Clark Hall lemmatizes the noun as *timber* ‘timber’ and also records *timbor* ‘timber’ as a variant spelling (Clark Hall 1960, 294). The Old English form is thus an ordinary citation noun, not a selected oblique cell or a reconstructed convenience form.

Development to Old English

With the consonantal frame **timbr-* ‘timber’ in place, the rest of the development is straightforward. Loss of final *-ą* leaves **tímbr* ‘timber’, and epenthetic *e* in the final cluster yields *timber* ‘timber’. Ringe and Taylor’s *timra* > *timbr* > OE *timber* and the handbook treatment of this epenthetic vowel point to the same Old English result (Ringe and Taylor 2014, 327; Campbell 1959, §§ 463–464).

Formation comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing consonantal input.

Formation or notation	Candidate form	OE relation	Result
Kroonen’s comparative citation	**timbra-*	already matches the consonantal frame of OE *timber* ‘timber’	closest comparative support for the derivational input
Ringe-Taylor citation line	**timra* > **timbr* ‘timber’	reaches the same OE noun through early cluster expansion	compatible comparative background
modeled input	**tímbrą*	regular output: *timber* ‘timber’	Old English-facing input

8.30 wake — OE *wacan*

Derivation: citation reconstruction **wakēną*; form followed here **wákaną* > *wacan* (early analogy).

Derivation trace

Proto input: **wákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wækaną*
[no change]	OE A Restoration	**wakaną*
	OE Heavy Syllable Nasal Apocope	**wakan*
Early Anglo-Frisian	OE Secondary Nasalization	**wakąn*
[no change]	OE Weak Tail Reduction	**wakan*

Old English form: *wacan*

Reconstruction and comparative evidence

Kroonen gives the strong verb as **wakan-* with Old English *wacan* ‘wake’ (Kroonen 2013, 568). Ringe and Taylor separately derive Old English *wacian* ‘wake’ from weak **wakai-* ~ **wakja-* (Ringe and Taylor 2014, § 3.3.2).

The difference is therefore lexical and class-based, not graphic. Strong *wacan* ‘wake’ ‘wake up, arise’ and weak *wacian* ‘wake’ ‘be awake, watch’ belong to related but distinct histories.

Old English evidence

Clark Hall lists *wacan* ‘wake’ and *wacian* ‘wake’ as separate headwords (Clark Hall 1960, 338). Bosworth-Toller cautions under *wacan* ‘wake’: the simplex infinitive itself does not occur, its place seeming to be taken by *wæcnan* ‘wake’ (Bosworth and Toller 1898, 226).

The target *wacan* ‘wake’ is therefore best understood as a normalized strong headword for the verb family, not as a directly quoted simplex infinitive. It still remains the correct Old English comparison form for the strong branch.

Development to Old English

With strong **wákaną* ‘wake’, Anglo-Frisian brightening first gives a form of the **wækaną* ‘wake’ type. A-restoration then returns *a*, and the ordinary tail reductions yield *wacan* ‘wake’. The weak verb *wacian* ‘wake’ belongs to a different prehistory and is not the expected outcome of this input.

Class comparison

The comparison below sets the relevant forms side by side. It separates the strong and weak verb lines.

Formation / class	Candidate input	OE outcome or comparison	Result
weak class-III / class-II branch	**wakēną*, **wakai-* ~ **wakja-*	OE *wacian* ‘wake’ and related weak forms	related lexeme, but not the target of this entry
strong class-VI branch	**wákaną*	regular output: *wacan* ‘wake’	Old English-facing input
strong normalized headword	*wacan* ‘wake’	dictionary comparison form beside attested strong-family forms	correct Old English comparator, though not a directly quoted simplex infinitive

8.31 water — OE *wæter*

Derivation: citation reconstruction **wátną*; form followed here **wátōr* > *wæter* (early analogy).

Derivation trace

Proto input: **wátōr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**wátær*
PWGmc Final Or Lowering	**wátar*	EAF Brightening	**wætær*
Early Anglo-Frisian		OE Unstressed AE Merger	**wæter*
[no change]			

Old English form: *wæter*

Reconstruction and comparative evidence

Kroonen reconstructs a heteroclitic noun **watar-* ~ **watan-* and states that the Proto-Germanic material points to **watōr* ‘water’ and **watenaz* ‘water’ (Kroonen 2013, 616). Ringe and Taylor likewise start from singular **wator* ‘water’ before the Old English branch (Ringe and Taylor 2014, § 3.1.4).

The generalized comparative label is therefore broader than the singular form that actually corresponds to Old English *wæter* ‘water’. The relevant comparator is the inherited nominative-accusative singular **wátōr* ‘water’.

Old English evidence

Bright gives the noun as *wæter* ‘water’ with the regular paradigm *wæteres* ‘water (gen.)’, *wætere* ‘water (dat.)’, *wæter(u)*, *wætera* ‘water (gen.pl.)’, *wæterum* ‘water (dat.pl.)’ (Bright 1971, 29). Ringe and Taylor add the dialectal contrast between West Saxon *weeter* ‘water’ and Mercian *weter* ‘water’ (Ringe and Taylor 2014, § 6.5.2).

The target is therefore an attested Old English citation form within a normal paradigm. The complication lies on the comparative side of the lexeme, not in Old English attestation.

Development to Old English

From **wátōr* ‘water’, pre-final *ō* becomes *a*, yielding **watar* ‘water’ (Ringe and Taylor 2014, § 3.1.4). Anglo-Frisian brightening then gives **wætær* ‘water’, and merger of unstressed *æ*/*e* yields *wæter* ‘water’.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the generalized lexeme label from the singular input that matches the Old English citation form.

Stage or notation	Candidate form	OE relation	Result
generalized comparative label	**wátnǫ*	broader lexeme shorthand, not the singular that corresponds directly to *wæter* ‘water’	useful background only
heteroclitic stem notation	**watar-* ~ **watan-*	source-faithful comparative reconstruction	explains why a singular comparator is needed
inherited singular input	**wátōr*	regular output: *wæter* ‘water’	Old English-facing input

8.32 whale — OE *hwæl*

Derivation: citation reconstruction **wálaz*; form followed here **xwálaz* > *hwæl* (early analogy).

Derivation trace

Proto input: **xwálaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xwál*
EAF Final Z Deletion	**xwála*	EAF Brightening	**xwæl*
Early Anglo-Frisian			
[no change]			

Old English form: *hwæl*

Reconstruction and comparative evidence

The comparative sources are not uniform. Orel gives **xwalaz* ‘whale’ and notes some mixed **xwaliz* ‘whale’ evidence (Orel 2003, 197). Kroonen instead cites **hwali-* (Kroonen 2013, 262).

Both notations agree on inherited initial *hw-/xw-*, but they differ in stem label. The a-stem-like input followed here is closer to Orel’s notation than to Kroonen’s citation form.

Old English evidence

Clark Hall lemmatizes the noun as *hwal* ‘whale’, and Bosworth-Toller preserves plural *hwalas* ‘whales’ (Clark Hall 1960, 170; Bosworth and Toller 1898, 326). The comparison form is normalized here as *hwæl* ‘whale’ for the singular citation form with Anglo-Frisian fronting.

The plural *hwalas* ‘whales’ supplies control evidence. It shows the same lexeme with *a* in an open syllable beside singular *hwæl* ‘whale’ in a closed monosyllable.

Development to Old English

From **xwálaz* ‘whale’, final *-z* disappears and bare final *-a* is lost. Anglo-Frisian fronting then yields *æ* in the closed monosyllable, and Old English orthography writes *hwæl* ‘whale’.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative notations from the normalized Old English singular.

Comparative line	Candidate form	OE relation	Result
Orel’s citation	**xwalaz*	same stem notation as the modeled singular line	closest comparative support for the derivational input
Kroonen’s citation	**hwali-*	same initial cluster, different stem label	important comparative rival, but not the notation followed here
modeled input	**xwálaz*	regular output: *hwæl* ‘whale’	Old English-facing input
plural control	*hwalas* ‘whales’	attested open-syllable plural beside singular *hwæl* ‘whale’	confirms that the lexeme also preserves an *a*-vocalism branch

8.33 whine — OE *hwīnan*

Derivation: citation reconstruction **wainōjanǫ*; form followed here **xwḯnanǫ* > *hwīnan* (early analogy).

Derivation trace

Proto input: **xwḯnanǫ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xwḯnan*
[no change]	OE Secondary Nasalization	**xwḯnǫn*
	OE Weak Tail Reduction	**xwḯnan*
Early Anglo-Frisian		
[no change]		

Old English form: *hwīnan*

Reconstruction and comparative evidence

The citation reconstruction preserved in the header belongs to the lament-family verb seen in German *weinen* ‘whine’ and Old English *wānian* ‘wane’. Kroonen instead separates Old English *hwīnan* ‘whine’ under **hwinan-* (Kroonen 2013, 267). Orel likewise distinguishes strong **xwinanan* ‘whine’ from weak **wainōjanan* ‘whine’ (Orel 2003, 201). Ringe and Taylor make the same split at the Northwest Germanic level, linking Old Norse *hvina* ‘whine’ and Old English *hwinan* ‘whine’ to the same strong verb (Ringe and Taylor 2014, 130).

The two families also differ phonologically and morphologically. The lament family has initial *w-*, diphthongal *ai*, and weak-II morphology, whereas the verb behind Old English *hwīnan* ‘whine’ has initial *hw-/xw-*, long *ī*, and strong-verb inflection. The derivational input **xwḯnanǫ* ‘whine’ therefore represents a competing comparative identification rather than a hidden cell of **wainōjanǫ* ‘whine’.

Old English evidence

Clark Hall records *hwinan* ‘whine’ with the gloss ‘to hiss, whizz, whistle’ (Clark Hall 1960, 171). Seebold keeps the verb among the strong verbs and notes that only a present-tense attestation is directly preserved (Seebold 1970, 280).

The Old English form is normalized here as *hwīnan* ‘whine’. That normalization adds the usual vowel length marking to the dictionary spelling *hwinan* ‘whine’; it does not turn an unattested verb into a reconstructed one.

Development to Old English

Once the strong-verb input **xwḯnanǫ* ‘whine’ is selected, the path to Old English is straightforward. The compact trace shows heavy-syllable nasal apocope, secondary nasalization, and weak-tail reduction, after which the form surfaces as *hwīnan* ‘whine’.

No special paradigm maneuver is needed for this verb. The comparison is between two different Germanic verb families: the Old English form belongs with the strong verb **hwīnan-*, not with the weak lament verb.

Verb-family comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative labels that stand behind the inherited Old English forms.

Verb family / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lament-family weak verb	**wainōjaną*	comparative continuation in OE *wānian* ‘wane’	*wānian* ‘wane’	competing citation reconstruction, but not the source of *hwīnan*
selected strong verb	**xwī́naną*	regular output: *hwīnan* ‘whine’	*hwīnan* ‘whine’	exact match between derivational input and OE verb
comparative North Germanic cognate	Northwest Germanic strong verb behind ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	*hwīnan* ‘whine’	supports the strong-verb identification

8.34 withy — OE *wīþiġ*

Derivation: citation reconstruction **wáiθiz*; form followed here **wī́θagą* > *wīþiġ* (early analogy).

Derivation trace

Proto input: **wī́θagą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**wī́θægą*
[no change]	OE Heavy Syllable Nasal Apocope	**wī́θæg*
Early Anglo-Frisian	OE Velar Palatalization	**wī́θæj*
[no change]	OE Unstressed AE Merger	**wī́θej*
	OE Late Unstressed Ag Suffix	**wī́θij*
	OE Palatal Fricative Merger	**wī́θij*

Old English form: *wīþiġ*

Reconstruction and comparative evidence

The comparative evidence groups the word with Germanic forms of the **wīþja* ‘withy’ (or **wīþjō-* variant) or **wiþ-* type (Orel 2003, 503). These forms establish the cognate set but do not by themselves explain the Old English suffix of *wīþiġ* ‘withy’.

For Old English, the relevant point is the suffix history. Campbell’s account of OE *-ig*, including forms such as *hunig*, supports an analysis in which the *-iġ* of *wīþiġ* ‘withy’ continues a derivational **-ag-* sequence rather than a heavy ja-stem **-ij-* (Campbell 1959, §§ 275, 376). The derivational input **wī́θagą* ‘withy’ is thus a formation choice rather than a mere respelling of the comparative headword.

Old English evidence

Clark Hall records the noun as *wiðig* ‘withy’ (Clark Hall 1960, 358). The form used here, *wīþiġ* ‘withy’, is a normalized Old English spelling with macrons and palatal ġ marked explicitly.

The relevant comparison form is therefore not a reconstructed dictionary convenience but an established Old English noun. What requires explanation is why the selected Proto-Germanic input is **wíθagą* ‘withy’ rather than a comparative headword of the **wīþja* ‘withy’ type.

Development to Old English

From **wíθagą* ‘withy’, the retained unstressed-fronting component gives the suffixal front vowel. Later raising recognizes the singleton palatal fricative, distinct from gg/ng stops; this late suffix history is not prehistoric i-mutation (Ringe and Taylor 2014, 334–35; Fulk 2018, 130–32). Post-mutation merger and native realization supply the final *ġ*, reaching *wīþiġ* ‘withy’.

This derivation is regular for the form compared here. The central claim of the entry is therefore morphological: Old English *wīþiġ* ‘withy’ belongs with an **-ag-* derivative, whereas the comparative **wīþja* ‘withy’ label belongs to a different way of presenting the cognate family.

Formation comparison

The comparison below sets the relevant formations side by side. It distinguishes the comparative headword from the Old English-facing formation that actually yields the attested noun.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative family label	**wáiθiz*	broader cognate-set headword	OE family context	useful lexeme label, but not the direct source of *wīþiġ*
heavy ja-stem analysis	**wīþja* ‘withy’ type	Campbell/Adamczyk-style heavy ja-stem *-e* / zero outcome	*wīþiġ* ‘withy’	does not account cleanly for the OE suffix
**-ag-* derivative followed here	**wíθagą*	regular output: *wīþiġ* ‘withy’	*wīþiġ* ‘withy’	exact match between formation and target

8.35 world — OE *weorold*

Derivation: citation reconstruction **wíra-àldiz*; form followed here **wír-àldu* > *weorold* (early analogy).

Derivation trace

Proto input: **wíràldu*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Inter Stress Raising	**wéruldu*
PNWGmc I Lowering	**wéràldu*	OE Back Mutation	**wéoruldu*
Early Anglo-Frisian		OE High Vowel Apocope	**wéoruld*
		OE Med Unstressed U Lowering	**wéorold*
[no change]			

Old English form: *weorold*

Reconstruction and comparative evidence

The word is the old compound ‘age of men’. Orel reconstructs an already mid-vowelled compound, and Ringe and Taylor discuss the form **weraldiz* ‘world’ and its pre-Old-English chain **weraldu* ‘world’ > **weruld* ‘world’ (Orel 2003, 462; Ringe and Taylor 2014, 326). Kluge-Seebold gives the compound **wira-aldō* ‘world’ and explicitly includes Old English *weorold* ‘world’ (Kluge 2011, 981).

The derivational input **wír-àldu* ‘world’ therefore combines the older **wir-* vowel of the comparative headword with the early shift of the compound into the ō-stems that Ringe and Taylor note for this lexeme (Ringe and Taylor 2014, 326). The early analogical step lies in that stem-class reassignment; the later phonological developments can then run regularly.

The form compared here is a transponent combining an earlier first-element vowel with a later inflectional formation, not an independently quoted single historical stage. An already **wer-* compound requires no lowering in its own derivation. Accordingly the selected transponent cannot prove that SC015 PNWGmcILowering historically preceded SC036 OEInterStressRaising. The adopted consonantal lowering has no following-vowel trigger for interstress raising to remove. The two processes remain compatible in this derivation without that supposed ordering argument (Orel 2003, 462; Ringe and Taylor 2014, 326; Hill 2017, 153–56).

Old English evidence

Old English does not preserve a single isolated form. Ringe and Taylor give West Saxon *weorold* ‘world’ ~ *worold* ‘world’, Mercian *weoruld* ‘world’, Northumbrian *woruld* ‘world’, and Kentish *wiarald* ‘world’ (Ringe and Taylor 2014, 326). Sievers-Brunner and Bright present the same wider set, including the syncopated *world* ‘world’ and later rounded *wurold* ‘world’ (Brunner 1965, § 113; Bright 1971, 465).

The Old English form used here is the West Saxon form *weorold* ‘world’. It is an attested Old English form within that broader variant cluster, not the only form the lexeme ever shows.

Development to Old English

From the derivational input **wír-àldu* ‘world’, Northwest Germanic *i*-lowering gives **wér-àldu* ‘world’. Inter-stress raising then changes the medial *a* to *u*, producing **wér-uldu* ‘world’. In the Old English branch that unstressed *u* lowers to *o*, and back mutation yields **wéor-oldu* ‘world’; final high-vowel apocope then gives *weorold* ‘world’.

This sequence matches the comparative background in Ringe and Taylor’s literature-stage chain for the word while preserving the **wir-* notation of the comparative label (Ringe and Taylor 2014, 326). The modeled Old English form therefore stands at the meeting point of an early stem-class reshaping and later regular sound change.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing stage chosen for the derivation.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
comparative compound with older first-element vowel	**wíra-àldiz*	citation reconstruction / lexeme label	preserves the older **wir-* tradition of the compound
literature-stage lowered compound after early stem-class shift	**weraldiz* > **weraldu* > **weruld*	Ringe-Taylor background chain to OE *weorold* ‘world’ ~ *worold* ‘world’	explains the older comparative literature cited for the word

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
Old English-facing input	**wír-àldu*	regular output: *weorold* ‘world’	exact match for the West Saxon form used here
broader OE variant cluster	—	*worold* ‘world’, *weoruld* ‘world’, *woruld* ‘world’, *wiarald* ‘world’, *world* ‘world’	real attested comparanda that remain outside that West Saxon line

8.36 youth — OE *ġeoguþ*

Derivation: citation reconstruction **júgunθiz*; form followed here **júgunθ* > *ġeoguþ* (early analogy).

Derivation trace

Proto input: **júgunθ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**júguθ*
EAF Nasal Spirant Lengthening	**júgūnθ*		
EAF Nasal Spirant Loss	**júgūθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *ġeoguþ*

Reconstruction and comparative evidence

The wider etymological tradition reconstructs an earlier form of the word as **ju(w)unþi-*. The comparative label **júgunθiz* ‘youth’ already stands at a later Germanic stage with *g*, and the derivational input **júgunθ* ‘youth’ is later again: it represents the form after final *-i* has been lost.

Ringe and Taylor explicitly give the sequence **jugunþi* ‘youth’ > **juguþ* ‘youth’ > OE geoguþ ~ iuguþ (Ringe and Taylor 2014, 141). The derivational input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final *-i*.

Old English evidence

The Old English noun is attested with varying spellings. Ringe and Taylor cite *geoguþ* ‘youth’ ~ *iuguþ* ‘youth’ (Ringe and Taylor 2014, 141). The form is normalized here as *ġeoguþ* ‘youth’: the initial palatal is written with *ġ*, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun *ġeoguþ* ‘youth’ itself.

Development to Old English

The decisive early step is the loss of final *-i* before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted *y*-type vowel instead of the attested form (Ringe and Taylor 2014, 141).

From the derivational input **júgunθ* ‘youth’, the later development is regular: nasal-spirant lengthening and loss give **júgūθ* ‘youth’ (Fulk 2018, 109); unstressed long-vowel shortening then produces **júguθ* ‘youth’. The root vowel remains phonological *u* through medial-vowel lowering; late West Saxon spelling, not early palatal fronting, yields *ġeoguþ* ‘youth’ (Ringe and Taylor 2014, 5; Brunner 1965, 64–65, §92.1). Campbell preserves *u* after accented *u* in forms such as *duguþ* ‘troop’ and *munuc* ‘monk’ (Campbell 1959, 156, §373). Brunner likewise cites *iuzuð* ‘youth’ and *munuc* ‘monk’ under the same development (Brunner 1965, § 150.3).

Stage comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the later stages relevant to the Old English noun.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
earlier etymological headword	**ju(w)unþi-*	comparative family background	older comparative reconstruction of the lexeme
later g-bearing comparative label	**júgunθiz*	citation reconstruction / lexeme label	preserves the later Germanic stage behind the selected entry
Old English-facing input	**júgunθ*	regular output: *ġeoguþ*	exact match for the Old English form used here
full *-i* stage retained too long	**jugunþi*	expected over-umlauted *y*-type result	negative control showing why early *-i* loss must precede the OE umlaut stage

Chapter 9

Late analogy and paradigm-cell selection

These Old English forms continue a particular paradigm cell or a later analogical remodeling, not the unaltered citation form. The phonology may be regular once the proper morphological history has been identified.

9.1 ban — OE *bannes*

Derivation: citation reconstruction **bánną*; form followed here **bánnas* > *bannes* (late analogy).

Derivation trace

Proto input: **bánnas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**bánnæs*
[no change]	OE Unstressed AE Merger	**bánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *bannes*

Reconstruction and comparative evidence

Orel cites a bann-noun under **bannan* ‘ban’, while Seebold distinguishes bann-stems of both masculine and neuter type and gives Old English *gebann* ‘ban’ as the noun reflex (Orel 2003, 35; Seebold 1970, 89). The citation reconstruction **bánną* ‘ban’ names the lexeme, but the comparison here turns on the genitive singular **bánnas* ‘ban’.

The analysis therefore depends on medial, not final, gemination.

Old English evidence

Old English lexicographic evidence securely supports the noun itself. Bosworth-Toller records the noun under nominative *ge-bann* ‘ban’, with oblique usage such as *gebanne* ‘ban’ (Bosworth and Toller 1898, 303). The exact unprefixed genitive *bannes* ‘ban’ is less directly cited in the dictionaries, so it is best treated here as the regular genitive form used for comparison rather than as a dictionary headword.

Development to Old English

From **bánnas* ‘ban’, the geminate remains medial before the case ending and the unstressed vowel develops regularly to give *bannes* ‘ban’. The paradigm comparison therefore sets the genitive against nominative *ban* ‘ban’, the ordinary nominative form of the same noun, rather than against a directly cited genitive headword.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows why the genitive singular is the conservative cell used for the entry.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bánną*	regular output: *ban*	ban	regular nominative outcome, but not the Old English form here
genitive singular	**bánnas*	regular output: *bannes*	*bannes*	direct match for the conservative genitive

9.2 berry — OE *berġes*

Derivation: citation reconstruction **bázją*; form followed here **bázjas* > *berġes* (late analogy).

Derivation trace

Proto input: **bázjas*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**bárjæs*
EAF Rhotacism	**bárjas*	EAF Brightening	**bærjæs*
Early Anglo-Frisian		OE I Umlaut	**berjæs*
		OE Unstressed AE Merger	**berjes*
[no change]			

Old English form: *berġes*

Reconstruction and comparative evidence

Kroonen reconstructs the berry noun as **basja- ~ *bazja-* (Kroonen 2013, 54). The derivational input **bázjas* ‘berry’ is therefore not a rival lexeme headword, but a specific genitive singular cell drawn from that paradigm.

The relevant point is that **rj* did not geminate in Proto-West Germanic. Ringe and Taylor’s *here* ‘army’, *herges* ‘army’s’ comparison shows the same *rj* environment in an Old English paradigm without any hidden gemination repair (Ringe and Taylor 2014, 181).

Old English evidence

Campbell cites feminine *berige* ‘berry’ and notes that *-j-* is retained after *r* in this type (Campbell 1959, 250). The reviewed evidence therefore supports the citation form more directly than the exact genitive *berġes* ‘berry’, which is best read here as the regular genitive comparison form rather than as a dictionary headword.

Development to Old English

Citation **bázją* ‘berry’ gives *bere* ‘berry (variant)’, not the Old English form here. The genitive singular **bázjas* ‘berry’, however, gives *berġes* ‘berry’, with medial *-rġ-* preserved in the same way that Ringe and Taylor cite *herges* ‘army’s’ beside *here* ‘army’ (Ringe and Taylor 2014, 181). This points to paradigm choice rather than to an extra phonological rule.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bázją*	regular output: *bere*	*berige* / *berġe*	useful citation-form background, but not the Old English form here
genitive singular	**bázjas*	regular output: *berġes*	*berġes*	exact match for the conservative cell

9.3 bore (3sg) — OE *boraþ*

Derivation: citation reconstruction **burōną*; form followed here **búrōθi* > *boraþ* (late analogy).

Derivation trace

Proto input: **búrōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Late O Shortening	**bóroθ*
PWGmc Early I Apocope	**búrōθ*	OE Final Unstressed O Lowering	**bóraθ*
PNWGmc U Lowering	**bórōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *boraþ*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* ‘bore’ among its reflexes (Kroonen 2013, 85). The form compared here isolates the finite 3sg cell **búrōθi* ‘bores’ rather than the infinitive.

Campbell’s class-II pattern lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor’s account of stable class-II *a* in finite forms make this 3sg cell the relevant comparison form for the ending (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Old English evidence

Clark Hall lemmatizes the verb as *borian* ‘bore’; the imperative singular *bora* and present-third-singular *boraþ* ‘bores’ are the relevant comparison forms (Clark Hall 1960, 48).

Campbell's dialect survey allows secondary *-e-* and *-o-* spellings in 2sg and 3sg class-II forms, but the basic ending remains *-aþ* (Campbell 1959, § 757). The regular comparison form for this non-*j* 3sg cell is *boraþ* 'bores'.

Development to Old English

From **búrōθi* 'bores', loss of final *-i* first gives **búrōθ* 'bores'. Northwest Germanic lowering then produces **bórōθ* 'bores', and late shortening of unstressed *ō* yields **bóraθ* 'bores'. *Boraþ* 'bores' is therefore the regular comparison form for this non-*j* 3sg cell.

Campbell's class-II ending evidence and Ringe and Taylor's discussion of stable *a* in finite non-*j* cells support this sequence (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the bore-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjanǫ*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
3sg present	**búrōθi*	regular output *boraþ* 'bores'	*boraþ* 'bores'	exact match
imperative singular companion	**búrô*	related finite form *bora* 'bore'	*bora* 'bore'	useful control, but not the target

9.4 bore (iptv.2sg) — OE *bora*

Derivation: citation reconstruction **burōną*; form followed here **búrô* > *bora* (late analogy).

Derivation trace

Proto input: **búrô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**bóra*
PNWGmc U Lowering	**bórô*		
Early Anglo-Frisian			
[no change]			

Old English form: *bora*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* 'bore' among its continuants (Kroonen 2013, 85). Ringe and Taylor give the class-II weak imperative singular ending as -a < **-ō* (Ringe and Taylor 2014, 314).

The derivational input **búrô* 'bore (iptv.2sg)' is therefore an imperative cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *borian* ‘bore’ (Clark Hall 1960, 48). The comparison form here is the normalized imperative singular *bora* ‘bore (iptv.2sg)’, used beside the headword and the related 3sg form *boraþ* ‘bores’.

The imperative is thus a paradigm form rather than a replacement for the dictionary lemma. It is the most direct Old English comparator for the non-*j* finite cell represented by **búrô* ‘bore (iptv.2sg)’.

Development to Old English

Northwest Germanic lowering first gives **bórô* ‘bore (iptv.2sg)’ from **búrô* ‘bore (iptv.2sg)’, and late shortening of the unstressed long vowel then yields **bóra* ‘bore (iptv.2sg)’, whence *bora* ‘bore’.

Ringe and Taylor’s class-II imperative singular -a < **-ō* points to exactly this type of outcome (Ringe and Taylor 2014, 314). The form compared here therefore isolates the regular finite-cell development more cleanly than the remodelled infinitive does.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the bore-family lemma and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
imperative singular	**búrô*	regular output *bora* ‘bore’	*bora* ‘bore’	exact match
3sg present companion	**búrōθi*	related finite form *boraþ* ‘bores’	*boraþ* ‘bores’	useful control, but not the target

9.5 bow — OE *bēag*

Derivation: citation reconstruction **béuganą*; form followed here **báug* > *bēag* (late analogy).

Derivation trace

Proto input: **báug*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**báeug*
[no change]	OE Diphthong Leveling	**bēag*
Early Anglo-Frisian		
[no change]		

Old English form: *bēag*

Reconstruction and comparative evidence

The inherited verb belongs to the class-II strong-verb family **béuganą* ‘bow’ (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The derivational input **báug* ‘bow’ is the singular preterite cell, whereas the citation form **béuganą* is the infinitive.

Campbell’s account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag* ‘bowed’, whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* ‘to bow’ (Ringe and Taylor 2014, 55).

Old English evidence

Bosworth-Toller and Clark Hall both record *bēag* ‘bowed’ as a preterite form of *būgan* ‘to bow’ (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan* ‘to bow’, but the relevant comparison form for this entry is the singular preterite *bēag* ‘bow, ring’ ‘bowed’. That is the paradigm cell in which the inherited **au* grade is preserved most directly.

Development to Old English

From **báug* ‘bow’, Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* ‘bow, ring’ (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan* ‘bow, bend’. The derivational input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**béuganą*	inherited present-stem history behind *būgan*	*būgan*	establishes the lexeme, but not the Old English form here
singular preterite	**báug*	regular output: *bēag*	*bēag*	exact match between input, output, and attested cell
past participial branch	participial **bugan-* type	later participial outcomes	bogen-type evidence	relevant to the paradigm, but not the clearest match for this entry

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag* ‘bow, ring’, while the citation form *būgan* ‘bow, bend’ belongs to a paradigm whose present stem has already undergone later leveling.

9.6 cow — OE *cȳ*

Derivation: citation reconstruction **kōz*; form followed here **kūi* > *cȳ* (late analogy).

Derivation trace

Proto input: **kūi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**kȳi*
[no change]	OE High Vowel Apocope	**kȳ*
Early Anglo-Frisian		
[no change]		

Old English form: *cȳ*

Reconstruction and comparative evidence

Kroonen reconstructs a root noun with the inherited alternation **kō- ~ *ku-*, explicitly nom. *kōz ‘cow’, obl. **kū-* (Kroonen 2013, 299). The citation form therefore belongs to the nominative singular, whereas the derivational input **kūi* ‘cow’ belongs to the oblique stem.

Ringe and Taylor also posit a later PNWGmc nominative **kūaz* ‘cow’ > **kūz* ‘cow’ behind Old English *cū* ‘cow’ (Ringe and Taylor 2014, § 3.1.3). That nominative history accounts for the headword, whereas the form cȳ ‘cow’ depends on the oblique **kū-* stem and a following **i*.

Old English evidence

Clark Hall lemmatizes the noun under *cū* ‘cow’ and records gen.sg. *cū* ‘cow’ (or *cū(e)*), *cȳ* ‘cow’, *cūs* ‘cow (gen.)’, dat.sg. *cȳ* ‘cow’, nom.-acc.pl. *cȳ* ‘cow’, and dat.pl. *cūm* ‘cow’ (Clark Hall 1960, 67). Ringe and Taylor likewise state that the root-noun *cū* ‘cow’ exhibits dat.sg., nom.-acc.pl. *cȳ* ‘cow’ < **cūi* ‘cow’, dat.pl. *cūm* < **cūm*, and apparently gen.sg. *cā* ‘cow (gen.)’ < **cūiz* ‘cow (gen.)’ (Ringe and Taylor 2014, § 6.6.1).

The dictionary headword is therefore *cū* ‘cow’, but *cȳ* ‘cow’ is an established Old English paradigm form rather than a convenient reconstruction. It serves as dative singular and also as nominative-accusative plural. The genitive singular is less stable, with *cā* ‘cow (gen.sg.)’, *cȳ* ‘cow’, *cū(e)*, and *cūs* ‘cow (gen.sg.)’ all appearing in the local source record.

Development to Old English

**kūi* ‘cow’ is the dative singular of the oblique **kū-* stem. The following **i* triggers i-umlaut, so *ū* becomes *ȳ*, and loss of the final high vowel leaves *cȳ* ‘cow’. The development is **kūi* ‘cow’ > **kȳi* ‘cow’ > **kȳ* ‘cow’ > cȳ.

This is the regular oblique-cell path recognized by Ringe and Taylor’s paradigm statement (Ringe and Taylor 2014, § 6.6.1). It also explains why *cȳ* ‘cow’ is the cleanest comparison form for the present entry, even though the ordinary headword is *cū* ‘cow’.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*kōz	OE headword *cū* ‘cow’ belongs to the nominative history of the lexeme	*cū*	useful background, but not the chosen comparison for *cȳ* ‘cow’
later generalized nominative	PNWGmc *kūaz* > kūz	inferred nominative *cū* ‘cow’	*cū*	explains the leveled headword, not the oblique target
dative singular oblique	*kūi	regular output: *cȳ* ‘cow’	*cȳ*	exact match between input, output, and paradigm cell
genitive singular oblique	*kūiz	Ringe-Taylor: apparently *cā* ‘cow (gen.sg.)’; Hall: *cū(e)*, *cȳ*, *cūs*	gen.sg. variable	too unstable to control the entry

The dative singular is the relevant comparison form. It gives a regular path to attested *cȳ* ‘cow’, while the broader Old English paradigm shows how far the oblique **kū-* grade spread beyond that one cell.

9.7 find — OE *fundene*

Derivation: citation reconstruction **fínθanąˀ*; form followed here **fúnðanǭ* > *fundene* (late analogy).

Derivation trace

Proto input: **fúnðanǭ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Fronting Early	**fúndœnǭ*
PWGmc Dental Hardening **fúndanǭ*	OE Unstressed Long Vowel Shortening	**fúndœnæ*
Early Anglo-Frisian	OE Unstressed AE Merger	**fúndene*
[no change]		

Old English form: *fundene*

Reconstruction and comparative evidence

The inherited verb is the strong verb **fínθanąˀ* ‘find’, continued by Old English *findan* ‘find’ (Ringe and Taylor 2014, 344). The form followed here, **fúnðanǭ* ‘find’, belongs to the past-participial paradigm rather than to the infinitive. It represents an oblique singular form of the participle.

The familiar dictionary form *funden* ‘found’ is not the form compared here. The derivational input instead models an attested participial form directly, rather than treating the infinitive or the ordinary dictionary headword as primary. It therefore reaches *fundene* ‘found (p.p.)’ in the form where the trace and the attested evidence match directly.

Old English evidence

Bosworth-Toller records *fundene* ‘found (p.p.)’ ‘found’ under *findan* ‘find’, citing the form in *Beón þá herigeata swa fundene* (Bosworth and Toller 1898, 219). Clark Hall likewise preserves the participial stem in forms such as *funden* ‘found’ and *tō-fundennes* ‘finding’ (Clark Hall 1960, 124).

The ordinary dictionary headword for the participle is *funden* ‘found’, but the relevant comparison form for this entry is the attested oblique *fundene* ‘found (p.p.)’ ‘found’. It is an Old English form in its own right, not a merely convenient probe.

Development to Old English

From **fúnðanǭ* ‘find’, the participial oblique develops through regular loss and weakening of the final ending, yielding *fundene* ‘found (p.p.)’. In that cell both the consonantism and the medial vowel history remain regular.

The broader participial paradigm fixes the interpretation. The more familiar nominative *funden* ‘found (p.p.)’ is the ordinary dictionary form, whereas the oblique form *fundene* ‘found (p.p.)’ is the attested form compared here.

Paradigm comparison

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**fínθanǫ*	inherited verb *findan*	*findan*	establishes the lexeme, but not the form compared here
nominative participial line	**fúnðanaz*	ordinary dictionary *funden* type	*funden*	important paradigm background, but not the form compared here
oblique participle compared here	**fúnðanǭ*	regular output: *fundene*	*fundene*	exact match between input, output, and attested cell

The oblique participle is the relevant comparison form. It matches the derivational input and Old English form directly, while the nominative participial headword remains a different presentation cell within the same paradigm.

9.8 fright — OE *fyrhte*

Derivation: citation reconstruction **furxtīn*; form followed here **fúrxtīnaz* > *fyrhte* (late analogy).

Derivation trace

Proto input: **fúrxtīnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fúrxtīn*
EAF Final Z Deletion	**fúrxtīna*	OE I Umlaut	**fyrxtīn*
Early Anglo-Frisian		NWGmc In Stem N Loss	**fyrxtī*
		OE Unstressed Long Vowel Shortening	**fyrxti*
[no change]		OE Med Unstressed I Lowering1	**fyrxte*

Old English form: *fyrhte*

Reconstruction and comparative evidence

The noun belongs to the inherited in-stem abstract **furxtīn* ‘fright’, the same family as Gothic *faurhtei* ‘fear, fright’ (Orel 2003, 120). The derivational input **fúrxtīnaz* ‘fright’ is not a different lexeme but an oblique singular cell within that in-stem paradigm.

Ringe and Taylor treat the later nominative forms with *-u* or *-o* as analogically remodeled (Ringe and Taylor 2014, 395–96). I therefore use the oblique in-stem form, which continues the older formation rather than the remodeled nominative.

Old English evidence

Bosworth-Toller records *fyrhte* ‘fright’ with textual attestation, and it also records nominative forms such as *fyrhtu* ‘fright’ and *fyrhto* ‘fright’ (Bosworth and Toller 1898, 160). Clark Hall lists adjective and verb material separately under *fyrht* / *fyrhtan* (Clark Hall 1960, 141).

The relevant comparison form is therefore the attested oblique *fyrhte* ‘fright’. The nominative lemma forms remain part of the Old English evidence, but the Old English form here of this entry is the oblique cell.

Development to Old English

From **fúrxtīnaz* ‘fright’, the oblique in-stem develops through the loss and weakening of the final ending, yielding *fyrhte* ‘fright’. The form compared here therefore follows the ordinary Old English reduction of the abstract ending in this paradigm.

The later nominative forms with *-u* or *-o* belong to a subsequent analogical reshaping of the paradigm. The Old English form here is earlier in that sense: it is the attested OE cell in which the inherited in-stem development remains most transparent.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested oblique form from the later remodeled nominative line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation in-stem headword	**furxtīn*	broader noun-class label	wider family context	useful lexeme label, but not the cell compared here
remodeled nominative line	nominative in-stem forms	*fyrhtu* ‘fright’ / *fyrhto* ‘fright’ type lemma forms	*fyrhtu* ‘fright’ / *fyrhto* ‘fright’	genuine OE evidence, but later remodeled

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
selected oblique singular	**fúrxtīnaz*	regular output: *fyrhte* ‘fright’	*fyrhte* ‘fright’	exact match between input, output, and attested cell

The oblique in-stem form is the relevant comparison form. It yields attested *fyrhte* ‘fright’ directly, while the more familiar nominative forms belong to a later analogical layer.

9.9 hammer — OE *hameres*

Derivation: citation reconstruction **xámaraz*; form followed here **xámaras* > *hameres* (late analogy).

Derivation trace

Proto input: **xámaras*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**xámœræs*
[no change]	OE Unstressed AE Merger	**xámeres*
Early Anglo-Frisian		
[no change]		

Old English form: *hameres*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **xámaraz* ‘hammer’, reflected in Old English citation forms such as *hamor* ‘hammer’ and *hamer* ‘hammer’ (Kroonen 2013, 206; Orel 2003, 197; Clark Hall 1960, 160). The derivational input **xámaras* ‘hammer’ is the genitive singular of that same noun rather than a different lexeme.

The citation tradition is already mixed in its unstressed vowel, while the genitive singular gives a closer comparison form. This is a cell choice within one paradigm, not a change of stem class.

Old English evidence

Bosworth-Toller directly records *hameres* ‘hammer’ in an Old English genitival phrase (Bosworth and Toller 1898, 78). Clark Hall preserves the simplex headword as *hamer* ‘hammer’ / *hamor* ‘hammer’ (Clark Hall 1960, 160).

Sievers-Brunner gives a paradigm line *hamor* ‘hammer’ — *hamores* ‘hammer’, which shows that the oblique tradition itself was not entirely uniform (Brunner 1965, § 245). The relevant comparison form here is the attested genitive singular *hameres* ‘hammer’.

Development to Old English

From **xámaras* ‘hammer’, Anglo-Frisian brightening and the subsequent merger of unstressed *æ* with *e* yield *hameres* ‘hammer’. The derivation of that oblique form is straightforward once the genitive singular cell is selected.

The noun as a whole retains a mixed citation tradition in *hamor* ‘hammer’ and *hamer* ‘hammer’, and the selected oblique cell avoids making that variation carry the argument of the entry.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested genitive singular from the less stable citation tradition.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**xámaraz*	regular citation form *hamer* / *hamor*	*hamor* / *hamer*	good lexical background, but not the Old English form here
genitive singular	**xámaras*	regular output: *hameres*	*hameres*	exact match between input, output, and attested cell
later oblique tradition	oblique a-stem forms	*hamores* type evidence	*hamores*	attested background variant, but not the chosen comparison form

9.10 have — OE *hæfeþ*

Derivation: citation reconstruction **xabēną*; form followed here **xábēθi* > *hæfeþ* (late analogy).

Derivation trace

Proto input: **xábēθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xæbēθ*
PWGmc Early I Apocope	**xábēθ*	OE Velar Fricative Palatalization	**çæbēθ*
Early Anglo-Frisian		PGmc B Allophony	**çæβēθ*
		OE Unstressed Long Vowel Shortening	**çæβeθ*
[no change]			

Old English form: *hæfeþ*

Reconstruction and comparative evidence

The verb belongs to the inherited class-III weak paradigm usually cited under **xabēną* ‘have’ and Old English *habban* ‘have’ (Kroonen 2013, 237; Ringe and Taylor 2014, 93). Within that paradigm, however, the infinitive and the singular present indicative do not continue the same stem. Ringe and Taylor distinguish a *-ja-* stem in the infinitive from a non-geminating -ai- / *-ē-* stem in the 2sg and 3sg present forms (Ringe and Taylor 2014, 93).

The derivational input **xábēθi* ‘have’ is therefore the 3sg present cell rather than a rephrasing of the infinitive. For the present analysis, that finite cell is the cleaner comparator for the inherited non-geminating stem.

Old English evidence

The ordinary Old English headword is *habban* ‘have’ (Clark Hall 1960, 157). Campbell’s Anglian paradigm includes unsyncopated 3sg forms of the *hæfed* ‘has’ type, and the present paradigm

therefore shows forms of the *hæf-* type that support the normalized target *hæfeþ* ‘has’ (Campbell 1959, § 762).

The target form is therefore a normalized finite cell rather than the ordinary dictionary lemma. It represents the inherited non-geminating present stem more directly than *habban* ‘have’ does.

Development to Old English

From **xábēθi* ‘have’, the finite form yields *hæfeþ* ‘has’ regularly. Ringe and Taylor discuss this non-geminating present stem under *habban* ‘have’ (Ringe and Taylor 2014, 364). Campbell’s Anglian paradigms include unsyncopated 3sg forms of the *hæfeþ* ‘has’ / *hæfed* ‘has’ type (Campbell 1959, § 762).

The wider lexeme is less straightforward only because the infinitive *habban* ‘have’ shows later leveling. The 3sg present cell is therefore the closer comparison form.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the analogically leveled citation form from the regular 3sg present line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*-ja-* stem of **xabēną*	citation form *habban*	*habban*	important headword, but shaped by later leveling
3sg present	**xábēθi*	regular output: *hæfeþ*	*hæfeþ*	exact match between input, output, and finite form compared here
syncopated finite tradition	same present stem	*hæfþ* type evidence	*hæfþ*	genuine later OE finite form, but not the normalized target used here

9.11 knee — OE *cneowe*

Derivation: citation reconstruction **knéwą*; form followed here **knéwai* > *cneowe* (late analogy).

Derivation trace

Proto input: **knéwai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**knéowē*
PNWGmc Unstressed Ai Monophthongization	**knéwē*	OE Unstressed Long Vowel Shortening	**knéowe*
Early Anglo-Frisian			
[no change]			

Old English form: *cneowe*

Reconstruction and selection of the dative

The citation reconstruction is the neuter noun **knéwą* 'knee'. Ringe and Taylor expressly give the singular **knewą* 'knee' and distinguish its endingless development from the inflected stem (Ringe and Taylor 2014, 387, §7.2.4). The acute in the selected notation marks root stress; it does not make the vowel long.

The derivational input **knéwai* 'knee (dat.sg.)' belongs to another cell of the same inherited paradigm. It combines the stem with the conventional dative ending. Fulk regards Proto-Germanic *-ai* as the most likely source of the West Germanic dative singular, while considering the competing *-ē* analysis and the uncertainties of the wider Germanic comparison (Fulk 2018, 147, §7.8). The whole-word input is therefore an explicit paradigm construction under that defended working account, not a verbatim reconstruction quoted from Hogg or Campbell. It is neither a later Old English transponent nor a nominative from which the dative has been generated by an invented sound law.

The distinction between citation reconstruction and derivational input is essential. Selecting the dative does not deny the existence of the familiar nominative *cnēow* 'knee', nor replace the inherited neuter reconstruction with a different lexeme. It selects the cell in which the regular pre-ending development can be compared without importing the nominative's analogically restored consonant.

Old English quantity and paradigm evidence

The citation and dative are different cells, not rival reconstructions of one cell. The source comparison accordingly adds an uncertainty about the inherited ending, not another dispute over the already documented short dative. Fulk's preferred *-ai* account licenses the selected paradigm construction without establishing a uniquely recoverable complete ancestor (Fulk 2018, 147). The detailed quantity and levelling evidence below remains independently necessary.

Hogg explicitly uses *cneowe* 'knee (dat.sg.)' as an example of short *eo*. His adjacent long-diphthong example, *cnēowe* 'know (past subjunctive)', is a verb form, not an alternative knee dative. In the discussion of breaking he again names the knee dative as a regular example before *w*, while assigning the final *w* of nominative *cnēo(w)* to analogy [Hogg (2011), pp. 21–22, §2.33; p. 86, §5.22 and note 7]. The quantity contrast is an explicit grammatical analysis, not an inference from an unmarked manuscript digraph.

Campbell's regular paradigm similarly has genitive *cneowes* 'knee (gen.sg.)' and dative *cneowe* beside long endingless *cnēo(w)*. He then allows the long diphthong to extend into the inflected forms, giving the secondary *cnēowes* type (Campbell 1959, 232–33, §584). Luick independently gives *cneowe(s)* as the genitive and dative products of short-vowel diphthongization before *w*, with the following-*i* exclusion (Luick 1914--1940, 139, §134). These accounts support the selected short dative. They do not establish that every occurrence of an inflected knee form necessarily retained the original quantity.

Ringe and Taylor give the same basic opposition: long endingless *cnēo* and short *cneow-* before syllabic endings. They cautiously consider subsequent extension of long quantity into knee's inflected stem, noting the difficulty of finding decisive verse evidence [Ringe and Taylor (2014), pp. 187–188, §6.2.4; p. 387, §7.2.4]. Fulk describes the wider two-way leveling pattern of this stem class: the endingless diphthong spreads into inflected forms, and their *w* spreads into uninflected forms. His explicit illustrative example is the servant paradigm, not an independent metrical attestation for knee (Fulk 2018, 154, §7.12).

The regular selected development

Before a retained syllabic ending, the root vowel remains short. The singleton *w* conditions short *e* to *eo*, unless followed by a high front vocalic. Hogg's knee dative and Luick's corresponding examples belong to this regular pre-ending domain (Hogg 2011, 86; Luick 1914--1940, 139; Ringe and Taylor 2014, 187–88). The weakening of the selected dative ending leaves Old English *-e*; the medial *w* has not become a word-final offglide and does not need to be restored.

The modeled short diphthong is supplied by SC044 OEBreaking, not by assigning long quantity at SC033 OEEwLongDiphthong. The latter now excludes this singleton path and retains only the separately disclosed j-created representation discussed with *hīew* 'hue'. The inherited-ending and short-diphthong developments are independently motivated; arriving at the right final spelling is not itself their historical justification.

Why the nominative is a comparison, not the selected target

The endingless development follows a different route. Ringe and Taylor derive long *cnēo* from the inherited singular through an offglide-bearing intermediate, and explain the *w* in *cnēow* as transfer from the inflected forms (Ringe and Taylor 2014, 174, 387). Campbell likewise distinguishes endingless diphthong formation from the later addition of *w* from the obliques (Campbell 1959, 46–47, 232–33). Luick's older account distinguishes the endingless knee/tree *eo* and *eu* sources and their subsequent leveling (Luick 1914--1940, 118, §101). Agreement on the short/long paradigm opposition does not make all these detailed early reconstructions identical.

The nominative is thus not an appropriate obligation for a purely phonological derivation that retains *w* and lengthens the root before its ending disappears. Such a derivation reaches a real Old English form by conflating two histories. The selected dative avoids that conflation without a knee-specific lengthening rule, an analogical sound change, or grammatical conditioning of a phonetic law. It also avoids choosing a long oblique which itself reflects the quantity leveling under investigation.

Paradigm comparison and confidence

Cell or interpretation	Reconstruction / form	Historical role
PGmc citation singular	*knéwą	Preserved lexeme-level reconstruction
Selected PGmc dative singular	*knéwai	Explicit inherited-cell construction under the preferred *-ai analysis
OE selected dative singular	cneowe	Regular short pre-ending diphthong
OE regular genitive singular	cneowes	Independent short-stem comparison
OE endingless singular	cnēo	Regular long offglide-derived form
OE familiar nominative	cnēow	Endingless form with restored w
Later long obliques	cnēowe, cnēowes	Possible quantity leveling; not the selected regular comparison

Confidence is strong in the regular paradigm distinction: Campbell, Luick, Hogg and Ringe–Taylor independently identify the short pre-ending stem. Confidence is more qualified in the exact inherited dative-ending reconstruction and in how widely later long quantity spread through knee's attested obliques. Neither uncertainty authorizes an unrestricted short-*ew* to long-*ēow* sound law.

The authors' exact inputs are not silently harmonized. Hogg's breaking footnote prints **knewaz* 'knee', whereas Ringe and Taylor give the neuter *knewą* (Hogg 2011, 86, note 7; Ringe and Taylor 2014, 387). The latter supplies the retained citation reconstruction. This discrepancy in the quoted ending is separate from their agreement on short pre-ending quantity; it is not evidence for replacing the selected dative with a nominative or for claiming identical whole-word inputs.

This entry belongs to late analogy and paradigm-cell selection because the conventional citation paradigm has undergone later reshaping. That placement does not mean that the selected dative itself requires an analogical operation. It follows the same principle as the regular selected obliques used for cow, night and meed: distinguish the history of the lexeme's paradigm from the sound-law path of the chosen cell.

9.12 learn (3sg) — OE *leornaþ*

Derivation: citation reconstruction **liznōjanǫ*; form followed here **líznōθi* > *leornaþ* (late analogy).

Derivation trace

Proto input: **líznōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornōθ*
PWGmc Early I Apocope	**líznōθ*	OE Late O Shortening	**léornoθ*
PNWGmc I Lowering	**léznōθ*	OE Final Unstressed O Lowering	**léornaθ*
EAF Rhotacism	**lérnōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *leornaþ*

A textually attested third singular

The selected Old English *leornaþ* 'learns' corresponds to Miller's printed *leornað* 'learns'. In the Bede passage, *swa hwylc swa* 'whoever' governs coordinated *rædeð* 'reads' and *leornað*. The singular indefinite-relative subject establishes the person and number; the ending alone would not suffice. Final eth is represented by thorn without changing the vowel or cell (Miller 1891, 380, ll. 26–27).

The input **líznōθi* 'learns' is a finite third-singular selection, not the infinitive citation label. Its inherited **lizn-* root remains distinct from the later lowered intermediate. The wider family has a class III prehistory and subsequent conjugational remodelling; the English class II form does not recover that complete earlier paradigm (Kroonen 2013, 340; Orel 2003, 247; Jasanoff 1973, 865, 868).

SC015 PNWGmcILowering lowers the selected root before **zn*. Rhotacism, English breaking, ending reduction and dental spelling then give *leornaþ*. This explanation leaves the inherited input high-vowelled. It does not select an **e*-grade just because the Old English word contains *eo* (Hill 2017, 153; Ringe and Taylor 2014, 38, 84).

The high-vowel *liorn- tradition remains important, but the infinitive's spelling cannot establish the prehistory of this specific cell. Jasanoff's proposed lost **liznip* and Hill's later-mutation account identify possible morphological sources of high vowels; neither yields a fully certain inherited paradigm. The family classification accordingly retains conjugational remodelling without claiming that the selected vowel itself requires analogical replacement (Jasanoff 1973, 865, 868; Hill 2017, 153, n. 25; Tarsi and Laker 2026, 42–43).

9.13 learn (iptv.2sg) — OE *leorna*

Derivation: citation reconstruction **liznōjanǫ*; form followed here **líznô* > *leorna* (late analogy).

Derivation trace

Proto input: **líznô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornô*
PNWGmc I Lowering	**léznô*	OE Unstressed Long Vowel Shortening	**léorna*
EAF Rhotacism	**lérnô*		
Early Anglo-Frisian			
[no change]			

Old English form: *leorna*

The imperative singular

Old English *leorna* ‘learn!’ is the second-person singular imperative, not an abbreviated infinitive. Deuteronomy14:23 prints the command followed by *þæt þu ondræde Drihten* ‘that you fear the Lord’. The addressee is singular, and the form is read directly in the edited biblical text (Crawford 1922, 351).

The derivational input **líznô* ‘learn!’ is the imperative cell, distinct from the lexeme-level infinitive label. The inherited **lizn-* family is supported by the comparative dictionaries; its historical weak class III formation and later remodelling remain uncertain (Kroonen 2013, 340; Orel 2003, 247; Jasanoff 1973, 865, 868).

Under SC015 PNWGmcILowering, lowering before preconsonantal **z* produces **léznô*; rhotacism gives **lérnô*, and English breaking and ending reduction lead to *leorna*. The companion infinitive *leornian* and present third singular *leornaþ* are different cells of the same family. Their agreement does not multiply the independent etymological evidence (Hill 2017, 153; Crawford 1922, 71, 351; Miller 1891, 380).

The attested alternative *liorna* ‘learn!’ remains relevant to the *liorn-/leorn- distribution. Mutation in an earlier paradigm, later levelling and dialect history must be considered rather than identifying every high vowel as unchanged inheritance. The retained family classification marks conjugational remodelling, not finite-cell selection or an extra replacement needed for this imperative (Campbell 1959, 48, 59, 125; Jasanoff 1973, 865, 868).

9.14 lick (3sg) — OE *liccaþ*

Derivation: citation reconstruction **likkōną*; form followed here **líkkōθi* > *liccaþ* (late analogy).

Derivation trace

Proto input: **líkkōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Late O Shortening	**líkkoθ*
PWGmc Early I Apocope	**líkkōθ*	OE Final Unstressed O Lowering	**líkkaθ*
Early Anglo-Frisian			
[no change]			

Old English form: *liccaþ*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **ekkōn* ‘lick (3sg.)’ continuing as Old English *liccian* ‘lick’, Old Saxon *likkon* ‘lick’, and Old High German *lecchon* ‘lick’ (Ringe and Taylor 2014, 50). Orel gives

the fuller weak-verb reconstruction **likkōjanan* ‘lick’ with the same Old English continuation (Orel 2003, 285).

The form compared here in this entry is the non-*j* present third singular **líkkōθi* ‘licks’, not the remodeled infinitive. Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, § 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80).

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* ‘lick’ (Bosworth and Toller 1898, 614). The same consonantal frame appears in Campbell’s and Brunner’s grammatical citations of *liccian* ‘lick’ (Campbell 1959, § 398.1; Brunner 1965, § 45 Anm. 3). The Old English headword is therefore clear even though the entry here is not about the citation form.

The form treated here is the present third singular *liccaþ* ‘licks’. It is a selected paradigm form beside the lemma *liccian* ‘lick’ and the related imperative *licca* ‘lick’, not a separately lemmatized headword.

Development to Old English

**líkkōθi* ‘licks’ first loses final *-i*, giving **líkkōθ* ‘licks’. Campbell’s class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support the regular 3sg outcome *-aþ* (Campbell 1959, § 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). Because this ending never contains *-j-*, the form does not pass through an i-umlauted *-eþ* stage.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* ‘lick’	*liccian* ‘lick’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**líkkô*	regular output *licca* ‘lick’	*licca* ‘lick’	useful family control, but not the target of this entry
present third singular	**líkkōθi*	regular output *liccaþ* ‘licks’	*liccaþ* ‘licks’	exact match between the derivational input and the Old English form here

9.15 lick (iptv.2sg) — OE *licca*

Derivation: citation reconstruction **likkōną*; form followed here **líkkô* > *licca* (late analogy).

Derivation trace

Proto input: **líkkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**líkka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *licca*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **ekkōn* ‘lick (3sg.)’ continuing as Old English *liccian* ‘lick’, Old Saxon *likkon* ‘lick’, and Old High German *lecchon* ‘lick’ (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* ‘lick’ with the same Old English continuation (Orel 2003, 285).

Campbell’s weak class-II discussion gives present forms such as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, § 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The form treated here is therefore not that remodeled infinitive but a finite cell in bare trimoric **-ō*.

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* ‘lick’ (Bosworth and Toller 1898, 614). Campbell cites *liccian* ‘lick’ among Old English forms with preserved geminate *cc* (Campbell 1959, § 398.1). Brunner likewise cites *liccian* ‘lick’ (Brunner 1965, § 45 Anm. 3). The Old English evidence therefore establishes the verbal headword and its consonantal frame securely.

The Old English form here in this entry is the imperative singular *licca* ‘lick’. It is a paradigm form chosen beside the headword *liccian* ‘lick’ and the related present *liccaþ* ‘licks’, not a separately lemmatized citation word.

Development to Old English

With the stem *licc-* established, the remaining development is brief. Campbell’s class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support late *-a* in this finite cell (Campbell 1959, § 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). The same stem consonantism that appears in *liccian* ‘lick’ is preserved here, giving *cc* throughout the finite form.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* ‘lick’	*liccian* ‘lick’	ordinary dictionary headword of the verb, but not the finite form compared here

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
imperative singular	**líkkô*	regular output *licca* ‘lick’	*licca* ‘lick’	exact match between the derivational input and the Old English form here
present third singular companion	**líkkōθi*	regular output *liccaþ*	*liccaþ*	useful family control, but not the target of this entry

9.16 live — OE *lifeþ*

Derivation: citation reconstruction **libēną*; form followed here **líbēθi* > *lifeþ* (late analogy).

Derivation trace

Proto input: **líbēθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PGmc B Allophony	**líβēθ*
PWGmc Early I Apocope	**líbēθ*	OE Unstressed Long Vowel Shortening	**líβeθ*
Early Anglo-Frisian			
[no change]			

Old English form: *lifeþ*

Reconstruction and comparative evidence

The inherited verb belongs to the class-III weak family cited by Kroonen under **libēn-*, reflected in Old English *libban* ‘live’ (Kroonen 2013, 336). Ringe and Taylor show that the paradigm also contained a separate 3sg present stem, continued in late Northumbrian *lifed* ‘lives’, which they treat as an archaism (Ringe and Taylor 2014, 364).

The derivational input **líbēθi* ‘live’ therefore represents a finite present cell rather than the citation infinitive. The later lemma tradition also includes remodeled forms such as *lifian* ‘live’.

Old English evidence

The ordinary lemma tradition centers on *libban* ‘live’ and, in later remodeling, *lifian* ‘live’. For this entry, however, the relevant comparison form is the archaic 3sg present attested as *lifed* ‘lives’, here normalized as *lifeþ* ‘lives’ (Ringe and Taylor 2014, 364; Campbell 1959, § 762).

The target is thus a normalized finite form, not the ordinary dictionary lemma. It preserves the older present-stem history more clearly than the remodeled lemma tradition does.

Development to Old English

From **líbēθi* ‘live’, regular reduction of the final syllable and later weakening of the unstressed vowel yield *lifeþ* ‘lives’. The attested spelling *lifed* ‘lives’ belongs to the same finite form in late Northumbrian orthography (Campbell 1959, § 762; Ringe and Taylor 2014, 364).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the archaic finite cell from the ordinary infinitival and later remodeled lemma lines.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive line	**libēną*	OE *libban* headword tradition	*libban*	establishes the lexeme, but not the Old English form here
3sg present	**líbēθi*	regular output: *lifeþ*; attested *lifed*	*lifed*, normalized here as *lifeþ*	selected archaic finite cell
later remodeled present tradition	later class-II-type forms	*lifian* and related finite remodeling	*lifian*	genuine OE development, but secondary to the cell compared here

9.17 make (3sg) — OE *macaþ*

Derivation: citation reconstruction **makōną*; form followed here **mákōθi* > *macaþ* (late analogy).

Derivation trace

Proto input: **mákōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mækōθ*
PWGmc Early I Apocope	**mákōθ*	OE A Restoration	**makōθ*
Early Anglo-Frisian		OE Late O Shortening	**makoθ*
		OE Final Unstressed O Lowering	**makaθ*
[no change]			

Old English form: *macaþ*

Reconstruction and comparative evidence

Kroonen derives the Old English verb from **makōjan-* on the make-family base **maka-* (Kroonen 2013, 350). Ringe and Taylor likewise derive Old English *macian* ‘make’ from PWGmc **makon* ‘make’ through **mekojan* ‘make’ (Ringe and Taylor 2014, 191).

The derivational input **mákōθi* ‘makes’ is therefore a finite 3sg cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *macian* ‘make’ (Clark Hall 1960, 193). The relevant comparison form here is normalized present-third-singular *macaþ* ‘makes’, set beside the dictionary headword and the related imperative singular *maca* ‘make’.

Campbell’s class-II paradigm gives ordinary 3sg ending *-aþ*, while his dialect survey allows secondary *-e-* spellings in some traditions (Campbell 1959, § 356.4; 1959, § 757). *Macaþ* ‘makes’ is thus the regular comparison form for the non-*j* 3sg cell.

Development to Old English

After early loss of final *-i*, **mákōθi* ‘makes’ → **mákōθ* ‘makes’ after loss of final -i. Anglo-Frisian brightening gives **mækōθ* ‘makes’, but Campbell lists *macian* ‘make’ among class-II verbs with restored *a*, so the stem returns to *mak-* before the ending is reduced (Campbell 1959, § 159).

The ending then follows ordinary class-II 3sg development: **makōθ* ‘makes’ > **makaθ* ‘makes’ > macaþ_. Campbell’s_lufas, -aþ (< -ōsi, -ōþi)_and Ringe and Taylor’s discussion of stable class-II_a_in finite non-_j‘ cells support this sequence (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the make-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**makōjaną*	dictionary headword *macian* ‘make’	*macian* ‘make’	family background, but not the cell compared here
3sg present	**mákōθi*	regular output *macaþ* ‘makes’	*macaþ* ‘makes’	exact match
imperative singular companion	**mákô*	related finite form *maca* ‘make’	*maca* ‘make’	useful control, but not the target

9.18 make (iptv.2sg) — OE *maca*

Derivation: citation reconstruction **makōną*; form followed here **mákô* > *maca* (late analogy).

Derivation trace

Proto input: **mákô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**mækô*
[no change]	OE A Restoration	**makô*
	OE Unstressed Long Vowel Shortening	**maka*
Early Anglo-Frisian		
[no change]		

Old English form: *maca*

Reconstruction and comparative evidence

The make-family belongs to the Old English class-II weak verbs. Campbell cites *lapian, macian* ‘make’ among verbs with restored *a* (Campbell 1959, § 159). Ringe and Taylor place the Germanic verb in the same class, comparing West Germanic continuants such as Old Frisian *makia* ‘make’, Old Saxon *makon* ‘make’, and Old High German *mahhon* ‘make’ (Ringe and Taylor 2014, 191).

The derivational input **mákô* ‘make (iptv.2sg)’ is not the citation form of the lexeme but a finite paradigm cell. Ringe and Taylor give the class-II weak imperative singular as -a < **-ō*, which makes this cell the relevant comparison point for the Old English form treated here (Ringe and Taylor 2014, 314).

Old English evidence

The dictionary headword is *macian* ‘make’ (Clark Hall 1960, 193). The form compared here in this entry is therefore not the lemma but the imperative singular *maca* ‘make’, chosen as a paradigm form beside the headword *macian* ‘make’ and the related finite form *macaþ* ‘makes’.

The lexical history is still that of *macian* ‘make’, but the finite cell isolates the regular outcome of trimoric **ō* more cleanly than the citation form does.

Development to Old English

From **mákô* ‘make (iptv.2sg)’, Anglo-Frisian brightening first gives **mækô* ‘make (iptv.2sg)’. Campbell cites *macian* ‘make’ among the class-II verbs with restored *a* (Campbell 1959, § 159). Ringe and Taylor’s class-II weak imperative singular -a < **-ō* then supports the later finite ending (Ringe and Taylor 2014, 314).

The same development explains why earlier fronted forms of the *mæċa* ‘make’ type do not control the entry. Once trimoric **ô* is treated as a back-vocalic trigger for restoration, the imperative singular falls into line with the broader *macian* ‘make’ family.

Paradigm comparison

A paradigm comparison identifies which finite cell of the make-family matches the Old English form chosen here. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**mákōjanǫ*	comparative continuation *macian* ‘make’	*macian* ‘make’	ordinary headword of the verb, but not the finite form compared here
imperative singular	**mákô*	regular output: *maca* ‘make’	*maca* ‘make’	exact match between input, output, and selected paradigm form
present third singular companion	**mákōθi*	comparative companion *macaþ*	*macaþ*	useful family control, but not the target of this entry

9.19 man — OE *mannes*

Derivation: citation reconstruction **mánnaz*; form followed here **mánnas* > *mannes* (late analogy).

Derivation trace

Proto input: **mánnas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**mánnæs*
[no change]	OE Unstressed AE Merger	**mánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *mannes*

Reconstruction and comparative evidence

The lexeme-level reconstruction is not uniform. Kroonen cites **mannan-*, and Orel has **mannz* ‘man’ (Kroonen 2013, 354; Orel 2003, 299). The derivational input **mánnas* ‘man’ belongs to a different level: it is the genitive-singular cell chosen for the Old English comparison.

The target is not the ordinary citation form. Its medial geminate precedes the genitive ending.

Old English evidence

Campbell gives the paradigm *mann* ‘man’, *man* ‘man’ / *mannes* ‘man’s / *menn* ’men’ (Campbell 1959, § 621). Sievers-Brunner likewise cites *man* ‘man’ *mannes* ‘man’s (Brunner 1965, § 226). He also explains that word-final simplification underlies forms such as *man* ’man’ beside inflected *monnes* ’man’s (Brunner 1965, § 231). Clark Hall keeps the dictionary headword under *mann* (Clark Hall 1960, 197).

The relevant comparison form is therefore the attested genitive singular *mannes* ‘man’s, not the citation lemma *mann* ’man’.

Development to Old English

Campbell’s paradigm *mann*, *man* / *mannes* / *menn* ‘men’ confirms the selected genitive singular *mannes* (Campbell 1959, § 621). In the present analysis, **mánnas* ‘man’ develops through Anglo-Frisian brightening and later unstressed merger to *mannes*. In this cell the geminate remains medial before *-es*. The citation form behaves differently because word-final gemination was simplified in Old English (Brunner 1965, § 231).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the citation-form line from the genitive singular.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mannăz*	expected citation-form outcome *man*	*mann* / *monn*	establishes the lexeme, but not the Old English form here
accusative singular	**mannǫ*	expected *man*	*man*	same word-final simplification as the nominative
dative singular	**mannăi*	expected *manne*	*manne*	preserves medial *nn*, but not the chosen cell

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
genitive singular	**mánnas*	regular output: *mannes*	*mannes*	exact match between input, output, and attested comparator

9.20 meed — OE *meorde*

Derivation: citation reconstruction **mizdō*; form followed here **mízdai* > *meorde* (late analogy).

Derivation trace

Proto input: **mízdai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**méordē*
PNWGmc Unstressed Ai Monophthongization	**mízdē*	OE Unstressed Long Vowel Shortening	**méorde*
PNWGmc I Lowering	**mézdē*		
EAF Rhotacism	**mérdē*		
Early Anglo-Frisian			
[no change]			

Old English form: *meorde*

Reconstruction and comparative evidence

The lexeme-level reconstruction is **mizdō* ‘meed’, but the derivational input **mízdai* ‘meed’ is a dative-singular cell rather than the citation form. The Old English evidence for the *meord* ‘meed, reward’ side is oblique.

The wider history of competing *mēd* ‘meed, reward’ remains disputed. Kroonen and Fulk explain it through some form of *z*-loss and compensatory lengthening (Kroonen 2013, 370; Fulk 2018, 52), while Orel keeps a doublet analysis (Orel 2003, 272). The comparison here concerns the attested oblique line *meorde*.

Old English evidence

The directly attested forms are obliques: *meorde* ‘meed’ as a dative singular and *meorda* ‘meed’ as a genitive plural (Bright 1971, 328; Bosworth and Toller 1898, 647). Lexicographers reconstruct a bare nominative *meord* ‘meed’ from those obliques, while West Saxon prose more commonly shows the competing doublet *mēd* ‘meed’ (Clark Hall 1960, 214; Bosworth and Toller 1898, 647).

The target of this entry is therefore the attested oblique *meorde* ‘meed’, not the reconstructed lemma *meord* ‘meed’ and not the better-known West Saxon citation form *mēd* ‘meed’.

Development to Old English

Ringe and Taylor give the broader noun history as **mizdo* ‘reward’ > **mizdu* ‘reward’ > OE meord ~ méd (Ringe and Taylor 2014, 84). The fuller oblique-cell path modeled here spells out the intermediate monophthongization, lowering, rhotacism, breaking, and unstressed-shortening steps needed for the selected dative-singular comparison.

This entry therefore follows the attested oblique line within the broader development. It does not depend on a full decision about the history of the competing *mēd* ‘meed, reward’ tradition.

The consonantal distinction is substantial. Retained *z can become a rhotic, producing the consonantal group that conditions English breaking in *meord-. Loss of the sibilant with compensatory lengthening instead leads toward the long-vowel* mēd* tradition. These are alternative histories of the consonantal sequence, not two interchangeable spellings of the selected dative (Kroonen 2013, 370; Crist 2002, 3–6).

SC015 PNWGmcILowering places lowering before either continuation: its condition is the original preconsonantal *z, not the later rhotic or a particular following vowel. The selected *mízdai* therefore supplies a positive application with a historically relevant consonant group. It does not independently establish the entire history of the rival nominative or the absolute date of every lowering environment (Hill 2017, 153; Crist 2002, 3–6).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the attested oblique target from the broader lemma history.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*mizdō	inferred lemma outcome *meord* ‘meed’	*meord* ‘meed’	useful background, but the bare lemma is reconstructed rather than directly attested
selected dative singular	*mízdai	regular output: *meorde* ‘meed’	*meorde* ‘meed’	exact match between derivational input and attested target
genitive singular	*mizdōz	regular output: *meorde* ‘meed’	*meorde* ‘meed’	converges on the same attested string, but the dat.sg. has the clearest direct support
genitive plural control	plural oblique line	attested *meorda* ‘meed’	*meorda* ‘meed’	confirms the broader oblique tradition, but not the chosen singular target

9.21 night — OE *niht*

Derivation: citation reconstruction *náxtz; form followed here *náxti > niht* (late analogy).

Derivation trace

Proto input: *náxti*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**nœxti*
[no change]	OE Breaking	**neaxti*
	OE I Umlaut	**niexti*
Early Anglo-Frisian	OE Ws Palatal Umlaut	**nixti*
[no change]	OE High Vowel Apocope	**nixt*

Old English form: *niht*

Reconstruction and comparative evidence

Ringe and Taylor cite gen.sg. **nahtiz* ‘night’, dat.sg. **nahti* ‘night’, and nom.pl. **nahtiz* ‘night’ for the high-vowel side of the paradigm, and derive West Saxon *niht* ‘night’ from that side (Ringe and Taylor 2014, 240). The citation reconstruction **náxtz* ‘night’ therefore belongs to the nominative-like headword, while the derivational input **náxti* ‘night’ represents the dative-singular cell.

The word later became the model for endingless datives. Ringe and Taylor explicitly explain forms such as *dæg* ‘day’ by analogy with dat.sg. niht < **nahti* ‘night’ (Ringe and Taylor 2014, 380).

Old English evidence

Clark Hall lemmatizes *niht* ‘night’ and cross-references forms such as *neaht* ‘night’, *neht* ‘night’, and *nieht* ‘night’ (Clark Hall 1960, 215). Campbell likewise preserves the fluctuation between *neaht* ‘night’ and *niht* ‘night’, giving genitive *nihte, nihtes*, dative *niht, nihte*, nominative plural *niht* ‘night’, and the contrasting plural-side forms represented by *neahtas* ‘night’ (Campbell 1959, § 628.3).

The comparison form used here is therefore an attested Old English *niht* ‘night’, not a reconstructed substitute. The broader lexical record still preserves the non-umlauted side of the paradigm in *neaht* ‘night’-type forms.

Development to Old English

Ringe and Taylor derive West Saxon *niht* ‘night’ from **nahti* ‘night’ via **nehti* ‘night’ and **neahti* ‘night’ (Ringe and Taylor 2014, 240). Campbell and Brunner preserve the contrasting non-umlauted *neaht* ‘night’-type forms elsewhere in the paradigm (Campbell 1959, § 628.3; Brunner 1965, § 284).

The modeled path is therefore **náxti* ‘night’ > **nœxti* ‘night’ > **neaxti* ‘night’ > **niexti* ‘night’ > **nixti* ‘night’ > niht.

Paradigm comparison

The comparison below sets the relevant forms side by side. It identifies the inherited cell that matches the attested Old English form.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**náxtz*	expected non-umlauted outcome *neaht*	*neaht*	useful background, but not the comparison used for *niht*
selected dative singular	**náxti*	regular output: *niht* ‘night’	*niht* ‘night’	exact match between input, output, and paradigm cell

9.22 rest — OE *ræste*

Derivation: citation reconstruction **rastō*; form followed here **rástōz* > *ræste* (late analogy).

Derivation trace

Proto input: **rástōz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Surviving Bimoric O Unrounding	**rástā*
EAF Final Z Deletion	**rástō*	EAF Brightening	**ræstā*
Early Anglo-Frisian		EAF Brightening Long Final	**ræstǣ*
		OE Unstressed Long Vowel Shortening	**ræstæ*
[no change]		OE Unstressed AE Merger	**ræste*

Old English form: *ræste*

Reconstruction and comparative evidence

Kroonen treats the noun as a feminine ō-stem **rastō-*, continued by Old English *ræst* ‘rest’ (Kroonen 2013, 445). The derivational input **rástōz* ‘rest’ therefore does not replace the lexeme-level headword. It identifies one oblique singular cell on the side of the paradigm that yields *ræste* ‘rest’.

The source tradition used here labels that cell specifically as genitive singular, but the broader local synthesis of the ō-stem paradigm shows that the oblique singulars converge on the same front-vocalic *ræste* ‘rest’ side, in contrast to a nominative singular that would remain *rast* ‘rest’.

Old English evidence

The ordinary Old English citation form is *ræst* ‘rest’ (Kroonen 2013, 445). Clark Hall likewise gives *ræst* ‘rest’ (Clark Hall 1960, 239). Bosworth-Toller also preserves oblique uses of *ræste* ‘rest’, including prepositional examples such as *on ræste* ‘at rest’ and *tó ræste* ‘to rest’ (Bosworth and Toller 1898, 121).

The comparison form used here is therefore an attested oblique *ræste* ‘rest’, not a reconstructed surrogate. The dictionary headword *ræst* ‘rest’ remains an equally real part of the Old English record.

Development to Old English

Once final **z* is lost, the derivational input moves through the front-vocalic oblique side of the paradigm rather than the back-vocalic nominative side. In the modeled derivation, the surviving final long vowel is first exposed, unrounded, fronted, shortened, and then reduced to the final *-e* of *ræste* ‘rest’.

The key point is the paradigm split. Nominative **rastō* ‘rest’ yields a regular *rast* ‘rest’, whereas the selected oblique input yields *ræste* ‘rest’. The later citation form *ræst* ‘rest’ is best explained as leveling from that oblique *ræst-* stem.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the nominative citation form from the oblique singular chosen here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**rastō*	expected regular outcome *rast*	*ræst* ‘rest’	useful background, but not the cell that matches attested oblique *ræste*
selected oblique singular	**rástōz*	regular output: *ræste* ‘rest’	*ræste* ‘rest’	exact match between derivational input and attested OE oblique form

9.23 shoulder — OE *sċuldrum*

Derivation: citation reconstruction **skuldrō*; form followed here **skúldramiz* > *sċuldrum* (late analogy).

Derivation trace

Proto input: **skúldramiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Sk Palatalization	**fúldrum*
PNWGmc A To U Before M	**skúldrumiz*		
PWGmc Early I Apocope	**skúldrumz*		
EAF Final Z Deletion	**skúldrum*		
Early Anglo-Frisian			
[no change]			

Old English form: *sċuldrum*

Reconstruction and comparative evidence

The handbooks do not agree on the reconstruction of the Germanic word. Orel gives **skuldr(j)ō* as a feminine ō-/jō-stem and explicitly notes that Old English *sculdor* ‘shoulder’ is masculine beside OFrisian *skulder* ‘shoulder’, Middle Low German schulder ‘shoulder’, and Old High German *scultra* ‘shoulder’, *scultirra* ‘shoulder’ (Orel 2003, 345). Kroonen reconstructs **skuldra-*, a masculine a-stem, and derives the Old High German feminine forms from **skuldrjōn-* (Kroonen 2013, 478). Ringe and Taylor cite PWGmc **skuldru* ‘shoulder’ for the Old English branch (Ringe and Taylor 2014, 142).

These forms imply different stem classes and different expectations for the Old English inflection. The question is which inflectional cell best aligns with the Old English evidence.

A dative/instrumental plural form **skúldramiz* ‘shoulder’ aligns with the inherited plural ending that later yields Old English *-um*, and it corresponds directly to the attested dative plural discussed below.

Old English evidence

The ordinary Old English headword is *sculdor* ‘shoulder’. Clark Hall lemmatizes *sculdor* ‘shoulder’ as the normal dictionary form (Clark Hall 1960, 257). Bosworth-Toller also preserves the dative plural *sculdrum* ‘shoulder’ (Bosworth and Toller 1898, 85).

Bosworth-Toller's Supplement records a weak-feminine *sculdra* 'shoulder' (Bosworth and Toller 1898, 699), so *sculdra* 'shoulder' belongs to the Old English record beside the stronger masculine paradigm headed by *sculdor* 'shoulder'. Brunner and Luick also record later spellings such as *sceoldor* 'shoulder' and the i-mutated dative plural *scyldrum* 'shoulder', which reflect secondary phonological and analogical reshaping within Old English (Brunner 1965, § 92.2.a; Luick 1914--1940, 230).

The singular and plural evidence point to different parts of the paradigm. The relevant comparison form here is the attested dative plural *sċuldrum* 'shoulder (dat.pl.)'. The spelling with *sċ-* is a normalized representation of the same Old English initial cluster.

Development to Old English

Proto-Germanic **skúldramiz* 'shoulder' can be interpreted as a dative/instrumental plural form. In this environment the post-tonic *a* before *m* is raised to *u*, giving a form of the **skúldrumiz* 'shoulder' type. Unstressed *u* is regularly preserved before *m*, especially in the dative plural ending *-um*: Campbell states this explicitly, and Hogg formulates the same condition for the dative plural inflexion (Campbell 1959, § 373; Hogg 1992b, § 3.3.1.3). Brunner points in the same direction by excluding *m* from the environments in which medial *o* became general in West Saxon (Brunner 1965, § 44 Anm. 7).

Subsequent reduction of the ending removes the final **i* and **z*, so that the inflectional ending appears in Old English as *-um*. The initial cluster is written here as *sċ-*, and the development is **skúldramiz* 'shoulder' > **skúldrumiz* 'shoulder' > **skúldrum* 'shoulder' > *sċuldrum* 'shoulder'.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
singular-oriented citation input	**skúldrō*	probe output: *sċoldor*	*sculdor* 'shoulder'	fails: the singular output has root *o*, not the attested *u*
serious plural-based singular alternative	**skúldru*	probe output: *sċuldor*	*sculdor* 'shoulder'	close formally, but it compares a plural-stage input with a singular form
dat./inst.pl. input	**skúldramiz* 'shoulder'	regular output: *sċuldrum* 'shoulder (dat.pl.)'	*sculdrum* 'shoulder'	matches both the output and the dative plural comparison form
later weak-feminine singular	—	OE *sculdra* 'shoulder'	*sculdra* 'shoulder'	secondary doublet, useful as a control rather than the inherited target

The dative plural line is decisive because it matches both the output and the paradigm cell of Old English *sculdrum* 'shoulder'. Singular-oriented candidates either lower the root vowel or compare unlike cells.

9.24 shove — OE *sċēaf*

Derivation: citation reconstruction **skéubaną*; form followed here **skáub* > *sċēaf* (late analogy).

Derivation trace

Proto input: **skáub*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**skáeub*
[no change]	OE Diphthong Leveling	**skēab*
	PGmc B Allophony	**skēaβ*
Early Anglo-Frisian	OE Sk Palatalization	**ʃēaβ*
[no change]		

Old English form: *sċēaf*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **skeuban-* ~ **skūban-* and cites Old English present forms *scēofan* ‘shove’, *scūfan* ‘shove’ (Kroonen 2013, 444). Those present-system forms belong to the same verb family, but the comparison here uses the singular preterite **skáub* ‘shove’, not the infinitive.

Old English evidence

The ordinary dictionary verb is *scūfan* ‘shove’/*scēofan* ‘shove’, but the preterite itself is well attested. Bright gives the principal parts *scufan* ‘shove’, *sceaf* ‘shove’, *scufon* ‘shove’, *scofen* ‘shove’ (Bright 1971, 347). Sweet gives the same paradigm (Sweet 1953, 29). The normalized form here is *sċēaf* ‘sheaf’, regularizing the attested spellings *sceaf* ‘shove’ and prefixed *āsceaf* ‘shove’.

Development to Old English

From **skáub* ‘shove’, the development is straightforward. **au* fronts and levels to *ēa*, final **b* becomes a fricative and is written *f*, and initial **sk-* undergoes the usual Old English palatalized spelling in this environment. The derivation therefore gives **skáub* ‘shove’ > **skáeub* ‘shove’ > **skēab* ‘shove’ > **skēaβ* ‘shove’ > *sċēaf* ‘sheaf’.

Paradigm comparison

A paradigm comparison is needed here because the ordinary citation verb and *sċēaf* ‘sheaf’ belong to different cells of the same strong paradigm. The comparison below is manual.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation infinitive	**skéubaną*	inherited infinitive line *sċēofan*; present system also leveled *scūfan* ‘shove’	*scēofan* ‘shove’ / *scūfan* ‘shove’	necessary background, but not the comparison used for *sċēaf* ‘sheaf’
1/3 sg. preterite	**skáub*	documented regular output: *sċēaf* ‘sheaf’	*sċēaf* ‘sheaf’	direct match for the singular preterite

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
preterite plural	**skúbun*	later leveled plural *scufon* ‘shove’ beside expected *sċufun* under the corrected cascade	*scufon* ‘shove’	poorer comparison for the singular-preterite target
past participle	**skúbanaz*	attested participial line *scofen* ‘shove’	*scofen* ‘shove’	valid alternative cell, but not the form compared here

9.25 show (3sg) — OE *sċēawaþ*

Derivation: citation reconstruction **skawōną*; form followed here **skáwōθi* > *sċēawaþ* (late analogy).

Derivation trace

Proto input: **skáwōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Aw Long Diphthong	**skḗawōθ*
PWGmc Early I Apocope	**skáwōθ*	OE Sk Palatalization	**ʃḗawōθ*
Early Anglo-Frisian		OE Late O Shortening	**ʃḗawoθ*
[no change]		OE Final Unstressed O Lowering	**ʃḗawaθ*

Old English form: *sċēawaþ*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* ‘show’ and cites Old English *sceáwian* ‘show’ beside Old Frisian *skawia* ‘show’, Old Saxon *skawōn* ‘show’, and Old High German *scouwōn* ‘show’ (Orel 2003, 337). This is infinitive-family evidence. CAPR assembles the present third singular **skáwōθi* ‘show (3sg)’, a finite class-II cell with stable *a* in the 3sg ending (Ringe and Taylor 2014, 80).

Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* in his late-shortening discussion (Campbell 1959, 148). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The relevant comparison is therefore the 3sg cell itself, not an i-umlauted alternative.

Old English evidence

The held fourth edition of Bright’s reader contains the simplex infinitive *scēawian* ‘show’ and imperative *scēawa* ‘show’ in the Blickling homily (Bright 1917, 71). The Wanderer contains prefixed *geondscēawað* ‘surveys’ (Bright 1917, 161). This supports the finite ending pattern, not a separately quoted unprefixed third singular or a passage in the separate Cassidy/Ringler edition.

The form written here as *sċēawaþ* ‘shows’ is the normalized simplex comparison form for that weak class-II pattern. It is therefore not a dictionary headword but a finite comparison form aligned with the attested *scēaw-* evidence and the directly cited prefixed *geondscēawað* ‘surveys’ ending pattern.

Development to Old English

The infinitive entry and SC034 OEAwLongDiphthong explain the dictionary-singleton versus handbook-geminate root premise. The same computational proxy remains here without claiming that output fit resolves it (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49).

The complete trace of **skáwōθi* ‘show (3sg)’ has already lost final *i* before that proxy: the actual checkpoint is **s*k*á*w*ō*θ*. Thus absence of *j* alone is not the explanation of the compiled lack of mutation. Campbell’s late-shortening account and Ringe and Taylor’s stable-*a* discussion independently support *-aþ*, not *-eþ* (Campbell 1959, 148; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	regular output *sċēawian*	*scēawian* ‘show’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**skáwô*	regular output *sċēawa* ‘show’	*scēawa* ‘show’	useful family control, but not the target of this entry
present third singular	**skáwōθi*	regular output *sċēawaþ* ‘shows’	normalized *sċēawaþ* ‘shows’; source-side pattern *sceawað*	exact match for the finite form compared here

9.26 show (iptv.2sg) — OE *sċēawa*

Derivation: citation reconstruction **skawōną*; form followed here **skáwô* > *sċēawa* (late analogy).

Derivation trace

Proto input: **skáwô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skéawô*
[no change]	OE Sk Palatalization	**ʃéawô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**ʃéawa*
[no change]		

Old English form: *sċēawa*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* ‘show’ and cites Old English *sceáwian* ‘show’ beside Old Frisian *skawia* ‘show’, Old Saxon *skawōn* ‘show’, and Old High German *scouwōn* ‘show’ (Orel

2003, 337). This is infinitive-family evidence, not a quotation of the selected imperative. CAPR assembles imperative singular **skáwô* ‘show (iptv.2sg)’, using the finite class-II cell with imperative -a < **-ō* (Ringe and Taylor 2014, 314).

The imperative singular provides the direct comparison with the Old English form. The lexical history still belongs to the *sceáwian* ‘show’ verb, but the cell compared here isolates the finite *-a* outcome more clearly than the citation form does.

Old English evidence

The held fourth edition of Bright’s reader contains *scēawian* ‘show’ and the imperative singular *scēawa* ‘show’ in the Blickling homily (Bright 1917, 71). The form treated here is therefore an attested finite paradigm form, not a reconstructed convenience form; it is not attributed to an unverified passage in the separate Cassidy/Ringler edition.

The spelling used in this entry is normalized *sċēawa* ‘show’, while Bright’s cited text gives source spelling *scēawa* ‘show’. The ordinary Old English headword remains *scēawian* ‘show’; *sċēawa* ‘show’ is the imperative singular chosen beside it.

Development to Old English

The infinitive entry and SC034 OEAwLongDiphthong distinguish the dictionary singleton stem from Campbell and Brunner’s inherited-geminate analysis. The same computational proxy is retained here; successful finite output does not resolve that prehistoric root history (Orel 2003, 337; Campbell 1959, 45; Brunner 1965, 49).

Ringe and Taylor’s class-II weak imperative singular -a < **-ō* supports the finite ending that yields *sċēawa* ‘show’ (Ringe and Taylor 2014, 314). The ending comparison is therefore justified independently of the unresolved root quantity history. The inherited *late_analogy* corpus classification is not changed by this root-only adjudication; it does not make the proxy a demonstrated analogical sound law.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjanǫ*	regular output *sċēawian*	*scēawian* ‘show’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**skáwô*	regular output *sċēawa* ‘show’	*scēawa* ‘show’ / normalized *sċēawa* ‘show’	exact match between the derivational input and the Old English form here
present third singular companion	**skáwōθi*	regular output *sċēawaþ* ‘show’	*sċēawaþ* ‘show’	useful family control, but not the target of this entry

9.27 span — OE *spanne*

Derivation: citation reconstruction **spannō*; form followed here **spánnai* > *spanne* (late analogy).

Derivation trace

Proto input: **spánnai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**spánne*
PNWGmc Unstressed Ai Monophthongization	**spánnē*		
Early Anglo-Frisian			
[no change]			

Old English form: *spanne*

Reconstruction and comparative evidence

Seebold gives Old English *spann* ‘span’ under this noun family (Seebold 1970, 450). The form followed here, **spánnai* ‘span’, is therefore not a rival headword, but the specific dative singular form compared on the model of the feminine ō-stem paradigm (Brunner 1965, § 252; 1965, § 255.2).

Old English evidence

The reviewed lexicographic evidence more directly supports the citation noun *spann* ‘span’ than the exact form *spanne* ‘span’. Clark Hall gives *spann* ‘span’ (Clark Hall 1960, 286), and *spanne* ‘span’ is accordingly treated as the regular dative singular form compared here rather than as a dictionary headword.

Development to Old English

Citation **spannō* ‘span’ yields *span* ‘span’. The oblique cell **spánnai* ‘span’ therefore supplies the conservative comparison form: it preserves the medial geminate and yields *spanne* ‘span’, while citation **spannō* gives the nominative background form.

Paradigm comparison

The comparison below sets the citation form beside the dative singular form compared here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**spannō*	regular output: *span* ‘span’	*spann* ‘span’	useful citation-form background, but not the form compared here
dative singular compared here	**spánnai*	regular output: *spanne* ‘span’	*spanne* ‘span’	exact match for that conservative form

9.28 straw — OE *strēaw*

Derivation: **stráwą* yields regular **strēa*; the Old English form here is *strēaw* (late analogy).

Derivation trace

Proto input: **stráwą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**stráeu*
PWGmc Endingless Aw	**stráu*	OE Diphthong Leveling	**strēa*
Early Anglo-Frisian			
[no change]			

Regular outcome: **strēa*

Old English form: *strēaw*

Reconstruction and comparative evidence

Kroonen cites the neuter stem **strawa-* ‘straw’ and Old English *strēa(w)* ‘straw’ (Kroonen 2013, 484). The selected **stráwą* ‘straw’ is CAPR’s accented nominative/accusative singular, not Kroonen’s verbatim full-word quotation. His derivation of the noun from the strew formation concerns genealogical direction; it does not explain the later restoration of the glide (Kroonen 2013, 484).

Old English evidence

The comparison remains attested *strēaw* ‘straw’, not reconstructed **strēa* ‘straw’. Campbell discusses restoration of the glide and the compound *strēaberige* ‘strawberry’ (Campbell 1959, 46–47, 232–33). Brunner explicitly treats the regular free simplex and short obliques as unattested; his paradigm note limits the West Saxon straw evidence to nominative/accusative singular and a doubtful plural. The compound preserves an endingless pattern, not an independently attested free noun or a complete surviving regular paradigm (Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 173, 387).

Development to Old English

The regular endingless branch loses the final vowel and vocalizes final **w*: **stráwą* ‘straw’ > **stráu* ‘straw’. The narrow fragment SC111 PWGmcEndinglessAw combines those operations; its computational holding site does not establish their order against unrelated changes. Its product feeds SC030 OEAuBrightening, giving **stráeu* ‘straw’, and SC032 OEDiphthongLeveling, giving regular **strēa* ‘straw’ (Campbell 1959, 46–47; Ringe and Taylor 2014, 172–73).

Attested *strēaw* ‘straw’ combines that long diphthong with a glide restored from inflected forms. Reciprocally, the long diphthong spreads into the formerly short oblique stem. This mutual levelling explains the long-plus-glide form without inventing a sound law that preserves final singleton **w* in the regular endingless branch (Campbell 1959, 232–33; Brunner 1965, 204–5; Ringe and Taylor 2014, 387). The cascade stops before that levelling: its reconstructed output remains an explicit mismatch, while the attested comparison and derivational input are unchanged.

Paradigm comparison

The table distinguishes source-supported historical comparisons from the one selected compiled derivation; it is not a set of newly compiled oblique inputs.

Cell or evidence	Regular development	Attested comparison and qualification
selected nominative/accusative singular	**stráwą* > **strēa*	*strēaw* ‘straw’: restored glide; regular free simplex unattested
short oblique stem, exemplified by genitive singular	**strawes* ‘of straw’; corresponding short dative also reconstructed	Brunner’s regular short obliques are unattested; long quantity is extended from the endingless cells, not demonstrated by a surviving short straw paradigm
remodelled singular and limited plural evidence	long diphthong plus restored glide	West Saxon *strēaw* ‘straw’; Brunner records plural *strewu* ‘straws’ with a question mark, not as a secure complete paradigm
compound witness	endingless pattern	*strēaberige* ‘strawberry’, not an attested free **strēa* ‘straw’

These qualifications follow Brunner’s paradigm and attestation discussion and Ringe and Taylor’s explicit mutual levelling account (Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 173, 387). The noun/verb genealogical direction remains a separate qualified question; this entry adopts neither a new ancestry nor an analogical transducer rule.

9.29 thistle — OE *þistles*

Derivation: citation reconstruction **θéstilaz*; form followed here **θístilas* > *þistles* (late analogy).

Derivation trace

Proto input: **θístilas*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**θéstilæs*
PNWGmc I Lowering	**θéstilas*	OE I Umlaut	**θistilæs*
Early Anglo-Frisian		OE L Adjacent Syncope	**θistlæs*
[no change]		OE Unstressed AE Merger	**θistles*

Old English form: *þistles*

Reconstruction and comparative evidence

Orel prints **þe(x)stilaz* ‘thistle’ for the lexeme (Orel 2003, 458). The comparative label **θéstilaz* ‘thistle’ therefore remains in view as the lexeme-level headword, while the derivational input **θístilas* ‘thistle’ is a specific genitive singular cell.

Old English evidence

The ordinary simplex headword tradition is broken *þistel* ‘thistle’ / *ðistel* ‘thistle’. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The Old English form here here is the genitive singular *þistles* ‘thistle’, which preserves the same stem in an oblique form where the cluster is medial.

Development to Old English

Campbell's discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn* 'token, sign', *wépn* 'weapon', and *botm* 'bottom' beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The genitive singular **θístilas* 'thistle' therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *þistles* 'thistle', while the simplex nominative belongs to the broken headword tradition *þistel* 'thistle'.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**θéstilaz*	computed output: *þistl*	*þistel* 'thistle'	useful citation-form background, but not the Old English form here
genitive singular	**θístilas*	computed output: *þistles* 'thistle'	*þistles* 'thistle'	exact match for the conservative cell

Chapter 10

Reconstructed Old English comparators

Direct attestation does not supply the required comparator in these entries. The target is an explicitly reconstructed Old English form and carries the corresponding evidential burden.

10.1 knob — OE **cnobba*

Derivation: citation reconstruction **knúppaz*; form followed here **knúbbô* > **cnobba* (reconstructed Old English comparator).

Derivation trace

Proto input: **knúbbô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**knóbba*
PNWGmc U Lowering **knóbbô*		
Early Anglo-Frisian		
[no change]		

Old English form: **cnobba*

Reconstruction and comparative evidence

The wider knob-family is not uniform. Kroonen's discussion of the Germanic n-stems points to related voiced and voiceless branches within this group (Kroonen 2011, 297). The citation reconstruction **knúppaz* 'knob' represents the broader cognate-set headword, while the form followed here, **knúbbô* 'knob', represents the voiced weak-noun branch treated here.

The Old English record is uneven. The better attested OE material belongs to the voiceless branch, but the present entry represents the reconstructed OE form that would continue the voiced branch behind later English knob.

Old English evidence

Clark Hall preserves Old English evidence of the *cnoppa* 'knob' type (Clark Hall 1960, 79). Those forms are genuine Old English evidence, but they belong to the voiceless branch of the family.

The target **cnobba* ‘knob’ is a reconstructed Old English form, not a directly attested one. I use it for the voiced branch because attested *cnoppa* ‘knob’ continues the voiceless branch and therefore represents a different prehistory.

Development to Old English

From the weak-noun form followed here, **knúbbô* ‘knob’, the regular Old English outcome is **cnobba* ‘knob’, with Proto-Germanic *kn-* represented in Old English as *cn-* and with the expected weak-noun ending.

The entry therefore does not claim that **cnobba* ‘knob’ is attested. Its claim is different: if the voiced weak-noun branch is the one to be represented, then **cnobba* is the regular Old English form corresponding to that branch.

Reconstruction status

Form	Status	Relevance to this entry
*knúbbô > *cnobba	reconstructed OE form; regular derivation	reconstructed Old English form compared here
cnopp / *cnoppa*	attested OE branch	important control form, but belongs to the voiceless branch
cnæp	attested OE form from another family	not part of the present lexeme line

This remains the most review-sensitive item here, because the choice between reconstructed **cnobba* ‘knob’ and attested *cnoppa* ‘knob’ is still a comparator-policy question rather than a settled point of OE attestation.

10.2 reek — OE **rēac*

Derivation: citation reconstruction **ráukiz*; form followed here **ráukaz* > **rēac* (reconstructed Old English comparator).

Derivation trace

Proto input: **ráukaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**ráeuka*
EAF Final Z Deletion	**ráuka*	OE Diphthong Leveling	**rēaka*
Early Anglo-Frisian		PWGmc Final Bare A Loss	**rēak*
[no change]			

Old English form: **rēac*

Reconstruction and comparative evidence

The wider noun family is represented by **ráukiz* ‘reek’ / **rauki-*, with Old English *rēc* ‘reek’ as the attested noun reflex in the comparative dictionaries (Kroonen 2013, 446; Orel 2003, 338). The derivational input **ráukaz* ‘reek’ is therefore not the lexeme-level headword, but the form used here for the Old English derivation.

Old English evidence

The attested noun is *rēc* ‘reek’, not **rēac* ‘reek’. Clark Hall records *rēc* ‘reek’ as the noun and also preserves related forms such as *rēcels* ‘incense’; Kroonen likewise gives OE *rēc* ‘reek’ under the noun family (Clark Hall 1960, 255; Kroonen 2013, 446). Clark Hall and Seebold also record verbal *rēac* ‘reek’ as the preterite of *rēocan* ‘reek’, but that verbal form is separate from the noun treated here (Clark Hall 1960, 254; Seebold 1970, 380).

The Old English form here **rēac* ‘reek’ is therefore a reconstructed West Saxon noun form, not a directly attested manuscript headword.

Development to Old English

From **ráukaz* ‘reek’, the regular West Saxon development gives **rēac* ‘reek’. The attested noun *rēc* ‘reek’ belongs to the same lexical family, but reflects a later smoothed surface form rather than the regular noun target represented here.

Form note

The distinction here is between an attested noun headword *rēc* ‘reek’ and a reconstructed regular West Saxon target **rēac* ‘reek’. The latter is treated as the modelling target, while the former remains philological background.

10.3 strew — OE **strīeġan*

Derivation: **stráwjaną* > **strīeġan* (reconstructed Old English comparator).

Derivation trace

Proto input: **stráwjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Awwj Resolution	**stráujaną*
PWGmc J Gemination	**stráwwjaną*	OE Au Brightening	**stráeujaną*
Early Anglo-Frisian		OE Diphthong Leveling	**strēajaną*
		OE Heavy Syllable Nasal Apocope	**strēajan*
[no change]		OE Secondary Nasalization	**strēająn*
		OE I Umlaut	**strījąn*
		OE Weak Tail Reduction	**strījan*
		OE J Strengthening After Front Diphthong	**strīʒan*

Old English form: **strīeġan*

Reconstruction and comparative evidence

Kroonen cites the inherited weak verb as **straujan-* and gives Old English *streowian* ‘strew’ as its dictionary continuation (Kroonen 2013, 483). Ringe and Taylor separate the inherited class-I branch from the remodelled class-II branch: Anglian *strēgan* ‘strew’ continues the inherited verb, while West Saxon *streowian* ‘strew’ is remodelled (Ringe and Taylor 2014, § 6.1 n. 27).

Luick groups **strauwjan* ‘strew’ with the same set as **hauwja-* and **kauwjan* ‘chew’, yielding Anglian *strēzan* ‘strew’ beside West Saxon forms of the *hīez* ‘hew’, *ciezan* ‘choose’ type (Luick 1914--1940, § 98). Fulk likewise allows an early West Saxon **striegan* ‘strew’ directly from Proto-Germanic **straujana* ‘strew’ (Fulk 2018, § 4.10 n. 1).

Old English evidence

The attested inherited Old English form is *strēgan* ‘strew’ in Anglian. The attested West Saxon citation forms are *strewian* ‘strew’, *streowian* ‘strew’, and *strēawian* ‘strew’, which belong to the remodelled class-II branch (Ringe and Taylor 2014, § 6.1 n. 27; Campbell 1959, § 753.7).

The target **strīeġan* ‘strew’ is therefore a reconstructed Old English form, not an attested manuscript lemma. It represents the inferred West Saxon reflex of the inherited class-I branch.

Development to Old English

From **stráwjaną* ‘strew’, the inherited West Saxon line passes through **straujaną* ‘strew’, then a fronted **strēajan-* stage, then **strījan* ‘strew’ by i-umlaut, with retained/strengthened glide after the front diphthong to yield reconstructed **strīeġan* ‘strew’.

This differs from Anglian *strēgan* ‘strew’, where smoothing removes the diphthongal sequence, and from West Saxon *strewian* ‘strew’ / *streowian* ‘strew’ / *strēawian* ‘strew’, where the verb has already been remodelled into class II (Fulk 2018, § 4.10 n. 1; Campbell 1959, § 753.7).

Reconstruction status

Form or branch	Status	Relevance to this entry
strēgan ‘strew’	attested Anglian inherited class-I form	proves that the inherited verb survived into Old English
**strīeġan* ‘strew’	reconstructed West Saxon inherited class-I form; trace-supported	Old English form here
strewian / *streowian* ‘strew’ / *strēawian*	attested remodelled West Saxon class-II forms	genuine OE evidence, but not the inherited branch modeled here

Chapter 11

Known but unmodelled developments

The historical development is sufficiently supported by comparative and philological evidence, but the current transducer cascade does not yet implement the phonological or morphological process required to derive the target. Naming the gap explains the status without pretending that no historical account exists.

11.1 fire — OE *fȳre*

Derivation: **fūri* yields regular **fȳr*; the Old English form here is *fȳre* (known but unmodelled development).

Derivation trace

Proto input: **fūri*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**fȳri*
[no change]	OE High Vowel Apocope	**fȳr*
Early Anglo-Frisian		
[no change]		

Regular outcome: **fȳr*

Old English form: *fȳre*

Reconstruction and comparative evidence

Kroonen places the lexeme in a heteroclitic family **fōr* ‘fire’ ~ **fun-* and explains the front-mutated West Germanic forms from an oblique form of the **fu(w)eri* type (Kroonen 2013, 151). The derivational input **fūri* ‘fire’ is CAPR’s oblique-input proxy supplying the *i* needed for i-umlaut. The account does not independently certify that exact complete word, its long *ū*, or its cell.

The Old English target combines a regular inherited form *fȳr* ‘fire’ with an attested oblique surface form *fȳre* ‘fire’, interpreted by CAPR as analogically restored.

Competing ancestry and the legitimacy of the input

Kroonen distinguishes the dative **fu(w)eni*, its *r*-remodelled **fu(w)eri*, and metathesized **feuri*. His preferred pre-Germanic nominative is **péh₂-ur*; the **ph₂-uōr* proposal reported from Simms is rejected, not endorsed (Kroonen 2013, 151). Ringe instead supplies a reaccented zero-grade collective, printed **ph2uōŕ*, and separately considers post-Proto-Germanic oblique dissimilation and *n* replacement by *r* (Ringe 2017, 147, 309). These are consequential formation and accent alternatives, not interchangeable quoted dative inputs.

Ringe and Taylor reconstruct apparently disyllabic West Germanic **fuir* with *r* levelled into the oblique stem, but doubt that the endingless dative evidence establishes inherited **fuiri*. A front-rounded intermediate in the history of *fȳr* is plausible; the intervening stages remain uncertain (Ringe and Taylor 2014, 119, 225). Neither passage endorses CAPR's exact long-*ū* proxy. Equal citation and input fields select the Proto-Germanic stage by project convention, not by a new scholarly quotation. Quantity, cell and historical input legitimacy require separate scientific assessment; no input migration is made here.

Old English evidence

Toller's Supplement records *fyr* 'fire' as a noun in its *fyr. Add* entry and separately preserves oblique *fyre* 'fire', including *in fyre* (Toller 1921, 276). These attestations establish the forms, not the date or mechanism of restoration. The first is the regular output of the phonological development from the selected input; the second is interpreted by CAPR as restoration of final *-e* within the paradigm, rather than demonstrated as such by the dictionary.

The entry therefore distinguishes a regular path from an adopted input proxy, the comparative legitimacy of that proxy, and an attested surface form whose additional inflection CAPR treats as morphological remodeling.

Development to Old English

From **fūri* 'fire', i-umlaut changes *ū* to *ȳ* (Hogg 1992b, § 3.3.3.1). Subsequent loss of the final high vowel after a heavy syllable yields *fȳr* 'fire' (Campbell 1959, § 345). The selected phonological path is complete at that point.

CAPR interprets the final *-e* of *fȳre* 'fire' as analogical restoration in the Old English paradigm rather than retention of the derivational input's ending. The form therefore remains a known but unmodelled remodelling: the deterministic phonology is regular, but the attested surface form includes later morphological rebuilding.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the lexeme-level etymological background from the inherited cell that actually produces the front-mutated form and from the later analogical surface result.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
lexeme-level heteroclitic headword	**fōr* 'fire' ~ **fun-*	comparative background only	fire family	explains the wider lexeme, but not the selected oblique input
selected oblique-input proxy	**fūri*	regular output: *fȳr* 'fire'	*fȳr* 'fire'	regular output; exact ancestral cell remains qualified

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
later analogical surface form	—	attested *fȳre* ‘fire’ with restored *-e*	*fȳre* ‘fire’	genuine OE target, but not the direct phonological output

11.2 tap — OE *tæppa*

Derivation: **táppô* yields regular **tappa*; the Old English form here is *tæppa* (known but unmodelled development).

Derivation trace

Proto input: **táppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**tæppô*
[no change]	OE A Restoration	**tappô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**tappa*
[no change]		

Regular outcome: **tappa*

Old English form: *tæppa*

Reconstruction and comparative evidence

Orel gives the noun under **tappòn* ‘tap’ and already connects it with Old English *tæppa* ‘tap’ (Orel 2003, 402). The derivational input **táppô* ‘tap’ is therefore the inherited noun itself; the entry does not depend on a different lexeme-level proto or a different inherited noun cell.

Old English evidence

The Old English noun family is well attested. Orel gives *tæppa* ‘tap’, and Clark Hall records *tæppa* ‘tap’ together with derivatives *tæppere* ‘tapster’ and *tæppestre* ‘tapster’ (Orel 2003, 402; Clark Hall 1960, 305). The target is therefore a real Old English noun form, not a reconstructed convenience spelling.

Development to Old English

From **táppô* ‘tap’, the regular inherited noun path gives *tappa* ‘tap’. The attested target *tæppa* ‘tap’ therefore stands outside that regular phonological development.

The mismatch is historically intelligible, but it is not solved here by a new inherited input. A related j-verb pathway would give *teppan* ‘tap’, not the noun target *tæppa* ‘tap’. The entry accordingly remains a known but unmodelled case.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**táppô*	regular output: *tappa* ‘tap’	regular output, but not the target
attested OE target	—	*tæppa* ‘tap’	genuine target form, but analogically remodelled in the present classification
related j-verb background	**táppjaną*	*teppan* ‘tap’	related formation, but not the noun target

Chapter 12

Unexplained or deliberately unmodelled exceptions

No sufficiently supported account yet reconciles the regular output with the Old English form. An ad hoc sound law would conceal rather than explain the mismatch.

12.1 buck — OE *bucc*

Derivation: **búkkaz* yields regular **bocc*; the Old English form here is *bucc* (unexplained exception).

Derivation trace

Proto input: **búkkaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**bókk*
PNWGmc U Lowering	**bókkaz*		
EAF Final Z Deletion	**bókka*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **bocc*

Old English form: *bucc*

Reconstruction and comparative evidence

Kroonen and Orel both reconstruct the word with a geminate stop, **bukkaz* ‘buck’ (Kroonen 2013, 121; Orel 2003, 61). Orel also preserves parallel n-stem material behind Old English *bucca* ‘buck’ (Orel 2003, 62). The derivational input therefore remains identical with the lexeme label: no alternative inherited cell accounts for the form.

Old English evidence

Old English preserves a mixed lexical picture. Campbell cites *bucca* ‘buck’ in the exception set for this phonological environment (Campbell 1959, § 115). Clark Hall and Bosworth-Toller show that Old English has both *bucca* ‘buck’ and *bucc* ‘buck’ (Clark Hall 1960, 53; Bosworth and Toller 1898, 122). The a-stem citation form *bucc* ‘buck’ is the target treated here; *bucca* ‘buck’ supplies genuine philological background from the same lexical family.

Development to Old English

From **búkkaz* ‘buck’, the regular inherited path gives *bocc* ‘buck’. That is the form expected under the ordinary lowering pattern in this environment. *bucc* ‘buck’ therefore remains outside the deterministic phonology.

No accepted inherited cell repairs the mismatch. A high-vowel alternative would introduce i-umlaut and produce a *byċċ* ‘buck’-type form rather than the target. *bucc* ‘buck’ is therefore best treated as a documented exception, not as a regular paradigm-cell survival.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**búkkaz*	regular output: *bocc* ‘buck’	regular output, but not the target
attested OE target	—	*bucc* ‘buck’	genuine target form, but unexplained in the present classification
parallel OE lexical background	—	*bucca* ‘buck’	related n-stem form, not the present target

12.2 fowl — OE *fugol*

Derivation: **fúglaz* yields regular **fogol*; the Old English form here is *fugol* (unexplained exception).

Derivation trace

Proto input: **fúglaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fógl*
PNWGmc U Lowering	**fóglaz*	OE Epenthetic Vowel	**fógol*
EAF Final Z Deletion	**fógla*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **fogol*

Old English form: *fugol*

Reconstruction and comparative evidence

The noun is the ordinary Germanic a-stem **fúglaz* ‘fowl’, continued by forms such as Old Norse *fugl* and Old High German *fogal* ‘bird, fowl’ (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the derivational input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

Old English evidence

Old English dictionaries record the noun as *fugol* ‘fowl’, with variant spelling *fugel* ‘fowl’ (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fu-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

Development to Old English

From **fúglaz* ‘fowl’, the regular cascade yields *fogol* ‘fowl’: the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol* ‘fowl’, not *fogol* ‘fowl’. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914--1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

Expected and attested forms

The comparison below sets the relevant forms side by side. It distinguishes the regular prediction from the attested Old English noun.

Form	Status	Relevance to this entry
fogol	computed regular output from **fúglaz*	establishes the expected inherited outcome
fugol	attested Old English form	Old English form here; preserves unexplained root *u*
fugel	attested variant spelling	secondary spelling variant of the attested noun

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from **fúglaz* ‘fowl’ to attested *fugol* ‘fowl’.

12.3 rust — OE *rust*

Derivation: **rústaz* yields regular **rost*; the Old English form here is *rust* (unexplained exception).

Derivation trace

Proto input: **rústaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**róst*
PNWGmc U Lowering	**róstaz*		
EAF Final Z Deletion	**rósta*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **rost*

Old English form: *rust*

Reconstruction and comparative evidence

Orel gives both **rustō* ‘rust’ and **rustaz* ‘rust’, with masculine/feminine alternatives and Old English *rust* ‘rust’, Old Saxon rost ‘rust’ and Old High German rost ‘rust’ (Orel 2003, 309). The conventional feminine *ō*-stem and masculine *az*-stem are distinct morphological alternatives within his account. Ringe independently reconstructs **rustaz* ‘rust’; his discussion of alternative suffixal and cluster histories for *-st-* concerns the formation, not the explanation of retained *u* (Ringe 2017, 247–48, especially p. 248).

The masculine Old English noun favors the masculine citation reconstruction, selected here as **rústaz* ‘rust’ with the project’s acute stress notation. This is a morphological and comparative choice, not an exclusive disagreement between Orel and Ringe or a claim that Orel’s *ō*-stem is unsupported. The comparative lowered forms provide context, but output agreement does not choose a stem class: both formations regularly predict **rost* ‘rust’, not the attested Old English noun.

Old English evidence

The noun is attested, not reconstructed. Clark Hall gives *rūst* ‘rust’ (m.) (Clark Hall 1960, 245), and Toller’s supplement to Bosworth-Toller records *rúst* ‘rust’, beside *rust*, with a qualification about quantity (Toller 1921, 691). Their lexicographical signs are retained as source spellings; the selected comparator remains normalized *rust*, without a newly adopted quantity interpretation. The evidence establishes the noun and masculine gender, not an independent explanation of its retained vowel.

Development to Old English

SC017 PNWGmcULowering regularly lowers the root *u* of **rústaz* ‘rust’ before the following non-high vowel, giving **róstaz* ‘rust’. Final-*z* loss and subsequent regular developments complete the predicted **rost* ‘rust’, not the attested comparator *rust*. Campbell describes the lowering and its unresolved retained-*u* residues (Campbell 1959, 43, §115); Ringe and Taylor discuss the remaining exceptions without establishing a general consonantal exemption (Ringe and Taylor 2014, 27–34).

The control **frústą* ‘frost’ regularly gives *forst* ‘frost’, so *-st-* cannot be exempted generally. Selecting the masculine stem is not a phonological repair: *rust* remains an unexplained lexical retention and the mismatch total does not fall. No exception is inserted into the sound law.

Paradigm comparison

Fresh production and isolated-wrapper diagnostics give the following results. The oblique-type constructions test conditioning; they are not newly attested Proto-Germanic paradigm cells.

Form	Role	Result
**rústaz*	Selected masculine citation form	**rost*, mismatch
**rústō*	Orel’s alternative *ō*-stem	**rost*, mismatch
**rústas*	Genitive-singular-type probe	**rostes*
**rústai*	Dative-singular-type probe	**roste*
**rústi*	High-front-vowel diagnostic	**ryst*
**rústu*	Marginal high-vowel diagnostic construction	**rust*

The hypothetical instrumental-type **rústu* ‘rust’ is a diagnostic construction, not an attested Proto-Germanic citation form or an adopted historical solution. Its matching output does not establish a source for the ordinary citation noun.

12.4 wolf — OE *wulf*

Derivation: **wúlfaz* yields regular **wolf*; the Old English form here is *wulf* (unexplained exception).

Derivation trace

Proto input: **wúlfaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wólf*
PNWGmc U Lowering	**wólfaz*		
EAF Final Z Deletion	**wólfa*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **wolf*

Old English form: *wulf*

Reconstruction and comparative evidence

The inherited noun is an a-stem: Kroonen gives **wulfa-*, and Ringe and Taylor list PGmc **wulfaz* ‘wolf’ among the inherited words that preserve *u* in Old English beside Old High German *wolf* ‘wolf’ (Kroonen 2013, 638; Ringe and Taylor 2014, 47). Campbell accordingly names Old English *wulf* as an exception to the regular lowering of stressed *u* before a following non-high vowel (Campbell 1959, § 115).

The older literature often notices that the exceptional words cluster near labials. Bülbring lists *full*, *wulle*, and *wulf* together, but he also says that the ordinary rule still gives *o* in comparable forms such as *folc* and *bolt* (Bülbring 1902, § 116). Luick rejects a categorical labial blocker on the same grounds and prefers a lexical or analogical account instead (Luick 1914--1940, 148).

Old English evidence

Campbell treats *wulf* ‘wolf’ as part of the exceptional *u* set (Campbell 1959, § 115). Sievers-Brunner notes that oblique *wulfe* ‘wolf (dat.)’ continues *wulfi* ‘wolf (dat.)’ or older *wulfai* ‘wolf (dat.)’ (Brunner 1965, § 160).

The surviving oblique forms do not supply a regular route back to bare *wulf* ‘wolf’. They belong to the same lexeme, but they do not remove the explanatory problem presented by the citation form.

Development to Old English

Under the ordinary Northwest Germanic lowering of stressed *u* before a following non-high vowel, the citation-form input would point toward *o*-vocalism (Ringe and Taylor 2014, 42–44). The compact trace shows exactly that path: **wúlfaz* ‘wolf’ > **wólfaz* ‘wolf’ > **wólfa* ‘wolf’ > wolf.

A high-vowel oblique input would behave differently. There the following high vowel would block the lowering of *u*, but the same environment would also trigger i-umlaut, so the regular control result would be *wylf* ‘wolf’ or *wylfe*, not bare *wulf*. The attested noun therefore remains unexplained at the citation-form level.

Expected and attested forms

The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**wúlfaz* > *wolf*	computed regular output from the citation form	shows the regular development expected from the inherited a-stem
OHG wolf	comparative regular cognate	confirms that the *o*-vocalism is the ordinary outcome
**wúlfi* / **wúlfis* > *wylf* / *wylfe*	expected high-vowel control forms	shows why oblique high-vowel cells do not solve the noun's vowel history
wulf	attested Old English noun	attested Old English form; the preservation of *u* remains unexplained

12.5 wool — OE *wull*

Derivation: **wúllō* yields regular **woll*; the Old English form here is *wull* (unexplained exception).

Derivation trace

Proto input: **wúllō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**wóll*
PNWGmc U Lowering	**wóllō*		
PNWGmc Final Long O Raising	**wóllu*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **woll*

Old English form: *wull*

The inherited paradigm

The selected **wúllō* 'wool' represents the feminine nominative singular. It must be distinguished from accusative **wúllǭ*, whose final vowel was nasalized. Ringe explicitly reconstructs both cells; Orel and Kroonen discuss the inherited formation and its comparative history (Ringe 2017, 171–73, 241–44; Orel 2003, 474; Kroonen 2013, 598–99). Attested *wull* 'wool' and *wulle* 'wool' preserve a high root vowel. The latter is not, merely by its shape, evidence that the whole noun was originally a weak n-stem (Campbell 1959, 43, §115; Bülbring 1902, 45–46, §116).

Regular development and the alternative explanation

CAPR provisionally follows Fulk's lowering-before-final-raising chronology (Fulk 2018, 56). Both oral **ō* and nasal **ǭ* condition root lowering before their distinct subsequent developments:

Inherited cell	Preferred chronology	Raising before lowering
nominative **wúllō*	**wóllō* → **wóllu* → regular **woll*	**wúllu* → regular **wull*
accusative **wúllǭ*	**wóllǭ* → regular **wolle*	**wóllǭ* → regular **wolle*

These are sound-law predictions, not additional attestations. Earlier raising supplies a genuine nominative **wullu* with high root vowel and an independently conditioned low-root oblique.

Generalization of the nominative vowel could then produce the attested high-root paradigm: this is Ringe and Taylor's plausible but unproved explanation, not an impossible escape cell (Ringe and Taylor 2014, 32). Old Frisian *wolle* 'wool' and Old High German *wolla* 'wool' support the comparative reality of a lowered root, but do not attest that particular early English accusative (Kroonen 2013, 598–99).

The main comparison of the two chronologies, the evidence from other ō-stems and the costs of the proposed analogies is in SC019 PNWGmcFinalLongORaising. The preferred ordering does not regularly derive the attested high root here. The derivational input, target and classification therefore remain unchanged: regular **woll* 'wool', not attested *wull*, is what the current sound laws produce. No lexical exception or analogical levelling has been inserted into those laws.

Luick discusses the alternations in §§77–79, especially §78 (printed pp. 105–106), not §§105–106. He rejects a proposed labial protection rule using regular low-root counterexamples (Luick 1914--1940, 104–6, §§77–79). The separate trimoric genitive **-ôz* must not be substituted for either cell above: its analysis differs from the bimoric genitive and its complete computational pathway remains unmodeled (Fulk 2018, 155; Ringe and Taylor 2014, 59, 299).

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